pixelactive

CITYSCAPE 1.8

REAL CITIES REAL FAST

User Manual

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PART 1: GETTING STARTED

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1.1 INTRODUCTION

1.1.1 About the CityScape Tool

PixelActive CityScape is an interactive tool for creating large 3D environments for realtime rendering. CityScape has all the capabilities that you need to create robust 3D environments for video games, simulations, GPS navigation, urban planning, visualizations, etc.

DEMO VS. FULL VERSIONS

The free Demo version of CityScape functions exactly the same as the Full version, with the following exceptions:

1. The Demo cannot export terrains.
2. When you run the Demo, a blank terrain automatically opens in association with the Demo project file for library items. In the Full version, you must open or create a terrain manually, and associate it with a project file of your choice (or creation).

LEARNING CITYSCAPE IN ONE DAY

After installing CityScape, get started fast by doing ”Chapter 2: Quick Start Tutorial” on page 7 with CityScape open in front of you.

1.1.2 About this Document

When complete, this *CityScape User Manual* will describe every aspect of installing, configuring, and using CityScape. This draft is in progress.

- **For IT Staff:** You can find instructions or installation and configuration in the “Single User Installation” section of this chapter.
- **For Artists:** Instructions for using every aspect of CityScape constitute the majority of this *User Manual*.
- **For Technical Artists and Programmers:** Instructions for exporting to standard formats such as Collada and OpenFlight will help you get data from CityScape into 3D tools such as Maya, Creator, and 3D Studio Max for further customization – or into 3D rendering engines. You can find exporter instructions in ”Chapter 22: Exporting Terrains” on page 217.

1.2 SINGLE USER INSTALLATION

The Single User version of CityScape allows you to install and use one or more copies of CityScape, according to the number of licenses you have. (The Demo version does not require a license or dongle.) A dongle controls the licenses of the Full version of CityScape via two common configurations:

- Install one standalone copy of the CityScape editor on a computer, and insert the dongle on the same machine.
- Install multiple copies of CityScape (one per computer) on the same network, up to the limit of your license. Then install the dongle on a server sharing the same network. With this configuration, the shared server functions as a "License Server" providing the licenses for all the copies of CityScape on the same LAN.

1.2.1 System Requirements

- You must have administrator rights to install PixelActive software on any computer.
- Although CityScape will run with an 64-bit OS, CityScape itself is a 32-bit application. The operating system on the target computer can be any of the following:
  - Windows XP (32-bit or 64-bit) with Service Pack 2 or higher
  - Windows Vista (32-bit or 64-bit)
  - Windows 7 (32-bit or 64-bit)
- Microsoft .NET Framework 2.0 or higher must be installed.
- A minimum of 200 MB of free space must be available on the computer's hard drive.
- The video card on the computer should support shader model 3.0 and DirectX 9, or higher.
- The computer should have at least 1 GB of RAM, and preferably 4 GB.
- The display resolution of the computer's monitor should be 1024 x 768 or higher.

1.2.2 Contents of Installation CD

The installation CD contains these files:

- The CityScape editor (in compressed MSI format).
- The *CityScape User Manual* which you are reading now (in PDF and/or HTML Help format).

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1.2.3 Installing CityScape

The Full version of CityScape requires you to install a dongle to provide licenses, as described below. The Demo version of CityScape does not need a dongle. Hence, Demo users can skip "Attach Dongle Hardware on License Server" and go directly to "Install CityScape Software on User Machines".

Attach Dongle Hardware on License Server

If you have a white dongle, you can simply insert the dongle into the machine you wish to become the PixelActive License Server, as explained below. If you have a green dongle, contact PixelActive for instructions.

1. **Standalone Licensing:** To provide a license for a standalone installation of CityScape, insert the white PixelActive dongle into a USB port on the same machine where CityScape will be installed. Then follow the instructions on the screen.

2. **Server Licensing:** To provide multiple licenses so CityScape can run independently on many machines, insert the white network dongle into a USB port of a server sharing the same LAN, and preferably the same subnet, as the user machines. Then follow the instructions on the screen.

Install CityScape Software on User Machines

To install the Demo version, or to install one or more licensed copies of CityScape on user machines sharing the same network, follow the instructions below:

Install the Demo Version

1. Download **PixelActiveCityScape_v1_8_Demo.msi** from PixelActive's web site.

2. After download, DOUBLE-CLICK this file locally to run it.

3. Follow the instructions in the installer dialog that appears.

Install the Full Version

1. For the full version, insert the PixelActive CityScape Installation disc in the target machine.

2. When AutoPlay displays the installation dialog, follow the instructions in the dialog. (If the installation dialog does not appear automatically, then manually run the **PixelActiveCityScape_v1_8.msi** file on the CityScape installation disc.) Notes:
   - The Installer will ask you where to install CityScape. This is the location of the CityScape executable and other files necessary to running CityScape). We recommend you select the default location: *C:\Program Files\PixelActive\CityScape 1.8*
   - The Installer will ask you where to install "common assets". These assets collect the key elements of a sample project under a Worlds folder, for use with the Quick Start Tutorial. These elements include a sample terrain (sample_terrain.cst), sample project (demo.csp), sample GIS data, and sample assets (props, textures, etc).

1.3 ADDITIONAL RESOURCES

To quickly learn how to run and use CityScape, do the tutorials in ”Chapter 2: Quick Start Tutorial” on page 7.

The other chapters in this *CityScape User Manual* provide in-depth reference information on every aspect of CityScape. To learn details about a specific tool, widget, or process, just look it up in this manual.

A growing library of *video tutorials* are available on PixelActive’s web site.

Exporting your terrains to the PixelActive Interchange format allows you to share terrains with those who don’t have CityScape: Using the *PixelActive Viewer*, anyone can view (but not edit) a terrain in PixelActive Interchange format. You can download the Viewer for free from PixelActive’s web site.

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2.1 START HERE

PixelActive CityScape is an interactive tool for creating large 3D environments for realtime rendering. CityScape has all the capabilities that you need to create and populate 3D environments ranging from wilderness terrains to large detailed cities – for use in video games, training simulations, GPS navigation, urban planning, and many more applications.

This “Quick Start Tutorial” provides a hands-on tour of CityScape to quickly familiarize you with the basics. For best results, follow the topics in the order they are presented, with a new terrain loaded in CityScape in front of you. To begin, follow the instructions below:

- Running CityScape
- Loading the Demo Project in the Full Version

2.1.1 Running CityScape

After CityScape is installed, a line entry for CityScape appears in the list of Windows Programs. You can run CityScape by selecting the line entry.

If you have the Demo version of CityScape, skip to Loading the Demo Project in the Demo Version. If you have the Full version, see Running CityScape for the First Time below.

RUNNING CITYSCAPE FOR THE FIRST TIME

If you have the Demo version, licensing is not required. The Full version of CityScape requires that you license your copy of CityScape the first time you run it. Here’s how:

1. The first time you run CityScape, the *Input KeyLok Computer* dialog may appear (see figure below). In this dialog, enter the **network name** (or IP address) of the License Server, and LEFT-CLICK **OK**. (For a standalone installation, with the dongle physically attached to the same machine as the CityScape editor, enter **localhost** in the dialog.).

2. CityScape will execute after it contacts the License Server. When running CityScape in the future, you will not have to re-enter the name of the License Server, unless the server changes.

Loading the Demo Project in the Demo Version

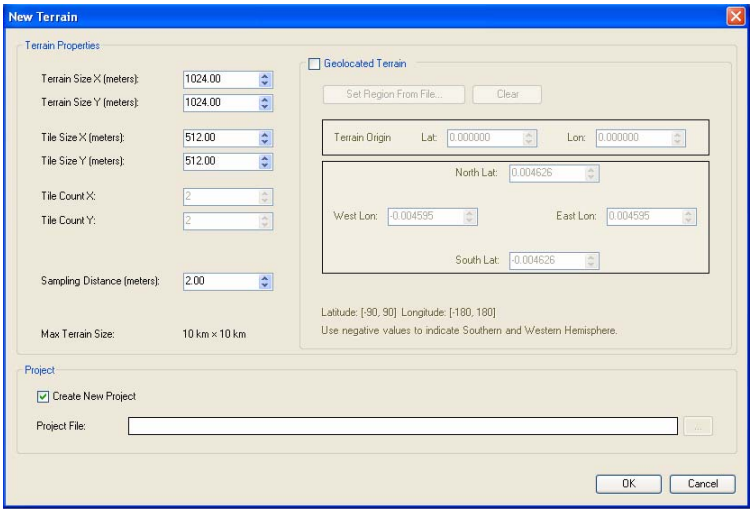
When you run the Demo version of CityScape, a blank terrain automatically loads. This blank terrain is associated with the Demo Project containing various 2D and 3D items (e.g. textured trees, buildings, road styles, etc.) that you can use while doing this Quick Start.

In the Demo version, you can create a new terrain at any time, and associate the new terrain with the Demo project, using the same procedure as the Full version (see the next section).

Loading the Demo Project in the Full Version

When you run the full version of CityScape, you have two options: manually create a new terrain or load an existing terrain. To do this Quick Start with the full version, create a new terrain as follows:

1. In the Main Menu, choose **File > New Terrain**. The New Terrain dialog appears.



2. Near the bottom of New Terrain dialog, in the Project area, you can choose the Demo project file as follows:
   i. LEFT-CLICK the browse button to the right of the *Project File* field.
   ii. In the file loader that appears, open the **demo.csp** file in the World's folder, which is installed in the following default location for Vista and Win 7 users:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds

or here for WinXP users:

C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds

3. In the New Terrain dialog box, LEFT-CLICK **OK** to give birth to a new terrain.

LOADING THE SAMPLE TERRAIN

If you want to see what a partly constructed terrain looks like, you can open the sample terrain file that comes with CityScape:

1. Select **File / Open Terrain (Ctrl-O)**.
2. In the Open dialog, load the **sample_terrain.cst** sample file, which is in your *Worlds* folder located here by default:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds\Terrains\SampleTerrain

or here for WinXP users:

C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds\Terrains\SampleTerrain

3. After you're done checking out the sample terrain, return to the blank terrain you created in the previous section to continue this Quick Start.

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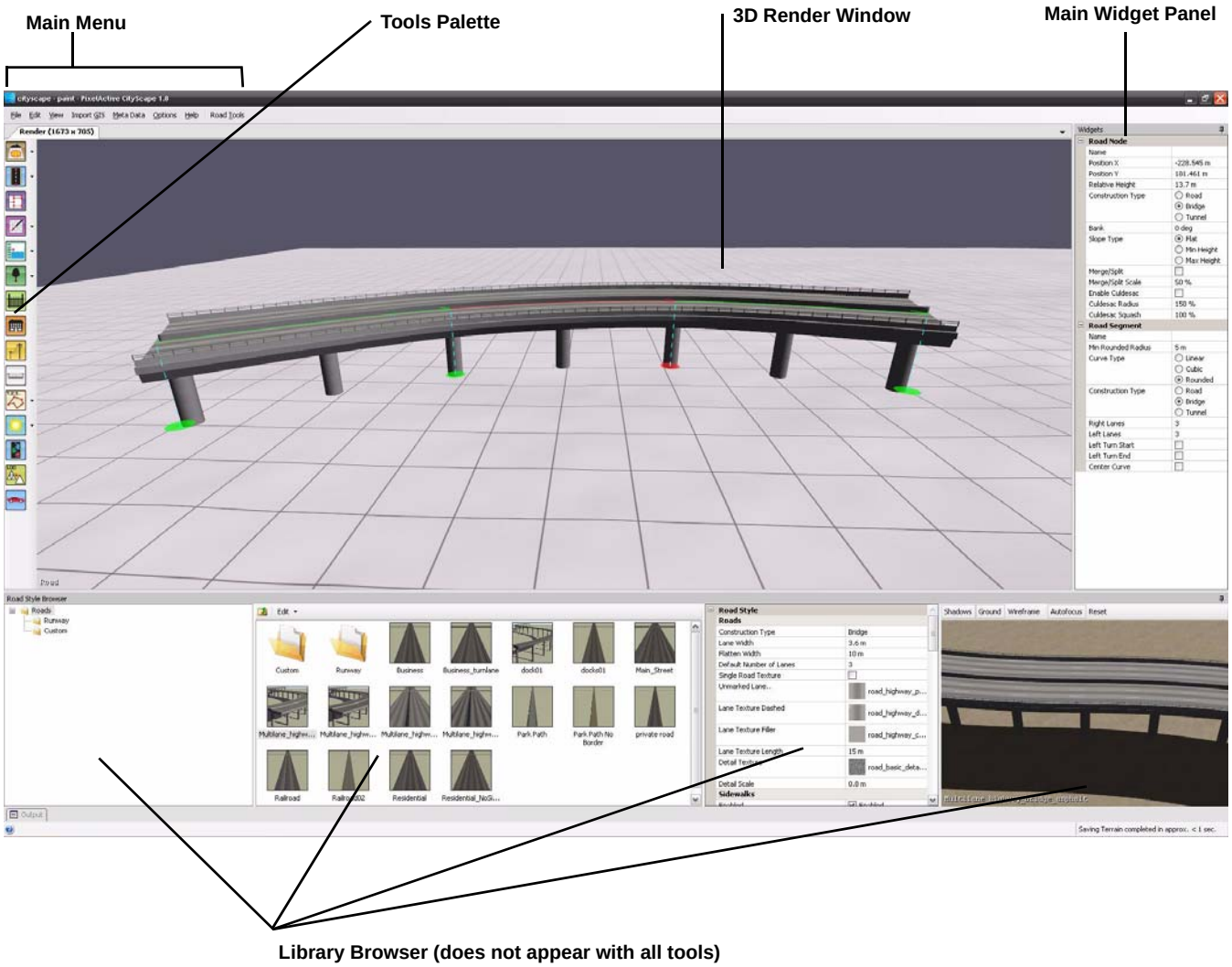
2.2 INTRODUCING THE UI AND CAMERA

The sections below introduce the user interface (UI) and camera. (For details on the UI, see “Chapter 3: User Interface” in the *CityScape User Manual*.)

- Overview of the User Interface next.
- “Using the Camera” on page 14

2.2.1 Overview of the User Interface

The following figure provides a quick look at the main elements in CityScape’s User Interface (UI). You can adjust the size of each element by dragging its borders with the mouse. You can also detach each element and move it around. To reset all the UI elements to the default, close CityScape and then hold the **SHIFT-KEY** while restarting CityScape.



Main Menu

The Main Menu provides access to much of the functionality in CityScape. A quick summary is below. For details on each menu, see “Chapter 3: User Interface” in the *CityScape User Manual.*

- **File:** includes the usual new, open, and save terrain commands – as well as export options. (Note: You cannot save or export terrains with the Demo version of CityScape.)
- **Edit:** includes standard cut / copy / paste and undo / redo commands, as well as the Enable Grid Snap option.
- **View:** includes options for cameras, real-time rendering, and visibility toggles (so you can show / hide stuff like overlay images).
- **Import GIS:** If your terrain is geo-located, then you can import a wide range of geo-specific data into the terrain.
- **Meta Data:** allows you to define and assign meta data (i.e. arbitrary ascii data)
- **Options:** collects miscellaneous options such as the ability to change key Terrain Settings (e.g. the size, geo-location, and tile breakdown of the open terrain).
- **Help:** gives you access to this *User Manual* and other help resources as they become available.
- **Road Tools:** When you activate any Road Tool, the Road Tools menu appears to provide you with additional functionality.
- **Water Tools:** When you activate any Water Tool, the Water Tools menu appears to provide you with additional functionality.
- **Fence Tools:** When you activate the Fence Tool, the Fence Tools menu appears to provide you with additional functionality.
- **Building Tools:** When you activate the Building Tool, the Building Tools menu appears to provide you with additional functionality.

Tools Palette

The Tools Palette gives you quick access to all the key editing tools in CityScape. Each icon on the Palette represents one or more closely related tools. For example, Geography Tools include the Geography Circle Tool, Geography Polygon Tool, and Geography Grid Tool among others.

To active a tool, LEFT-CLICK the icon of the tool in the Tools Palette, which appears on the left of the 3D Render Window by default. In the case of tool groups, LEFT-CLICK the drop-down menu next to the icon, and select the desired tool from the list.

3D Render Window

The 3D Render Window displays the currently loaded terrain via a simulated camera that you can move around, as described in “Using the Camera” later in this chapter.

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Main Widget Panel

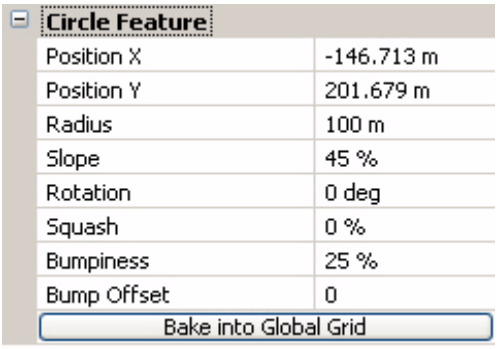
The Main Widgets give you quick access to functionality that complements the active tool. The Main Widgets change whenever you switch active tools or select different types of objects.

For the sake of simplicity, this Quick Start does not document the functionality of each and every individual widget. Later chapters on using specific tools document, in detail, every widgets for the tool of interest. However, going through this tour will teach you how to use enough widgets to accomplish key tasks. To get familiar with the various ways you can change the values in widgets, see Using the Widgets below.

Using the Widgets

You can change the values of the widgets in different ways, depending on the type of widgets. Regardless of the type and method of adjustment, the final value will always be clamped between predefined min and max values.

- To increase or decrease the value of a widget by a single step, LEFT-CLICK the UP/DOWN BUTTONS that appear to the right of the widget after you click in its value field.
- To type in an exact value, LEFT-CLICK the CURRENT VALUE in the widget, and then TYPE the NEW VALUE.
- To smoothly scroll the value of a specific widget, LEFT-CLICK-HOLD the TITLE of that widget and DRAG the MOUSE:
  - For example, to scroll the value of the “Slope” widget, LEFT-CLICK-HOLD on the word “Slope” in the Widget Panel (see figure below), and DRAG the MOUSE left or right. While you drag, the cursor disappears so it won’t hit the edge of the screen or get lost.



  - When scrolling, the value changes by a step size that is appropriate to that widget. For example, if the range of a value is from 0 to 1, the step size is typically fixed at 0.01. For a widget with a range -100 to +100, the step size would be a lot larger.
    - » HOLD the SHIFT-KEY while scrolling to increase the step size by 10x the default step.
    - » HOLD the CTRL-KEY while scrolling to decrease the step size to 1/10th the default.

LIBRARY BROWSER

Some tools involve the use of external items, such as the Prop Tools (i.e. you're working with imported 3D models with textures). These tools provide you with a version of the Library Browser. You use the Library Browser to import new items, modify their global properties, manage them, and select specific items to place into terrains.

The Library Browser takes slightly different forms with specific tools: With Road Tools, the Library Browser becomes the Road Style Browser and gives you access to road styles. With the Material Tool, the Library Browser becomes the Material Browser and gives you access to materials that can use imported textures. With the Prop Tool, the Library Browser becomes the Prop Browser and gives you access to imported 3D props. A few other tools also use a version of the Library Browser, while some tools do not use it at all.

For background information on the Library Browser, and the items it provides access to, see "Chapter 4: The Project System" in the *CityScape User Manual*. For details on using the library browser with a specific tool, see the chapter dedicated to that tool. (For example, for details on using the Road Style Browser with the Road Tools, see Chapter 6, section 2, "Using the Road Style Browser" in the CityScape User *Manual.)*

2.2.2 Using the Camera

The camera system is designed for you to quickly view the contents of terrains from any perspective. You must HOLD the ALT-KEY whenever you move the cameras. Press to toggle between Perspective and Orthographic Cameras:

- The Perspective Camera uses a polar model in which the camera orbits around the point of interest marked by a faint plus (+) in the center of the Render Window.
- The Orthographic Camera looks straight down on the terrain at all times (i.e. helicopter view). All sense of perspective is removed, so you have an orthogonal (flat) view down on everything that's visible in the viewport.

PERSPECTIVE CAMERA

As a type of polar camera, the Perspective Camera is described by its azimuth angle, inclination angle, and distance from the point of interest. You can switch to the Perspective Camera by selecting **View > Cameras > Perspective Camera**. You can toggle between the Perspective and Orthographic cameras with by pressing the **C-KEY**.

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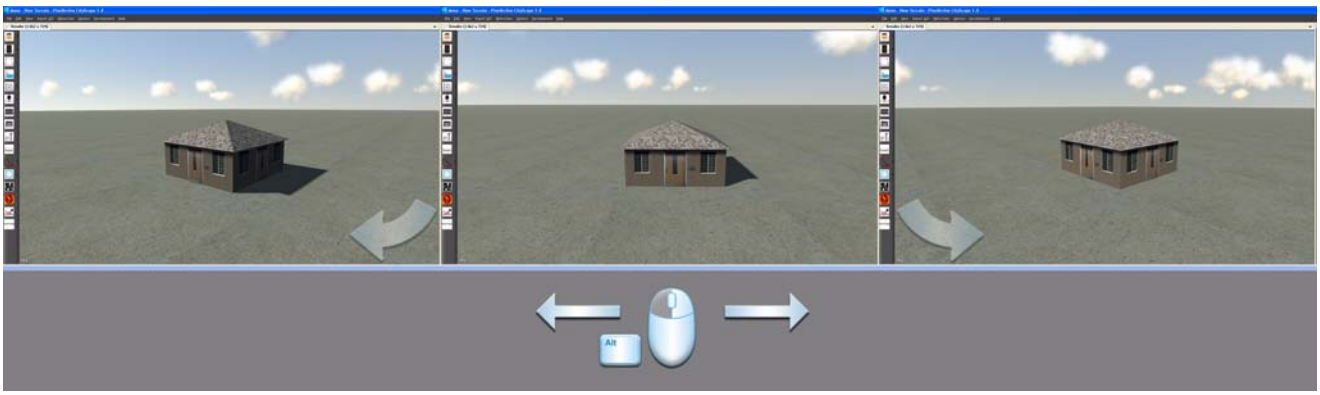
Rotate the Perspective Camera

When rotating the camera, the point of interest remains fixed. The camera orbits the point of interest without changing the distance between them.

1. To rotate the camera around the current point of interest, HOLD both the ALT-KEY and LEFT-MOUSE BUTTON, and then DRAG the MOUSE.



Zoom the Perspective Camera

When zooming, the camera moves closer to, or further from, the point of interest. The point of interest remains fixed.

1. To zoom the camera, HOLD both the ALT-KEY and RIGHT-MOUSE BUTTON, and then DRAG the MOUSE. If you have a mouse with a scroll wheel, you can also HOLD the ALT-KEY and rotate the wheel to zoom.

Note: *When you "zoom" a lens on a traditional camera, the magnification changes but the distance between the camera and point of interest do not change.*

In contrast, when you “zoom” the camera in CityScape, you are actually moving the camera itself closer to, or further from, the point of interest. We use the term "zoom" anyway because it’s widely understood, but a more precise technical term would be the verb form of “dolly,” as it’s used in the motion picture industry.

The distance between the camera and point of interest affects your perspective on the terrain. Helpful hints:

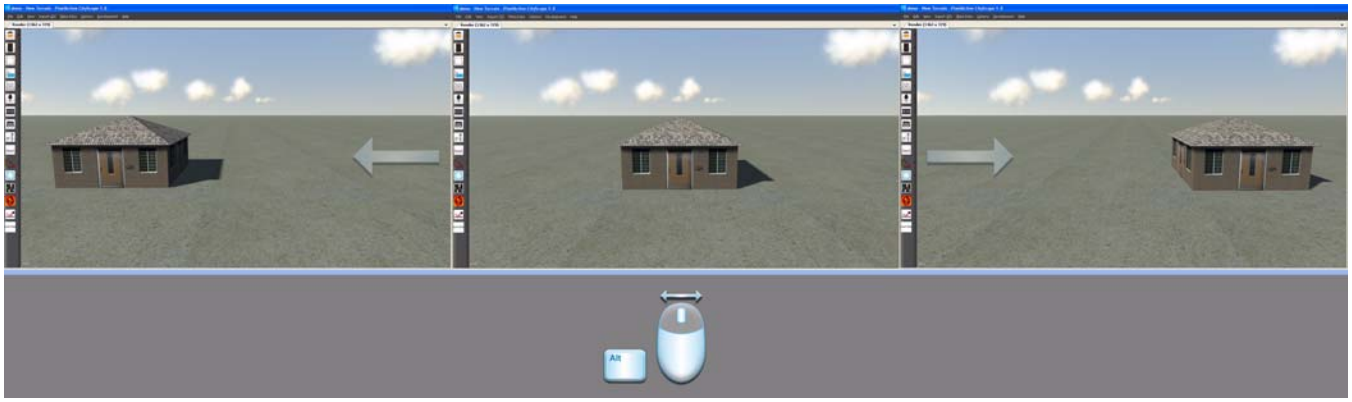
- To view from the point of view of a person standing on the terrain, zoom the camera all the way in until the camera stops. At this point, the position of the camera will be fixed approximately 2 meters above the terrain, so that rotating the camera will have the effect of turning your head while standing on the terrain.

- To view the terrain from an overhead perspective, try zooming the camera further back first, or switch to the Orthographic Camera.

Translate the Perspective Camera

Translating the camera moves the camera along with the point of interest, while preserving the distance between the two. The translation is limited to a plane that's perpendicular to the direction vector of the camera.

1. To translate the camera, HOLD the ALT-KEY, the LEFT-MOUSE BUTTON, and the RIGHT-MOUSE BUTTON at the same time. Then DRAG the MOUSE in any direction.



After you translate the camera, rotating and zooming the camera will operate relative to the new point of interest.

> ***Tip:*** *Translate the camera until the "+" in the middle of the Render View centers on the desired point of interest the terrain. Then zoom and rotate the camera to the desired perspective.*

ORTHOGRAPHIC CAMERA

The Orthographic Camera looks straight down on the terrain. All sense of perspective is removed, so you have an orthogonal (flat) view down on everything visible in the 3D Render Window. North is always towards the top of the screen.

You can switch to the Orthographic Camera by selecting **View / Orthographic Camera** from the Main Menu. You can also press the **C-KEY** to toggle between Perspective and Orthographic Cameras.

Zoom the Orthographic Camera

1. To zoom the Orthographic Camera, HOLD the ALT-KEY and RIGHT-MOUSE BUTTON, and then DRAG the MOUSE.

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Translate the Orthographic Camera

Since the look-at vector of the Orthographic Camera is always perpendicular to the ground, translating this camera moves it in a plane parallel to the ground. Put another way, translating the Orthographic Camera maintains the same elevation, like a helicopter flying level.

1. To translate the camera, HOLD the ALT-KEY and LEFT-MOUSE BUTTON, and then DRAG the MOUSE. The point of interest moves along with the camera.

2.3 CREATING AND EDITING WORLDS

You do most of the work in CityScape with editing tools. Examples:

- Use the Geography Tool to create and modify hills and valleys by combining circles, polygons, grids, and other geometric primitives. When you export the terrain, CityScape tessellates all geometric primitives into triangles.
- Use the Road Tool to create and modify roads in the terrain.
- CityScape includes many other tools for creating robust 3D worlds. This Quick Start covers the basics of using the most common tools:

- Using Geography Tools, next
- “Using Road Tools” on page 25
- “Using Water Tools” on page 39
- “Using Material Tools” on page 41
- “Using Prop Tools” on page 43
- “Using Sky Tools” on page 51
- “Using GIS Tools” on page 52

Initially, the Geography Tool is active by default. The name of the active tool is displayed at the bottom left corner of the 3D Render Window.

To activate a tool, LEFT-CLICK its icon in the Tools Palette on the left of your screen, by default. You can change tools at any time.

When you switch tools, the Main Widgets change to display the specific features and options that are available with the current tool. The Main Widgets also change when you select a different type of object in a terrain.

2.3.1 Using Geography Tools

When you initially run CityScape, the Geography Tools will be active by default. You can use Geography Tools to shape the underlying terrain. To accomplish this goal, you place and customize Circle Features, Polygon Features, and Grid Features that CityScape converts to 3D hills and valleys. You can also "paint" hills and valleys directly into the terrain, and use the Main Widgets to customize all geography features.

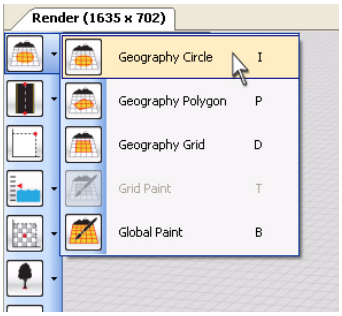
The Geography Circle Tool

A Circle Feature is an ellipse that causes the terrain within it to rise or lower to form a hill or valley. A Circle Feature starts life as a circle, but you can make it into an ellipse by changing the *Squash* parameter in the Main Widgets for Circle Features.

The Geography Circle Tool lets you place and customize Circle Features in the terrain. Like all geography features, the overlapping Circle Features are additive.

Create Circle Features

To place a Circle Feature in the terrain, follow these steps:

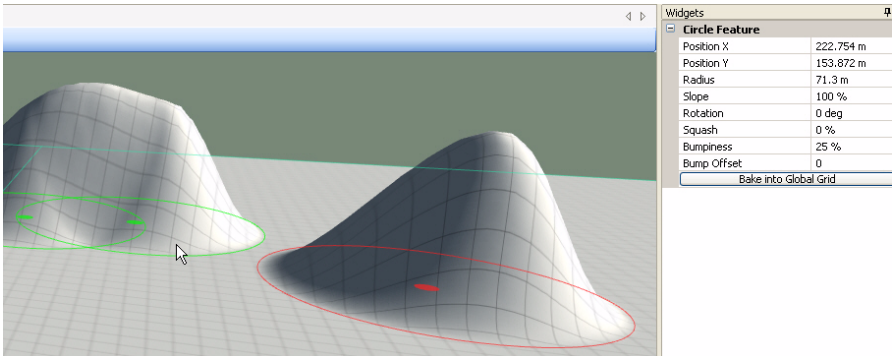


1. With a terrain loaded in CityScape, activate the **Geography Circle Tool** by CLICKING its ICON in the Tools Palette.
2. To place a Circle Feature in the terrain, RIGHT-CLICK at the target location in the terrain.
3. Try placing a few Circle Features – some separate, some overlapping.

Select Circle Features

Editing Circle Features requires that you first select one or more of them with the Geography Circle Tool:

- To select one existing Circle Feature, LEFT-CLICK its **center point**. It will change from green (unselected) to red (currently selected). In the figure below, the red Circle Feature in the Render Window has been selected, and the Circle Feature Widgets appear on the far right.



- To select several existing Circle Features at the same time, LEFT-CLICK-HOLD and DRAG THE MOUSE to draw a box around their center points.

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Edit Circle Features

After you create a geographic feature, you can always edit it later (unless you intentionally bake it into the terrain). To edit a Circle Feature, first select it and then proceed as follows:

- To move a selected Circle Feature, hover the mouse arrow over its center point, LEFT-CLICK-HOLD, and DRAG THE MOUSE.
- To delete a Circle Feature, select its center point and press the DELETE-KEY.
- Select some Circle Features and experiment with modifying the values of the Circle Feature Widgets. Examples:

  Slope Widget: *Controls the height and steepness of Circle Features. Big positive slopes make high hills, and negative slopes make valleys (that can optionally be filled with water later). Setting the Slope widget to 100% for hills, or -100% for valleys, makes the Circle Feature as high or low as it can be by itself. To go higher or lower, create multiple overlapping Circle Features.*

  Squash Widget: *Controls the eccentricity of the Circle Feature. If Squash is non-zero, the circle becomes an ellipse.*

THE GEOGRAPHY POLYGON TOOL

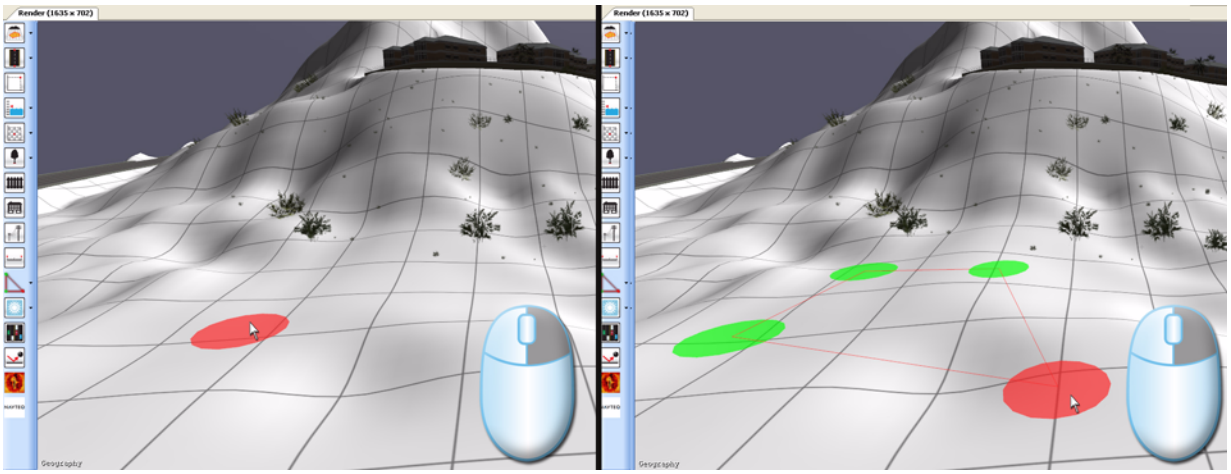
The Geography Polygon Tool lets you place and customize Polygon Features in the terrain. A Polygon Feature is a polygonal area defined by 3 or more points connected by line segments. You can use Polygon Features to make natural looking mountains and valleys with irregular contours.

Create Polygon Features

Follow these steps to place a Polygon Feature in a terrain:

1. With a terrain loaded in CityScape, activate the **Geography Polygon Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the drop-down menu next to the main Geography icon.)
2. RIGHT-CLICK at the location in the terrain where you want to place the first point in the polygon.

3. Continue RIGHT-CLICKING a few times to place new points and expand the polygon.

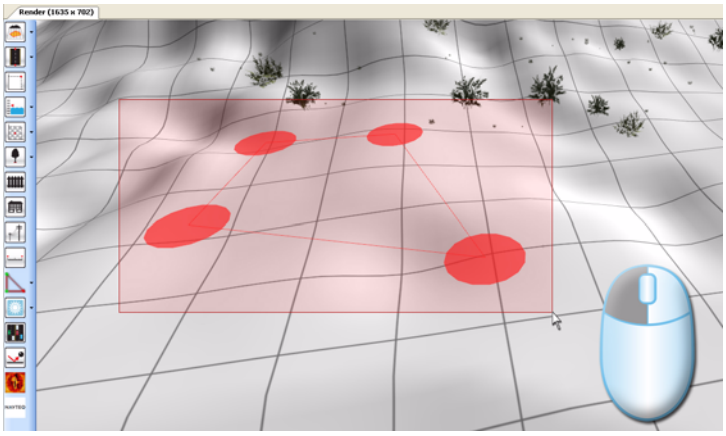


4. To start a new polygon, first select nothing by LEFT-CLICKING away from selectable objects in the terrain. Then repeat the steps above.

Select Polygon Features

Editing a polygon requires that you first select one or more points with the Geography Polygon Tool:

- To select one point in a Polygon Feature, LEFT-CLICK the point. The figure in the "Edit Polygon Features" section shows one actively selected point (red) and the other points that are not selected (green).
- To select multiple points (first method), HOLD the SHIFT-KEY and LEFT-CLICK each point in succession.
- To select multiple points (second method), LEFT-CLICK-HOLD and DRAG THE MOUSE to draw a pick box around them.



- To select an entire polygon, DOUBLE-LEFT-CLICK any point in it.

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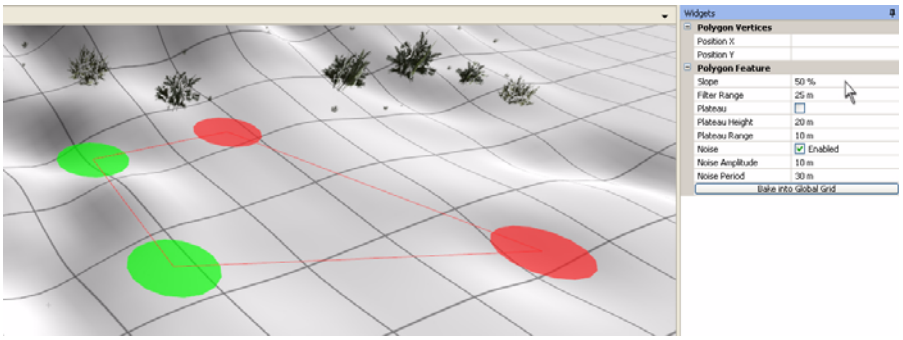
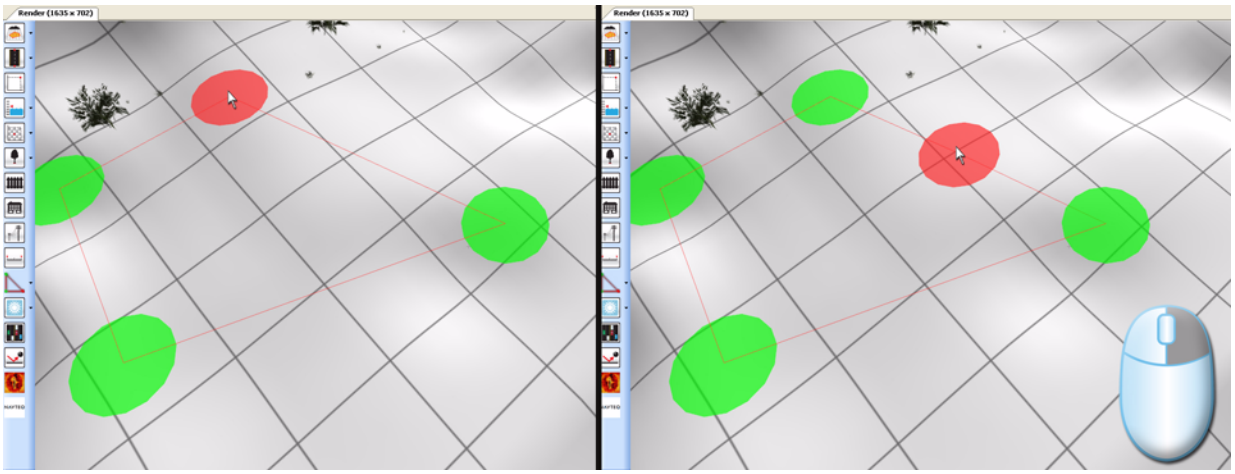
Edit Polygon Features

As with all geographic features, you can edit the properties of Polygon Features by selecting one or more points in an existing polygon, and then using the Main Widgets to modify the parameters controlling the polygon.



- To delete one point on an existing polygon, select the point and then press the DELETE-KEY.
- To move a point on an existing polygon, be sure the mouse is hovering over the selected point, and then LEFT-CLICK-HOLD the point and DRAG THE MOUSE.
- To move multiple points, first select the points as described in “Select Polygon Features”. Hover the mouse over one of the selected points, and then LEFT-CLICK-HOLD and DRAG THE MOUSE.
- To add a new point in an existing polygon, first select the appropriate node (i.e. the new node will be created clockwise relative to the selected node.) Then RIGHT-CLICK to add the new point to the polygon.



- Select a Polygon Feature and experiment with modifying the parameters of the Polygon Feature Widgets. Most of the parameters affect the entire polygon, even if you select only one point in it.
- To select the entire polygon, DOUBLE-CLICK any node in the polygon.

THE GEOGRAPHY GRID TOOL

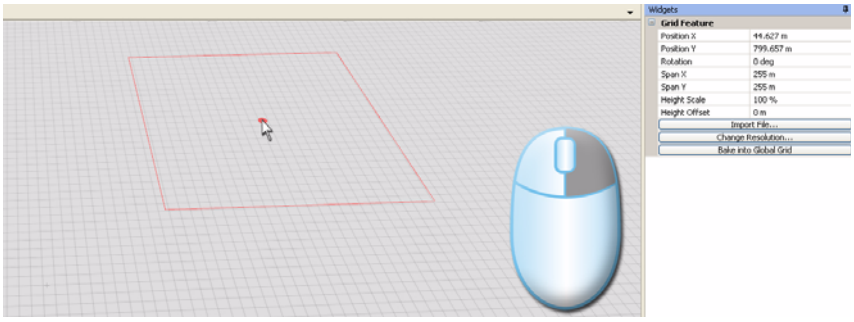
The Geography Grid Tool lets you place and customize a Grid Feature in the the terrain. A Grid Feature is a rectangular area controlled from its central point. You can raise and lower the terrain within a Grid Feature to form hills or valleys. Grid Features are also useful for importing localized height-maps based on 2D images, and hand-painting 3D terrain, as explained below.

Create Grid Features

To create a Grid Feature, follow these steps:

1. With a terrain loaded in CityScape, activate the **Geography Grid Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the drop-down menu next to the main Geography icon.)

2. To place a new Grid Feature, RIGHT-CLICK the location in the terrain where you want the grid to be centered. The figure below shows a Grid Feature actively selected in the terrain along with its Main Widgets.



Select and Edit Grid Features

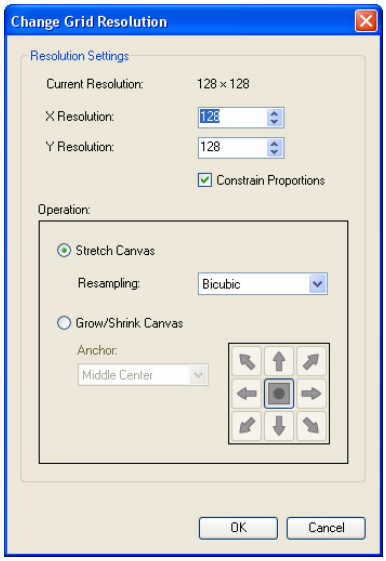
Use the Geography Grid Tool to select and edit a Grid Feature:

1. To select an existing Grid Feature, LEFT-CLICK its **center point**.

2. To move a selected Grid Feature, be sure the mouse arrow is hovering over its center point, and then LEFT-CLICK-HOLD and DRAG THE MOUSE.

3. To delete a Grid Feature, select it and press the DELETE-KEY.

4. When a Grid Feature is selected, you can change its properties using the Grid Feature Widgets. Examples:
   – To adjust the size of the selected grid, change the **Span X** and **Span Y** widgets.
   – To modify the resolution of the currently selected Grid Feature, LEFT-CLICK the **Change Resolution** button in the Widgets to display the Change Grid Resolution dialog box. In the dialog, change the values of the **X-Resolution** and **Y-Resolution** fields in the Change Grid Resolution dialog. (Note: Changing the resolution of a Grid Feature does not change its size.)



Import Non-GIS Height Maps

CityScape can convert almost any 2D image to a height field (i.e. hills and valleys) via the Geography Grid Tool. (For details on geo-located height maps, see “Chapter 21: Importing GIS Data” in the CityScape User *Manual.*) When you import an image, CityScape first converts it to grayscale. Then CityScape converts black to ground level, white to the highest elevation allowed by the **Height Scale** widget, and grays to intermediate elevations according to their brightness.

CityScape supports JPEG, TIFF, and most other common 2D image formats. To import an ordinary 2D image as a height map, follow these steps:

1. Activate the **Geography Grid Tool**, if necessary.

2. LEFT-CLICK the **center point** of an existing Grid Feature to select it. Or create a new Grid Feature.

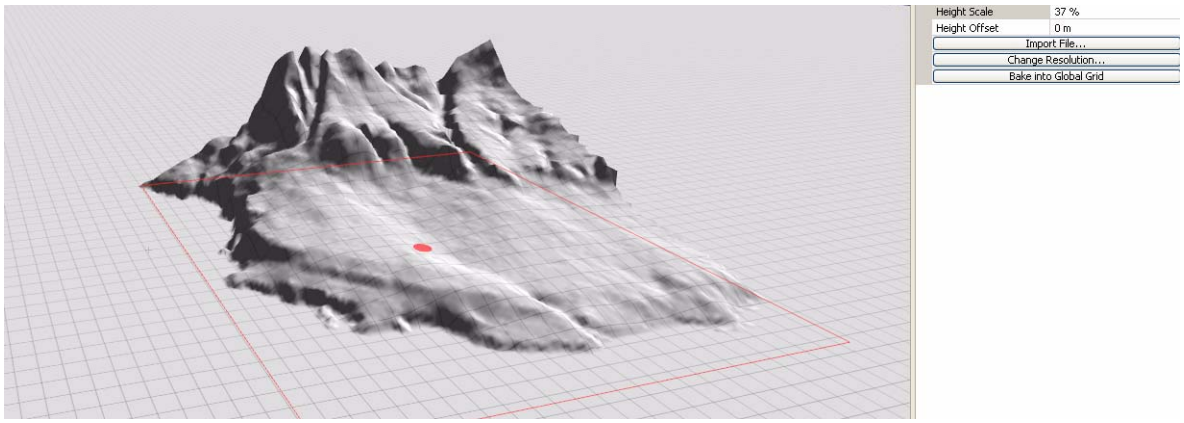
3. Press the **Import File** button in the Main Widgets for the Grid Feature. In the file browser that appears, locate and select the **SampleHeightmap.jpg** image in the following folder, by default:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds\heightmaps

or here for WinXP users:

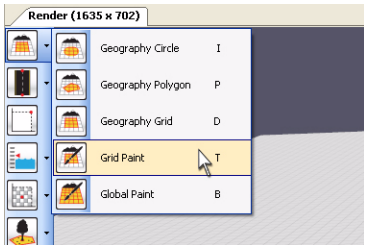
C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds\heightmaps

4. CityScape converts your image into 3D geography within the selected Grid Feature.



THE GEOGRAPHY GRID PAINT TOOL

To "paint" changes to 3D terrain within a Grid Feature, follow these steps:

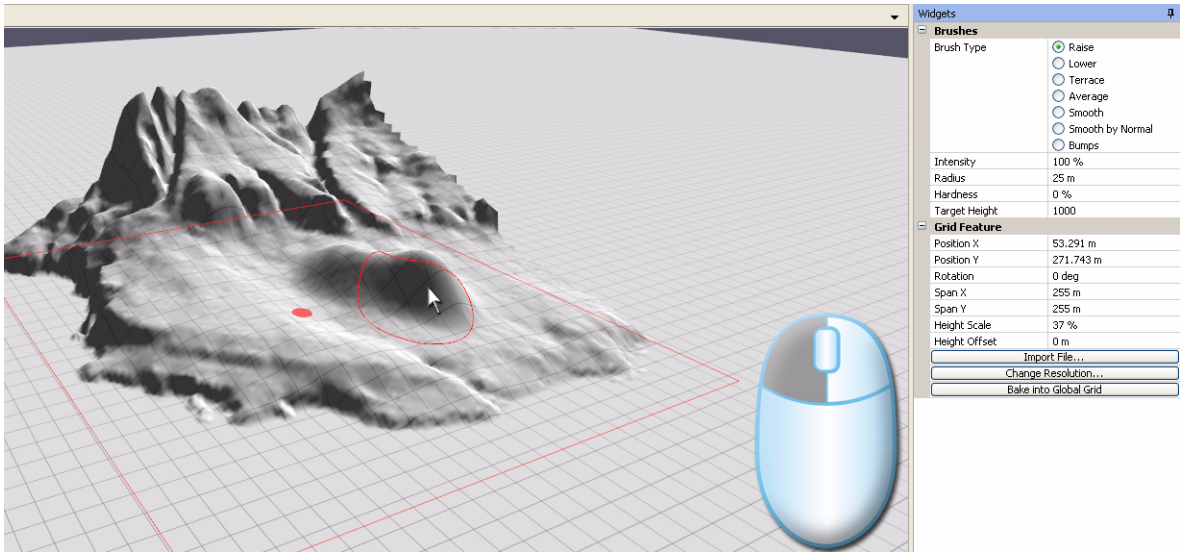


1. In CityScape, open a terrain with at least one Grid Feature, or create a new Grid Feature in the open terrain.

2. Select a Grid Feature in the terrain.

3. Activate the **Grid Paint Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the drop-down menu next to the main Geography icon.)

4. To paint 3D terrain on the selected Grid Feature, LEFT-CLICK-HOLD and DRAG the MOUSE within the Grid Feature. The resulting 3D terrain is part of the selected grid, and can be moved with the grid and edited with the Main Widgets for the Grid Feature, as shown below.



5. Experiment with modifying the values of the Main Widgets. Try changing brush types and adjusting their sub-parameters. The figure above shows the results of 3D painting with the Raise brush. Try painting with various brushes.

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The Geography Global Paint Tool

You can "paint" changes to 3D terrain anywhere in the open terrain in much the same way as painting within a Grid Feature. Here's how:

1. With a terrain loaded in CityScape, activate the **Global Paint Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the drop-down menu next to the main Geography icon.)

2. To paint changes to 3D terrain, LEFT-CLICK-HOLD and DRAG the MOUSE anywhere on the terrain. The resulting 3D terrain is part of the global grid, so this terrain cannot be moved or edited with widgets after you paint it.

3. Experiment with modifying the values of the Global Paint Widgets. Your changes will affect future painting only.

2.3.2 Using Road Tools

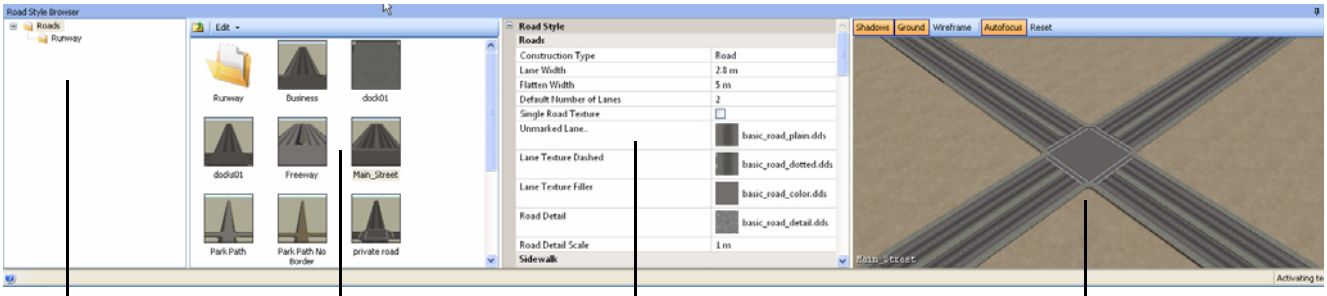
The Road Tools let you create and modify a network of roads and intersections. Some of these tools are in the Road Widgets; some are in the Road Style Browser or Road Tools Menu, which appear when any Road Tool is active.

This Quick Start introduces you to The Road Tool. Other tools in the Road Tools Group include the *Road Tangent Tool* and *Road Height Tool*, both of which are documented in "Chapter 6: Using Road Tools" in the CityScape User Man*ual.*

About The Road Style Browser

Whenever you select any of the Road Tools from the Tools Palette, the Road Style Browser appears in the lower area of the CityScape window, by default.

The Road Style Browser shows the contents of the Road Style Library, so you can modify and place specific road styles in the terrain. You can use road styles to visually differentiate between main streets, freeways, alleys, etc. You can also import new textures for road styles, and otherwise edit road styles.



To select a road style, simply LEFT-CLICK the ICON of the desired style in the Road Style Browser. With the Demo Project, try selecting the **Main_Street** road style. A sample of the style is previewed in the right sub-window.

About Road Style Sub-Widgets

The Road Style Sub-Widgets are part of the Road Style Browser at the bottom of the screen, by default. These Sub-Widgets allow you to edit the *global* properties of a road style, regardless of whether or not that road style is currently used in the open terrain.

The Prop Browser and Material Browser also have sub-widgets for editing the global properties of props and materials, respectively. In all cases, Sub-Widgets control the global properties of an object. Hence, changing these values will affect all instances of the object in all terrains sharing the same project file.

In contrast, the Main Widgets in the upper right of your screen, by default, let you edit only the *local* properties of objects that are actively selected in the open terrain.

For example, in the figure above, *Default Number of Lanes = 2* means that every new road using the selected **Main_Street** road style will begin life with 2 lanes in each direction. After you create some roads in a terrain, you can locally customize the number of lanes of each road using the Main Widgets.

THE ROAD TOOL

To use the Road Tool, you must understand road nodes and road segments: By definition, a “road node” is a point that defines the start or end of a single segment of road. An entire road consists of a series of nodes attached by segments. An intersection can optionally exist at a road node.

The usage of the Road Tool is organized into the following sections, which we recommend you read in order:

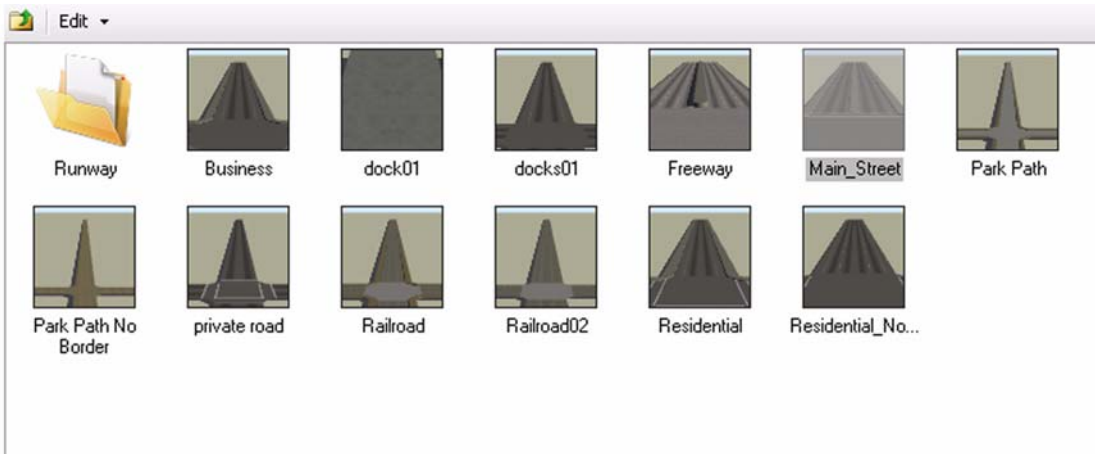
- Creating Your First Road
- Creating Additional Roads
- Continuing an Existing Road
- Smoothing Out a Road
- Close a Road Loop
- Select and Edit Road Nodes
- Select and Edit Road Segments
- Create a 2-way Intersection
- Create a 3-way Intersection
- Create a 4-way Intersection
- Link and Unlink Segments Across Intersections
- Create Merges and Splits
- Build Bridges and Overpasses

Creating Your First Road

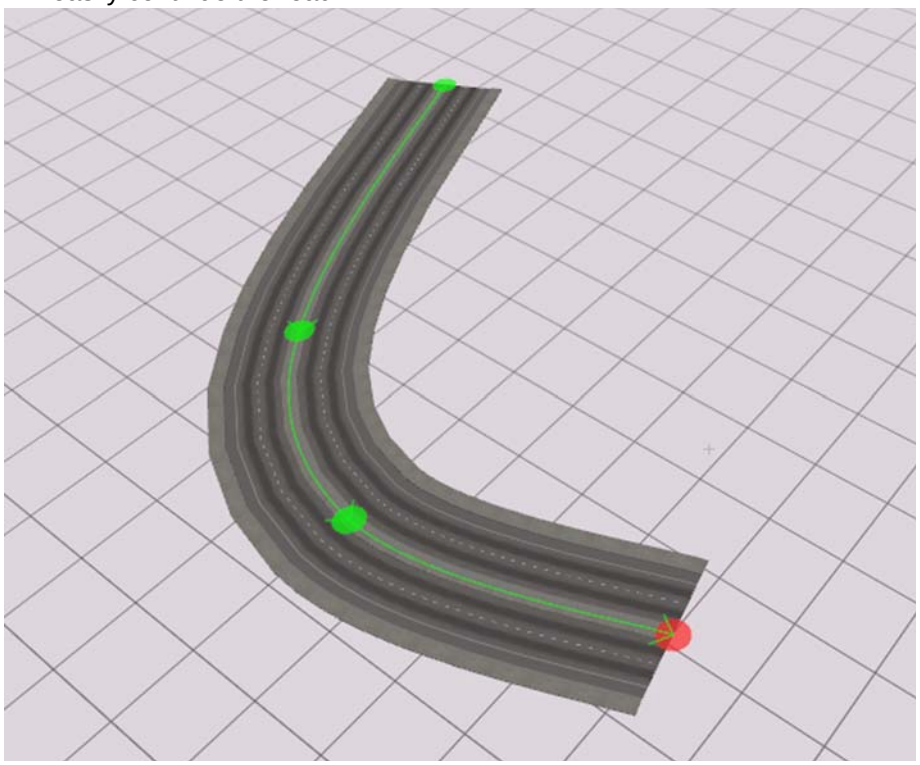
To create a road, proceed as follows:

1. Activate the **Road Tool** by CLICKING its ICON in the Tools Palette.

2. In the Road Style Browser, which appears when any of the Road Tools are active, select a style in the Road Styles Icons (the example icons below may not correspond to your road styles).



3. RIGHT-CLICK a few times on the terrain to lay down a few road nodes. Each time you click in an open area to place a new road node, CityScape adds a road segment connecting the previous node to the new node. The last road node you placed is always the current selection, so you can easily continue the road.



Creating Additional Roads

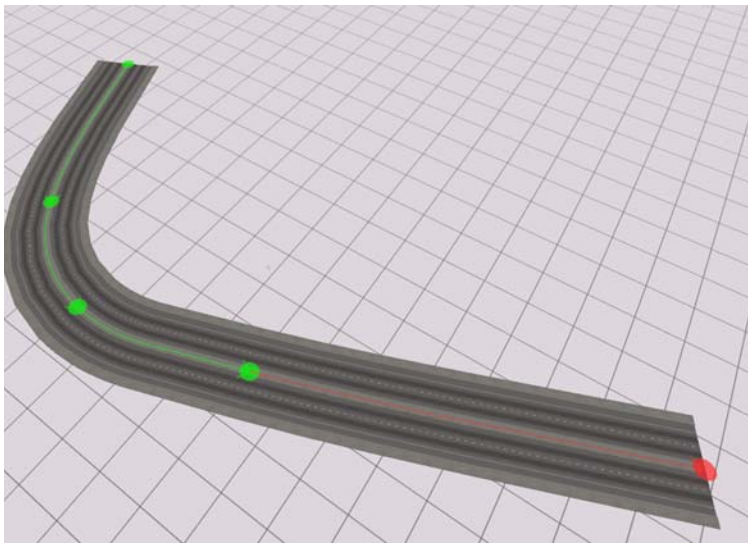
To start a new road that is not connected to the last road, proceed as follows:

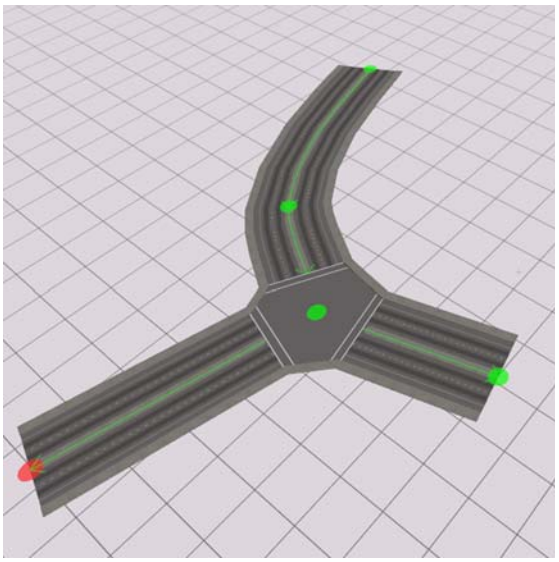
1. First LEFT-CLICK in an area without objects to deselect everything.
2. Then LEFT-CLICK the desired road style in the Road Style Browser to select it.
3. RIGHT-CLICK a few times to place new road nodes with interconnecting segments.

Continuing an Existing Road

If you deselect a road, and later want to resume work on that road, follow these steps:

1. To select the road node from which to resume, LEFT-CLICK the **target node** to select it.
   – If you resume from an end node, then placing new nodes will continue the same road, like the left image below. If you resume from any node other than an end node, then placing new nodes will cause a new road to branch out from an intersection at the selected node, like the right image below.





2. RIGHT-CLICK a few times to place some new nodes as usual.

Smoothing Out a Road

If a 2-way intersection appears in your road (like the left image below), but your desired goal is to have a smooth road without an intersection (like the right image below), follow these steps:

1. LEFT-CLICK-HOLD and DRAG the MOUSE to draw a pick box around the node in the intersection, as well as the segments on either side of the intersection, to select them all.

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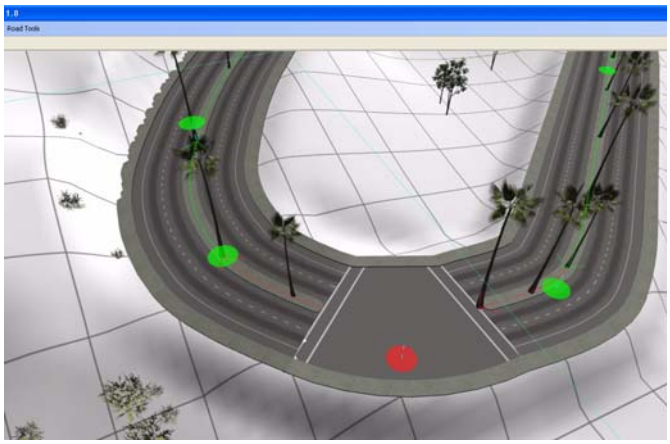
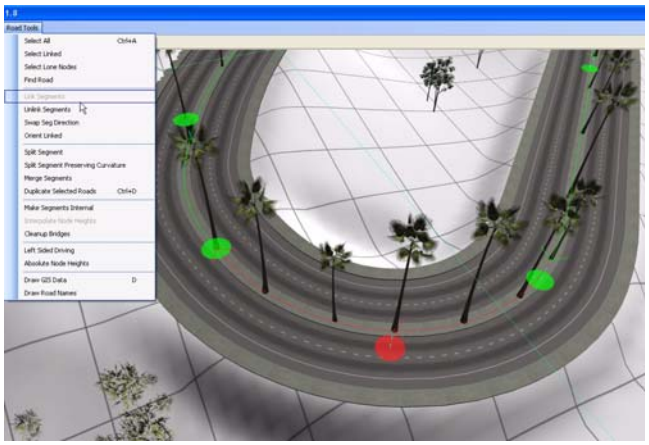
2. Select **Link Segments** from the Road Tools Menu in the Main Menu. (The Road Tools Menu appears only when a Road Tool is active.) The 2-way intersection will be replaced with a continuous road.



2-way intersection before linking segments on either side of the selected (red) node.

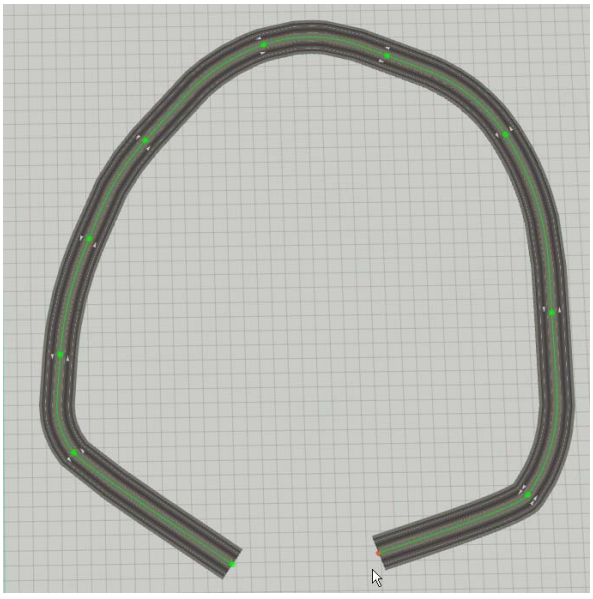


After linking segments, the 2-way intersection is replaced by smooth road through the selected node.

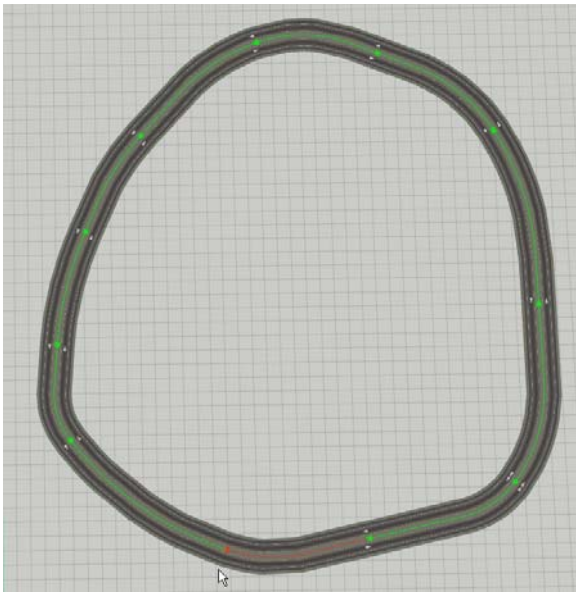
Close a Road Loop

Some features of CityScape call for a closed loop of road. There are many ways to create closed loops of road, but the simplest case is a single oval road without intersections:

1. Create an oval road, such as the one shown in the right figure below.



An open loop road with the end node selected.



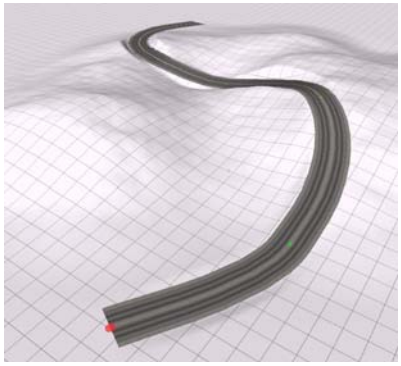
After linking the end node to the start node, the road becomes a closed loop.

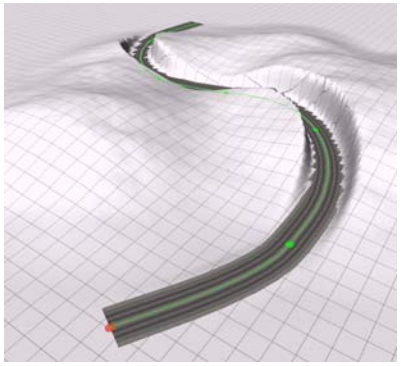
2. Select the end node. Then RIGHT-CLICK directly on the node that started the road. This action connects the end node to the start node, closing the loop of road. The right figure above shows the closed loop.

Select and Edit Road Nodes

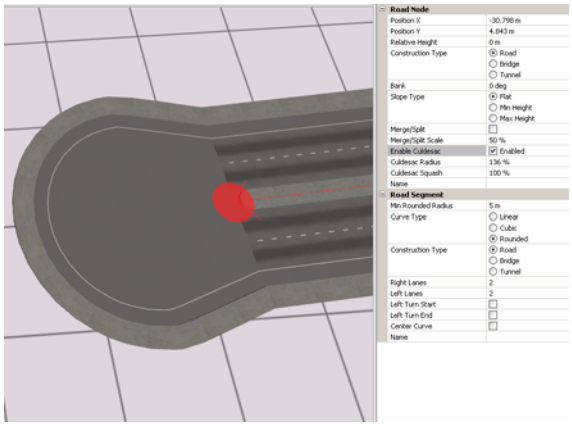
Some properties of roads are specific to road nodes, and other properties are specific to segments. This topic touches on some useful properties of nodes:

- To select one road node, LEFT-CLICK the target **node** (it will turn red).
- To select multiple nodes (first method), HOLD the SHIFT-KEY and LEFT-CLICK each node in succession.
- To select multiple nodes (second method), LEFT-CLICK-HOLD and DRAG THE MOUSE to enclose the target nodes in a pick box. This method will also select the segments within the pick box.
- To move a section of road, first select the node(s) in the section of road you wish to move. Then hover the mouse arrow over one of the selected nodes, and LEFT-CLICK-HOLD and DRAG THE MOUSE.
- To delete one or more nodes, select them and PRESS the **DELETE-KEY**. Deleting a node will also delete the segments attached to it. If your road was a closed loop before, deleting a node will open the loop.
- Use the Road Node Widgets to adjust the properties of currently selected nodes. Here are some useful properties that you can adjust with widgets:
  - **Absolute / Relative Height:** Absolute height controls the elevation of the node, relative to the absolute zero elevation. Relative height controls the elevation of the node, relative to the surrounding terrain.
    - » To switch between absolute and relative heights, toggle Absolute Node Heights in the Road Tools Menu of the Main Menu.
    - » The choice of absolute or relative can dramatically affect the way roads pass through mountains and valleys. The examples below show the same road with Relative Height = 0 (left image) and Absolute Height = 0 (right image).

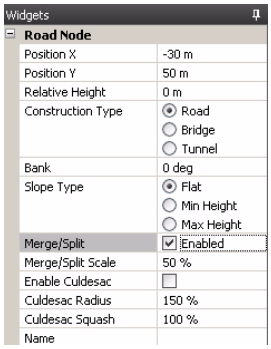


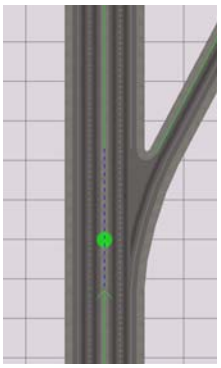


– **Enable Culdesac**: allows you to convert an end node of an open road into a cul-de-sac.



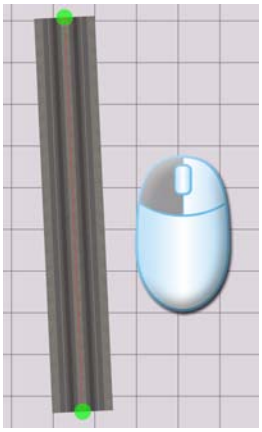
– **Merge / Split:** starts or ends a Y-branch without an intersection.
  » After enabling Merge/Split for node in an existing, placing additional nodes creates a new road stemming from the existing road, but without adding an intersection.
  » Similarly, you can merge one road into another without an intersection by enabling Merge/Split in a node of the target road (i.e. the road you're merging into).





Select and Edit Road Segments

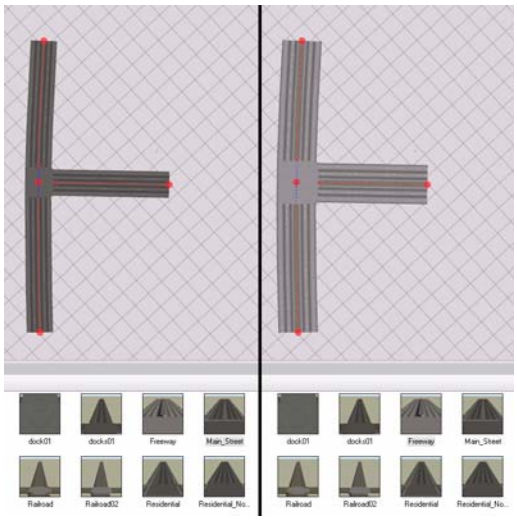
This topic addresses some of the properties of road segments:



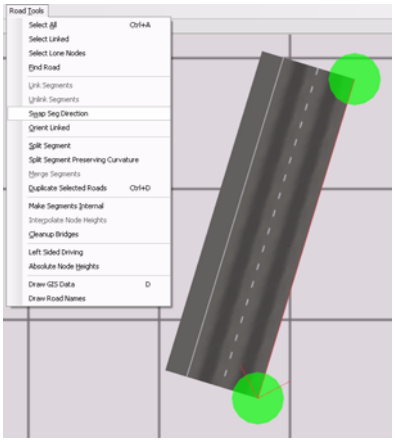
- To select one road segment, LEFT-CLICK the **green line** defining the segment. The line will turn red when it becomes the active selection, as shown in the figure to the left.
- To select multiple segments (first method), HOLD the SHIFT-KEY and LEFT-CLICK **each segment** in succession.
- To select multiple segments (second method), LEFT-CLICK-HOLD and DRAG THE MOUSE to partly or completely enclose the **segments** in a pick box. This method will also select the nodes within the pick box.
- To insert a new node within an existing segment, first select the target segment in an existing road. Then RIGHT-CLICK the **selected segment**.

- To delete one or more segments, select them and PRESS the DELETE-KEY. If your road was a closed loop before, deleting a segment will break the road into an open loop.

- To change the style of all or part of an existing road, select the **road segments** that you wish to change, and then DOUBLE-LEFT-CLICK the new road style in the Road Style Browser.



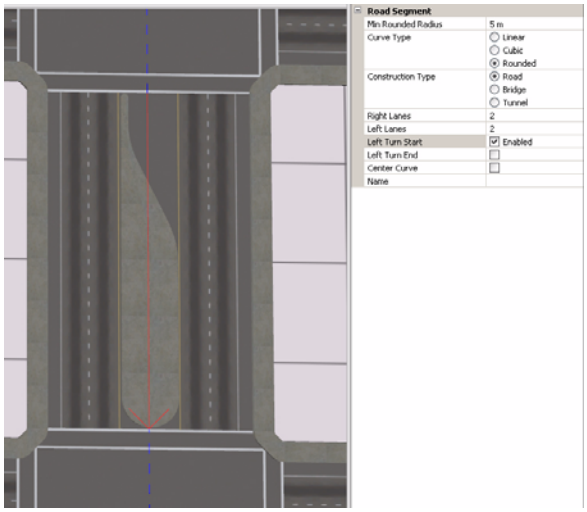
- Every road segment has a direction, which is indicated by an **arrow** at the end of each segment. References to right and left sides of a segment, as well as its start and end, are relative to its direction.
  – To switch the direction of actively selected segments, choose **Swap Seg Direction** in the Road Tools Menu. (The directions of road segments are different than the direction in which traffic will flow.)

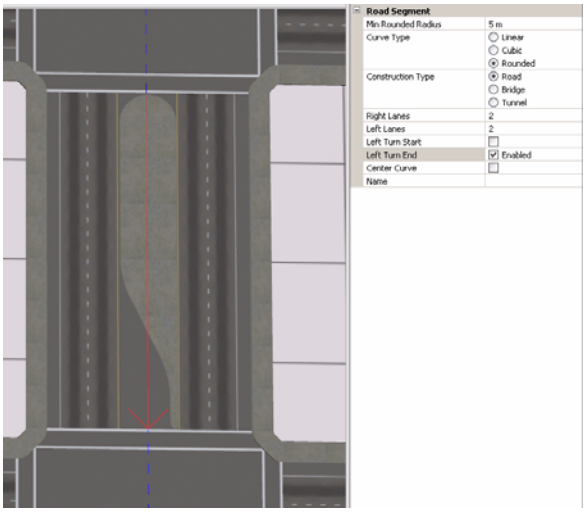


- To switch between right handed driving (default) and left handed driving, toggle **Left Sided Driving** in the Road Tools Menu. (The directions of road segments are different than the direction in which traffic will flow.)

- You can adjust the properties of currently selected segments using the Road Segment Widgets on the right of the screen. Some useful widgets:

- **Right Lanes** determines the number of lanes on the right side of the road, relative to the green arrow indicating the directionality of the road segment. Every road must have at least one (1) right lane.
- **Left Lanes** determines the number of lanes of the left side of the road. For a one-way road, set this widget to zero (0).
- **Left Turn Start** creates a left turn lane in the left side of the road, relative to the segment's direction (see left figure below).
- **Left Turn End** creates a left turn lane in the right side of the road, relative to the segment's direction (see right figure below).





Create a 2-way Intersection

A 2-way intersection might not seem logical, but it can be useful for creating precise 90-degree turns, for example. (Note: Creating a 2-way intersections is the inverse of replacing a 2-way intersection with a continuous road, as explained in "Smoothing Out a Road" on page 28.)

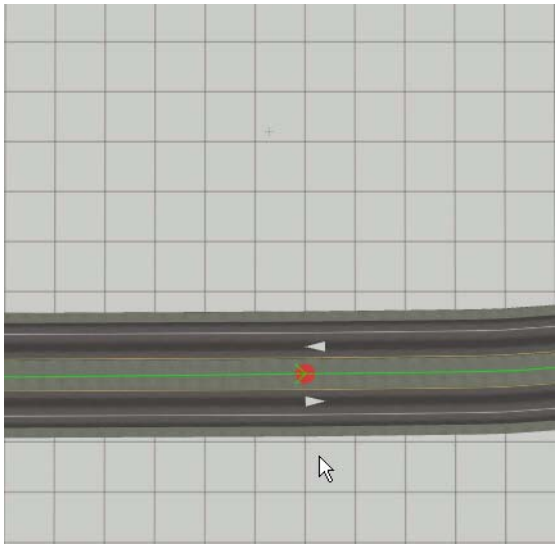
1. In an existing road, select the two segments on either side of the node you wish to change into a 2-way intersection. Selecting the node itself is optional.

2. Choose **Unlink Segments** from the Road Tools Menu in the Main Menu. (The Road Tools menu appears only when a Road Tool is active.) A 2-way intersection is created.

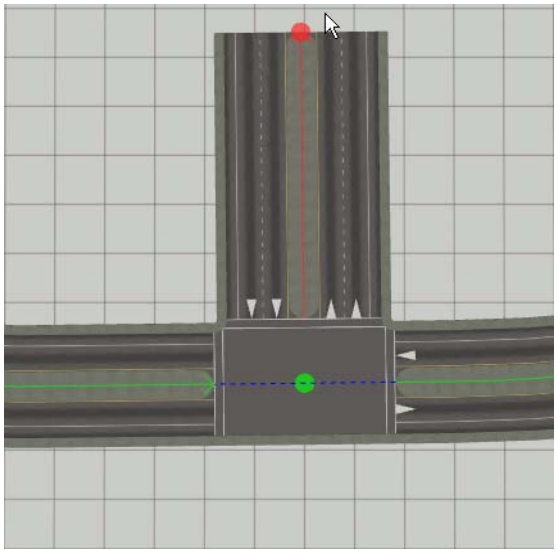
Create a 3-way Intersection

The following procedure is one of many ways to create a 3-way intersection. Such intersections are useful for branching a new road off an existing road, or creating a Y-intersection.

1. In an existing road, select an internal road node (i.e. not an end node) from which you want to branch a new road via a 3-way intersection. Upon being selected, the node turns red.



Steps 1 and 2: Node selected (red), but adjoining segments are not selected.



Step 3: A new road stems from 3-way intersection.

2. Ensure that the segments to either side of the selected node are NOT selected (i.e. green). See the left image in the figure above.

3. LEFT-CLICK again in an open area at roughly right angles to the existing road. A 3-way intersection is automatically created, and a new road branches off the intersection. See the right image in the figure above.

4. Extend the new road as usual.

Create a 4-way Intersection

There are many ways to create 4-way intersections. This section covers two common options.

OPTION 1

1. Create a 3-way T-intersection, as described in Create a 3-way Intersection.

2. Select the road node in the intersection. Make sure that the segments adjacent to this node are not selected.

3. RIGHT-CLICK in an open area at roughly right angles to the main road, in the opposite direction as the existing branch, as shown in the figure below. A new road branches off the intersection, and the 3-way intersection becomes a 4-way intersection.

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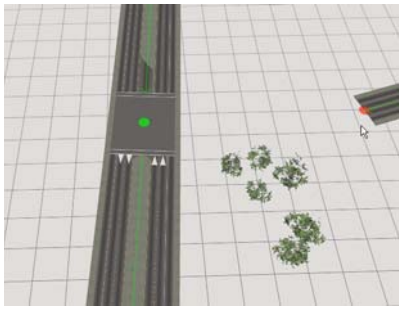
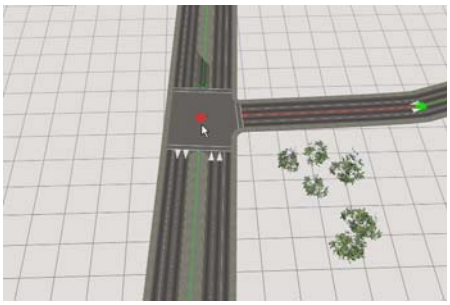
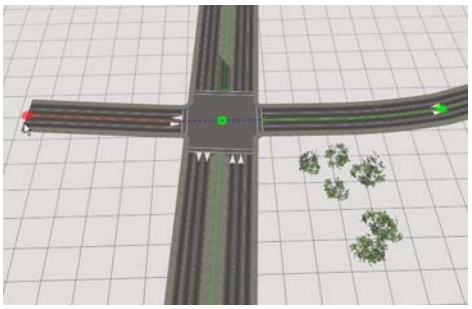
OPTION 2

1. Ensure that one road already exists (create it if necessary).

2. Create or extend a new road so it crosses the existing road. To form a 4-way intersection at their crossing, extend the new road by LEFT-CLICKING a node in the existing road.

Note: If you do not click a node at the crossing point of the two roads, an intersection will not form. You might not want an intersection in all cases, such as a bridge passing over or under another road.

Link and Unlink Segments Across Intersections

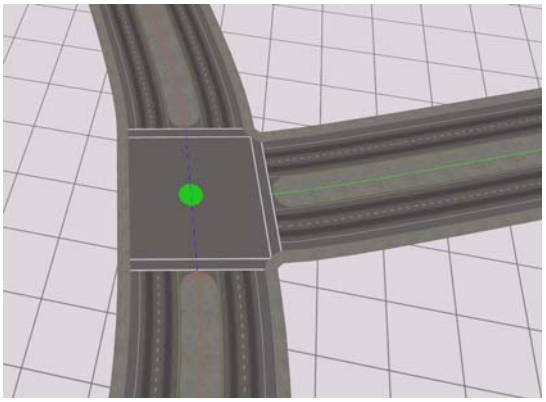
Linking and unlinking segments on either side of an intersection affects the exact positions of incoming and outgoing roads. Also linking could be used by AI software to determine the path of flowing traffic.

When you create any intersection, the segments around it may or may not be linked initially. To manually link or unlink segments across intersections, follow these steps:

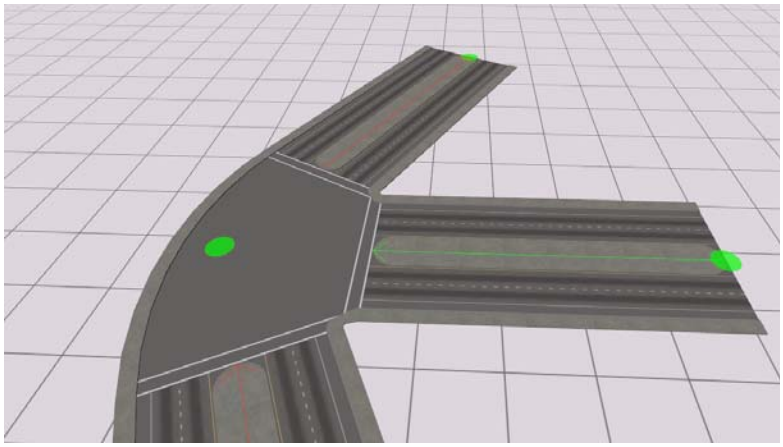
1. Start with an existing intersection, or create an intersection by following the instructions in the previous topics.

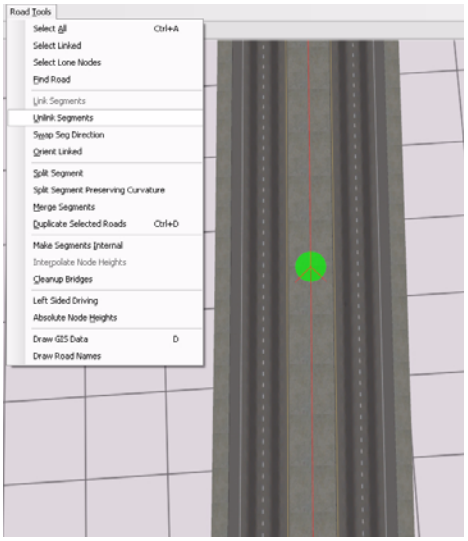
2. Select two road segments that you wish to link or unlink across an intersection. The two segments must share the node at the center of the intersection.

3. To link the two segments, choose **Link Segments** from the Road Tools Menu. A dashed blue line appears through the intersection to show the linking, as illustrated below.



4. To unlink the two segments, choose **Unlink Segments** from the Road Tools Menu. In the illustrations below, the left image shows 3 unlinked segments in a 3-way intersection, and the right image shows 2 unlinked segments in a smooth strip of road.



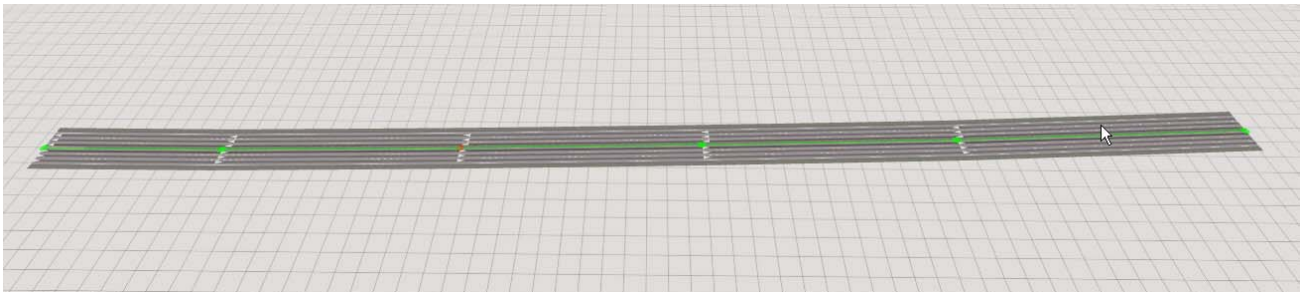


Create Merges and Splits

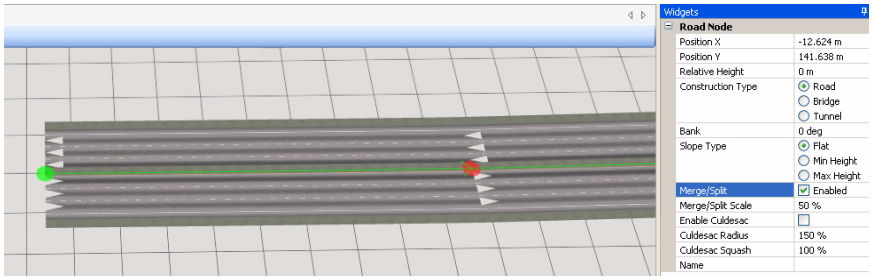
Many types of merges and splits for roads are possible with CityScape. In the exercises below, you will split a ramp off a freeway, and then merge the ramp back into the freeway.

1. In the Road Style Browser, select the **Freeway** road style.

2. In the terrain, lay down six new **road nodes** to form a standalone section of freeway with open ends.



3. Select the **node** located *second from the start*. In the Road Node Widgets, enable **Merge/Split**.



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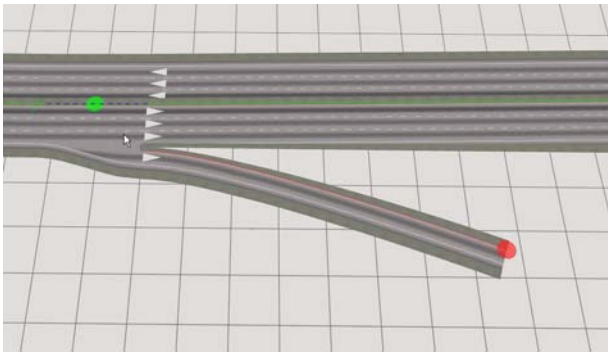
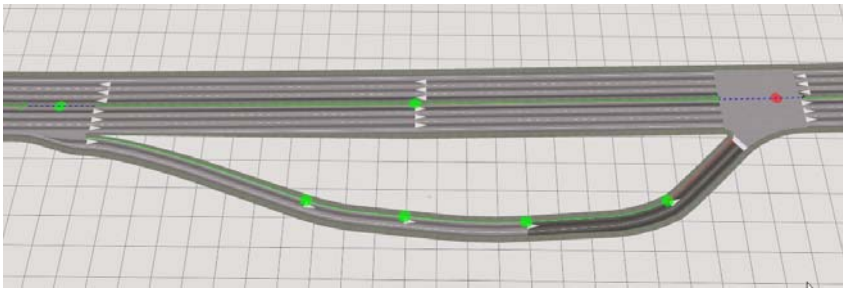
4. To place a **new node** for the ramp, RIGHT-CLICK to the right of the freeway, relative to the direction of the road segments in the freeway.



5. Lay down a few more nodes to continue the ramp. Then connect the ramp back to the freeway by clicking a node on the freeway. When you do this, an intersection might initially form instead of the "merge" you want (see figure below).



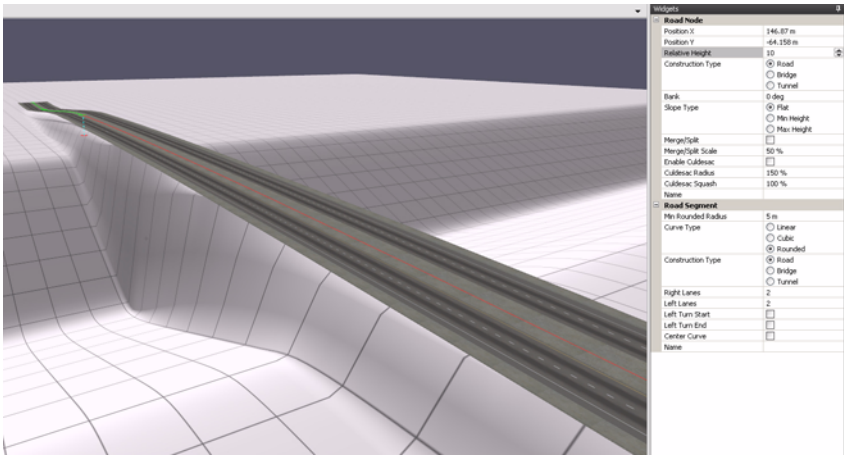
6. If an intersection forms, convert it to a merge: Select the **target node** in the intersection on the freeway, and then enable **Merge/Split** in the Road Node Widgets, as you did a few steps ago.

Build Bridges and Overpasses

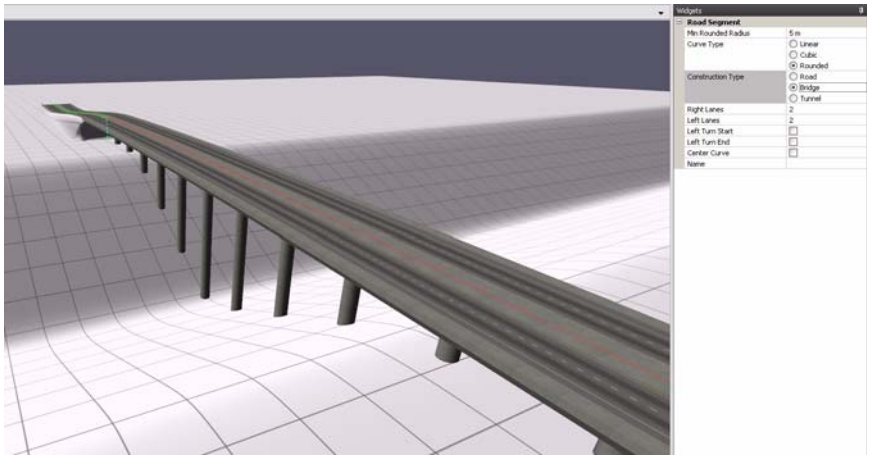
1. Start with the section of freeway, plus merge/split ramp, that you built in the previous section, Create Merges and Splits.

2. Select all the nodes and segments in the freeway, but do not select the nodes or segments unique to the ramp.

3. In the Road Node Widgets, change the *Construction Type* of the selected nodes to **Bridge**, and raise their *Absolute / Relative Height* to **10 meters**. This action elevates the road on a land fill, but does not add the pylons for the bridge until the road segments are also changed to the Bridge Construction Type.



4. In the Road Segment Widgets, change *Construction Type* of the selected segments to **Bridge**. The stretch of freeway becomes a long bridge elevated by pylons.



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More Information on Road Tools

Many additional features and subtleties within the road system are not covered in this short Quick Start. Examples include customizing road styles with imported textures, advanced editing of curves and tangents, creation of tunnels, and more. For details, see “Chapter 6: Using Road Tools” in the CityScape User Manual.

2.3.3 Using Water Tools

The Water Tools let you add global and/or local water to your terrains:

- Using The Global Water Tool, you can submerge everything in your terrain that’s lower in elevation than **Height** elevation in the Main Widgets. Global water is particularly useful for oceans and large lakes.
- Using The Water Polygon Tool, you can draw the shores of lakes at any elevation.

The Global Water Tool

Follow these steps to create and customize global water in your terrain:

1. With a terrain loaded in CityScape, activate the **Global Water Tool** by CLICKING its ICON in the Tools Palette.
2. To turn on water, enable the **Global Water** widget in the Global Water Widgets.
3. To set the elevation of the global water plane, adjust the **Height** widget. Try raising and lowering the water, and adjusting other water widgets.

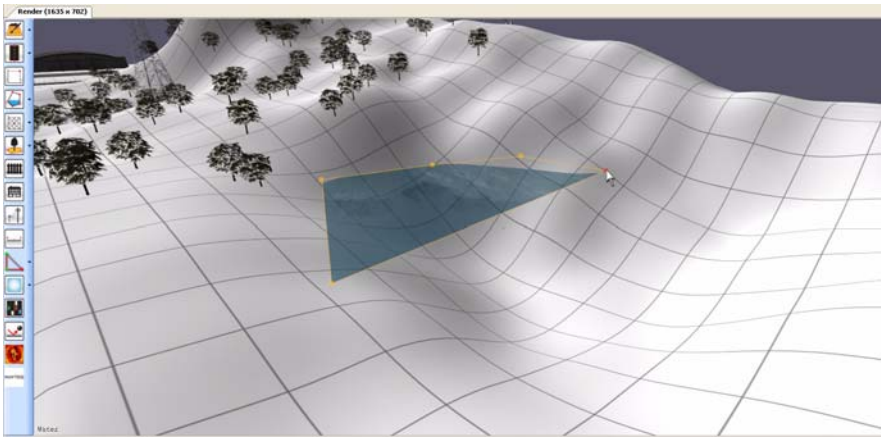
   Tip: *First create some valleys that drop below ground level, so the water has a place to pool. Then use a negative value in the Height widget so the water pools only in these valleys without flooding your entire terrain.*

The Water Polygon Tool

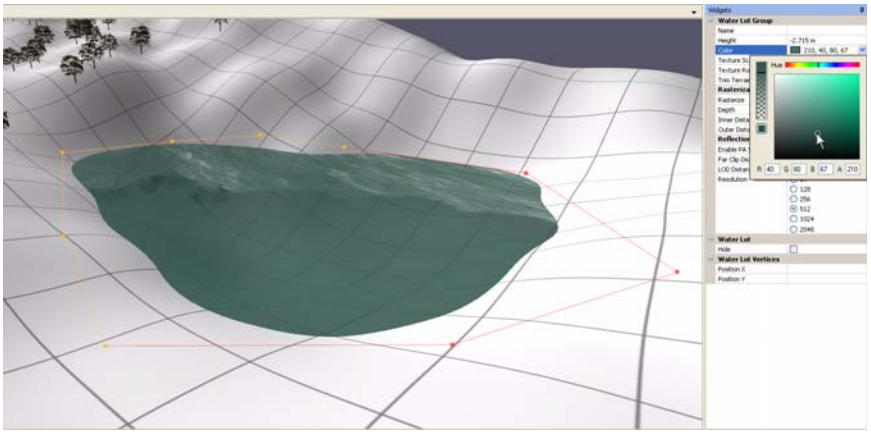
Using the Water Polygon Tool, you can draw the shores of lakes at any altitude in your terrain.

1. With a terrain loaded in CityScape, activate the **Water Polygon Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Water drop-down list to see the icon for the Water Polygon Tool.)

2. RIGHT-CLICK in the terrain a few times to define the shores of your lake. (Using the Water Polygon Tool is almost identical to using The Geography Polygon Tool, except water appears within the borders of water polygons, of course.)



3. To customize a Water Polygon, first select at least one point in the polygon, and then adjust the Main Widgets.



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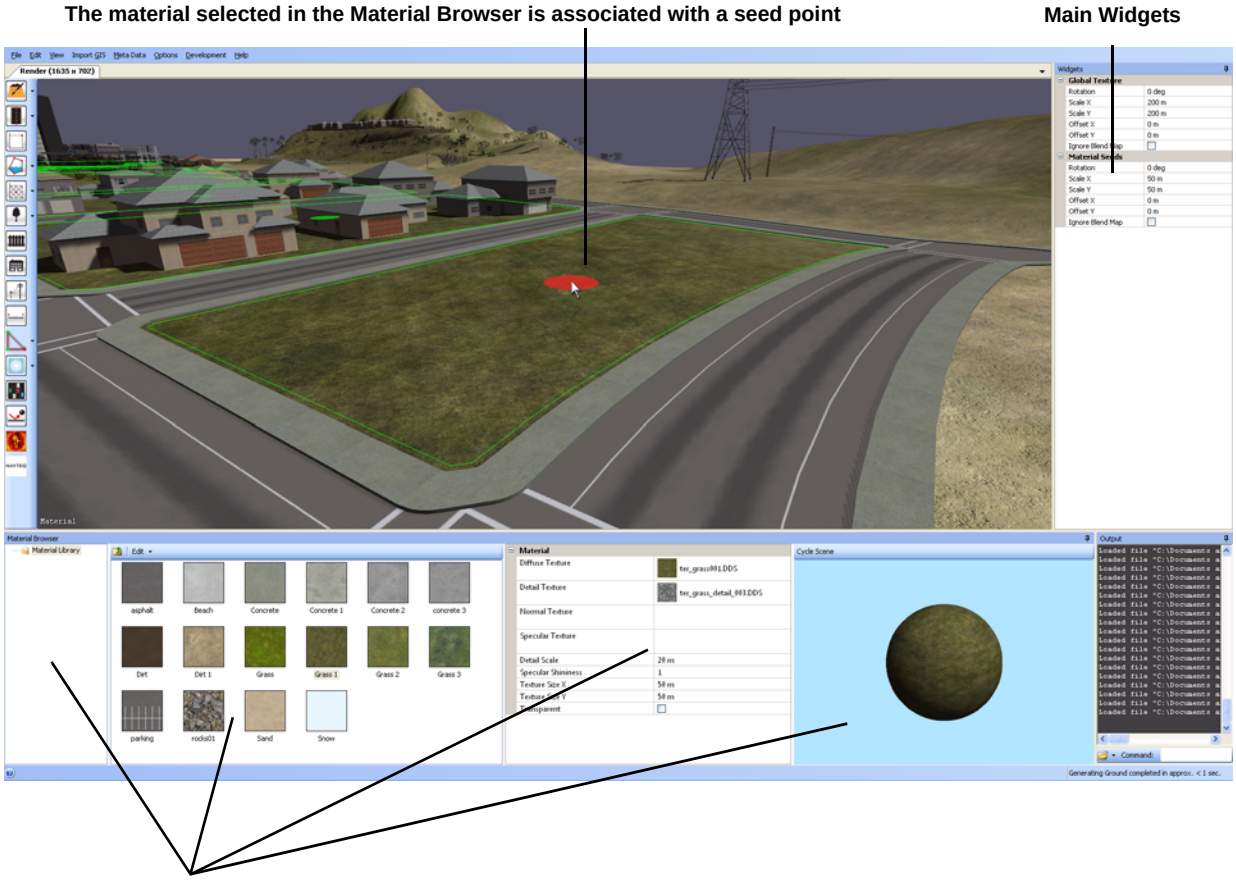
2.3.4 Using Material Tools

Using the Material Browser, you can assign a global material (such as dirt) to the entire terrain. Using the Material Seed Tool, you can assign a different a material (such as asphalt) to a "lot" or closed loop or road.

About the Material Browser

When you select any of the Material Tools from the Tools Palette, the Material Browser appears in the lower part of the CityScape window, by default (see figure below). You use the Material Browser to select materials to map onto geography; to create new materials; and to edit existing materials.



Material Browser showing (from left to right) the Material Tree, Material Icons, Material Sub-Widgets, and Material Preview

To select a material in the Material Library, simply LEFT-CLICK the ICON of the desired material in the Icon View of the Material Browser. A sample of the selected material will appear in the Preview Window.

About Material Sub-Widgets

As with Road Style Sub-Widgets, the Material Sub-Widgets allow you to edit the *global* properties of a material. Your edits will affect all instances of the material in the open terrain, as well as other terrains using the same material library.

The Material Seed Tool

You can use the Material Seed Tool to map any material in the Material Library to a “lot” in the open terrain. (A lot is any region completely enclosed by roads, or a lot polygon created with Lot Tools.) Materials assigned to lots override the global material.

1. A closed loop of road should already exist in your terrain as a result of Using Road Tools. If not, use the Road Tools to create it.

2. Activate the **Material Seed Tool** by CLICKING its ICON in the Tools Palette.

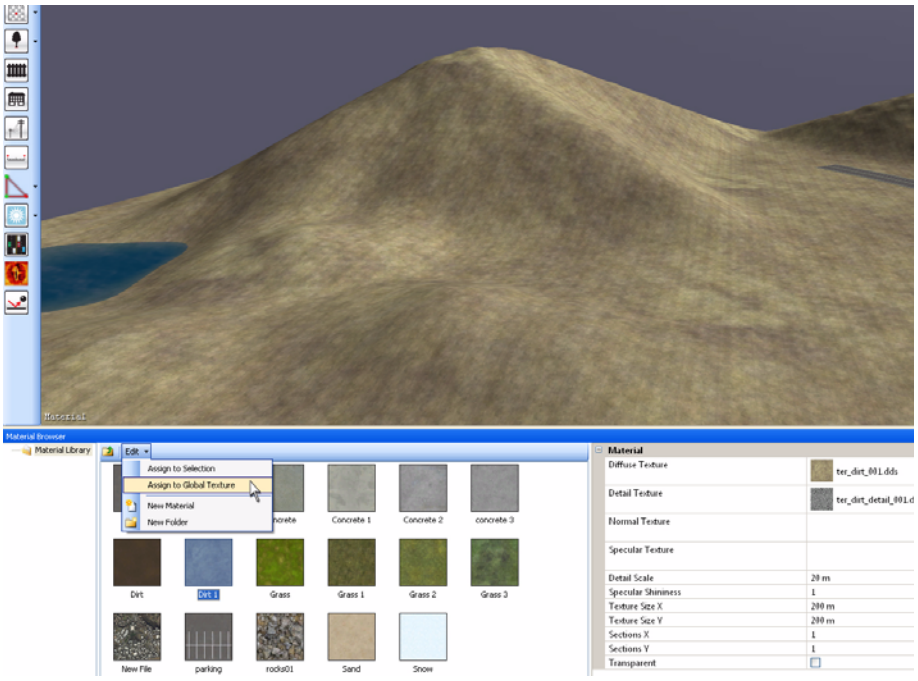
3. To assign a material to the “lot” formed by a closed loop of road, first select the material by LEFT-CLICKING the material’s **icon** in the Material Browser. Then RIGHT-CLICK anywhere in the **target lot**. This action places a material seed point within the lot, assigns the selected material to the seed point, and maps the material to the lot. For an example, see the image in About the Material Browser.

The Global Material Feature

When you map a global material to an entire terrain, the texture will appear on all geography (i.e. mountains, valleys, etc) – except for lots that have specific textures assigned to them. Textures assigned to lots override global textures.

1. Activate the **Material Seed Tool** by CLICKING its ICON in the Tools Palette.

2. Without placing any points on the terrain, LEFT-CLICK the icon of the desired Material in the Material Browser. Then select **Edit > Assign to Global Texture** in the Edit Menu of the Material Browser. (Shortcut: DOUBLE-CLICK the icon of the target material to use it globally in the open terrain.)



Edit Materials

To modify a material, follow these steps:

1. With a terrain loaded in CityScape, activate the **Material Seed Tool**.

2. To select a different material for a lot that already has an assigned material, first LEFT-CLICK the **seed point** in the target lot. Then DOUBLE-LEFT-CLICK the **icon** of the desired material in the Material Browser.

3. To modify the local properties of a material used in a terrain, first LEFT-CLICK the seed point of the material to select it, and then edit its parameters in the *Main Widgets*. Experiment with these widgets.

4. To modify the global properties of a material, LEFT-CLICK the **icon** of the target material in the Material Browser, and edit its parameters in the Material Sub-Widgets. Among other things, you can import textures that get blended to create a complete material.

2.3.5 Using Prop Tools

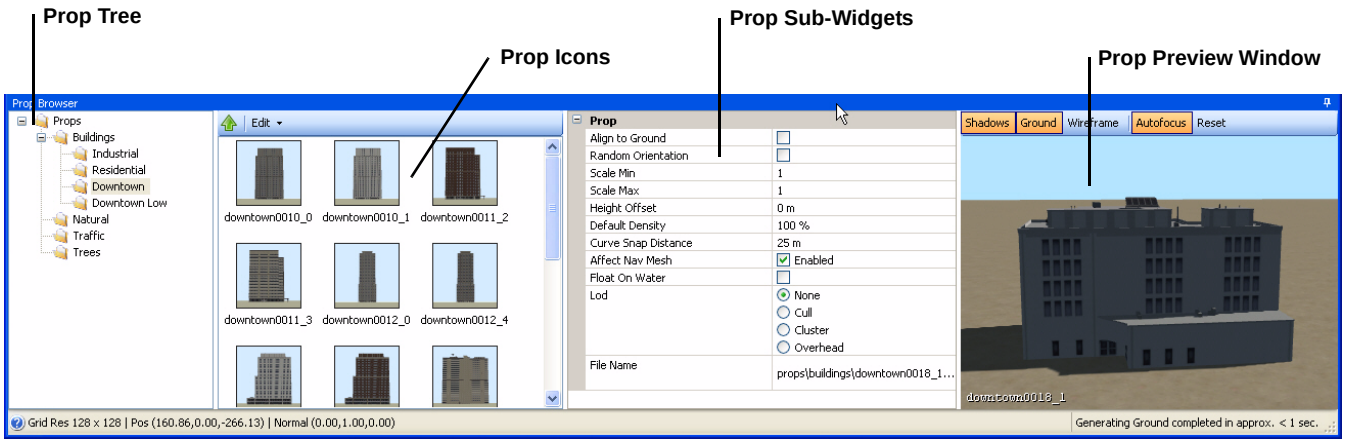
Props are 3D models that you can create and texture in external software such as 3D Studio Max, Creator, and Maya. Examples of props include buildings, trees, mailboxes, street lights, cars, and so forth. Using the Prop Tool Group, you can place props into a terrain and modify their properties.

About The Prop Browser

The Demo Project comes with a Prop Library that includes some buildings, cars, trees, etc. Using the Prop Browser in CityScape, you can select the props in the Prop Library; import, manage, and modify props; and select props to place into terrains.

1. With a terrain loaded in CityScape, activate the **Prop Point Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Point Tool). The Prop Browser appears along the bottom of the window.

2. Using the Prop Browser, explore the contents of the Prop Library in the Demo Project. When you click an icon in the Prop Icons, the Prop Sub-Widgets show its global properties (i.e. the properties shared by all instances of the prop). You will also see a sample of the prop in the Prop Preview Window. See the figure below for an example of the Prop Browser and its panels.



IMPORTING NEW PROPS

With the Full version of CityScape, you can import, manage, and edit new props, as well as save libraries and terrains. (The Demo Version does not allow you to save or export terrains or libraries.) This section gives you basic instructions for importing a prop. For simplicity, these instructions do not document all the secondary factors that pertain to importing complex assets with multiple LODs. For details on these factors, see “Chapter 10: Using Prop Tools” in the CityScape User Manual.

1. Export your prop from the external modeling software that created it.

   Note: *The **exported prop** should be a 3D model in Collada (DAE) format. If the prop includes textures, these textures need to be in a common 2D format such as JPG, and applied to the prop in your external modeling software. When exported from Maya or other creation software, the Collada prop file (DAE format) and the textures need to be exported to the same folder.*

2. Move the exported Collada file for the new prop, together with its textures, into a folder under the project file (CSP) associated with your target CityScape terrains.

3. With a terrain loaded in CityScape, active the **Prop Point Tool** by CLICKING its ICON in the Tools Palette. The Prop Browser appears near the bottom of the screen, by default.

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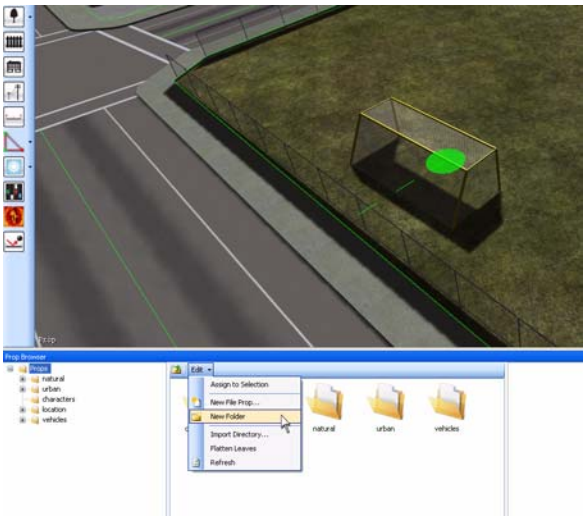
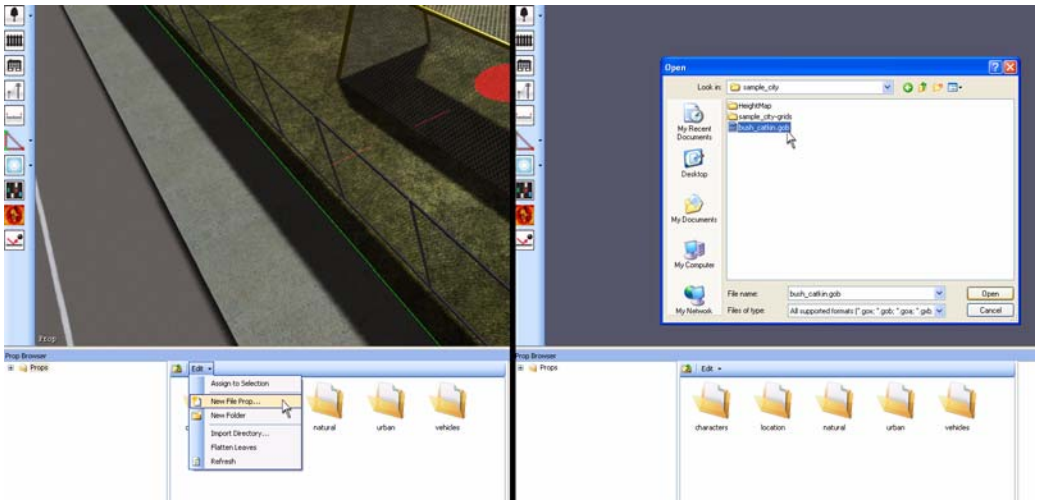
4. In the Prop Tree of the Prop Browser, LEFT-CLICK any **target folder** to contain the new prop, or create a new folder.



   – Note: The folders in the Prop Tree do NOT correspond to the folders on your hard drive, but using the same folder names and tree structure are good ways for you to avoid confusion.

5. Along the top of the Prop Icons Window in the Prop Browser, LEFT-CLICK **Edit > New File Prop**. The icon for a new blank prop appears in the Prop Icon window, and a file browser appears.



6. In the file browser, navigate to the **model file** (DAE) of your target prop and open it. The new prop and its textures will be converted to PixelActive formats and added to the Prop Library in the current project.

Once you've added a prop to the Prop Library, you can place the new prop into any terrain that references the same project file (CSP).

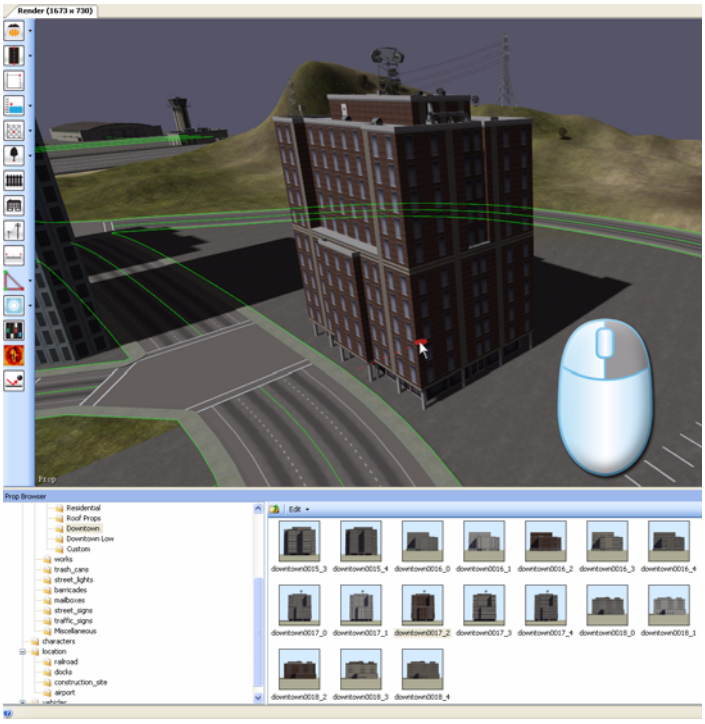
THE PROP POINT TOOL

Place individual props using the *Point Placement Tool*:

1. With a terrain loaded in CityScape, active the **Prop Point Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Point Tool).

2. In the Prop Browser, use the Prop Tree and Prop Icons to locate one of the buildings in the Prop Library. LEFT-CLICK the icon the desired building to select it.

3. To place the selected prop in a terrain, RIGHT-CLICK the target location in the terrain.



Before moving a prop point that already exists in the terrain, or editing its properties using the Main Widgets, you must first select it. To select a Prop Point in a terrain, hover the mouse pointer over the **placement point** of the prop, and then LEFT-CLICK-HOLD and DRAG the MOUSE.

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The Prop Circle Tool

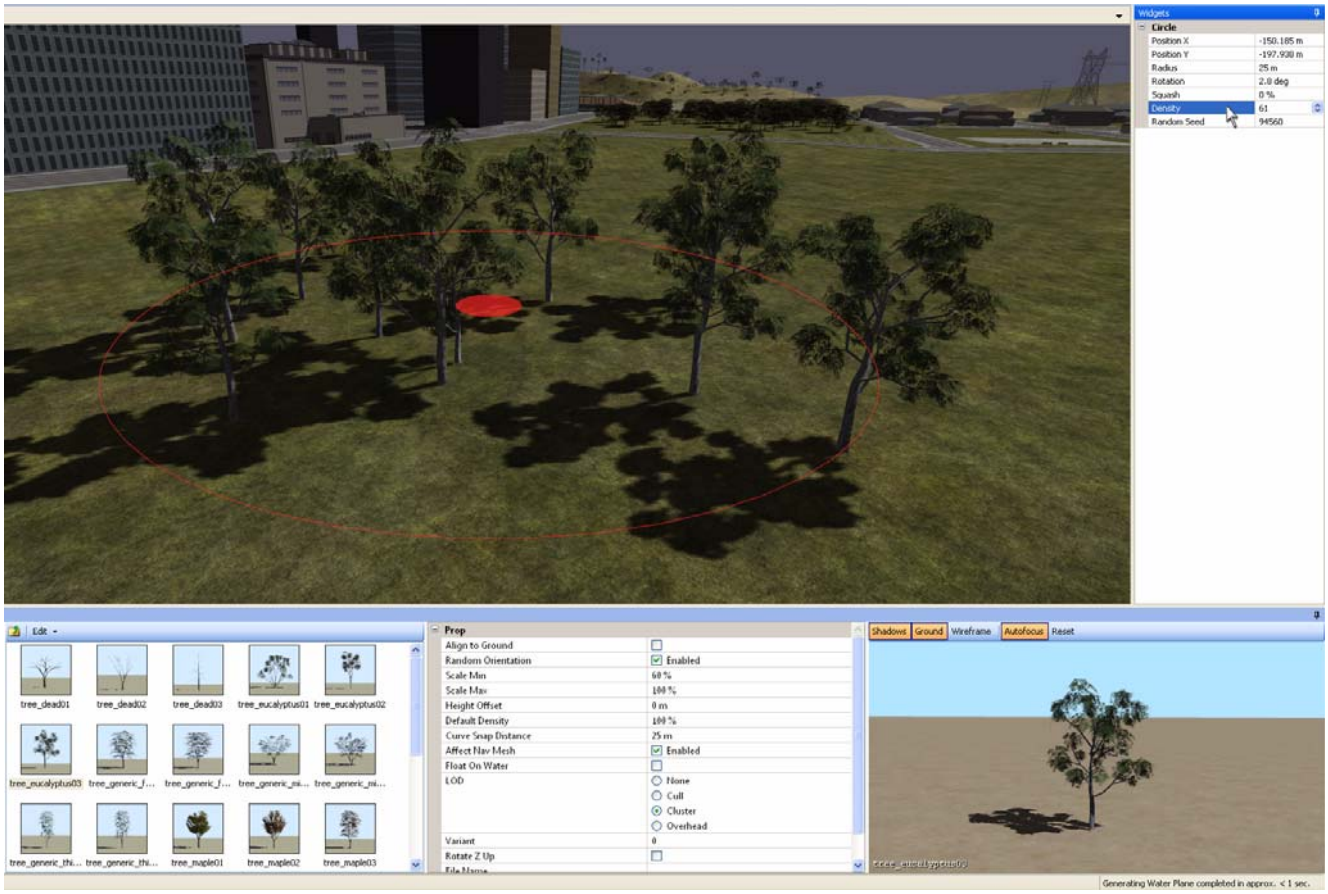
The Prop Circle Tool lets you distribute props in Circle Features, as described below:

1. With a terrain loaded in CityScape, activate the **Prop Circle Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Circle Tool).

2. Use the Prop Browser to locate and select any tree from the Prop Library.

3. RIGHT-CLICK in the terrain to place a Circle Feature in nearly the same way you did with the The Geography Circle Tool. Multiple instances of the tree prop appear within the Circle Feature.

4. LEFT-CLICK the center of the Prop Circle Feature to actively select it, and modify its parameters in the Main Widgets.



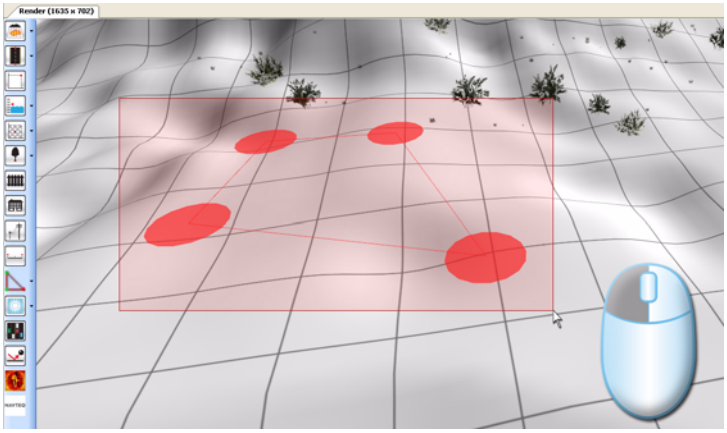
THE PROP POLYGON TOOL

The Prop Polygon Tool lets you distribute props in Polygon Features. Here's how you use this tool:

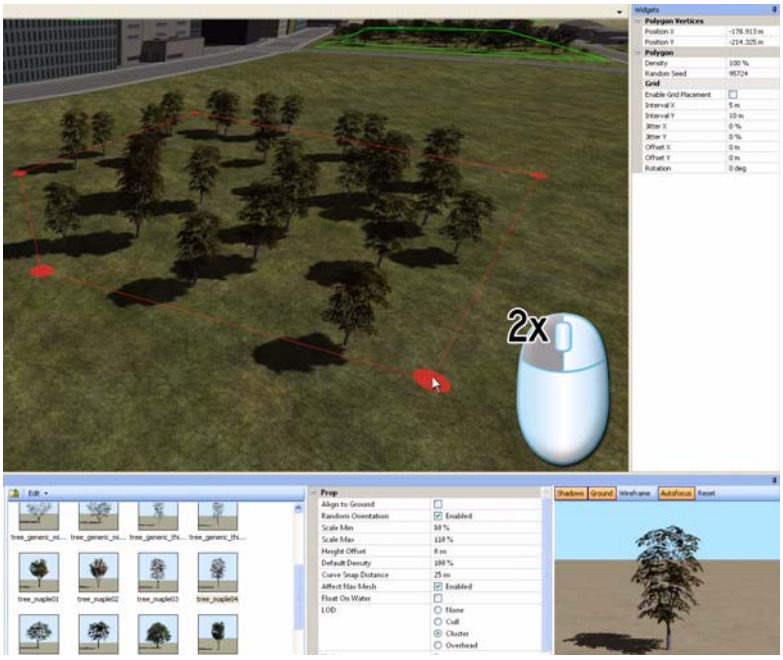
1. With a terrain loaded in CityScape, active the **Prop Polygon Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Polygon Tool).

2. Using the Prop Browser, locate and select any tree from the Prop Library.

3. Place points defining a polygon in the terrain, just as you would with The Geography Polygon Tool. Multiple instances of the selected prop appear within the polygon.



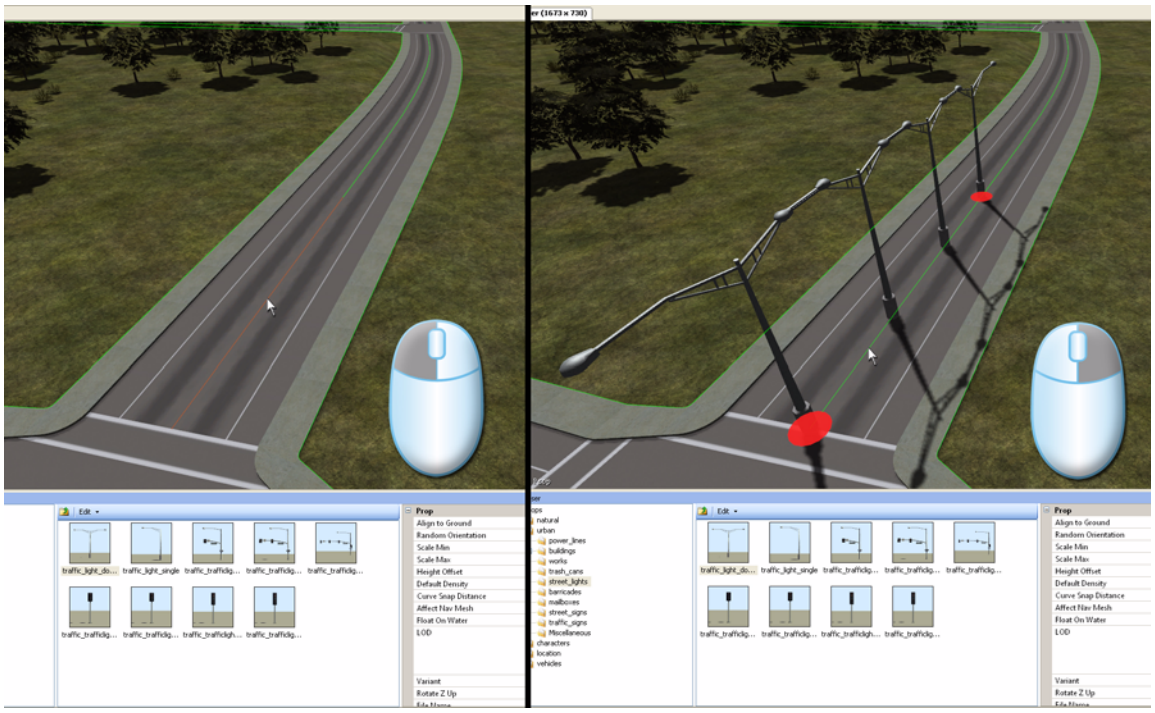
4. To select the entire Prop Polygon, DOUBLE-LEFT-CLICK any point in it. You can use the Main Widgets to modify the local parameters of the actively selected Prop Polygon, and use Prop Sub-Widgets to modify the global parameters of the props within in the Prop Polygon.



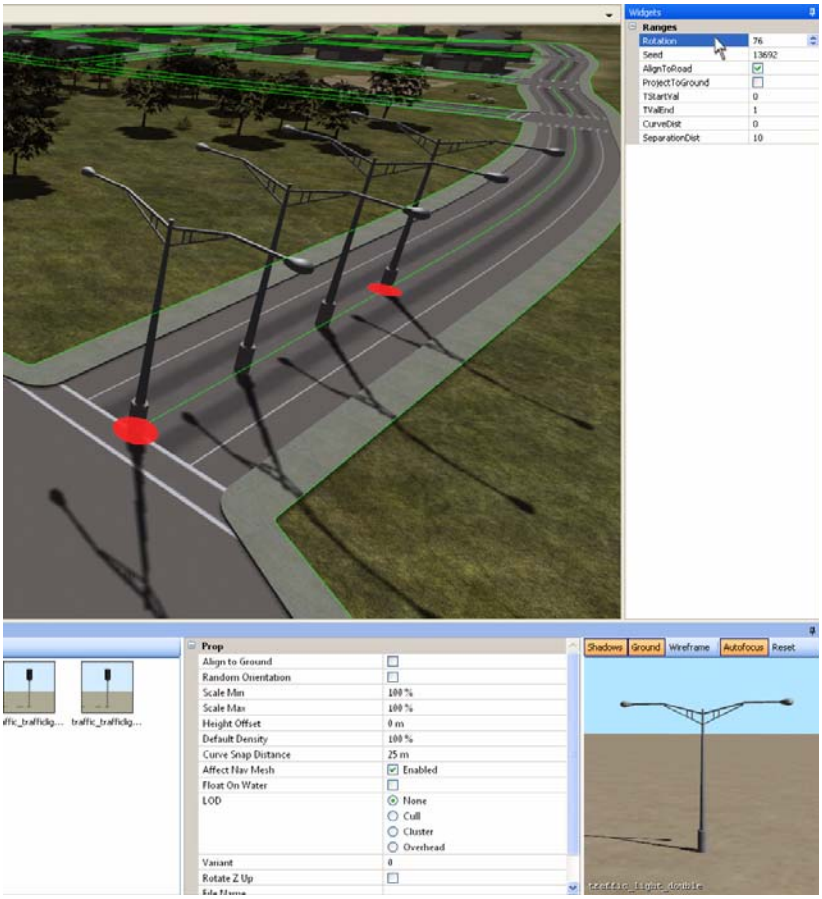
THE PROP RANGE TOOL

The Prop Range Tool is most useful for repeating props, such as trees or street lights, along the sides of roads.

1. Ensure your CityScape terrain contains at least one **road segment**.
2. Active the **Prop Range Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Range Tool).
3. In the Prop Browser, select any street light from the Prop Library.
4. LEFT-CLICK a road segment to select it, and then RIGHT-CLICK that same segment. The prop will appear repeatedly from the start to end nodes of the selected road segment.



5. To select the entire Prop Range Feature, LEFT-CLICK the start or end node of the road segment associated with the prop range. Use the Main Widgets to modify the local parameters of a Prop Range, and use Prop Sub-Widgets to modify the global parameters of the prop used in the Prop Range.



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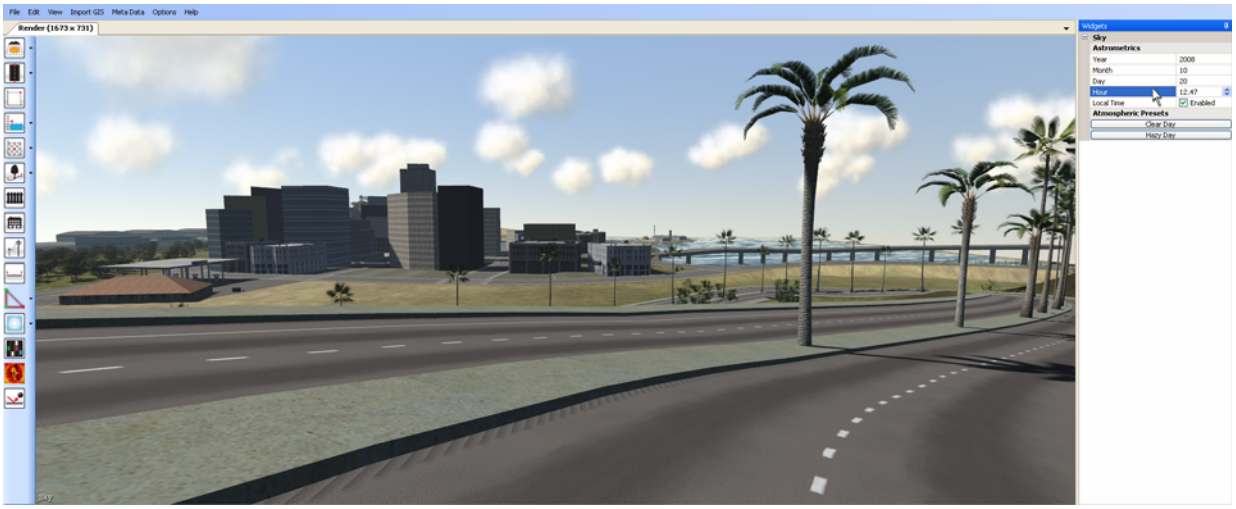
2.3.6 Using Sky Tools

Using the Sky Tools, you can edit atmospheric properties such as clouds and haze, and astronomical properties that affect natural lighting. Based on the date, time, and location you enter, CityScape will accurately determine the positions of the sun and moon, and illuminate the terrain accordingly.

The Sky Tool

1. Activate the **Sky Tool** by CLICKING its ICON in the Tools Palette.

2. Use the Main Widgets to customize the sky:
   – Specify date and time
   – Select between a clear or hazy day



If you want the settings in the Sky Widgets to position the sun accurately for a specific time, date, and location, the terrain must be geo-located using latitude and longitude. For details on geo-locating a new terrain, see Create a New Geo-Located Terrain. To geo-locate an existing terrain, follow the steps below:

1. Select **Options > Terrain Settings** in the Main Menu. The Terrain Settings dialog appears.

2. Fill in the **Geolocated Terrain** side of the Terrain Settings dialog just as you would if you were creating a new geo-located terrain.

For background information on geo-located terrains and related subjects, such as Shapefiles and elevation files, see “Chapter 21: Importing GIS Data” in the *CityScape User Manual*.

The Clouds Tool

1. Activate the **Clouds Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Sky drop-down list in the Tools Palette to see the Clouds Tool).

2. Experiment with the Main Widgets to customize the sky.

2.3.7 Using GIS Tools

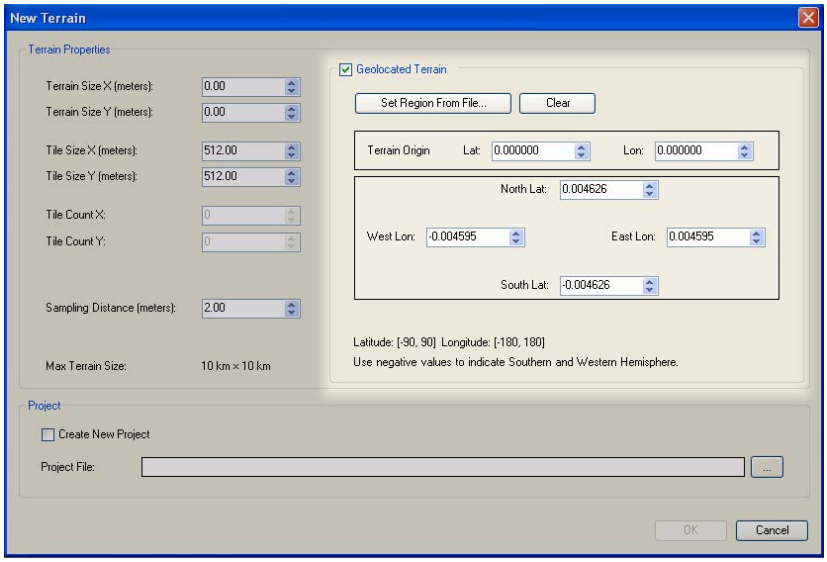
This section provides you with a simple case of importing GIS data into CityScape in the following phases:

1. Create a New Geo-Located Terrain
2. Import Elevation Data
3. Import Roads, Buildings, Water, and Props
4. Import an Orthographic Image

If you are unfamiliar with geo-specific data, read the background information in “Chapter 21: Importing GIS Data” in the *CityScape User Manual*.

Create a New Geo-Located Terrain

1. PRESS **Ctrl-N** to create a new terrain. The New Terrain dialog appears.
2. In the New Terrain dialog, LEFT-CLICK the **Geolocated Terrain** check box to enable the right side of the dialog, which pertains to the geo-location of the new terrain.



3. Select the **Demo Project** by LEFT-CLICKING the **Project File [...]** button near the bottom, and loading the target **demo.csp** file located in the Worlds folder, which is installed here by default:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds

or here for WinXP users:

C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds

4. Set the geo-location of the new terrain as follows:
   i. LEFT-CLICK the **Set Region from File** button.
   ii. In the dialog that appears, set *Files of Type* to **All Files**. Then locate and load the **lajolla_height.tif** elevation file in the *Sample GIS* folder under the main *Worlds* folder (see previous step).

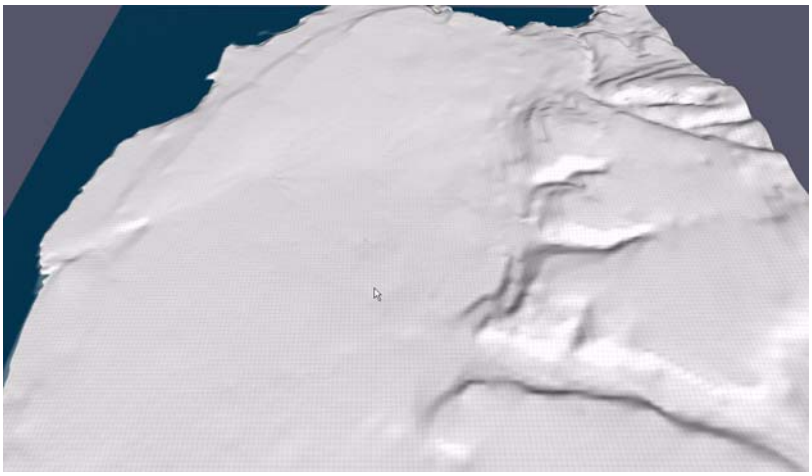
5. LEFT-CLICK **OK** to create a blank terrain that's geo-located. You can now import geo-specific data into this terrain.

Import Elevation Data

In a typical workflow, you import elevation files before importing any other types of geo-data:

6. To import an elevation file, select **Import GIS > Import Elevation > GeoTiff** from the Main Menu. (Note: CityScape can also import elevation files in DEM, DTED, and GridFloat formats).

7. In the Import Elevation file loader, locate and load the same **lajolla_height.tif** elevation file you just used to geo-locate the terrain. The geography in the terrain will be updated to match the height field you just imported.



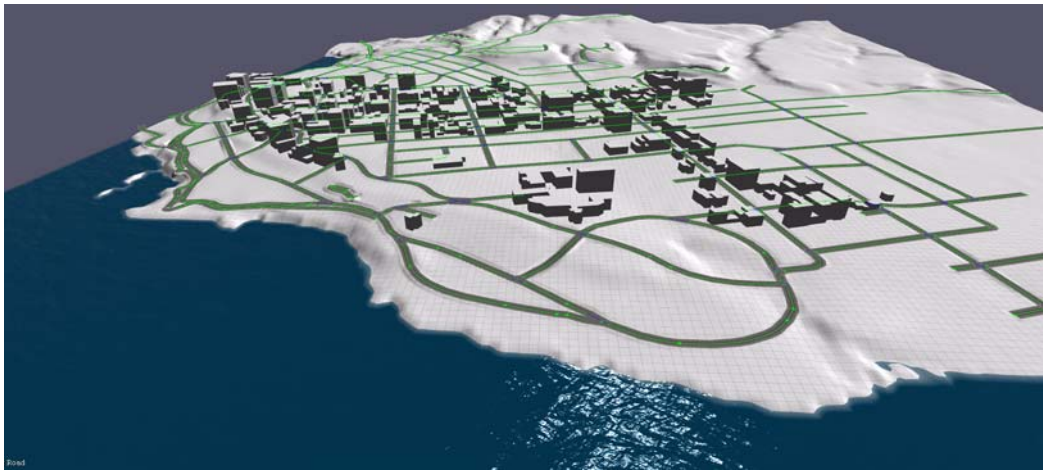
Import Roads, Buildings, Water, and Props

GIS data for roads, buildings, water, and props in the same region would require 12 different Shapefiles, or in one OpenStreetMap file. For simplicity, these instructions describe how to import an OpenStreetMap file:

8. To import an OSM data file, select **Import GIS > Import OpenStreetMap** from the Main Menu. The *Import OpenStreetMap* dialog appears.

9. In the *OpenStreetMap File* tab of the dialog:
   – Locate and load the **lajolla.osm** file in the same folder as the elevation file you just loaded.
   – Select the *Features to Import* (normally all).

10. In the *OpenStreetMap File* tab, LEFT-CLICK **OK** to finish the import.



IMPORT AN ORTHOGRAPHIC IMAGE

You can optionally import orthographic imagery in the same region, so as to compare a satellite photo with the 3D data you just imported:

11. Select **Import GIS > Import OrthoImagery** from the Main Menu. The *Import Ortho Image* dialog appears.

12. In the dialog, locate and load the **lajolla_ortho.tif** orthographic image file. The ortho image overlays your terrain. You can show or hide the ortho image with the **View > Show Overlay** toggle in the Main Menu.

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Chapter 3: User Interface

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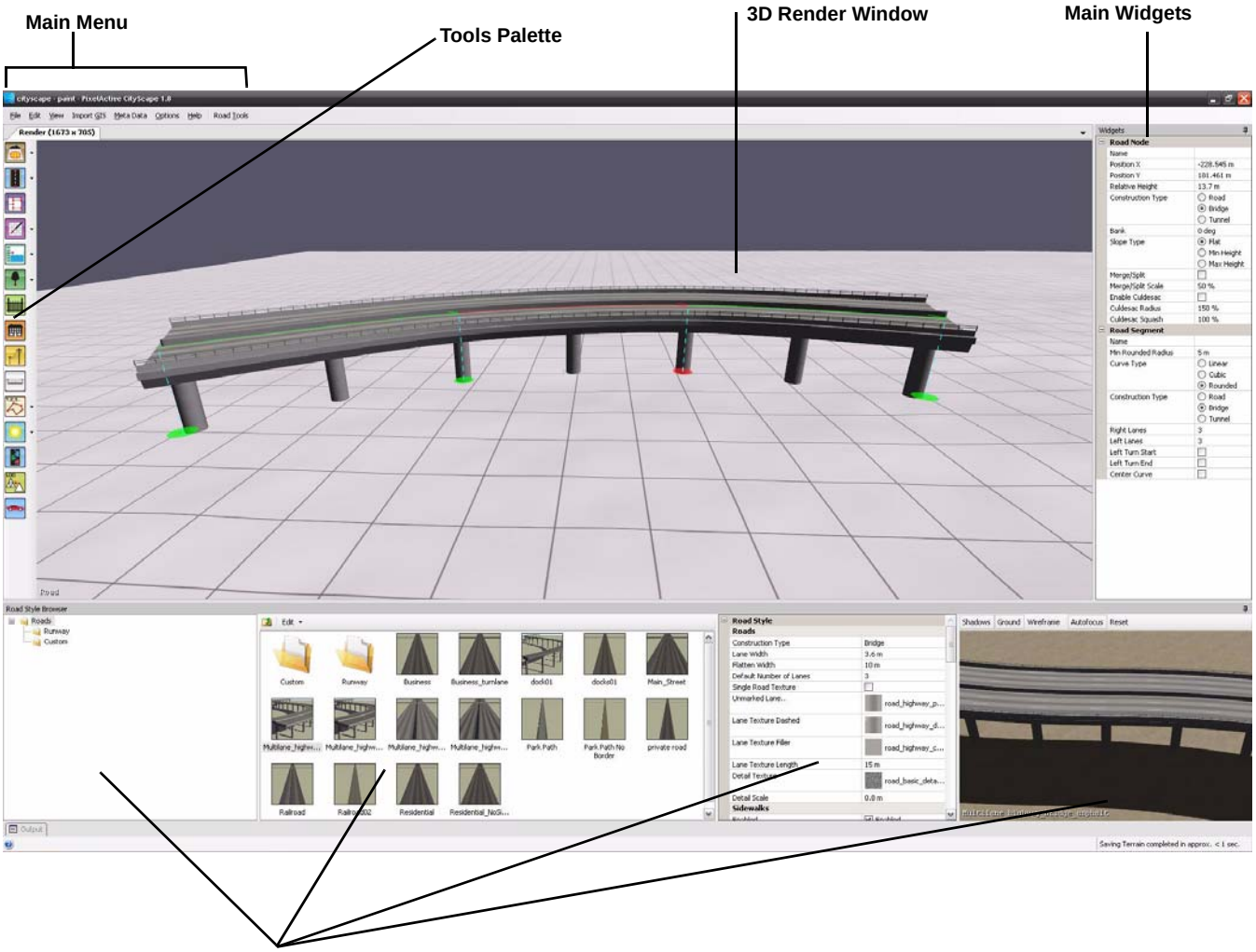
3.1 THE USER INTERFACE IN DETAIL

This chapter describes every element in CityScape's user interface (UI), and provides links to usage instructions elsewhere in this document. The information in this chapter assumes that you are using the default UI layout, as pictured below.



You can adjust the size of each UI element by dragging its borders with the mouse. You can also detach each element and move it around. To reset all the UI elements to their default sizes and locations, close CityScape and then hold the **SHIFT-KEY** while restarting CityScape.

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For details on the major elements in the UI, see the following sections:

- Main Menu provides global functions as well as tool-specific functions.
- Tools Palette provides quick access to the powerful tools in CityScape. Some closely related tools are grouped under a parent icon in the Tool Palette. LEFT-CLICK the main icon of a Tool Group to expand a drop-down list of individual tools.
- 3D Render Window shows the open terrain in real-time.
- Main Widgets Panel shows the editable properties of the selected item.
- Library Browser appears only when a tool is active that needs library items (e.g. Prop Tools, Road Tools, etc). Some tools do not use any library items, and therefore have no associated Library Browser.

3.1.1 Main Menu

The Main Menu bar resides across the top of the screen. The Main Menu includes the following top level pulldowns that, along with the tools, provide you with key functionality in CityScape:

- File
- Edit
- View
- Import GIS
- Meta Data
- Options
- Help
- Road Tools (appears only when Road Tools are active)
- Water Tools (appears only when Water Tools are active)
- Fence Tools (appears only when Fence Tools are active)
- Building Tools (appears only when Building Tools are active)

Although these sub-sections summarize each menu option, these summaries do not include details on how each option is used in the context of typical workflow.

FILE

The File Menu gives you the following options:

- **New Terrain (Ctrl-N):** Creates a new terrain. Allows you to define the terrain's size, number of tiles, geo-location, and association with an existing or new project file. For general usage, see "Create a New Terrain" on page 76. For GIS-specific usage, see "Geo-Locating Your Terrain" on page 201.
- **Open Terrain (Ctrl-O):** Opens an existing terrain of your choice.
- **Save Terrain (Ctrl-S):** Saves the terrain you have open by overwriting the terrain file (PixelActive CST format) and the project file (in PixelActive CSP format)
- **Save Terrain As:** Saves the terrain file you have open along with its associated project file. You can specify the filename and location of your choice for the terrain file.
- **Save Project:** Saves just the project file (in PixelActive CSP format) associated with the open terrain.
- **Change Library Asset Path:** If you move a project file to a new location, or move the source files for the items in the project's libraries, then CityScape will not be able to find the library items when you open a terrain using any of them. CityScape may prompt you to find the library items, or you can use the *Change Library Asset Path* option to update the path manually.
- **Export:** Provides options for exporting everything in a terrain, or just selected elements, to common formats such as Collada. For details, see "Chapter 22: Exporting Terrains" on page 217.
- **Recent Terrain Files:** Lists the last few terrains you have worked with. Selecting a terrain from this list opens that terrain.
- **Exit:** Quits CityScape.

EDIT

The Edit Menu provides common editing options such as undo, redo, cut (Ctrl-X), copy (Ctrl-C), paste (Ctrl-V), and delete (Delete Key). The undo option can take you back quite a few steps.

- **Enable Grid Snap:** When enabled, objects snap to the geography grid when you move them.

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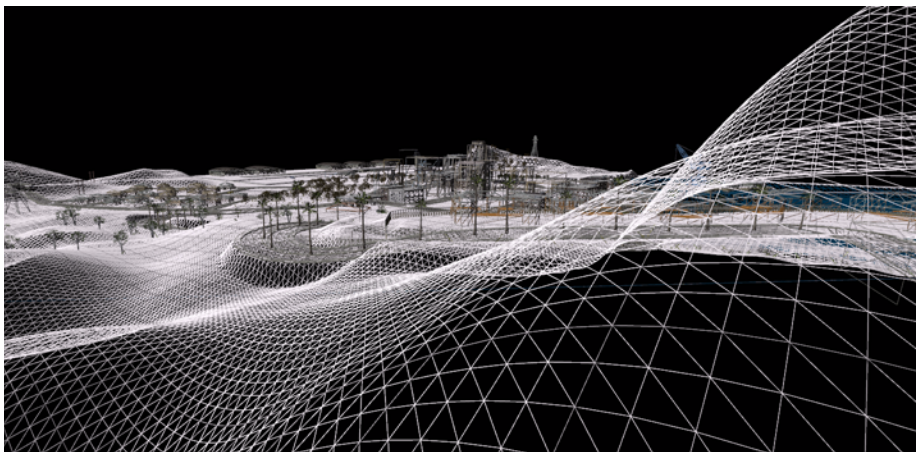
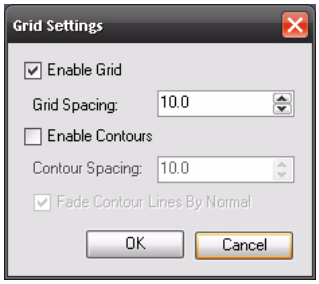
VIEW

The View Menu lets you select between cameras and toggle realtime rendering options that change slightly depending on the active tool.

- **Cameras:** Lets you select between the perspective and orthographic cameras. For details, see “Using the Camera” on page 14.
- **Wireframe:** Shows the terrain as naked 3D meshes without lighting, textures, or materials.



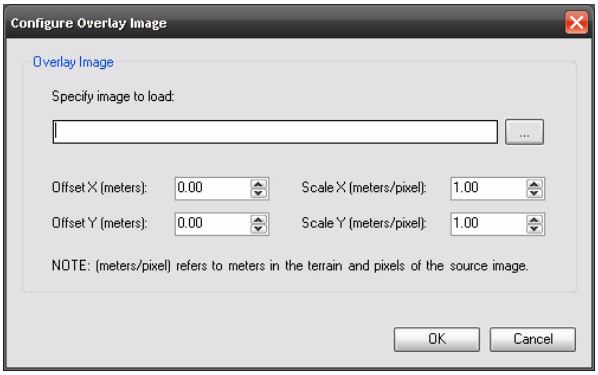
- **Shadows:** When any Geography Tool is active, CityScape can update the display of the geography more quickly if you turn off shadows.
- **Sky Shadows:** Toggles the ability of clouds to cast shadows on terrains.



- **Grid Settings:** Displays the Grid Settings dialog with options for defining and displaying the Geography Grid.
  - *Enable Grid:* Toggles grid lines at the spacing you specify.
  - *Enable Contours:* Toggles topographic lines showing altitude.
- **Terrain Boundary:** Toggles visibility of the edges of the terrain.
- **Show Overlapping Road Meshes:** Toggles the orange triangles that warn you when road geometry overlaps (due to a corner that’s too tight, or any other reason).
- **Show Overlay:** Toggles the visibility of an ordinary 2D image that you load via the *Configure Overlay* dialog (see below). When visible, the image is projected on the terrain.
  - To load geo-located orthographic photos, see ”Chapter 21: Importing GIS Data” on page 195.

- **Export Board:** Shows / hides visible borders and ID numbers of the tiles that your terrain will be divided into when you export it. Includes the magenta terrain boundary at the outer edges of the terrain.
  - Note: The outermost tiles will extend beyond the edges of your terrain in cases when the tile size does not divide evenly into the terrain size. The terrain boundaries use a magenta border, and the tiles boundaries use a light green border. Example: If the terrain size is 1000 square meters, and the tile size is 256 meters square, then CityScape generates a 4 x 4 grid of tiles with a total size of 1024 meters.

- **Configure Overlay:** Displays a dialog in which you can select any 2D image in a standard format, and CityScape will project the image onto your terrain. Parameters:
  - *Offset (X, Y)* controls the distance (in meters) between the center of the image and the center of the terrain. The default Offset = (0,0), which places the center of the overlay in the center of the terrain.
  - *Scale (X, Y)* controls the size of the image in meters/pixel, which refers to the proportion of meters in the terrain to pixels in the source image. The default Scale = 1 means each pixel in the source image is projected across one square meter in the terrain.



IMPORT GIS

This menu lets you import the following types of geo-specific data into the terrain. (For details, see "Chapter 21: Importing GIS Data" on page 195.

- **Import Elevation:** Use to import geo-located elevation files (sometimes called "height maps"). CityScape converts elevation files into 3D hills and valleys.
- **Import Shapefile:** Use to import geo-located data for roads, buildings, water, and props. (Note: Shapefiles and OpenStreetMap files are mutually exclusive.)
- **Import OpenStreetMap:** Use to import geo-located data for roads, buildings, water, and props. (Note: Shapefiles and OpenStreetMap files are mutually exclusive.)
- **Import Ortho Imagery:** Use to import geo-located orthographic imagery, such as satellite photos. CityScape projects the orthographic image onto the terrain, provided **View > Show Overlay** is enabled.

META DATA

The Meta Data Menu lets you define and assign meta data (i.e. arbitrary ascii data) to certain types of objects in the terrain. For details, see "Chapter 18: Using Meta Data & Abstract Data Tools" on page 171.

- **Edit Meta Data Templates:** lets you define custom types and default values of meta data, and organize your meta data into templates.
- **Assign Meta Data Templates:** lets you assign meta data templates to buildings, water polygons, power lines, and abstract objects created with the Abstract Data Tools. You can enter specific values for each meta data element separately per assigned object.

OPTIONS

The Options Menu contains miscellaneous options such as the ability to change key Terrain Settings (e.g. the size, geo-location, and tile breakdown of the open terrain).

- **Terrain Settings:** lets you change the size, geo-location, associated project file, and fundamental properties of the open terrain. For usage, see "Terrain Settings" on page 188.
- **Clear Dead References:** Pertains to meta data, among other things. For details, see "Fixing Dead References" on page 180.

HELP

The Help Menu gives you information about the specific build number of your version of CityScape, as well as links to this User Manual and the video tutorials on PixelActive's web site.

ROAD TOOLS

When you activate any Road Tool, the Road Tools Menu appears to provide you with additional road-specific options, as explained in "Road Tools Menu" on page 84.

WATER TOOLS

When you activate any Water Tool, the Water Tools Menu appears to provide you with the following options. (For additional information, see “Using Water Tools” on page 39.)

- **Select Water Lot Group:** actively selects all the nodes and connecting segments in the target water polygon, or group of polygons. Shortcuts:
  - DOUBLE-CLICK any node in a water polygon to select all the nodes in that polygon.
  - TRIPLE-CLICK any node in a group of water polygons to select all the nodes in the group.
- **Group Water Lots:** Groups all the water polygons you have actively selected in the open terrain. Internal polygons in a group can be islands.
- **Ungroup Water Lots:** Ungroups all the water polygons you have actively selected in the open terrain. (See *Group Water Lots* above.)

FENCE TOOLS

When you activate the Fence Tool, the Fence Tools Menu appears to provide you with the following options:

- **Link Segments:** Links two road segments that share the same node. (Even though the segments share a node, when unlinked they form a a sharp angle at the node. When linked, they form a smooth curve through the node. Linking also affects intersections.)
- **Unlink Segments:** Unlinks two road segments that share the same node. (See “Link Segments” above.)
- **Swap Seg Direction:** Reverses the *left* and *right* designations for the sides of the road.
- **Orient Linked Segments:** Corrects a situation where two segments sharing the same node have opposite direction vectors, such as with one-way streets pointed into each other.
- **Select All:** Selects all fence nodes and segments in the open terrain.
- **Select Linked:** Selects all segments in the open terrain that are linked to the segment you have manually selected in the terrain.
- **Select Lone Nodes:** Selects randomly placed nodes with no associated segments, so you can get rid of them.

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Building Tools

When you activate the Building Tool, the Building Tools Menu appears to provide you with the following options:

- **Select Entire Building:** actively selects all the nodes and connecting segments in the target building (shortcut: DOUBLE-CLICK any node in the building polygon). If the building is composed of a group of polygons, CityScape selects all the polygons in the group (shortcut: TRIPLE-CLICK any node in any building polygon in the group).
- **Group Polygons:** Groups all the building polygons you have actively selected in the open terrain. Notes:
  - Grouped polygons use the same building style (i.e. applying a style to one polygon applies the style to all the polygons in the group).
  - Internal polygons in a group can be courtyards or multi-level terraces.
- **Ungroup Polygons:** Ungroups all the building polygons you have actively selected in the open terrain. (See *Group Polygons* above.)

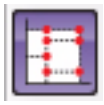
3.1.2 Tools Palette

The Tools Palette contains icons representing the major tools in CityScape. To active a tool, LEFT-CLICK the icon of the tool in the Tools Palette, which appears on the left of the 3D Render Window by default. In the case of tool groups, LEFT-CLICK the drop-down menu next to the icon in the Palette, and select the desired tool from the list.

The following Tool Groups are available in CityScape:

- **Geography Tools:** Activates tools for building 3D geography, such as potentially complex hills and valleys. For an introduction to usage, see “Using Geography Tools” on page 18 in the Quick Start. A future chapter will document usage in detail.

- **Road Tools:** Activates tools for building roads. Displays the Road Style Browser, as well as the *Road Tools Menu* with the options below. For an introduction to usage, see “Using Road Tools” on page 25 in the Quick Start. For details on usage, see “Using Road Tools” on page 83.



- **Lot Tool:** Activates tools for designating lots. A “lot” is an area of the terrain, such as a city block, that’s completely enclosed by roads, or by polygons that you draw with the Lot Tool. For an overview of using materials with lots, see “Using Material Tools” on page 41. A future chapter will document usage in detail.

- **Material Tool:** Activates tools for importing textures and mapping them to geography. Displays the Material Browser as well. For an overview of usage, see “Using Material Tools” on page 41 in the Quick Start. A future chapter will document usage in detail.

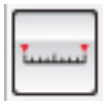
- **Water Tools:** Activates tools for configuring water in your terrain. For an overview of usage, see “Using Water Tools” on page 39 in the Quick Start. A future chapter will document usage in detail.

- **Prop Tools:** Activates tools for importing props, placing them in your terrain, and editing their properties. Displays the Prop Browser as well. (Props are 3D objects such as buildings, trees, and cars that you create in external 3D modeling software and import into CityScape.) For an overview of usage, see “Using Prop Tools” on page 43 of the Quick Start Tutorial. For details on usage, see ”Chapter 10: Using Prop Tools” on page 133.

- **Fence Tool:** Activates tools for building fences in your terrain, and opens the Fence Style Browser. A future chapter will document usage.

- **Building Tool:** Activates tools for creating simple buildings directly in your terrain, and opens the Building Style Browser. (The buildings you create with the Building Tool are different than props which take the form of buildings.) A future chapter will document usage in detail.

- **Powerline Tool:** Allows you to create a network of power lines. A future chapter will document usage in detail.



- **Measure Tool:** Lets you measure between any two points you place in the open terrain. Tells you the distance in 3D space between the points, the projected distance on the ground plane, and the net elevation change (i.e. vertical distance, slope, and vertical angle).

- **Abstract Data Tools:** Activates tools for importing or specifying arbitrary data and associating it with elements in CityScape terrains. For details, see “Using Meta Data & Abstract Data Tools” on page 171.

- **Sky Tools:** Activates tools for modifying the sky and lighting of your terrain. For an overview of usage, see “Using Sky Tools” on page 51 in the Quick Start. A future chapter will document usage in detail.

- **Traffic Tool:** Activates tools for configuring traffic on the roads in your terrain. A future chapter will document usage in detail.

- **LOD Tool:** Provides options for previewing and editing LOD transitions of tessellated terrain geometry. For details on usage, see “Setting LOD Transitions for Entire Terrains” on page 190.

- **Simulation Mode:** Lets you move around a terrain with a sample vehicle that you previously placed in Prop Mode. You can use one of the supplied vehicles in the Prop Library, or you can import your own just like any other prop. You must also attach a controller to your computer. We recommend using the Xbox 360 controller.

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3.1.3 3D Render Window

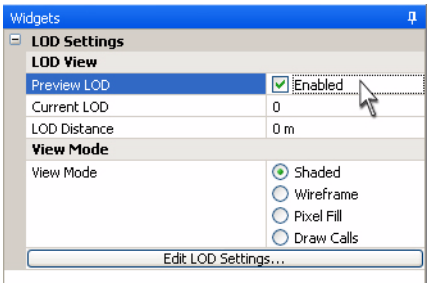
The large Render Window displays a 3D real-time view of the loaded terrain. In the View, you can change various options that affect the display of the terrain in the Render Window.

3.1.4 Main Widgets Panel

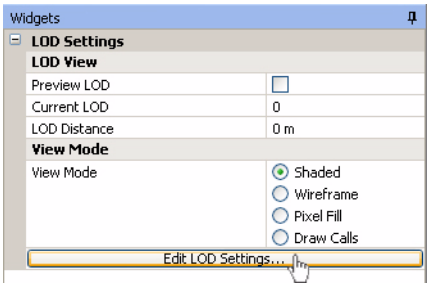
The Main Widget Panel appears on the right side of the screen by default. The Main Widgets give you quick access to functionality that complements the active tool. The parameters in the Main Widgets usually change whenever you switch active tools, or select different types of objects with the same tool.

The Main Widgets use a fairly standard layout and functionality. You can edit the following types of parameters Using the Slider Widgets.

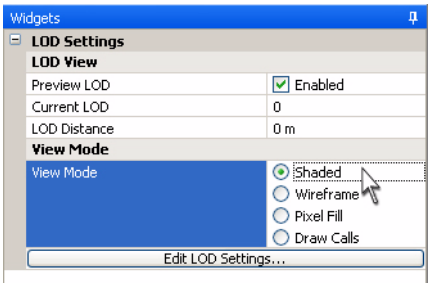
- **Check Boxes:** LEFT-CLICK the box to toggle a boolean value on and off.



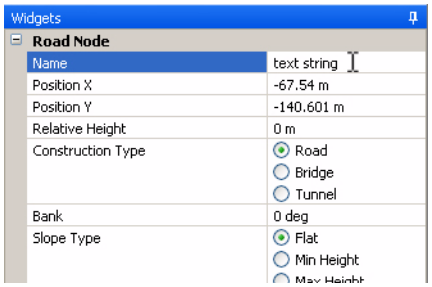
- **Buttons:** LEFT-CLICK the button to trigger an action or display a dialog box.



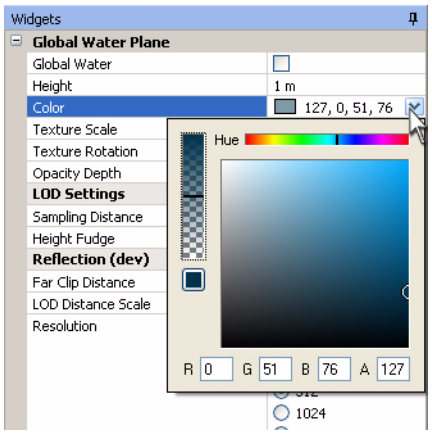
- **Radio Fields:** LEFT-CLICK the desired field to select from mutually exclusive options.



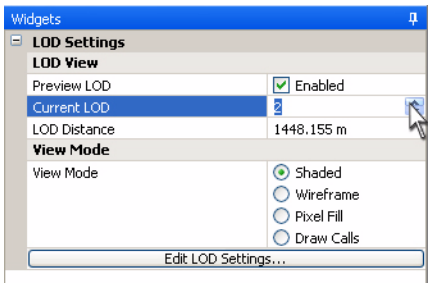
- **Text Strings:** LEFT-CLICK the field and type some text.



- **Colors:** LEFT-CLICK the drop-down arrow and select the desired color from the Hue dialog, or type in an RGBA value.



- **Integers and Real Numbers:** Edit using any of the following techniques. (Note that some widgets are clamped to a range of values that you cannot exceed.)



  - LEFT-CLICK the number and type in a new value.
  - LEFT-CLICK the widget to display up and down arrows, and then LEFT-CLICK the up-down arrows to raise and lower the value.
  - LEFT-CLICK-HOLD the name of the widget and drag the mouse left or right. This technique, called Slider Widgets, is explained in more detail below.

USING THE SLIDER WIDGETS

CityScape features Slider Widgets. You can change the values of these widgets in three different ways. Regardless of the method you use to adjust a widget, the final value will always be clamped between predefined min and max values. You can't adjust these min and max values.

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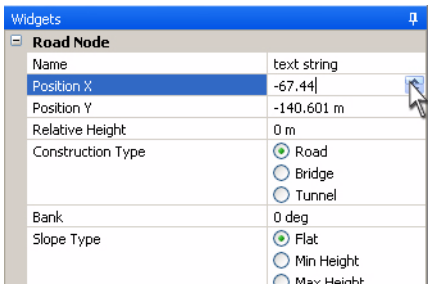
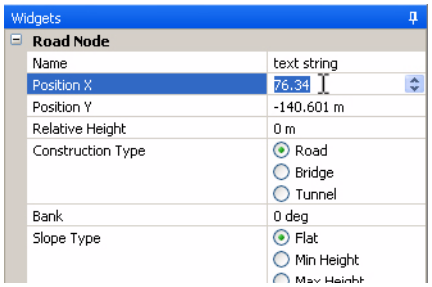
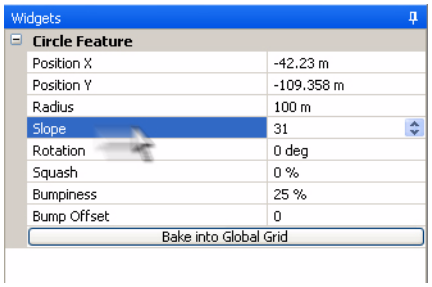
1. To increase or decrease the value of a widget by a single step, LEFT-CLICK the **Up/Down buttons** that appear to the right of the widget after you click in its value field.



2. To type in an exact value, LEFT-CLICK the **Current Value** of the widget, and then type the **New Value**.



3. To smoothly scroll the value of a specific widget, LEFT-CLICK-HOLD the **Name** of the widget and DRAG the MOUSE.



   – For example, to scroll the value of the “Slope” widget for a Circle Feature, LEFT-CLICK-HOLD on the word “Slope” in the Widget Panel (see figure below), and DRAG the MOUSE left or right. While you drag, the cursor disappears so it won’t stop at the edge of the screen.
   – When scrolling, the value changes by a step size that is appropriate to that widget. For example, if the range of a value is from 0 to 1, the step size is typically fixed at 0.01. For a widget with a range -100 to + 100, the step size would be a lot larger.
      » HOLD the SHIFT-KEY while scrolling to increase the step size by 10x the default step.
      » HOLD the CTRL-KEY while scrolling to decrease the step size to 1/10th the default.

3.1.5 Library Browser

When you activate a tool that uses the Library Browser, a tool-specific version of the library Browser appears at the bottom of your screen, by default. You can use the Library Browser to select items in the libraries for your project, and place selected in terrains. You can also the Library Browser to import new items, modify the global properties of items, and manage items.

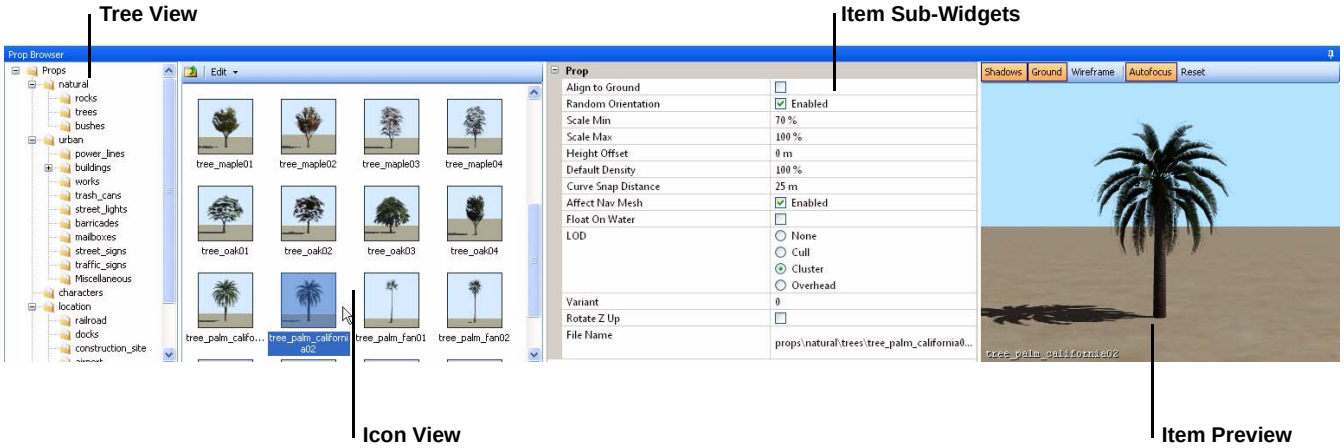
The Library Browser takes slightly different forms with specific tools. For example, when the Road Tools are active, the Road Style Browser gives you access to road styles; When the Material Tools are active, the Material Browser gives you access to textures and other materials; When the Prop Tools are active, the Prop Browser and gives you access to 3D props. Some other tools also use a version of the Library Browser, and some do not.

For general instructions on using the Library Browser to manage items, see ”Chapter 4: The Project System” on page 71. For instructions on using the Library Browser with a specific tool group, such as the Prop Tools, see the documentation of that tool.

Generic Components of the Library Browser

Although the functionality of the Library Browser varies somewhat between tools, the major components of the Library Browser are the same with most tools.

The Library Browser is made up of four smaller windows by default. From left to right, these include the Tree View, the Icon View, the Item Sub-Widget Panel, and the Item Preview window. These windows are explained below the figure showing the Prop Library as an example.



Tree View

The Tree View shows the folder structure within the Library Browser. These folders are used purely for managing items within CityScape, and are not related to the folders on your hard drive containing source files for items.

Icon View

The Icon View shows thumbnail images and names of items in the currently selected folder.

The Icon View also includes the **Edit Sub-Menu**, which provides tool-specific options for managing items in the active library.

Item Sub-Widgets

The Item Sub-Widget Panel displays widgets that globally affect the item you have selected in the Icon View, or placed in the terrain itself. The specific widgets depend on which tool is active, and what type of item you have selected.

Do not confuse the Item Sub-Widgets in the Library Browser with the Main Widgets in the upper right of the CityScape UI. Item Sub-Widgets let you modify global parameters that affect all instances of an item, in all terrains that use it. The Main Widgets let you modify local parameters that affect only the selected instances of items in the open terrain.

Item Preview Window

The Item Preview Window displays a rendered image of the selected item. You can control the Preview camera in the same way as the main camera in the Render Window (i.e. hold down the ALT-KEY, place the mouse pointer inside the Preview Window, hold down one of the mouse keys, and drag the mouse).

Across the top of the Item Preview Window are options for controlling how the item is previewed. For 3D items like props, the menu bar has options for toggling shadows, ground planes, and wireframes.

3.2 ABOUT THE CAMERA

Using the camera is easy. For instructions, see ”2.2.2 Using the Camera” on page 14.

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Chapter 4: The Project System

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Terrain Files 72
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4.1 OVERVIEW OF THE PROJECT SYSTEM

The CityScape project system assumes that all files in the same project are organized into folders under a master project folder. This organization allows CityScape to use relative paths to link terrain files, project files, and library items used in terrains. This chapter discusses CityScape tools and processes relating to the project system.

4.1.1 About Terrains, Projects, and Libraries

The key elements of a CityScape project are *terrain files*, and *items* (e.g. 3D models and 2D textures) organized in *libraries*, and *project files* that make library items available to terrains. For details, see the following sections:

- Terrain Files
- Project Files, Libraries, and Items

TERRAIN FILES

You use the tools in CityScape to build the contents of a terrain on rectangular tiles. You can build 3D geographical features (e.g. hills, valleys, lakes, etc), roads, buildings, etc. You can also import and place 2D and 3D items into terrains.

- A terrain file (CST extension) contains
  - descriptions of tiles that serve as the foundation of your terrain;
  - geometric primitives that CityScape uses to generate 3D geography, road splines, lakes, footprints to buildings, and certain other elements in terrains. (When CityScape exports the terrain, this geometry is tessellated into triangles at the resolution of your choice.)
  - the location, orientation, and scale of each item you've placed the terrain. "Item" means any 3D model, 2D texture, or other assets that was created in external software and imported into CityScape.
- The terrain file does not contain the source files for imported items. Nor does it contain references to these source files. The terrain file references a project file (CSP extension), and the project file references to the source files for items. See the next section for details.

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Project Files, Libraries, and Items

Using external software (e.g. Maya, Creator, 3D Studio Max, Photoshop, etc.), you can create 3D textured models of trees, buildings, cars, fire hydrants, street lights, and other props. You can also create 2D textures for roads, dirt, parking lots, and other surfaces. CityScape can import these items, so that you can place them in your terrains.

- When CityScape imports an item, that item is added to the appropriate library. Examples: the Prop Library for 3D props; the Road Style Library for road textures; the material library for diffuse, detail, normal, and specular textures; etc.

- The primary purpose of a project file (.CSP extension) is to reference the source files for items in libraries. Specifically, the CSP file uses relative path names, if possible, to reference 3D model files, 2D texture files, and other source files for items that have been imported into your current CityScape project. Hence the need to organize the source files for external items in folders under the master project folder.

- Each terrain file references exactly one project file, but the project file does not reference terrain files. This one-way reference solves certain workflow problems that might otherwise arise when many terrains share the same project file (and therefore the same library items), as is typically the case.

- When CityScape loads a terrain file, the associated project file also loads to provide the terrain with access to the source files for items that are placed (or could be placed) in the terrain. In short, the project file tells CityScape where to find non-native items like props that are part of the project file associated with the open terrain.

4.1.2 Summary of the Project-Centric Process

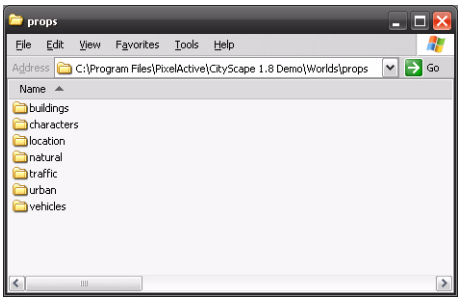
When you create a new project file, CityScape does not create a bunch of asset folders under it. To keep things portable, and organized as you prefer it, we recommend that you create a tree of folders under your master project folder. This tree of folders will contain the source files for all the items you will import into your project.

Here's the overall process:

1. Create a folder on your hard drive (or a network drive) to serve as the master project folder. CityScape uses relative paths, if possible. Hence, if everything related to your project is under this folder, you can move your entire project later without problems.

2. Under your project folder, create a *Worlds* folder to hold future project files, as well as sub-folders for library items.
   - Collections of terrains in the same greater region should probably share the same project file, so the same set of libraries / items will be available to all the terrains in that region.
   - Two different collections of terrains, each in a different region (e.g. tropical and arctic), should probably use different project files — but they don't have to.

3. Create folders under *Worlds* for each category of item you plan to eventually import into the project. Then branch down in the hierarchy of new folders. For example, create a *Props* folder under *Worlds*, and then under Props create folders for *Buildings*, *Plants*, *Signs*, etc.



4. When you create a new terrain (CST), you can associate it with either a new or a pre-existing project file (CSP). If you're establishing a new project, and need to create a new project file, just create a new terrain and opt to create a new project file with it. You can enter separate names and locations for the new terrain file and the new project file. (For details, see "Create a New Terrain" on page 76.)

5. Export items from your external authoring software to formats that CityScape can import. For a list of supported formats for 3D props, see "Importing New Props" on page 137.

6. Move the exported items into the appropriate folders you just created under the master project folder.

7. We recommend that you customize the folders in the Tree View to match the names and organization of the folders on your hard drive containing the source files for items.

   Important! The names of folders in the Tree View can be different the names of folders on your hard drive. The two are totally separate, in fact. The policy to use the same names will help avoid confusion, but it is not required to use CityScape.

8. In CityScape, open a terrain associated with the target project file, and import the items into the appropriate libraries.
   – To import props, select the **Prop Tool**, and then follow the directions in "Importing New Props" on page 137.
   – To import textures for new road styles, select the **Road Tool**, and then follow the directions in "Create New Road Styles" on page 88.
   – To import other items, select the appropriate tool, and then use the Library Browser in that tool to import the items. (For example, to import textures for new building styles, use the Building Tool.)

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4.2 OPENING TERRAINS

This section describes the nuances of opening existing terrains.

4.2.1 Open a Terrain

When you open any existing terrain in CityScape, be aware that the associated project file (CSP) also loads in order to make the items in its libraries available to the terrain. If the relative path between the terrain and project file is broken (such as with moving one file but not the other), the terrain may lose its reference to its associated project file. If CityScape cannot find the associated project file, you will be prompted to locate it.

Opening a terrain is easy:

1. Select **File > Open Terrain**, (shortcut: **Ctrl-O**).
2. In the Open dialog, navigate to the target terrain file, select it, and LEFT-CLICK **Open**.

When a terrain is open, the CityScape title bar displays the name of the project file followed by the name of the terrain file.

4.2.2 Access a Project File

You cannot open a project file directly. When you open any terrain, CityScape opens the associated project file too. Then you can use the tool-specific Library Browser to access the items in project's libraries. For example, you can access road styles via the Road Style Browser, which appears when you select the Road Tool.

The Demo Project comes with a **demo.csp** file that references some useful library items for you to play with. In fact, the "Quick Start Tutorial" on page 7 asks you to create a new terrain and associate it with the demo.csp project file, which is installed in the following folder, by default, for Vista and Win 7 users:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds

or here for WinXP users:

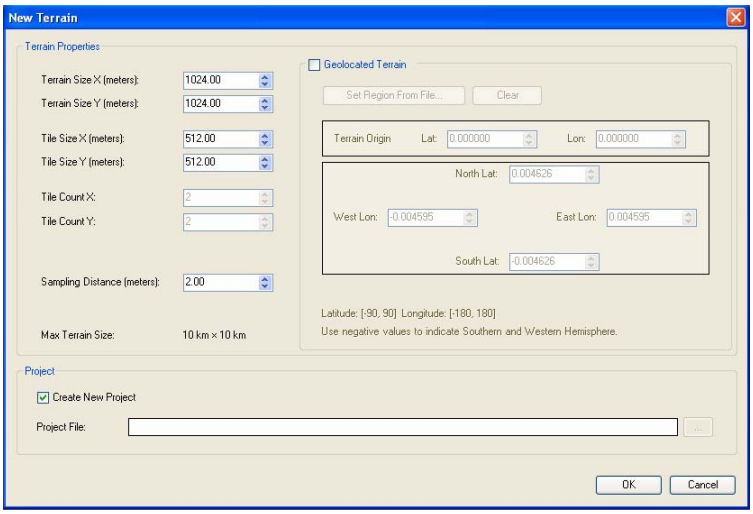
C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds

4.3 CREATING TERRAINS AND PROJECTS

4.3.1 Create a New Terrain

When you create a new terrain file, you must associate it with a project file. You can select an existing project file, or create a new project file. You do both while creating a new terrain. The following instructions are for creating a new terrain that is not geo-located. (For details on geo-located terrains, see "Chapter 21: Importing GIS Data" on page 195.)

1. Select **File > New Terrain** (shortcut: **Ctrl-N**). The New Terrain dialog box appears.



2. In the New Terrain dialog, configure your terrain as follows:
   - **Terrain Size (X, Y):** enter the size (in meters) of the entire terrain – up to the maximum size that CityScape allows.
   - **Tile Size (X, Y):** enter the size (in meters) of each tile. If you enter different values for X and Y, then the tiles becomes a rectangles instead of squares.
   - **Tile Count (X, Y):** CityScape determines the number of tiles that will fit along the X-axis and Y-axis, given you choices of Terrain Size and Tile Size. If the number of tiles does not divide evenly into the terrain, the outermost tiles will extend beyond the edges of the terrain.

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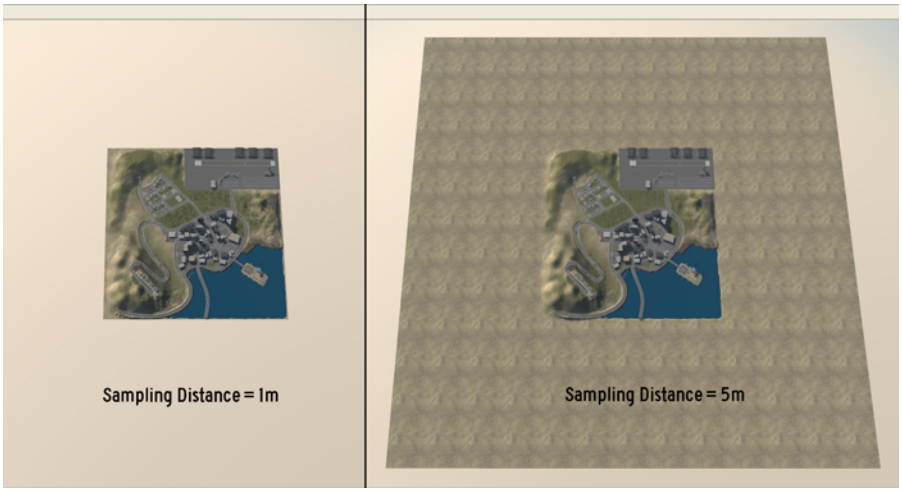
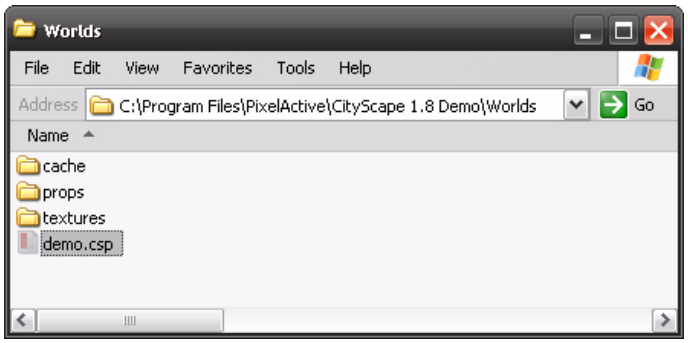
– **Sampling Distance:** enter the distance (in meters) between sample points in the terrain (i.e. points where CityScape samples the height of the terrain). Upon export, smaller global sampling distances cause the terrain to be tessellated into more polygons, which means more detail but slower rendering times.



– **Project**:
  » **Create New Project:** Check this box to create a new (empty) project file, and associate it with the new terrain. When you do the last step (below), CityScape will prompt you for the name and location of the new project file.
  » **Project File:** Use the Browse button [**...**] to locate and select an existing project file to associate with the new terrain. If you're looking for a project to experiment with, try the Demo Project that comes with CityScape. The *demo.csp* file is located here by default:



3. LEFT-CLICK **OK** in the New Terrain dialog. Your new terrain appears in the Render View. The CityScape title bar displays the name of the project file followed by the name of the terrain file.

4.3.2 Create a New Project File

You cannot create a project file apart from a terrain. As described in “Create a New Terrain” on page 76, you have the option to create a new project file when you create a new terrain. If you create a new project file, then it will be empty initially. You can then follow the procedures below to import items into your project, organize items and folders in libraries, etc.

4.4 USING THE LIBRARY BROWSER

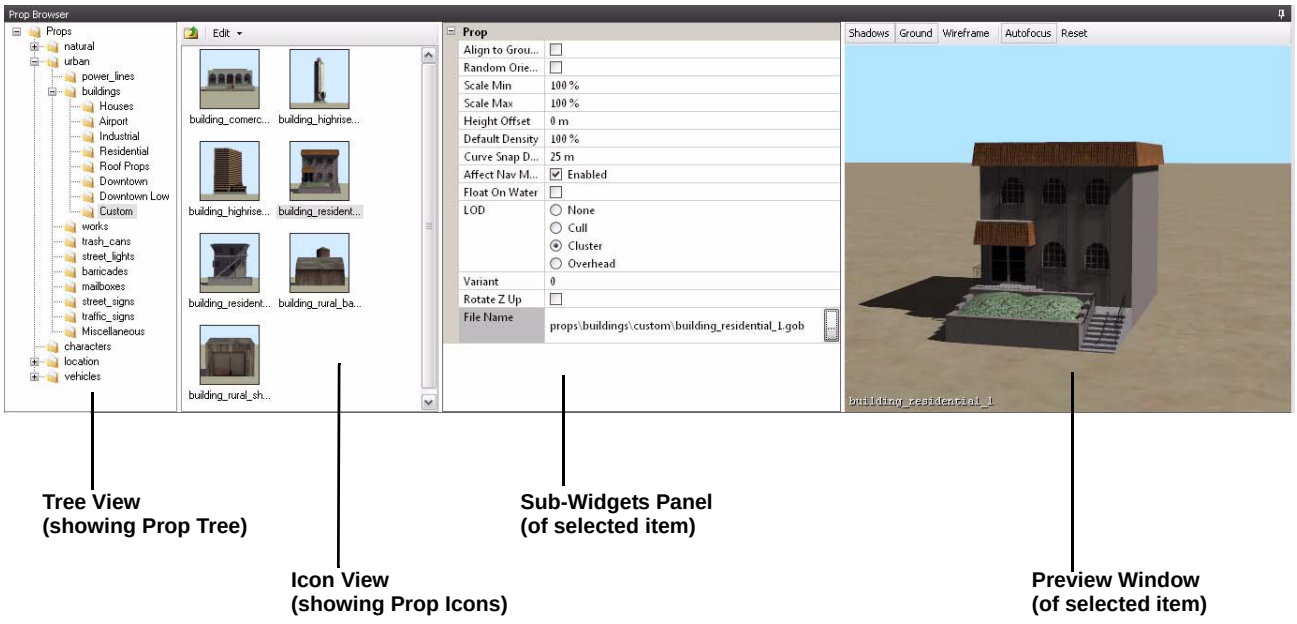
Using the tool-specific Library Browser, you can import 2D and/or 3D items into the libraries that are referenced by the current project file. (Different tools have different versions of the Library Browser.) Once imported, the items can be managed and configured using the Library Browser.

Most functionality is the same for every version of the Library Browser. These shared functions are documented here. The tool-specific functionality of the Library Browser is documented in the chapters about those tools:

- The tool-specific functionality of the Prop Browser is documented in ”Chapter 10: Using Prop Tools” on page 133.
- The tool-specific functionality of the Road Style Browser is documented in ”Chapter 6: Using Road Tools” on page 83.
- The other tool-specific versions of the Library Browser will be documented in future chapters on those tools.

4.4.1 Views

The Library Browser has 4 windows. From left to right (see figure below): Tree View, Icon View, Sub-Widgets, and the Preview Window.



TREE VIEW

The Tree View shows the folder structure in the Library Browser. You can create, rename, and move folders. (Note: The folders in the Tree View are completely separate from the directory structure on your hard drive where the source files for items are stored.)

ICON VIEW

The Icon View shows thumbnail images and names of items in the currently selected folder.

The Icon View also includes the **Edit Sub-Menu**, which provides tool-specific options for managing items in the active library.

SUB-WIDGETS PANEL

The Sub-Widget Panel displays widgets that globally affect the item you have selected in the Icon View or the terrain itself. The specific widgets depend on which tool is active, and what type of item you have selected.

Do not confuse Sub-Widgets in the Library Browser with the Main Widgets in the upper right of the CityScape UI, by default. Sub-Widgets let you modify global parameters that affect all instances of an item, in all terrains that use it. The Main Widgets let you modify local parameters that affect only the selected instances of items in the open terrain.

PREVIEW WINDOW

The Preview Window displays a rendered image of the selected item. You can control the Preview camera in the same way as the main camera in the Render Window (except you need to place the cursor in the Preview Window instead of Render Window).

Across the top of the Preview Window are options for controlling how the item is rendered in the Preview Window. The exact options depend on the type of item you have selected. For 3D items like props, the menu bar has options for toggling shadows, ground planes, and wireframes.

4.4.2 Create New Folders

1. In CityScape, activate any tool that uses a version of the Library Browser.
2. In the Tree View of the Library Browser, navigate to the parent folder in which you wish the new folder to be located in the folder tree.
3. RIGHT-CLICK inside the Icon View (but not on any icons) to display a context sensitive menu. In the menu, select **New Folder** to create a folder inside the current folder. Give the new folder a name. (Alternative: Select **Edit > New Folder** from the Icon *Edit* Menu.)

4.4.3 Rename Folders and Items

- To rename a folder, LEFT-CLICK the folder in the Tree View to select it. Wait a moment, and then LEFT-CLICK the selected folder a second time to enter editing mode. Type a new name.
- To rename an item, LEFT-CLICK its ICON in the Icon View. Wait a moment, and then LEFT-CLICK the thumbnail a second time to enter editing mode. Type a new name.

4.4.4 Move Folders and Items

- To move a folder, LEFT-CLICK-HOLD the folder in the Tree View, and DRAG the folder to its new location in the folder tree.
- To move an item to another folder, LEFT-CLICK the ICON of the item in the Icon View, and DRAG the item into the target folder in the Tree View.

4.4.5 Delete Folders and Items

- To delete a folder, LEFT-CLICK it in the Tree View, and press the DELETE-KEY.
- To delete an item, LEFT-CLICK its ICON in the Icon View, and press the DELETE-KEY.

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PART 2: USING THE EDITING TOOLS

Chapter 5: Using Geography Tools

This chapter is in progress and will be released soon.

Chapter 6: Using Road Tools

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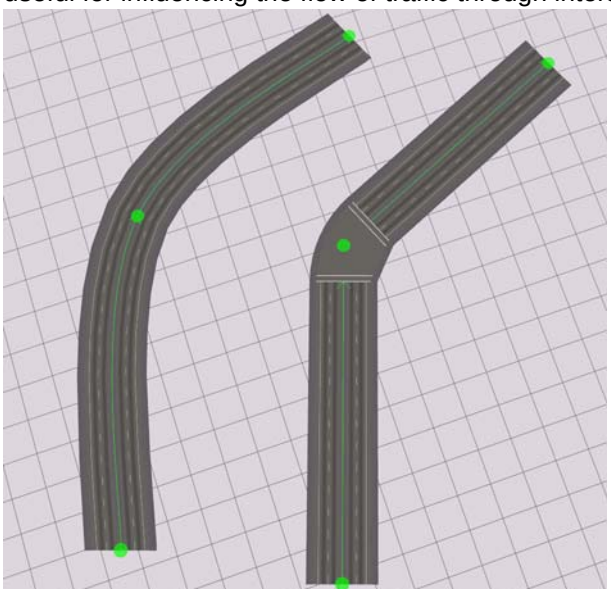
6.1 OVERVIEW OF ROAD TOOLS

You can use the Road Tools, in conjunction with the Road Style Browser, to build and customize road networks in terrains. When you create and move roads on hills and valleys, CityScape dynamically adds *cut and fill* to the terrain, so the road always sits on the ground in a realistic way. Similarly, if you move hills with Geography Tools, CityScape dynamically adjusts the roads so they integrate realistically with the altered terrain.

6.1.1 Road Tools Menu

When you activate any Road Tool, the Road Tools Menu appears to provide you with the following options. For details on usage, see the rest of this chapter.

- **Link Segments:** Links two road segments that share the same node. (Even though the segments share a node, when unlinked they form a a sharp angle at the node. When linked, they form a smooth curve through the node. Linking also affects intersections.) This function is particularly useful for influencing the flow of traffic through intersections.



- **Unlink Segments:** Unlinks two road segments that share the same node. (See “Link Segments” above.)

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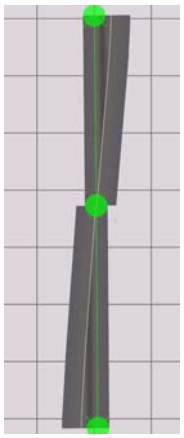
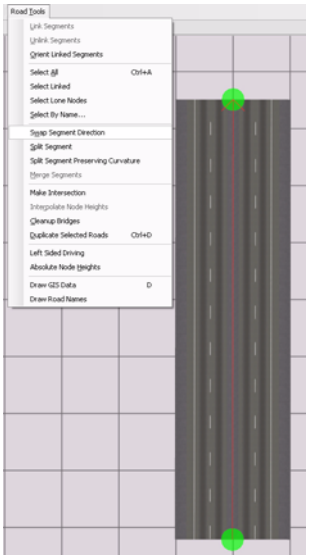
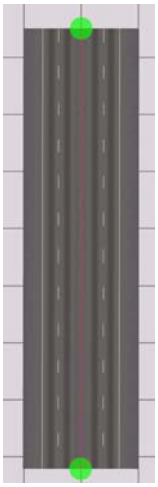


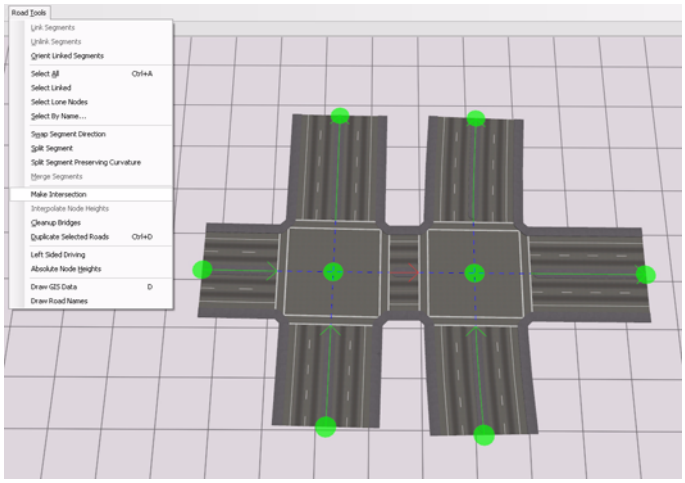
- **Orient Linked Segments:** Corrects the situation where two segments sharing the same node have opposite direction vectors, such as with one-way streets pointed into each other (see figure to the left.)
- **Select All:** Selects all road nodes and segments in the open terrain.
- **Select Linked:** Selects all segments in the open terrain that are linked to the segment you manually select in the terrain.
- **Select Lone Nodes:** Selects randomly placed nodes with no associated segments, so you can get rid of them.
- **Select by Name:** If you have previously given one or more road nodes and/or segments a name (using the *Name* parameter of the Main Widgets for road nodes and segments), then the Find Road function will automatically select (and zoom the camera into) all nodes/segments with the target name.
- **Swap Seg Direction:** Reverses the *left* and *right* designations of the sides of the selected road by reversing the direction of the selected segments in the road.

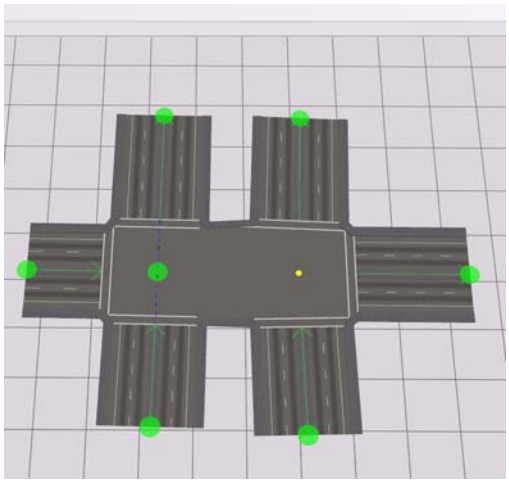




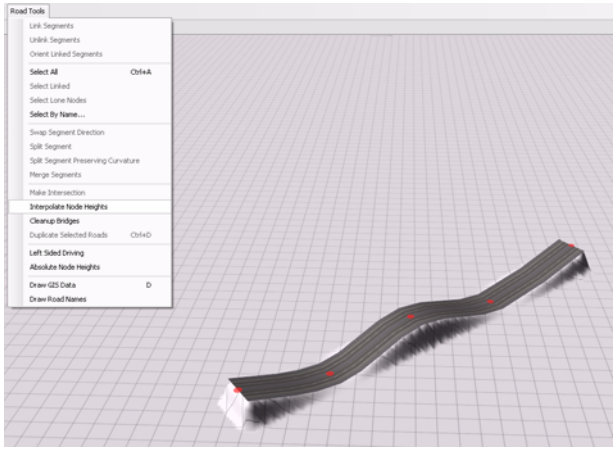
- **Split Segment:** Adds a node in the center of the selected segment.
- **Split Segment Preserving Curvature:** Adds a node in the center of the selected segment, and automatically modifies the tangents to preserve the shape of affected curves.
- **Merge Segments:** Removes the node shared by two selected segments, and unites the two into one longer segment.

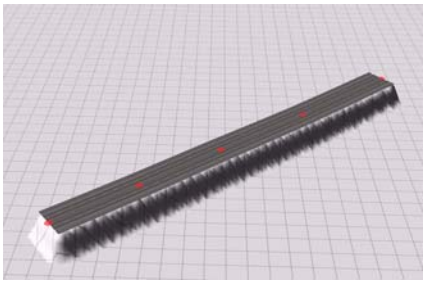
- **Make Intersection:** Converts roads with inappropriate scissor crossings (i.e. staggered crossings) into clean intersections.





- **Interpolate Node Heights:** Causes the heights (i.e. elevations) of the nodes in the middle of a selection to smoothly transition between the heights of nodes on the ends of the selection.





- **Straighten Nodes:** Straightens the selected stretch of road by aligning all nodes and segments.
- **Clean Up Bridges:** Changes all bridges that are too close to the ground to regular roads.
- **Duplicate Selected Roads:** Copies the selected road.
- **Left Sided Driving:** Causes traffic to drive on the left side of roads in the open terrain.
- **Absolute Node Heights:** Toggles between absolute and relative node elevations:
  - Absolute Node Heights are relative to zero elevation in the terrain.
  - Relative Node Heights are relative to the surrounding terrain.
- **Draw Road Names:** Displays the names that were previously entered into the *Main Widgets* > *Name* parameter for road segments.

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6.1.2 Visual Warnings

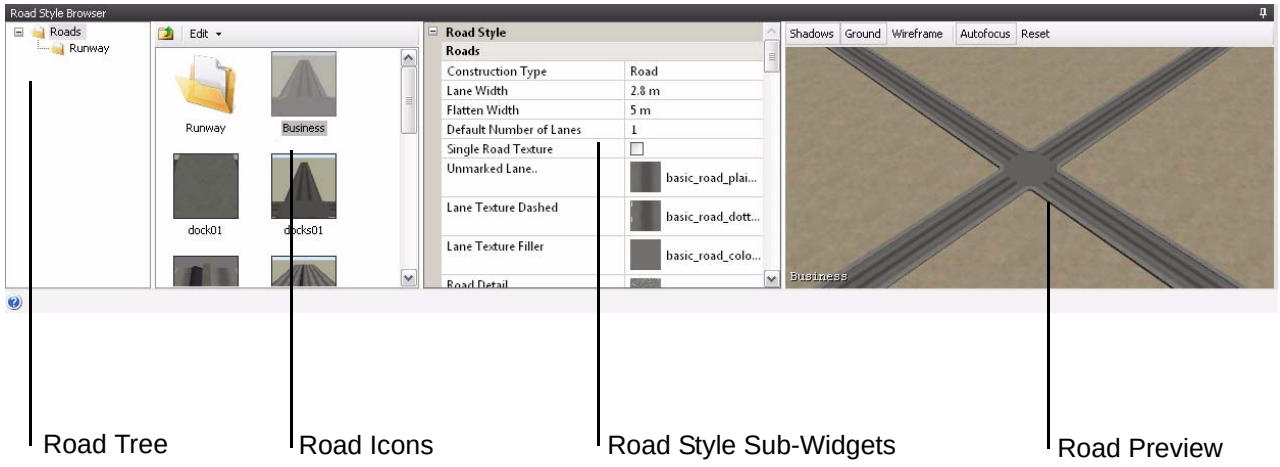
When constructing roads, CityScape warns you of two common problems:

- **Overlapping Road Meshes:** Orange triangles appear whenever a road corner is too tight, or in any other situation when road meshes are squeezed to the point of overlapping. The solution is to move the nearby nodes a bit, and/or to adjust the road tangents, until none of the meshes overlap.
- **Overlapping Intersections:** Exclamation points appear when two or more intersections are too close to each other. The solution is to either move the intersection nodes further apart, or merge the intersections with the *Make Intersection* command in the Road Tools menu.

6.2 USING THE ROAD STYLE BROWSER

A "road style" defines the appearance and properties of a type of road (e.g. small residential street vs. big freeway). A road style includes not only the road itself, but also related elements such as sidewalks and medians, among others. Using the Road Style Browser, which appears when any road tool is active, you can select styles to use in terrains via the Road Tools. You can also import, customize, and manage an extensive library of road styles.

In terms of doing standard stuff, like managing folders and files, the Road Style Browser works like any other version of the Library Browser. (For details on the standard features of the Road Style Browser, see "Using the Library Browser" on page 78.) For details on features that are unique to the Road Style Browser, read on.



6.2.1 Select Road Styles

The Road Icons represent the road styles in the selected folder of the Road Tree. LEFT-CLICK the ICON of a **road style** to select that style.

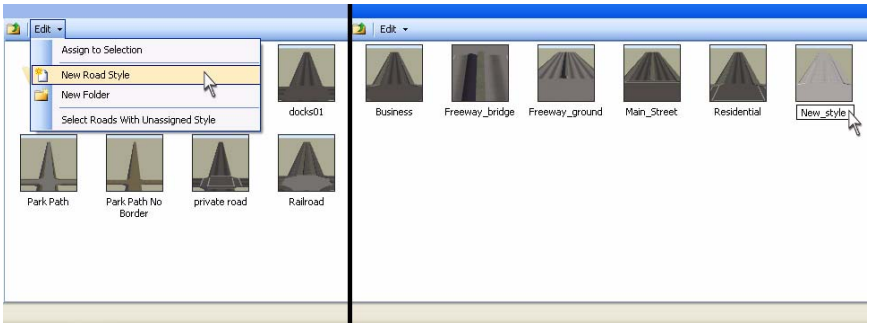
You can apply the actively selected road style to a road that exists in the terrain, or to the next road you create. For details, see "Apply a Road Style" on page 102.)

6.2.2 Create New Road Styles

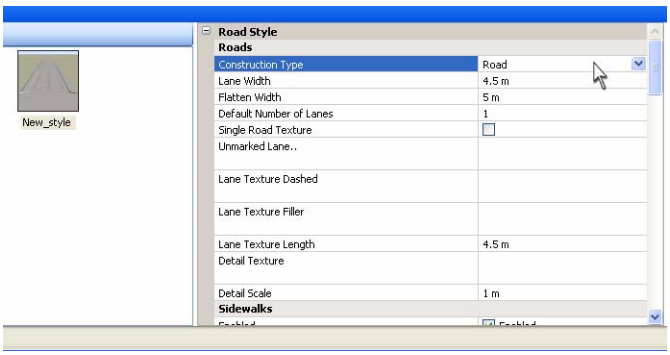
Using the Road Style Browser, you can create new road styles and modify existing styles. This section gives you basic instructions for creating a new road style – including the import of textures for new road styles.

The Overall Process

1. Verify the Road Tool is active.
2. Select **Edit > New Road Style** from the Road Icons Menu of the Road Style Browser. (*Shortcut:* RIGHT-CLICK inside the Road Icons View, and select **New Style** from the context sensitive drop-down menu.) A new icon appears in the Icon View.



3. Verify the icon for the new road style is actively selected, and then define values for its **parameters**, as explained in *"Road Style Sub-Widgets" on page 90*.



IMPORTING TEXTURES FOR ROAD STYLES

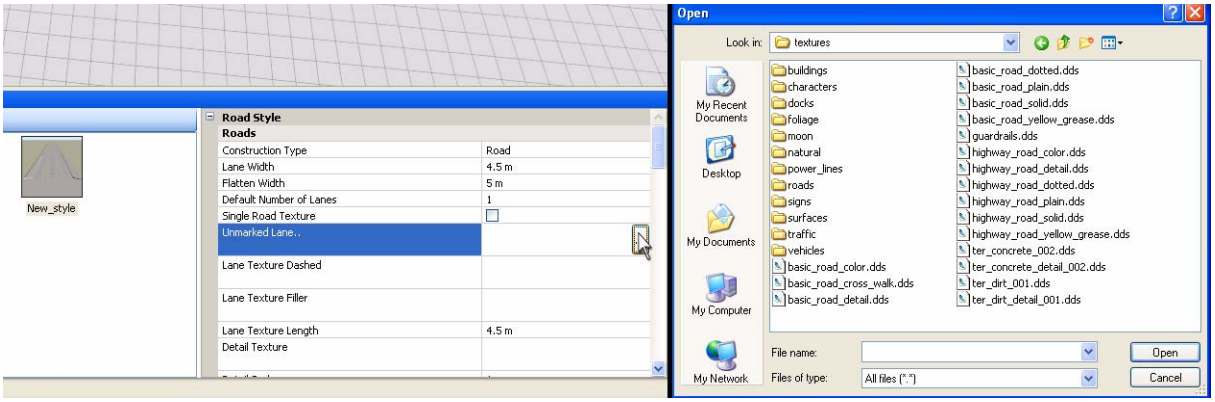
1. Create a new texture in external software such as Photoshop, and then export the texture to a common format such as JPEG or TIFF.

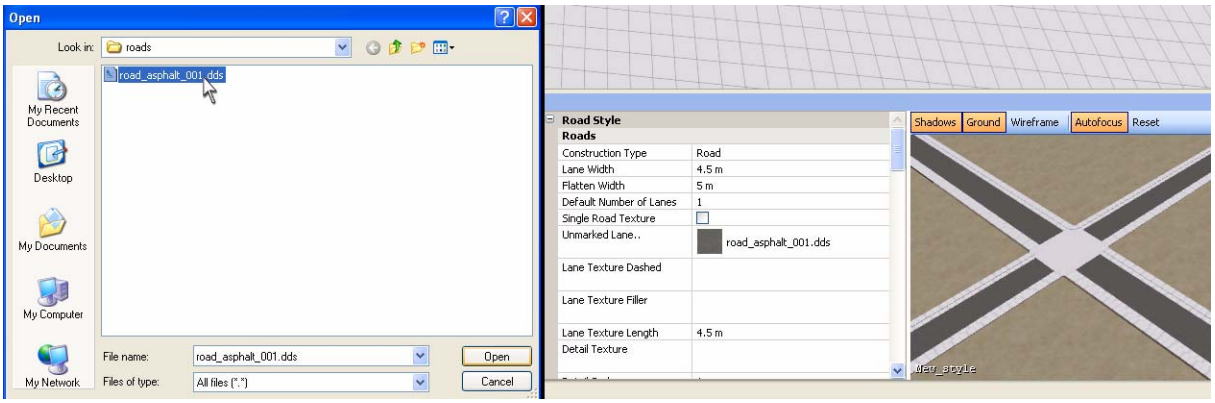
2. Move the new texture into a folder under the associated project file (CSP).

3. Verify the Road Tool is active, so that the Road Style Browser is displayed.

4. In the Road Style Browser, either create a new road style (as described in the section above), or select an existing road style. The parameters for the selected road style appear in the Sub-Widgets.

5. In the Road Style Sub-Widgets, LEFT-CLICK the texture parameter you wish to apply the new texture to. (For example, select the *Diffuse Texture* parameter.) Then LEFT-CLICK the browse button [...] to the right of the texture sub-widget, and navigate to the **texture file** you wish to import, and open it.
   – **Note:** CityScape supports the following formats for 2D textures: DirectDraw Surface (DDS), BMP, GIF, JPEG (and JPG, JPE, and JP2), PNG, PhotoShop (PSD), Silicon Graphics (SGI, BW, RGB, RGBA), TGA, TIF, and X Pixel Map (XPM).





6. Repeat this process for additional textures that you wish to apply to other parameters in the new road style.

6.2.3 Customize Road Styles

The Sub-Widgets in the Road Style Browser contain the global parameters of a road style that you have actively selected in the Road Icons, or in the terrain itself.

"*Global* parameters" means all terrains sharing a specific road style will automatically update when you modify any parameter in the Sub-Widgets, and then save the terrain file (and therefore the associated project file along with it). Before you make global changes to a road style, consider the consequences for all terrains, not just the terrain you currently have open in CityScape.

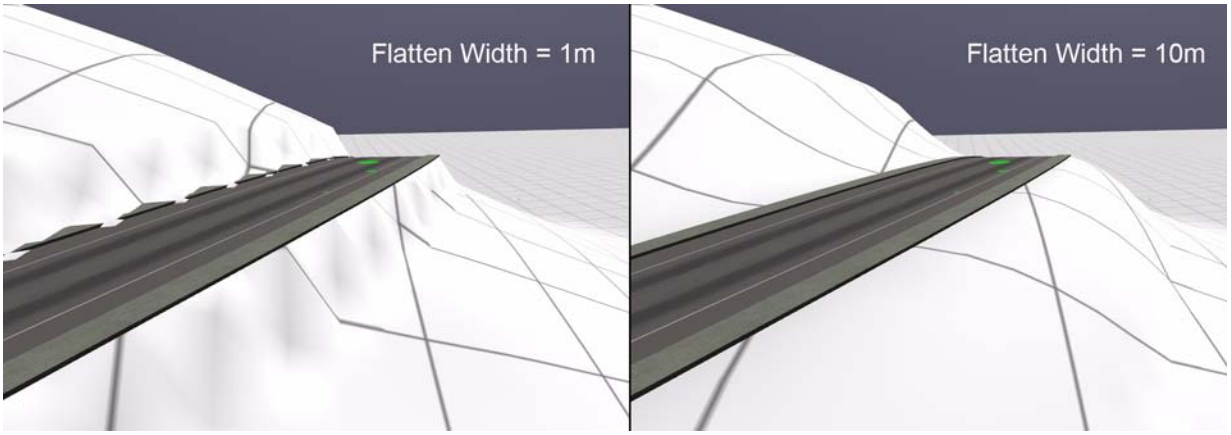
Some parameters in the Road Style Sub-Widgets are numerical; some are boolean (on / off); and some offer a choice from a list to control such things as the width of lanes and the height of curbs. Other parameters let you to link to textures for roads and related elements such as sidewalks, shoulders, medians, bridges, tunnels, intersections, and crosswalks. For details, see the next section below.

ROAD STYLE SUB-WIDGETS

You can customize some of these settings for individual nodes and/or segments of actual roads in terrains.

Roads

- **Construction Type:** designates the default road as a regular *road*, a *bridge*, or a *tunnel*. Notes:
- **Lane Width:** sets the width of each lane.
- **Flatten Width:** influences how the geometry for roads are "stitched" into the geometry for the underlying terrain. To clarify, consider the example of a road running along a hillside:



  – For simplicity, assume this road runs along a hillside. Further assume that *Relative Height = 0*; and *Bank = 0*; and *Construction Type = Road* in the Road Node Widgets (very different widgets than the Road Style Sub-Widgets).

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  - Then the nodes will be the same elevation as the surrounding hillside. The uphill side of the road will cut into the hillside, and the downhill side will need a fill under it, so that both sides connect directly to the surrounding terrain.
  - The Flatten Width parameter controls with width of the cut and fill. Put another way, this parameter controls the width of the transition area between the road and the surrounding terrain (see figure).

- **Default Number of Lanes:** sets the default number of lanes for new roads using this style. For example, a value of "2" means the road will have 2 right lanes and 2 left lanes, but default, when you initially create it. You can of course use the Main Widgets to override this default for road segments that are actively selected in the terrain.

- **Single Road Texture:** causes the lane texture to be stretched across all lanes in a road segment.

- **Unmarked Lane:** links to the texture for lane without lines.

- **Lane Texture Dashed:** links to the texture for lanes that need a visual separator. The appearance of this separator can be anything you want, such as a dashed white line.

- **Lane Texture Filler:** complex roads sometimes have small areas that aren't part of regular lanes, but are necessary fillers. The Lane Texture Filler parameter links to the texture for these miscellaneous areas.

- **Lane Texture Length:** determines the length (in meters) of the lane texture. (There is no "width" parameter because lane textures are always normalized to fit the lane width – or the road width if *Single Road Texture* is enabled.)

- **Detail Texture:** links to a detail texture, which blends with other road textures to provide more visual detail at close range.

- **Detail Scale:** adjusts the size of the *Detail Texture*.

Sidewalks

- **Enabled:** toggles the presence of sidewalks on the road. The other sidewalk parameters become active only if this parameter is enabled.

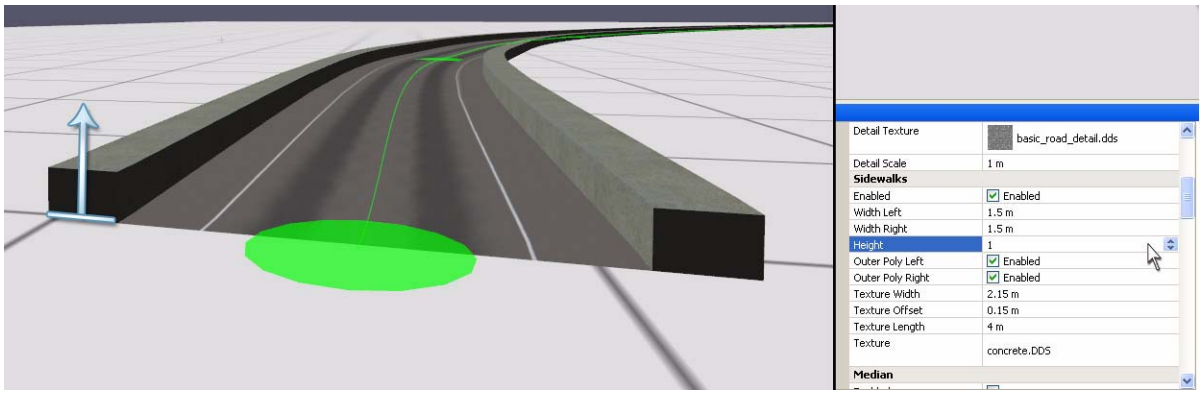
- **Width Left:** determines the width (in meters) of the sidewalk on the left side of the road.

- **Width Right:** determines the width (in meters) of the sidewalk on the right side of the road.

- **Height:** determines the height (in meters) of the sidewalk above the surrounding road. For best results, ensure the *Outer Poly* parameters are also enabled.

- **Outer Poly Left:** toggles the addition of vertical geometry for the outside edge of sidewalks on the left side of the road. The height of this geometry is determined by the *Height* parameter. If *Outer Poly Left* is enabled, then

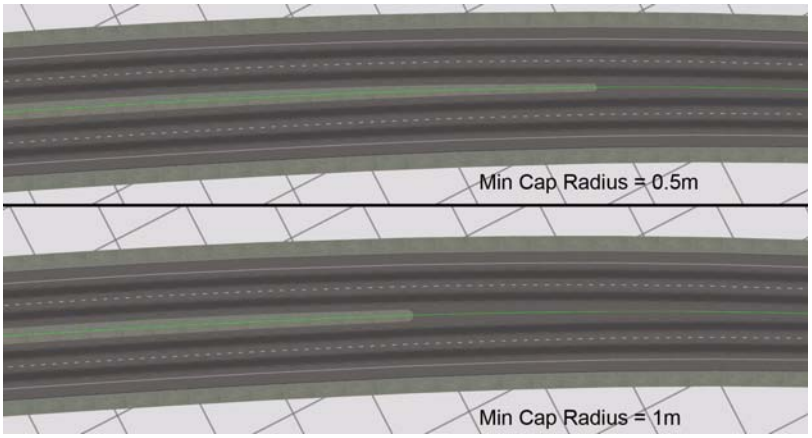
the terrain will connect to the bottom edge of the sidewalk. If disabled, then the terrain connects to the top edge of the sidewalk. In many city areas, it may be preferable to disable outer polygons.



- **Outer Poly Right:** same as above, except for sidewalk on the right side of the road.
- **Texture Width:** determines the width of the sidewalk texture.
- **Texture Offset:** offsets the sidewalk texture relative to its default mapping.
- **Texture Length:** determines the length of the sidewalk texture.
- **Texture:** links to the texture for sidewalks.

Median

- **Enabled:** toggles the presence of a median on the road. The other median parameters become active only if this parameter is enabled.
- **Width:** sets the width (in meters) of the median.
- **Height:** determines the height (in meters) of the median above the surrounding road.
- **Min Cap Radius:** sets the minimum radius (in meters) of the end caps for the median.



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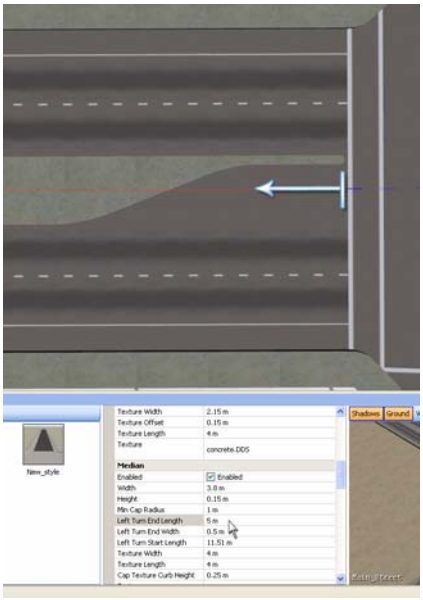
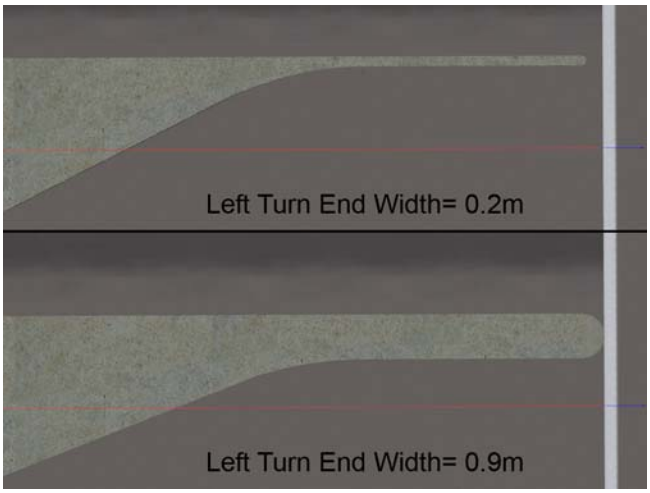
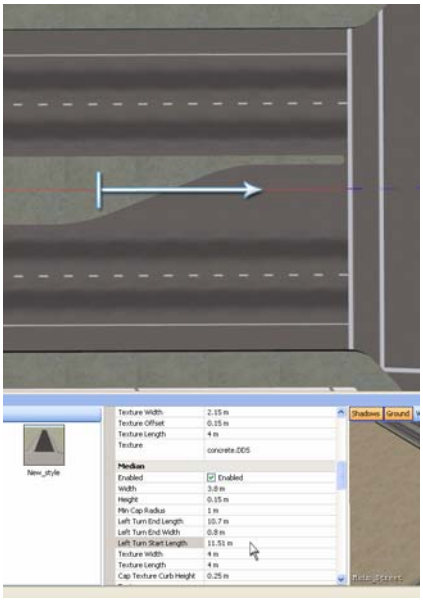
- **Left Turn End Length:** sets the distance (in meters) from the intersection to the point where a left turn lane begins to close.



- **Left Turn End Width:** sets the width (in meters) of the median in a left turn lane.



- **Left Turn Start Length:** sets the distance (in meters) from the point where the left turn lane begins to open from its parent road.



- **Texture Width:** sets the width of the Median Texture.
- **Texture Length:** sets the length of the Median Texture.
- **Cap Texture Curb Height:** sets the height (in meters) of the Cap Texture.
- **Texture:** links to the texture for medians (except the cap area).
- **Cap Texture:** links to the texture on the caps (ends) of the median.

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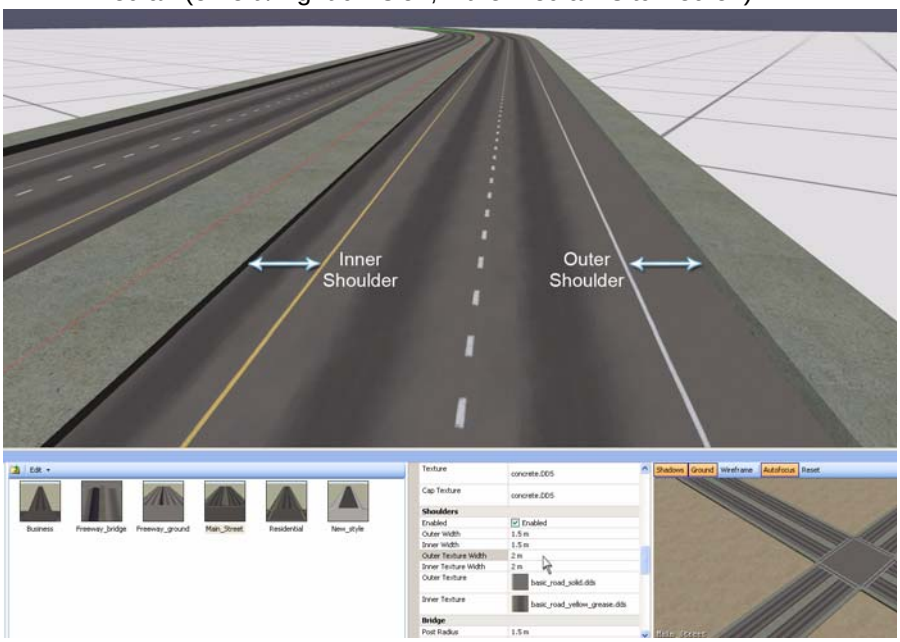
Shoulders

- **Enabled:** toggles the presence of a shoulder on the road. The other shoulder parameters become active only if shoulders are enabled. Notes:
  - The outer shoulder is the area between the outermost lane and the sidewalk (or underlying terrain, if the sidewalk is turned off).
  - The inner shoulder is the area between the innermost lane and the median (or left / right division, if the median is turned off).



- **Outer Width:** sets the width (in meters) of the outer shoulder.
- **Inner Width:** sets the width (in meters) of the inner shoulder.
- **Outer Texture Width:** sets the width (in meters) of the texture for the outer shoulder.
- **Outer Texture Length:** sets the length (in meters) of the texture for the outer shoulder.
- **Inner Texture Width:** sets the width (in meters) of the texture for the inner texture.
- **Inner Texture Length:** sets the width (in meters) of the texture for the inner shoulder.
- **Outer Texture:** links to the texture for the outer shoulder.
- **Inner Texture:** links to the texture for the inner shoulder.

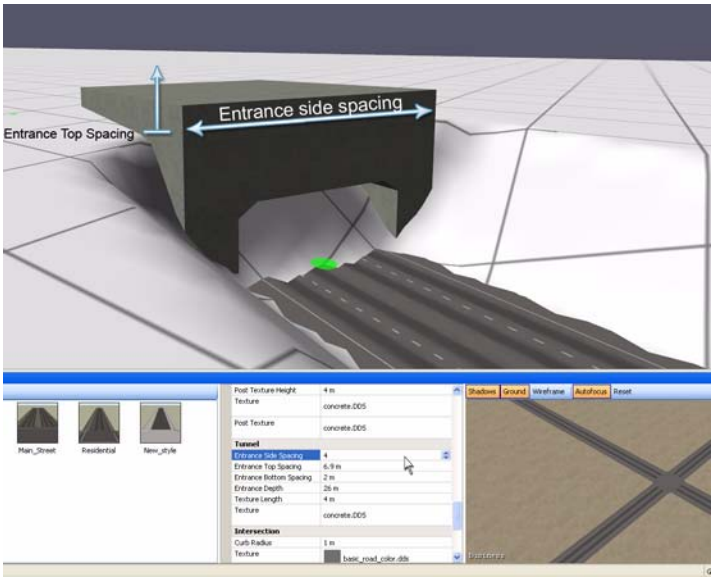
Bridge

CityScape directly connects the geometry of regular roads to the geometry of the ground terrain. Bridges are fundamentally different: CityScape forms a bridge by generating geometry for a stand-alone superstructure that is connected to the ground indirectly via posts that elevate the bridge.

- **Post Radius:** sets the radius (in meters) of the posts in the bridge.
- **Post Spacing:** sets the distance (in meters) between posts in the bridge.
- **Flatten Offset:** sets the minimum vertical distance (in meters) between a bridge and the ground. If the bridge is less than this distance above the surrounding terrain, then CityScape will lower the terrain under the bridge.
- **Texture length:** sets the length of the texture on the superstructure of the bridge.
- **Post Texture Height:** sets the height of the texture on the posts of the bridge.
- **Texture:** links to the texture for the superstructure of the bridge.
- **Post Texture:** links to the texture for the posts of the bridge.

Tunnel

- **Entrance Side Spacing:** sets the width (in meters) that the entrance extends below the road (see figure below).



- **Entrance Top Spacing:** sets the height (in meters) that the entrance extends above the road (see figure above).

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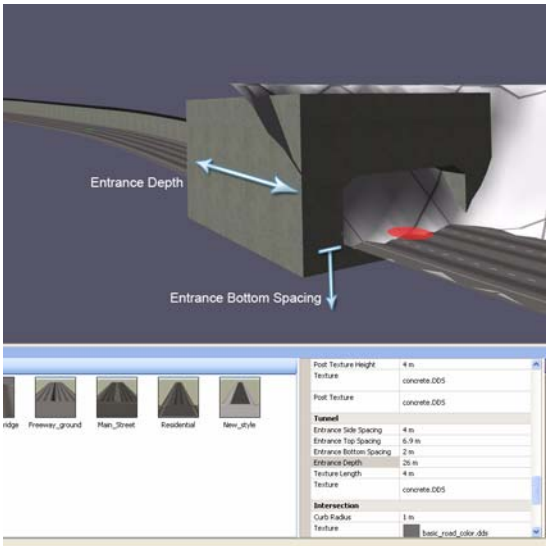
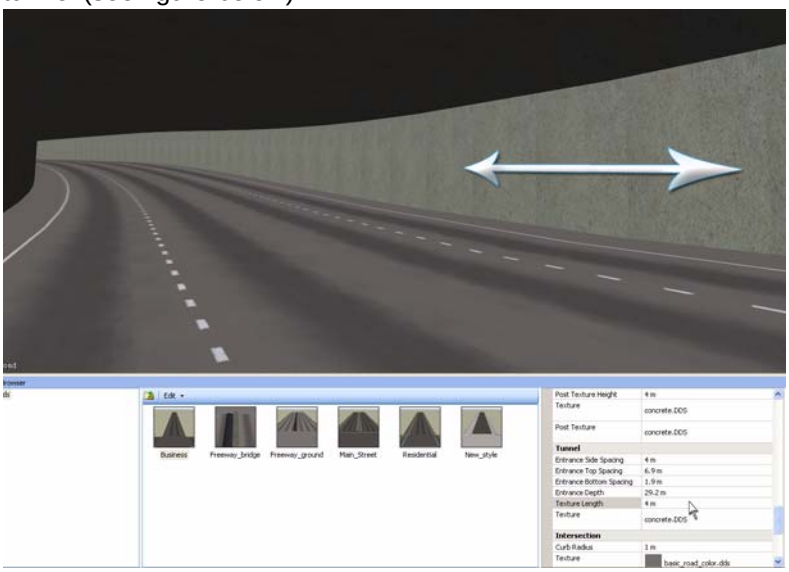
- **Entrance Bottom Spacing:** sets the height (in meters) that the entrance extends above the road (see figure below).



- **Entrance Depth:** sets the distance (in meters) that the entrance extends parallel to the road and into the hillside, typically (see figure above).

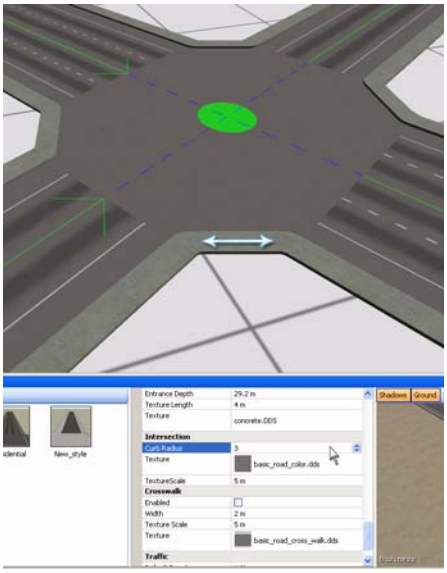
- **Texture Length:** sets the length of the texture for the entrance of the tunnel (see figure below).



- **Texture**: links to the texture for the entrance of the tunnel.

Intersection

- **Curb Radius:** influences the size of the intersection by setting the radius of the curbs inside the intersection.



- **Texture:** links to the texture for intersections.
- **Texture Scale:** scales the texture.

Crosswalk

- **Enabled:** toggles the presence of crosswalks. The other crosswalk parameters become active only if this parameter is enabled.
- **Width:** sets the width (in meters) of the crosswalk by scaling the geometry under the crosswalk. The width of the crosswalk Texture is automatically normalized to fit the width of its geometry.
- **Texture Scale:** scales the texture.
- **Texture:** links to the texture for crosswalks.

Traffic

The effects of the following parameters are visible only when the Traffic Simulator is activated:

- **Default Density:** sets the default density of traffic on the road.
- **Default Parking Density:** sets the default density of parked vehicles on the road.
- **Default Speed Limit:** sets the speed limit for the road. The vehicles in the Traffic Simulator will obey this speed limit.

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Guard Rails / Fences

- **Barrier / Fence:** gives you a choice of barriers along the median (including no barrier).
- **Right Rail / Fence:** gives you a choice of barriers along the right side of the road (including no barrier).
- **Left Rail / Fence:** gives you a choice of barriers along the left side of the road (including no barrier).

6.3 USING THE ROAD TOOL

Using the Road Tool in CityScape, you can create and customize sophisticated road networks. This section documents the basic use of the Road Tool to create roads and intersections. For details on the advanced use of the Road Tool, and the use of Road Tangent Tool and Road Height Tool, see “Editing Road Nodes” on page 108, and “Customizing Roads: Advanced Techniques” on page 118.

6.3.1 Creating Roads

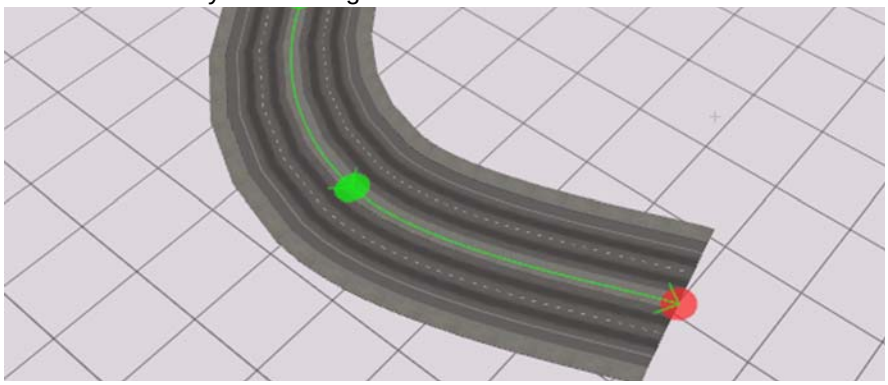
In CityScape, you create a road by placing a series of "nodes" in a terrain. Each node is a point that defines the start or end of a single segment of road. An intersection can optionally exist at a node. A complete road consists of a series of nodes and the segments connecting them.

This section documents how to create common types of roads.

CREATE A NEW ROAD

To create a road, proceed as follows:

1. Activate the Road Tool in the Tools Palette.
2. RIGHT-CLICK a few times in the terrain to place a few **nodes**. Each time you place a new node, CityScape adds a segment connecting the previous node to the new node. Notes:
   – The most recently placed node becomes the current selection automatically, so placing the next node will extend the current road.
   – If you wish to start a new road, deselect all nodes before placing the next node by left-clicking on the terrain.



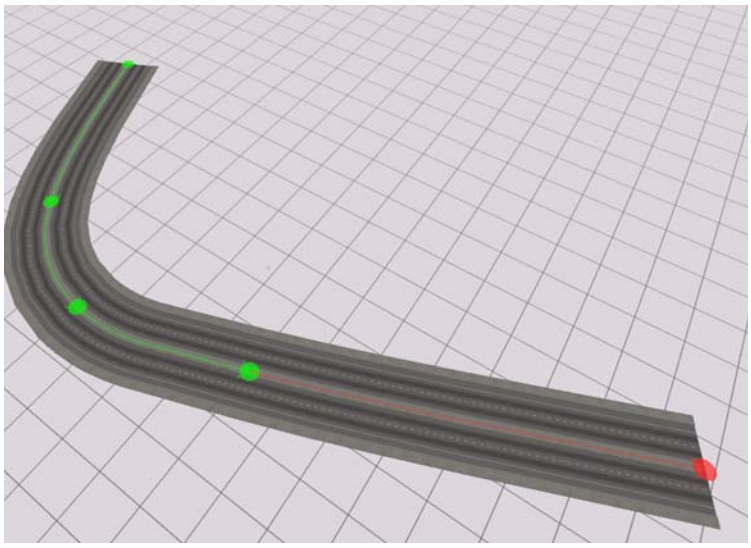
3. Optionally Apply a Road Style to the road.

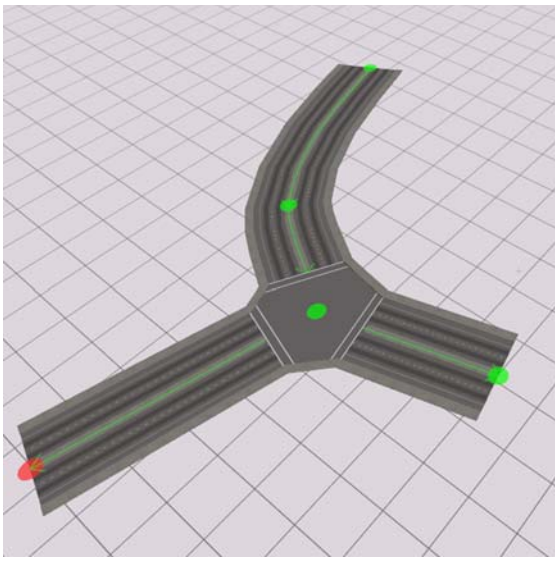
4. Optionally modify the number of lanes and other properties of the road, as explained in “Customizing Roads: Advanced Techniques” on page 118.

Extend an Existing Road

If you stop work on a road, and later want to resume work on it, follow these steps:

1. Verify the Road Tool is active, and then LEFT-CLICK the **road node** from which to resume:





   – If you resume from an end node, CityScape will extend the existing road.
   – If you resume from any node other than the start or end node, CityScape will create an intersection at the resume node, and allow you to digitize a new road from that intersection.

2. RIGHT-CLICK to place new nodes (and connecting road segments).

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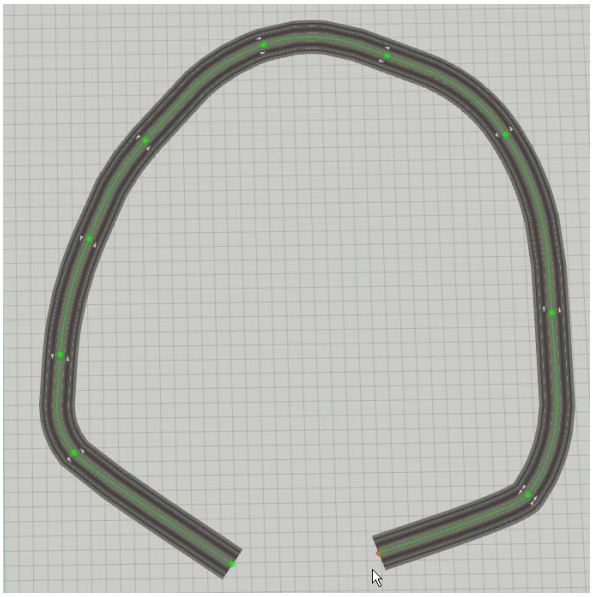
Close a Loop of Road

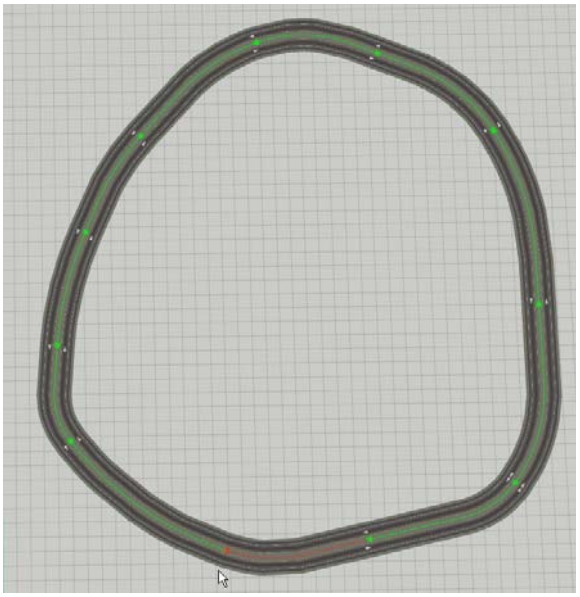
Certain situations, such as using roads to designate a "lot" for use with Lot and Material Tools, require a closed a loop of road:

1. Verify the Road Tool is active, and then LEFT-CLICK the end node to select it.

2. Then RIGHT-CLICK the start node to close the loop of road (see figure below).



An open loop road with the end node selected.



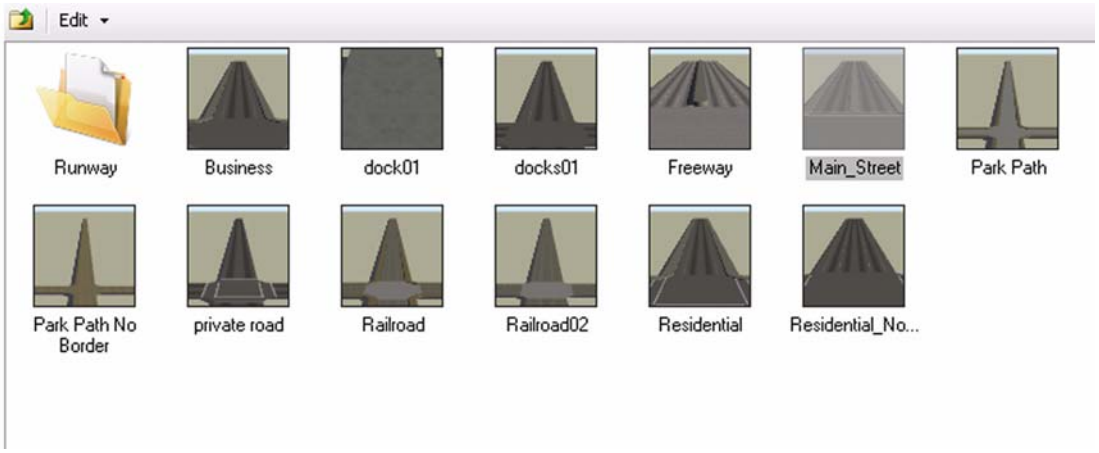
After linking the end node to the start node, the road becomes a closed loop.

Apply a Road Style

Use the Road Style Browser to select and apply styles to roads. You can select a road style before or after the target roads exist.

Selecting a Style Before Digitizing the Road

1. Verify the Road Tool is active. Then, in the Road Icons of the Road Style Browser, LEFT-CLICK the icon of the desired road style to select it.



2. When you Create a New Road, the style that's actively selected in the Road Icons will be applied to the new road as you create it.

Applying a New Style to an Existing Road

1. Verify the Road Tool is active. Then in the terrain select the road segments that you wish to apply the new style to.

2. In the Road Icons of the Road Style Browser, DOUBLE-CLICK the icon of the new style to apply it to the selected road segments. The new style replaces the old style, if any.

Editing a Road Style

Road styles control the textures and other global properties used on roads and related elements – including shoulders, sidewalks, medians, bridges, tunnels, crosswalks, and intersections. For details on editing an existing road style, see "Customize Road Styles" on page 90.

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6.3.2 Creating Intersections

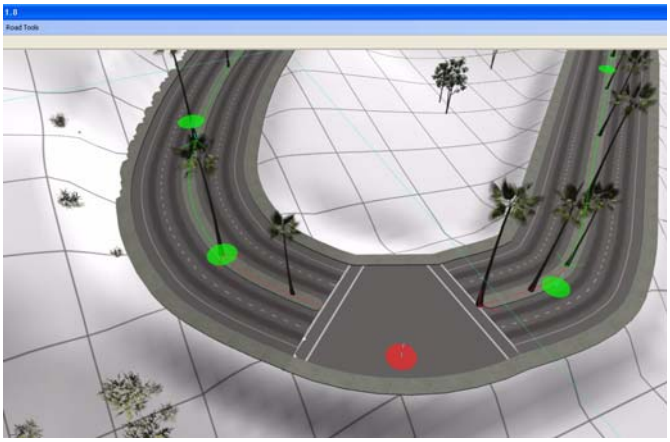
CityScape can do many powerful things with intersections, as documented in this section.

Create 2-way Intersections

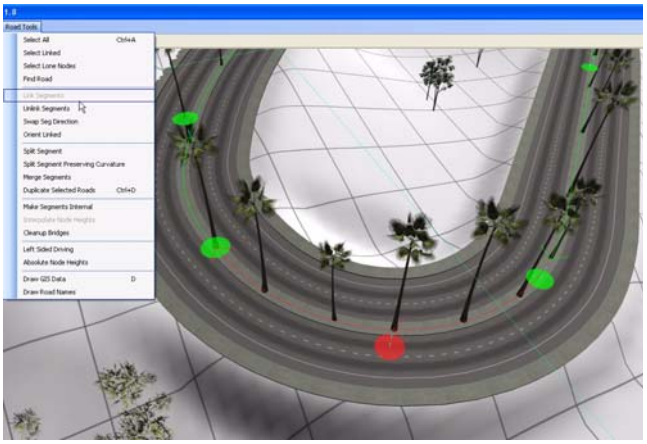
A 2-way intersection might not seem logical, but it can be useful for creating a turn with a fixed radius – with or without sidewalks, depending on the road style you are using.

1. Verify the Road Tool is active, and then select two segments on either side of the node you wish to change into a 2-way intersection.

2. Choose **Unlink Segments** from the Road Tools Menu in the Main Menu bar. (The Road Tools menu appears only when a Road Tool is active.) A 2-way intersection is created.



When the segments on either side of a node are *unlinked*, the result is a fixed radius turn through the intersection node. The option of sidewalks is controlled by the road style.



When the segments on either side of the node are *linked*, the result is a variable radius turn.

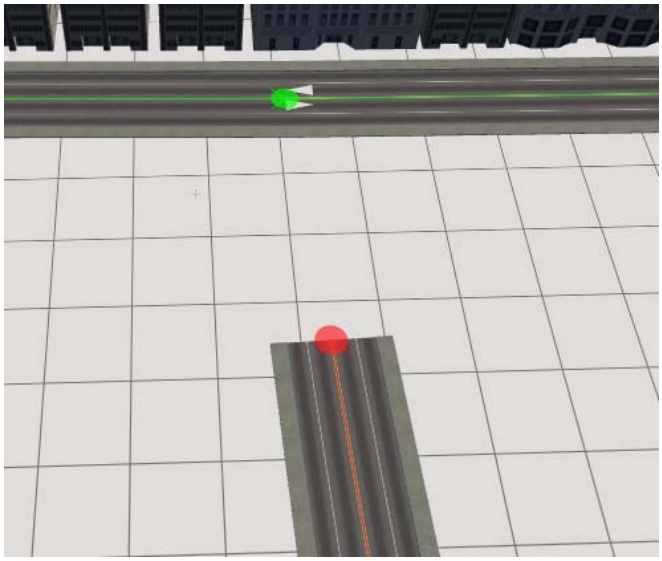
CREATE 3-WAY INTERSECTIONS

There are many ways to create 3-way intersections. This section shows you several common methods.

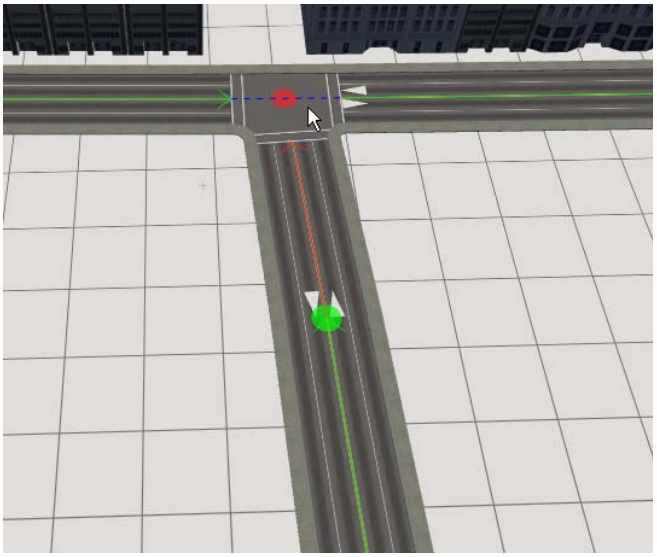
Option 1

1. Verify the Road Tool is active. Then extend a road as described in “Create a New Road” on page 99, until the new road is close to crossing an existing road.

2. Place the next node of the new road on top an existing node in the existing road. A 3-way intersection will be created automatically, and the node in the intersection is shared by both roads.



A new road is extended towards an existing road.



When the new road shares a node with the existing road, a 3-way intersection is born.

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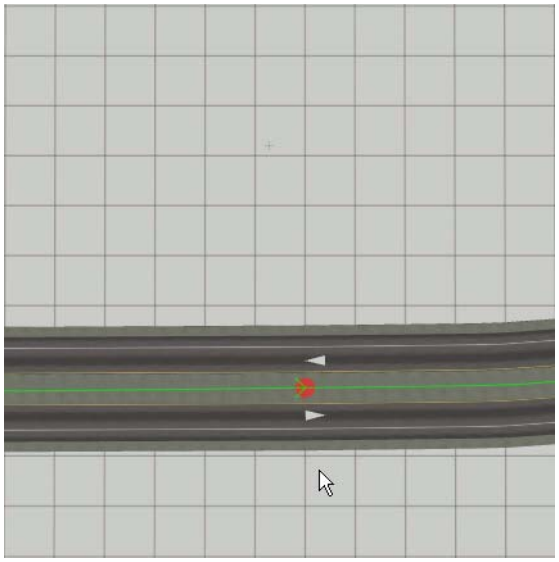
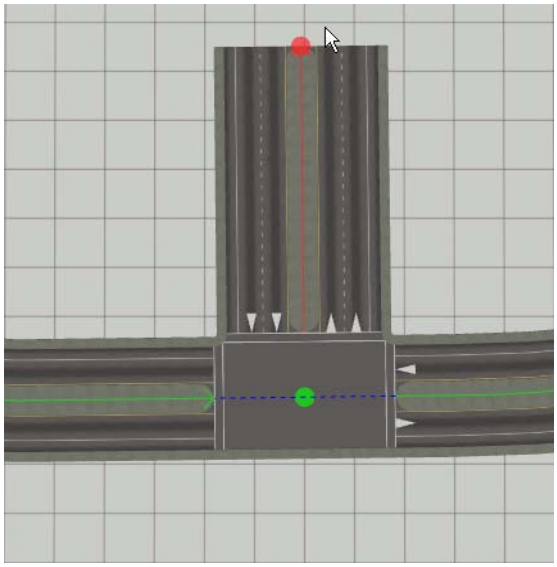
Option 2

1. Verify the Road Tool is active, and then select a node in an existing road, but don't select the segments to either side of the node. This node will become the center of the 3-way intersection.



The node in the original road is selected (red), but adjoining segments are not selected.



Placing a new node creates a new road that branches from the original road via a new intersection.

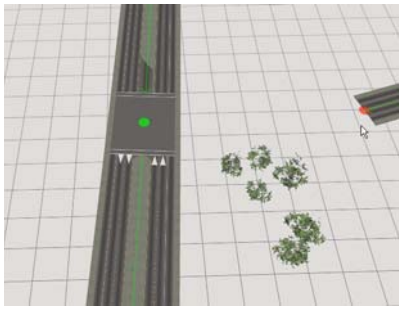
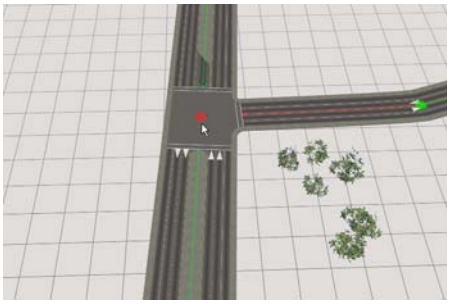
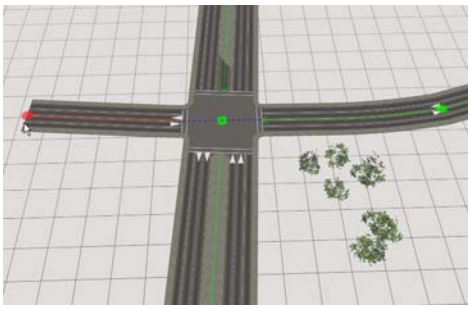
2. RIGHT-CLICK in an open area to the side of the original road. A 3-way intersection is automatically created, and a new road branches off the intersection. See the right image in the figure above.

3. You can optionally extend the new road, as described in "Create a New Road" on page 99.

Create 4-way Intersections

There are many ways to create 4-way intersections. This section covers one common method.

1. Verify the Road Tool is active, and then extend a road as described in “Create a New Road” on page 99, until the new road is close to crossing an existing road.
2. Place the next node of the new road on top an existing node in the original road. A 3-way intersection will be created automatically, and the node in the intersection will be automatically selected (red).
3. Place another node to extend the new road on the other side of the intersection, which in turn will convert the 3-way intersection into a 4-way.

Create Arbitrary Intersections

This section describes a generic method to create an intersection with an arbitrary number of roads connecting to it.

1. Verify the Road Tool is active, and then LEFT-CLICK a node in an existing road to select the node, but don’t select the segments to either side of the node.
2. RIGHT-CLICK in an open area near the original road. A new road branches off the node, and an intersection is formed or enlarged.
3. To add additional roads from this intersection, deselect all. Then repeat the previous 2 steps using the same node.

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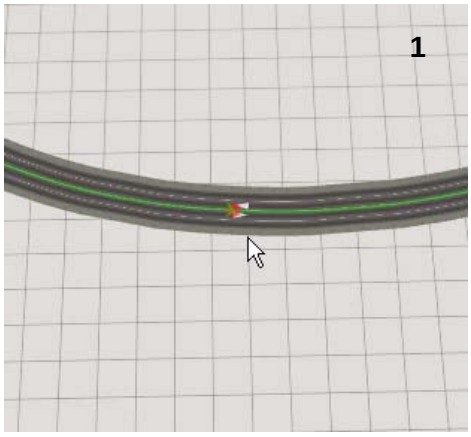
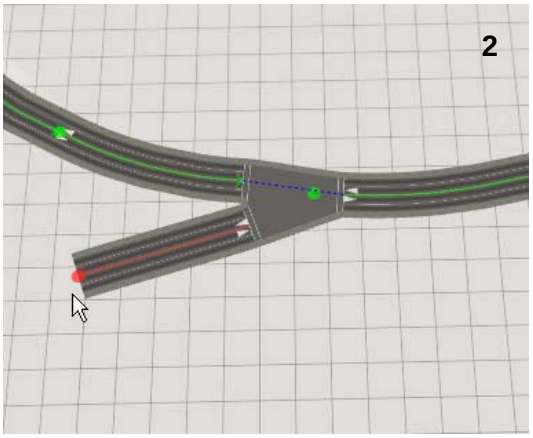
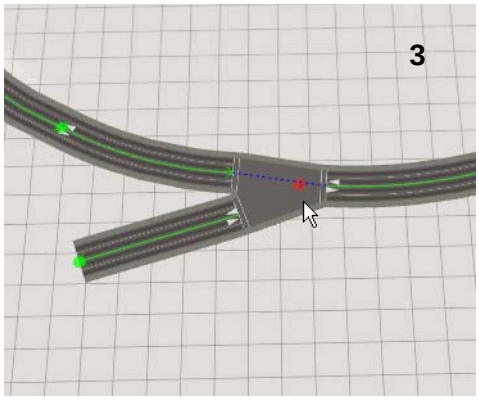
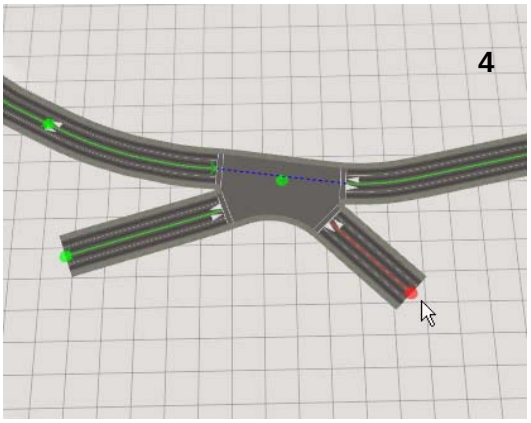
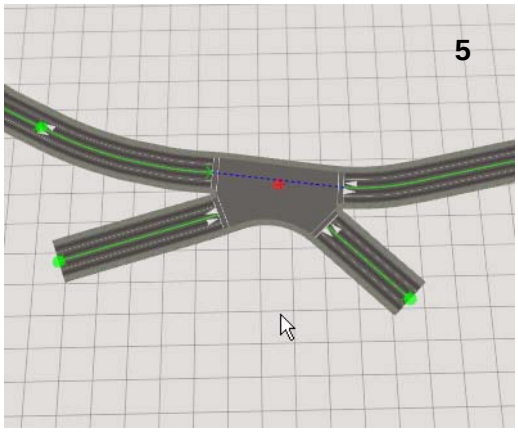
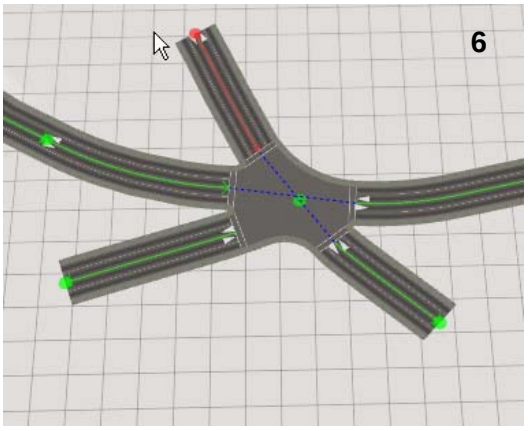

1


2


3


4


5


6

Delete Intersections

To delete an intersection, delete the node in the intersection, as described in “Move, Insert, and Delete Road Nodes” on page 108. Note that this action will also delete the road segments going into the intersection.

You can remove a 2-way intersection without removing any road nodes or segments. This process replaces the intersection with a smooth stretch of road:

1. Verify the Road Tool is active, and then select the road segments on either side of the node in the 2-way intersection.

2. Choose **Link Segments** from the Road Tools Menu in the Main Menu bar. The 2-way intersection is replaced by smooth road.

6.3.3 Editing Road Nodes

This section describes how to edit road nodes, which will modify the roads using those nodes.

Select Road Nodes

Before you can edit the properties of road nodes, or do much of anything with them, you must first select them while the Road Tool is active:

- To select one road node, LEFT-CLICK the **target node** (it will turn red).

- To select multiple nodes (first method), HOLD the SHIFT-KEY and LEFT-CLICK each node in succession.

- To select multiple nodes (second method), LEFT-CLICK-HOLD and DRAG THE MOUSE to enclose the target nodes in a pick box. This method will also select the segments within the pick box.

Move, Insert, and Delete Road Nodes

To adjust the exact position and shape of a road, and create / remove intersections, you will need to move, insert, and/or delete some nodes. To do so, ensure the Road Tool is active and proceed as follows:

- To move part of a road, just move the affected nodes. Here’s how:
  i. Select the nodes you wish to move.
  ii. Hover the mouse over one of the selected nodes.
  iii. LEFT-CLICK-HOLD and DRAG THE MOUSE to move the nodes, and the road with them.

- To insert a new node within an existing segment:
  i. LEFT-CLICK the **target segment** to select it.
  ii. RIGHT-CLICK the **target segment** to add a new *node* within it.

- To delete one or more nodes, select them and PRESS the **DELETE-KEY**. Deleting a node will also delete both segments on either side of it.

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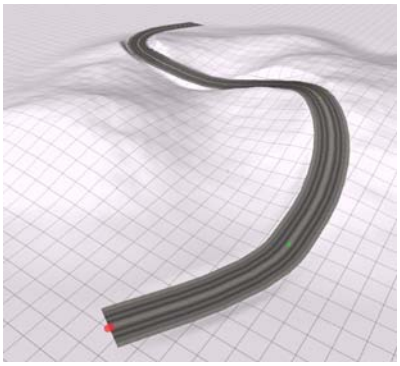
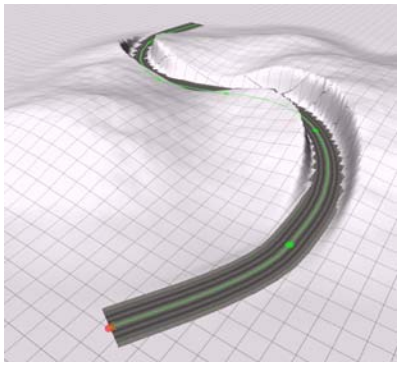
Main Widgets for Road Nodes

When the Road Tool is active, you can use it to select one or more road nodes, and then use the following Road Node Widgets to customize those nodes:

- **Name:** gives you the option of assigning a name to the selected node(s).
- **Position X, Y:** the position of the selected node, in meters, relative to the origin (center) of the terrain.
- **Absolute / Relative Height:** the elevation of the selected node. Depending on your selection of *Absolute Node Heights* in the Road Tools menu, the elevation of the selected node can be absolute or relative:
  - **Absolute Height** is measured in meters from the ground level to the selected nodes. (The ground level is set by CityScape.) A road with an absolute height of 10 meters, for example, would stay flat at 10 meters in elevation, even through a mountain range. CityScape would form a deep chasm through the mountains to accommodate such a road.
  - **Relative Height** is relative to the terrain surrounding the node. Nodes using a relative height of 10 meters would cause the surface of the road to sit 10 meters above mountainous terrain, which could be thousands of feet above ground level.
  - For an example of usage, see "Building Bridges and Overpasses" on page 122, as well as the images below.



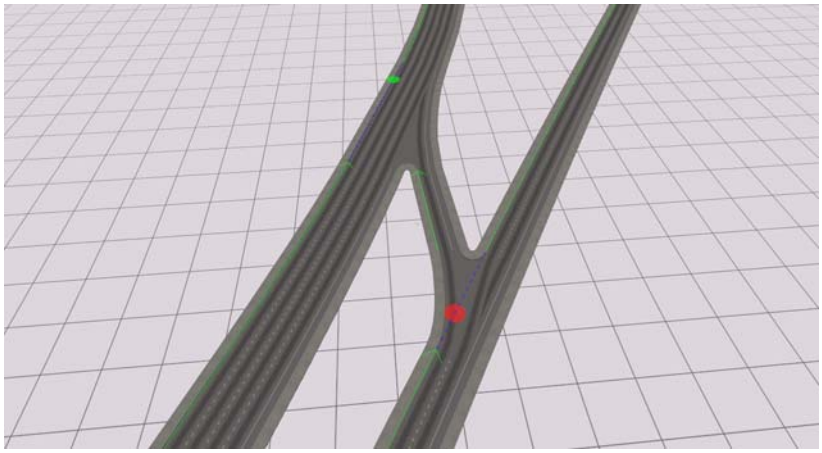


- **Construction Type:** designates the node as part of a regular road, a bridge, or a tunnel (see below). For an example of usage, see "Building Bridges and Overpasses" on page 122.
  - *Road:* designates the node as part of a regular road. CityScape cuts and fills the terrain to dynamically conform to the heights of the nodes in the road.
  - *Bridge:* designates the node as part of a bridge supported by pylons. For best results, ensure the nodes' Relative Height is significantly higher than the surrounding terrain, and all the segments between raised nodes are also designated as Bridge.
  - *Tunnel:* designates the node as part of a tunnel. For best results, ensure the nodes' Relative Height is significantly lower than the surrounding terrain, and all the segments between lowered nodes are also designated as Tunnel.

- **Bank:** Creates a sideways slope in the selected road by the specified degrees. Put another way, Bank creates a "camber" perpendicular to the direction of the road. Most roads are flat (Bank = 0 degrees), but some roads (particularly race tracks) use banked turns. For an example of usage, see “Banking and Sloping Roads” on page 118.

- **Slope Type:** Affects the slope of intersections on hillsides. For an example of usage, see “Banking and Sloping Roads” on page 118.
  - *Flat:* Forces an intersection on a hillside to be flat, regardless of the slope of the roads coming into and out of the intersection.
  - *Min Height:* Aligns the slope of the intersection with the slope of the connecting road which is closest to flat.
  - *Max Height:* Aligns the slope of the intersection with the slope of the connecting road which has the most extreme slope.

- **Merge / Split:** causes a road to split off the parent road, or merge into the parent road, without an intersection. For an example of usage, see “Creating Merges and Splits” on page 120.



- **Merge/Split Scale:** influences the length of the merge/split intersection.

- **Push Scale:** moves a merge/split relative to the node on the parent road controlling the merge/split (see figure).

- **Enable Culdesac:** makes the selected node into a cul-de-sac, provided the node is an end node (i.e. the last node in an open road). For an example of usage, see “Making Cul-de-Sacs” on page 122.

- **Culdesac Radius:** If Enable Culdesac is selected, you can set the Culdesac Radius as a percentage of the road’s width. For an example of usage, see “Making Cul-de-Sacs” on page 122.

- **Culdesac Squash:** A value of 100% creates a roughly circular cul-de-sac. Smaller values "squashes" the cul-de-sac. For an example of usage, see “Making Cul-de-Sacs” on page 122.

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6.3.4 Editing Road Segments

This section describes how to edit road segments, which will modify the roads using those segments.

Select Road Segments

Before you can edit the properties of road segments, or do much of anything with them, you must first select them while the Road Tool is active:

- To select one road segment, LEFT-CLICK the **green line** defining the segment. The line will turn red when it becomes the active selection.
- To select multiple segments (first method), HOLD the SHIFT-KEY and LEFT-CLICK **each segment** in succession.
- To select multiple segments (second method), LEFT-CLICK-HOLD and DRAG THE MOUSE to partly or completely enclose the **target segments** in a pick box. This method will also select the nodes within the pick box.

Delete Road Segments

- To delete segments, select them and PRESS the DELETE-KEY.

Applying a New Road Style

For details, see “Apply a Road Style” on page 102.

Main Widgets for Road Segments

You determine the exact positions of road nodes when you place them in a terrain. CityScape then calculates the shape of the road segments interconnecting the nodes you have placed.

You can use the Road Tool to select one or more road segments, and then use the following Road Segment Widgets to customize those segments:

- **Name:** allows you to assign a name to the selected segment(s). You can show or hide these names by toggling *Draw Road Names* in the Road Tools Menu.
- **Min Rounded Radius:** specifies the smallest radius, in meters, of curves generated by CityScape using the selected segments. For an example of usage, see “Tuning Curves” on page 118.
- **Curve Type:** changes the math that CityScape uses to calculate the exact shape of curves using the selected segments. For an example of usage, see “Tuning Curves” on page 118.
  - *Linear:* Forces the selected segments to be as straight as possible, given the shape of nearby segments.
  - *Cubic:* Causes the shape of a curved segment to be defined by four things: the positions of the two nodes at either end, and two tangent points that you can edit Using the Road Tangent Tools. Cubic turns usually have a variable radius.
  - *Rounded:* Attempts to make the selected segment a constant radius through a curve. If CityScape is unable to create a rounded curve, then it will create a Cubic curve.

- **Construction Type:** designates the selected segments as part of a regular road, a bridge, or a tunnel (see below). For an example of usage, see “Building Bridges and Overpasses” on page 122.
  - *Road:* designates the segment as part of a regular road. CityScape cuts and fills the terrain to dynamically conform to the heights of the segments in the road.
  - *Bridge:* designates the segment as part of a bridge supported by pylons. For best results, ensure the *Relative Height* of the segments is significantly higher than the surrounding terrain, and the nodes connecting raised segments are also designated as *Bridge*.
  - *Tunnel:* designates the segment as part of a tunnel. For best results, ensure the Relative Height of the segments is significantly lower than the surrounding terrain, and the nodes between lowered segments are also designated as *Tunnel*.

- **Right Lanes:** Determines the number of lanes on the right side of the road, relative to the direction of the selected segments. The minimum allowable value is *Right Lanes = 1*. For an example of usage, see “Defining Number of Lanes, One-Way Roads” on page 118.

- **Left Lanes:** Determines the number of lanes on the left side of the road, relative to the direction of the selected segments. You can create one-way roads via *Left Lanes = 0*. For an example of usage, see “Defining Number of Lanes, One-Way Roads” on page 118.

- **Left Turn Start:** Places a left turn lane at the starting node of the selected segment, provided the starting node is an intersection and there’s enough space in the median. For an example of usage, see “Creating Left Turn Lanes” on page 119.

- **Left Turn End:** Places a left turn lane at the ending node of the selected segment, provided the ending node is an intersection and there’s enough space in the median. For an example of usage, see “Creating Left Turn Lanes” on page 119.

- **Center Curve:** designates the center of the curve as either absolute (i.e. half-way between the left and right sides of the road), or relative (i.e. the line separating the right and left sides of the road). The effects of Center Curve are evident only when the left and right sides of the road have different number of lanes. For an example of usage, see “Tuning Curves” on page 118.

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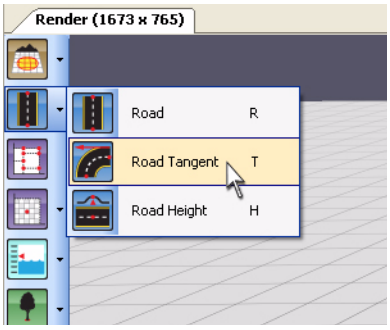
6.4 USING THE ROAD TANGENT TOOLS

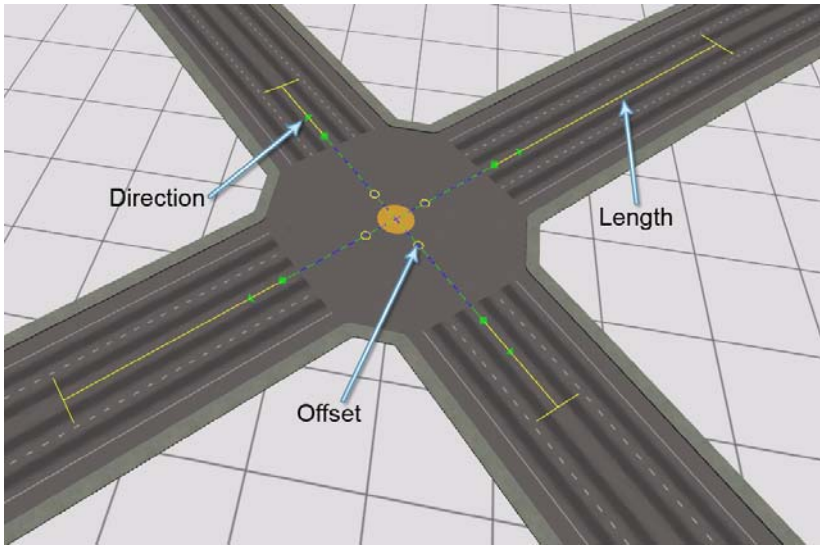
Using the Road Tangent Tool, you can fine-tune the exact shape of curves and intersections. Tangents use ***cubic curves***, which are defined by the following things: the positions of the nodes on either end of the affected segment, and the magnitude and direction of the tangents at the end of the segment.

To use the Road Tangent Tool, open a terrain with roads (or create some roads), and then do this:



1. Activate the **Road Tangent Tool**: LEFT-CLICK the Road Tools icon in the Tools Palette, and then LEFT-CLICK the Road Tangent Tool icon in the Sub-Tools Palette.

2. In the 3D Render View, LEFT-CLICK the **node in an intersection**. The following manipulators appear in the Render Window; see the linked sections for usage:
   - Direction Manipulator
   - Length Manipulator
   - Offset Manipulator



You can also use the Road Tangent Tool on a node in a curve, where no intersection is present – but an intersection node enables additional functionality in the Road Tangent Tool, so it serves as a more complete example.

6.4.1 Direction Manipulator

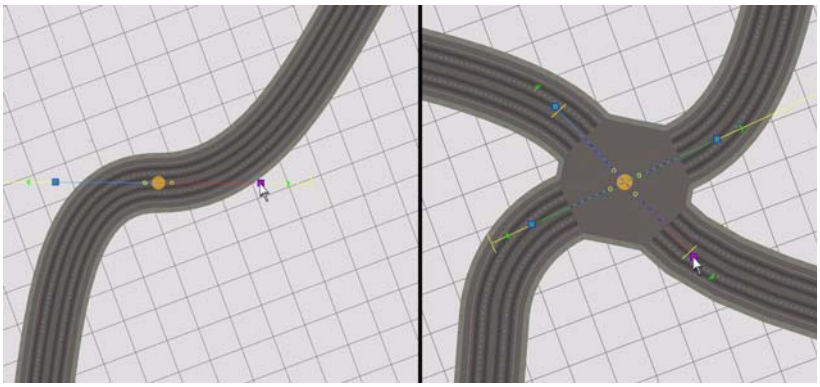
Each Direction Manipulator controls the exact direction from which a road segment enters an intersection. Direction Manipulators give you the means to adjust the directions from which roads enter an intersection. (See figure.)

Visually, a Direction Manipulator is a square box in a tangent line that appears when you select a road node with the Road Tangent Tool. For example, a 3-way intersection has 3 roads coming into it, and therefore 3 tangent lines (one for each incoming road) and 3 Direction Manipulators.

The following directions assume you have already activated the **Road Tangent Tool**, and selected a node in an intersection:

1. LEFT-CLICK-HOLD one of the **Direction Manipulators** to select it. The Direction Manipulator will turn from green (unselected and unmodified) to red (selected).

2. With the left mouse button still depressed, and hovering over the selected Direction Manipulator, drag the mouse perpendicular to the tangent line. Note the changes to the intersection.



3. Repeat the previous steps to move the other Direction Manipulators for the other roads coming into the intersection.

Notes:

- When you modify the position of a Direction Manipulator, it changes from a green square (unmodified) to a blue square (modified).

- To return a modified Direction Manipulator to its original position (i.e. to let CityScape determine its position), simply select the **Direction Manipulator** and press the **Delete-Key**. This will also reset the position of the associated Length Manipulator.

- When you move one Direction Manipulator, CityScape will dynamically move some or all of the other Direction Manipulators at the same intersection, unless you lock them down. You can lock down the positions of all the manipulators at an intersection by doing all the following things:
  - Unlink all the road segments through the intersection. When two road segments are linked through an intersection, moving the Direction Manipulator for one segment causes CityScape to move the Direction Manipulator for the other linked segment as well.
  - Move all the Direction Manipulators at the intersection (at least a little bit). When you change the default position of a Direction Manipulator, CityScape will attempt to preserve your custom setting.

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6.4.2 Length Manipulator

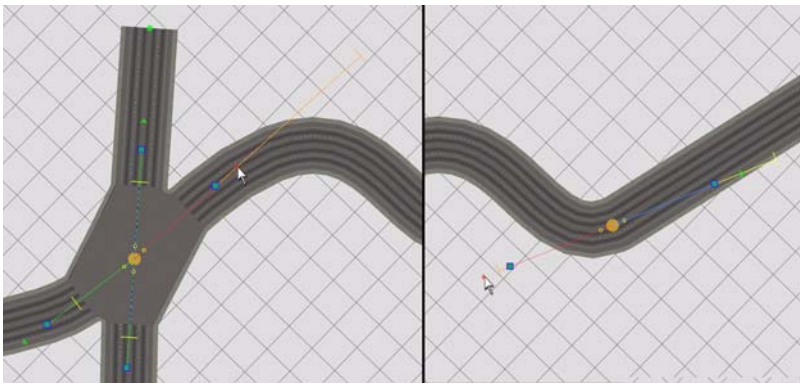
The magnitude of a tangent line affects the shape of the road segment associated with that tangent line. (For additional background, see the "Direction Manipulator" section.)

Visually, the Length Manipulator takes the form of an arrow along a tangent line. The arrow allows you to change the magnitude of its tangent line. This magnitude is visually represented by the length of the yellow line, with a "T" ending, that extends beyond the green tangent line itself.

The following directions assume you have already activated the Road Tangent Tool, and selected a node in an intersection:

1. LEFT-CLICK-HOLD one of the Length Manipulators (arrows) to select it. The Length Manipulator will turn from green (unselected) to red (selected). Also, the magnitude line turns from yellow (unselected) to orange (selected), and the associated Direction Manipulator changes from a green square (unmodified) to a blue square (modified).

2. With the left mouse button still depressed, and hovering over the selected Length Manipulator, drag the mouse parallel to the tangent line. The magnitude line will get longer or shorter to indicate its change in magnitude. Note the changes to the associated road.



3. Repeat the previous steps to adjust the other Length Manipulators for the other roads coming into the intersection.

Note: To return a modified Length Manipulator to its original length (i.e. to let CityScape determine its magnitude), simply select the **Length Manipulator** and press the **Delete-Key**. This will also reset the position of the associated Direction Manipulator.

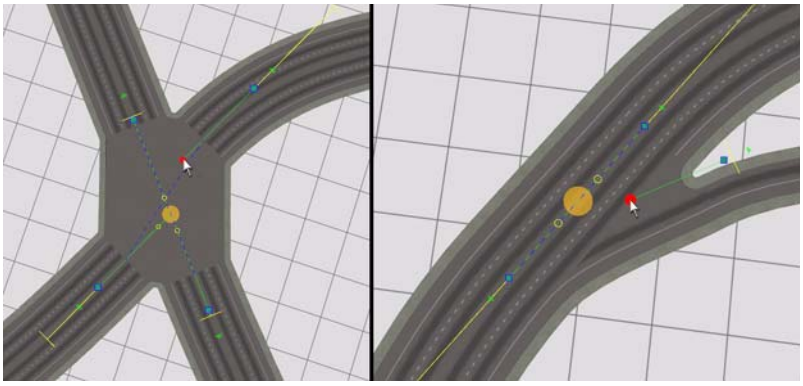
6.4.3 Offset Manipulator

By default, the tangent lines of the road segments entering an intersection point directly at the node in the intersection. This remains true even if you modify the tangents using the Direction and Length Manipulators. Using the Offset Manipulator, you can offset a tangent line relative to the node it would otherwise point at

Visually, an Offset Manipulator takes the form of a yellow circle. The following directions for using the Offset Manipulator assume you have already activated the Road Tangent Tool, and selected a node in an intersection:

1. LEFT-CLICK-HOLD one of the Offset Manipulators (circles) to select it. The Offset Manipulator will turn from yellow (unselected) to red (selected).

2. With the left mouse button still depressed, and hovering over the selected Offset Manipulator, drag the mouse to move the Offset Manipulator. Note that the modified Offset Manipulator changes from a hollow yellow circle to a solid yellow circle.



3. Repeat the previous steps to adjust the other Offset Manipulators for the other roads coming into the intersection.

Notes:

- To return a modified Offset Manipulator to its original position (i.e. pointing at the node in the intersection), simply select the **Offset Manipulator** and press the **Delete-Key**.

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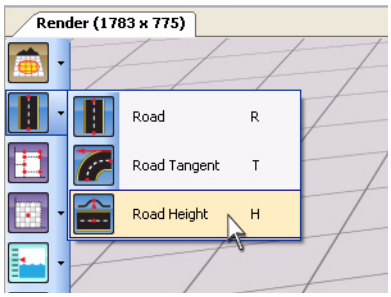
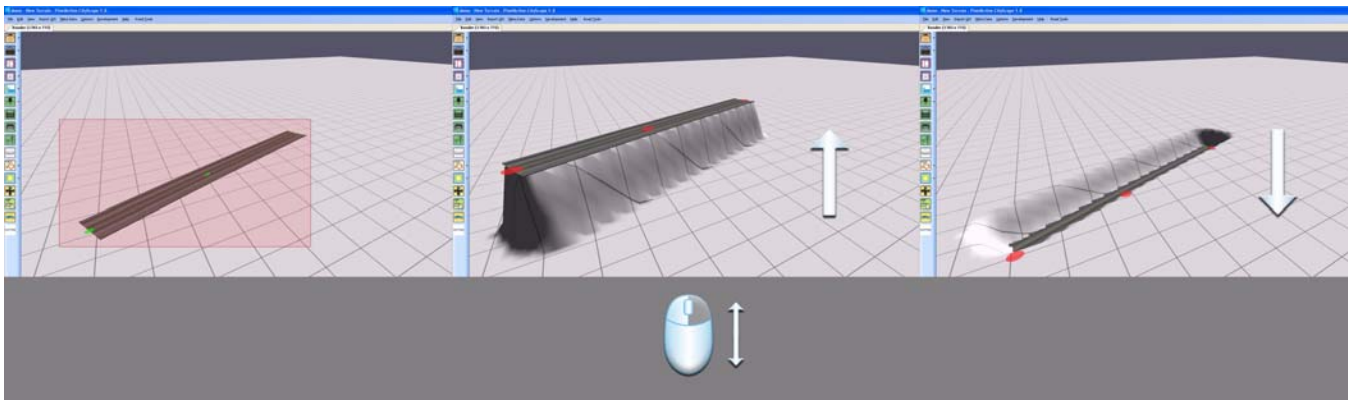
6.5 USING THE ROAD HEIGHT TOOL



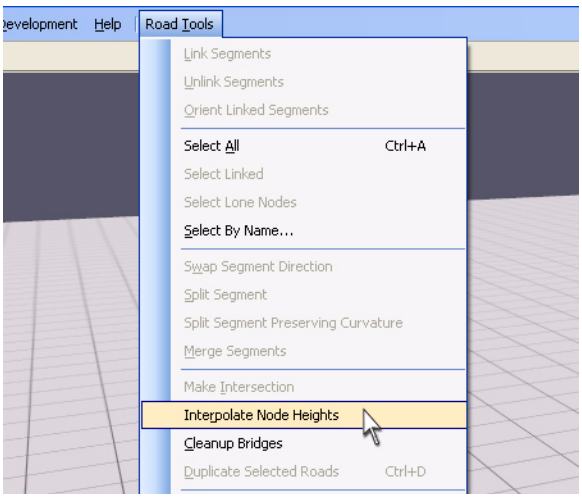
The Road Height Tool lets you visually raise and lower road nodes, so as to increase or decrease the elevation of a section of road. Usage is simple:

1. In CityScape, open a terrain with roads, or create some roads.
2. Select the **Road Height Tool** from the Tools Palette.
3. Select the node(s) you wish to raise or lower.
4. RIGHT-CLICK-HOLD and drag the mouse up to raise the selected nodes, and down to lower them.



You can also smooth transitions in road height, as follows:

1. Select the nodes that you want to raise / lower so as to spread elevation changes more evenly along the road. (Note: Selected nodes must form a connected strip. If they don't, then the Interpolate Road Heights option will be greyed-out (see next step).
2. Choose **Road Tools > Interpolate Road Heights** from the Main Menu.



6.6 CUSTOMIZING ROADS: ADVANCED TECHNIQUES

This section describes advanced techniques for customizing road networks in terrains. The information in this section assumes you are working with an existing network of road and intersections, which you can create by following the instructions in “Using the Road Tool” on page 99.

6.6.1 Banking and Sloping Roads

The bank of a road, and the slope of an intersection on a hillside, are controlled by the following parameters in the Main Widgets for Road Nodes:

- **Bank**
- **Slope Type**

6.6.2 Tuning Curves

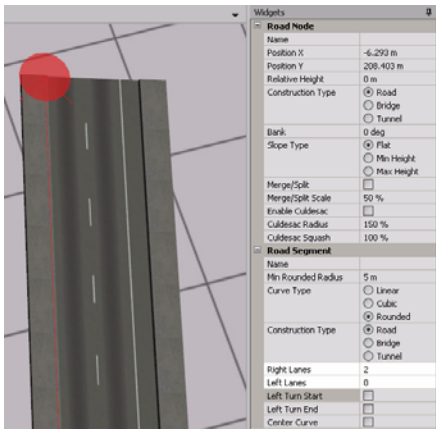
You can customize the shape of curves in roads by Using the Road Tangent Tools in conjunction with the following parameters in the Main Widgets for Road Segments:

- **Curve Types**
- **Center Curve**

6.6.3 Defining Number of Lanes, One-Way Roads

Every road segment has a direction, which is indicated by a green arrow on one end of the road segment. These directions determine which side of a given road is the "right" lane, and which is the "left" lane.

You can customize the number of lanes in each direction by editing the following parameters in the Main Widgets for Road Segments.



- **Right Lanes:** Must be 1 or greater.
- **Left Lanes:** Create one-way roads via Left Lanes = 0.

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6.6.4 Selecting Right or Left-Sided Driving

The direction that traffic moves is indicated by white arrows in the lanes of your roads. These white traffic arrows should not be confused with the green arrows that indicate the direction of road segments.

By default, CityScape uses right sided driving, which has a relative meaning in CityScape:

- **Right sided driving** means that cars on the right side of the road move in the same direction as the road segment.
- **Left sided driving** means cars on the left side move in the same direction as the road segment.

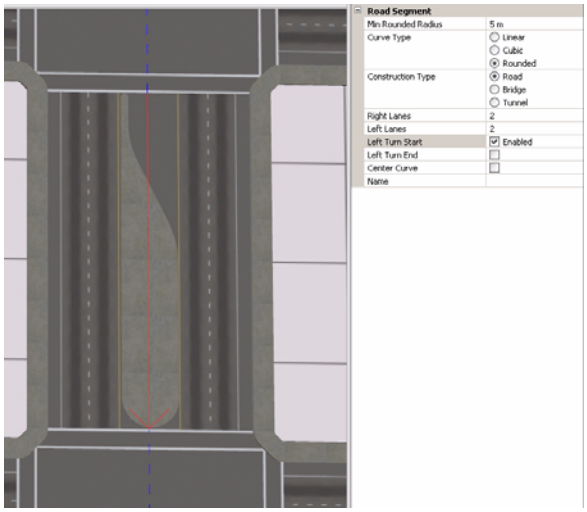
As implied by the *relative* definitions above, swapping the direction of road segments will NOT change *absolute* driving side. (For details, see "Changing the Direction of Road Segments" on page 126.) You can *globally* change the driving side on all roads in the open terrain, as follows:

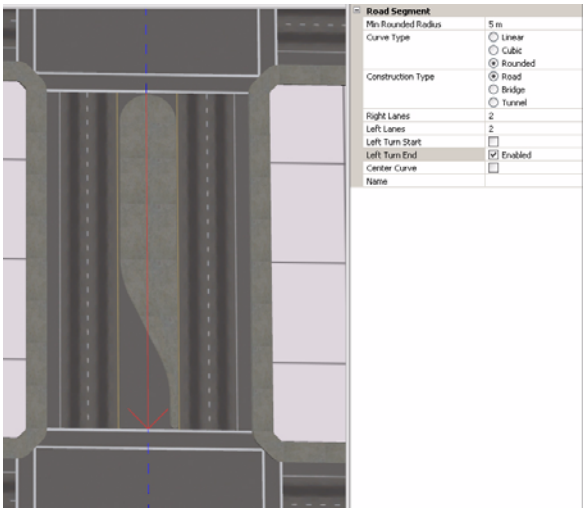
1. In the Road Tools Menu (which appears in the Main Menu whenever the Road Tool is active), select **Left Sided Driving**. The white arrows switch to indicate the direction of traffic has switched.
2. To switch back, select **Right Side Driving** in the Road Tools Menu.

6.6.5 Creating Left Turn Lanes

You can add a left turn lane approaching an intersection, provided that the road uses a style with a roomy median (left turn lanes use the space in the median). Just enable one of the following parameters, which are defined in the "Main Widgets for Road Segments" on page 111:

- **Left Turn Start** (left image below)
- **Left Turn End** (right image below)





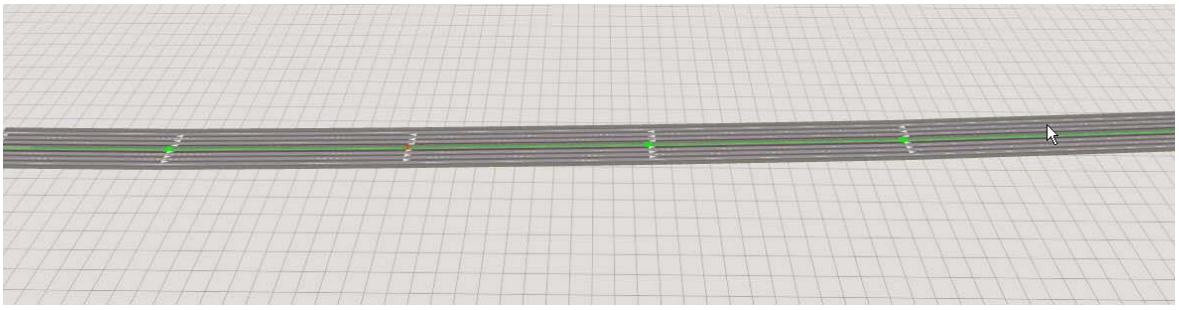
6.6.6 Creating Merges and Splits

A merge is when one road flows into another without an intersection, such as a freeway on-ramp. A split is when a new road flows out of the original road without an intersection. Many variations of merges and splits are possible with CityScape.

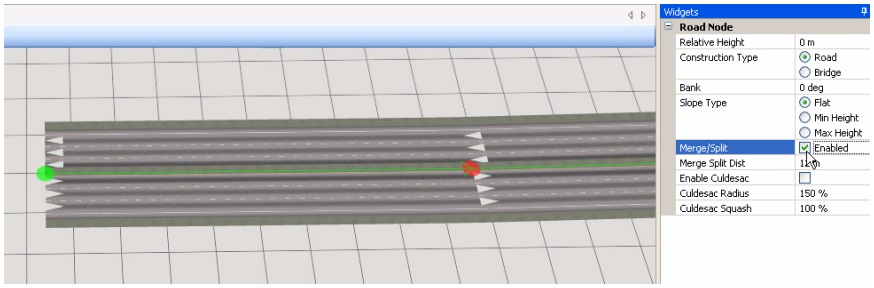
The example below shows you how to split a new road off an existing road, and then merge the new road back into the original road.

1. Activate the **Road Tool**, if necessary.

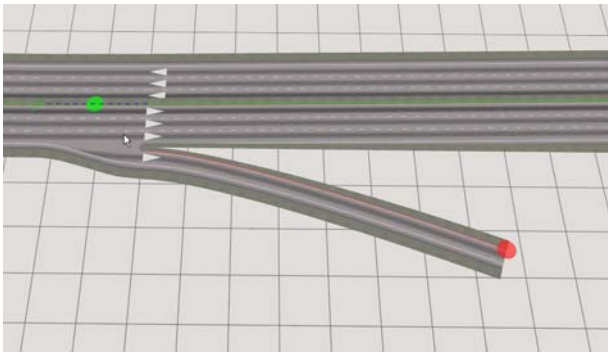
2. In the terrain, identify or create a stretch of wide road, such as a freeway.



3. On the wide road, select a **node** near one side. In the Road Node Widgets, enable the **Merge/Split** option, to prepare to split a new road off the selected node.



4. With the original node still selected, place a **new node** to create the ramp via a split (i.e. not an intersection).



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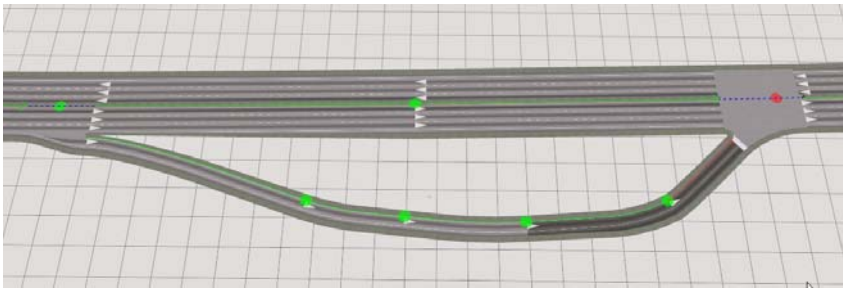
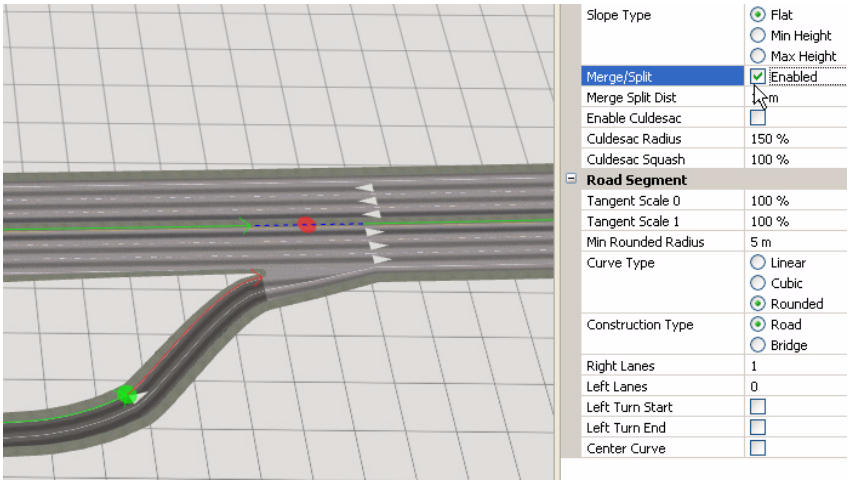
5. Place a few more nodes to continue the off-ramp. Then bend the ramp towards the freeway, and merge the ramp back into the freeway by selecting a node on the freeway. When you initially connect the ramp to the node on the freeway, an intersection might form (not a "merge" you would expect for a freeway ramp).



6. If an intersection forms, convert it to a proper merge by first selecting the **node on the freeway**, and then enabling **Merge/Split** in the Road Node Widgets.

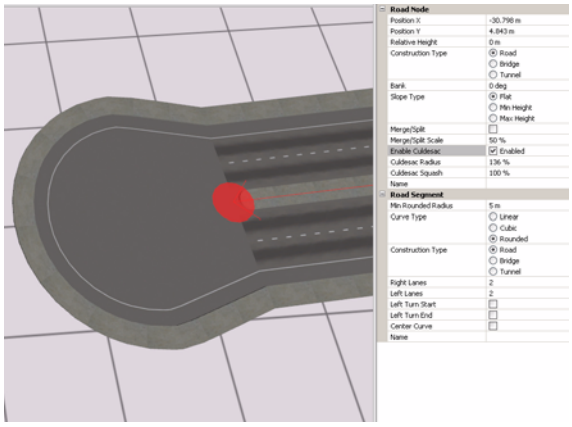


7. Adjust **Merge Split Scale** until you're happy with the offset distance between the freeway node and the point the ramp enters the freeway. (Note: Advanced users should not use Merge Split Scale, but rather use offset nodes to customize the shape of merge/splits. For details on offset nodes, "Using the Road Tangent Tools" on page 113.)

6.6.7 Making Cul-de-Sacs

You can create and customize a cul-de-sac using the Road Node Widgets:

1. Activate the **Road Tool**, if necessary.
2. In the terrain, select or create a road with a free end node.
3. With the end node selected, use the Road Node Widgets as follows:
   – **Enable Culdesac:** Active to create a cul-de-sac at the selected end node (and only an end node).
   – **Culdesac Radius:** Adjust to customize the radius of the cul-de-sac.
   – **Culdesac Squash:** Adjust below 100% create a "squashed" elliptical shape in the cul-de-sac.



6.6.8 Building Bridges and Overpasses

You can create and customize bridges / overpasses via the Construction Type parameters in both the Road Node and the Road Segment Widgets.

- **Construction Type:**
  – *Road:* Causes the selected strip of road to sit directly only the ground – although the ground immediately under the road could optionally be raised with the *Road Height* Road Widget.
  – *Bridge:* Causes the selected strip of road to elevate above the ground via pylons, much like an overpass or simple bridge. (Note: For the *Bridge* Road Widget to work as expected, the *Relative Height* Widget must be substantially positive.)

- **Road Height** is set by the Absolute / Relative Height widget for the affected nodes. Changing Road Height raises or lowers the selected stretch of road – relative to zero elevation in the terrain (Absolute Height), or to the terrain near the node (Relative Height).
  – To switch between absolute and relative road height, select **Road Tools > Absolute Node Heights** from the Road Tools Menu, which appears in the Main Menu when any Road Tool is active.
  – If *Construction Type* = *Road*, then the road will sit on a dirt fill for positive Road Height; the road will sit in a cut (i.e. an artificial canyon) for negative Road Height.

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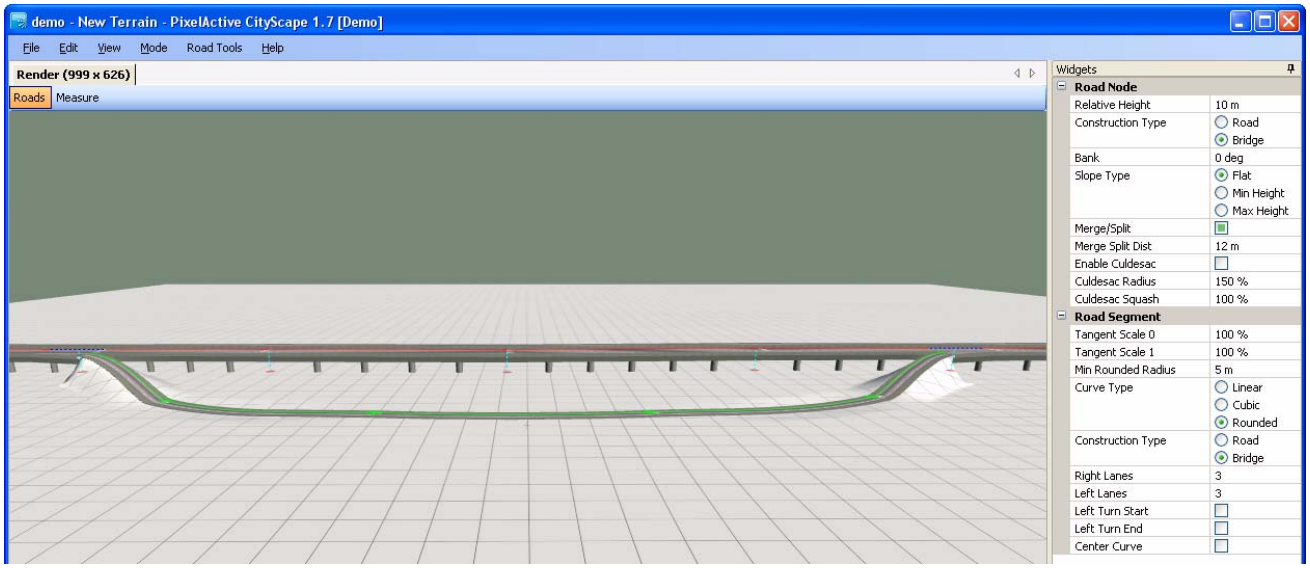
– If *Construction Type* = *Bridge*, then the road will sit on pillars or other support typical of bridges. (Note: Only positive Road Height works with Bridge. Negative Road Height has no effect with bridges.)

Although CityScape will let you use the "Road" construction type for nodes, and the "Bridge" construction type for segments (or vice versa), you would rarely want to do that. In most cases, you should use the same Construction Type for both nodes and associated segments.

In the example below, the ramp uses the Road Construction Type, and the Freeway uses the Bridge Construction Type.

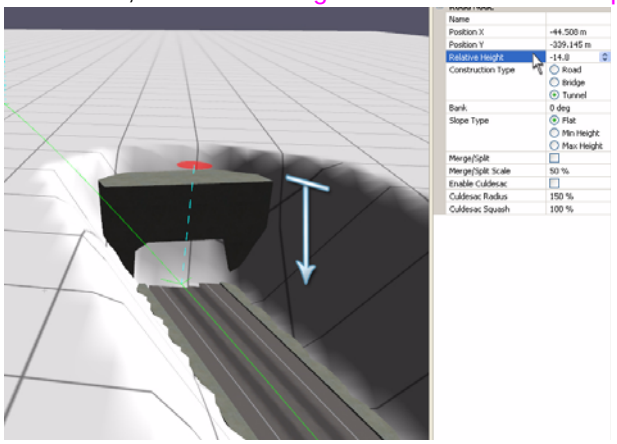


6.6.9 Building Tunnels

CityScape allows you to create entrances to tunnels at nodes only. To create a tunnel, select the road nodes and segments that will be in the tunnel, and then edit the Road Widgets, as described below.

Road Node Widgets

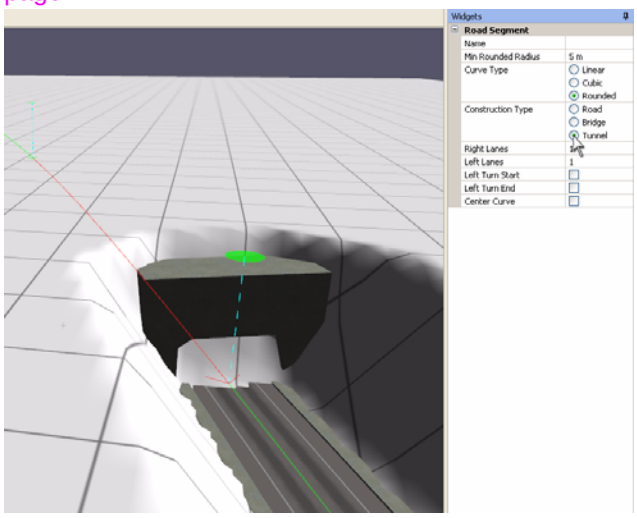
For details, see "Main Widgets for Road Nodes" on page 109



- **Absolute / Relative Height** must move the nodes far enough under the ground for the tunnel to be fully buried.
- **Construction Type:** must be set to *Tunnel*.

Road Segment Widgets

For details on these widgets, see "Main Widgets for Road Segments" on page 111.



- **Construction Type:** must be set to *Tunnel*.

Road Style Sub-Widgets

The Road Style Sub-Widgets are not critical to the basic existence of a tunnel. These Sub-Widgets fine-tune the appearance of the entrance to the tunnel. For details, see the Tunnel section of "Road Style Sub-Widgets" on page 90.

6.6.10 Linking and Unlinking Road Segments

If a 2-way intersection appears in your road, but your desired goal is to have a smooth road without an intersection (like the right image below), then you can "link segments" to solve your problem.

More generally, linking and unlinking segments on either side of an intersection affects the exact positions of incoming and outgoing roads.

To link or unlink segments across intersections, follow these steps:

1. Start with an existing intersection, or create such an intersection by following the instructions earlier in this chapter.
2. Select two segments that you wish to link or unlink across an intersection. The two segments must share the node at the center of the intersection.

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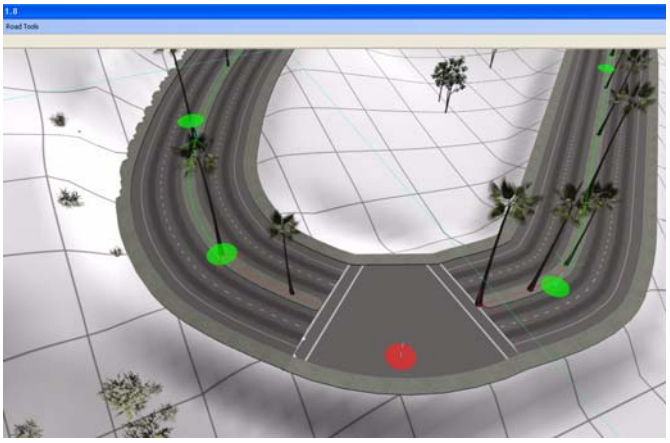
3. To link the two selected segments, choose **Link Segments** from the Road Tools Menu. (Shortcut: HOLD the SHIFT KEY and RIGHT-CLICK on empty terrain.)
   – With 2-way intersections, linking segments replaces the intersection with smooth road, as shown below.
   – With 3+ way intersections, a dashed blue line appears through the intersection to show the linking.

2-way intersection before linking segments on either side of the selected (red) node.

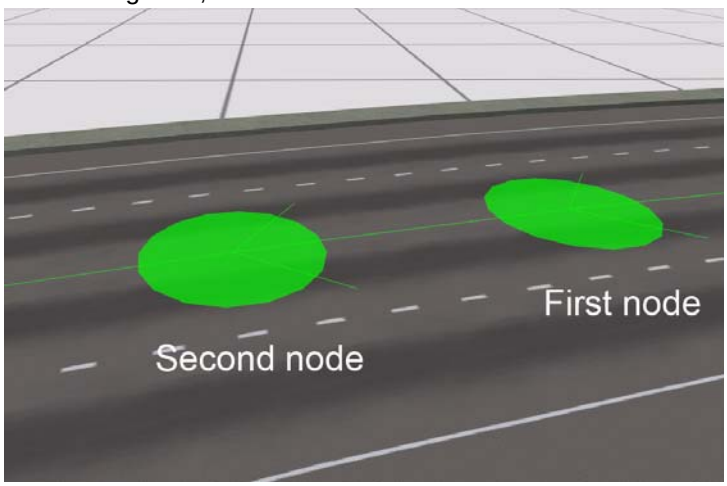


After linking segments, the 2-way intersection is replaced by smooth road through the selected node.

4. To unlink the two segments choose **Unlink Segments** from the Road Tools Menu.

6.6.11 Changing the Direction of Road Segments

Every road segment has a direction, which is indicated by a green arrow where the segment approaches the next node. The initial direction corresponds to the order in which the nodes were placed: The first node starts the first segment; the second node ends the first segment and starts the second segment; etc.



References to right and left sides of a road segment, as well as its start and end, are relative to the direction of the segment.

The directionality of segments has a relative role in determining the driving side (i.e. the direction of traffic in America verses the UK).

1. To switch the direction of segments, first select them.

2. Then choose **Swap Seg Direction** in the *Road Tools Menu*, which appears in the Main Menu whenever you activate any of the Road Tools.

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Chapter 7: Using Lot Tools

This chapter is in progress and will be released soon.

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Chapter 8: Using Water Tools

This chapter is in progress and will be released soon.

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Chapter 9: Using Material Tools

This chapter is in progress and will be released soon.

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Chapter 10: Using Prop Tools

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10.1 OVERVIEW OF PROP TOOLS

Props are models constructed from 3D meshes and optional 2D materials. Examples of useful props include buildings, trees, mailboxes, street lights, cars, and so forth.

You can create props in external software such as Maya, 3D Studio Max, Creator, and SketchUp; configure and export props from your external software to an intermediary format, such as Collada; and then import the props into CityScape, where the props will be stored in the prop library of your project so you can reuse the same props in many terrains.

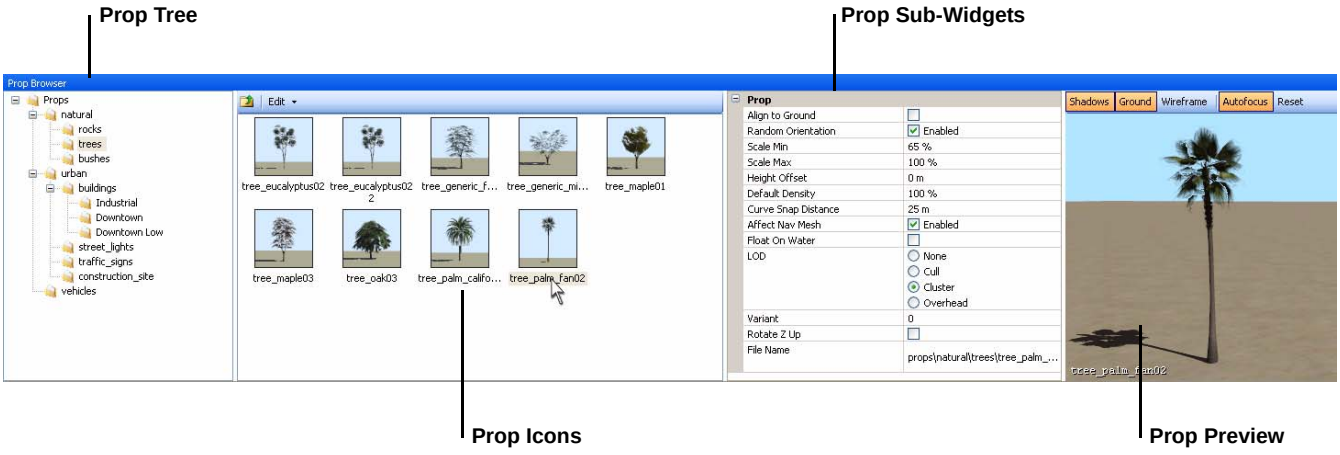
To get started, you can use the sample props in the prop library of the Demo Project. Ultimately, you will want to create your own prop library and import props into it. Using the Prop Browser in CityScape, you can use, modify, and import props in the prop library of your project.

This chapter fully documents the Prop Browser and Prop Tools.

10.2 USING THE PROP BROWSER

When you select any of the Prop Tools, the Prop Browser appears near the bottom of the CityScape window, by default. The Prop Browser lets you import and manage a potentially vast collection of props in a Prop Library for your project. Using the Prop Tools, you can place and configure props in terrains.

The Prop Browser is made up of four windows. From left to right, these windows include the Prop Tree View, the Prop Icon View, the Prop Sub-Widget Panel, and the Prop Render View. For details on using the standard features of the Prop Browser, such as creating and moving folders, see “Using the Library Browser” on page 78. For details on features that are unique to the Prop Browser, read on.



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10.2.1 Select Props

The Prop Icons represent the props in the selected folder of the Prop Tree. LEFT-CLICK the ICON of a **prop** to select that prop. The prop you have actively selected is previewed in the Prop Render View, and the global parameters for that prop appear in the Pro Sub-Widgets. You can use any of the Prop Tools to place an actively selected prop into the open terrain.

10.2.2 Customize Props Globally

The Sub-Widgets in the Prop Browser contain the global parameters of the prop you have actively selected in the Prop Icons, or in the terrain itself.

"*Global* parameters" means that modifications you make in the Prop Sub-Widgets will affect all instances of that prop in all terrains referencing the project file containing the modified prop. Before you make global changes to a prop, consider the consequences for all terrains using it, not just the terrain you currently have open in CityScape.

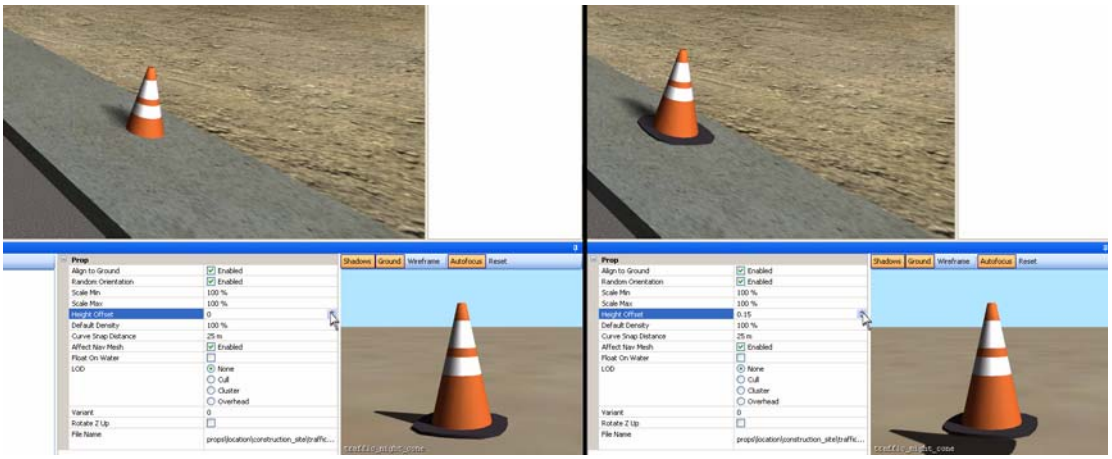
To modify a prop globally, change the desired Prop Sub-Widgets listed below.

PROP SUB-WIDGETS

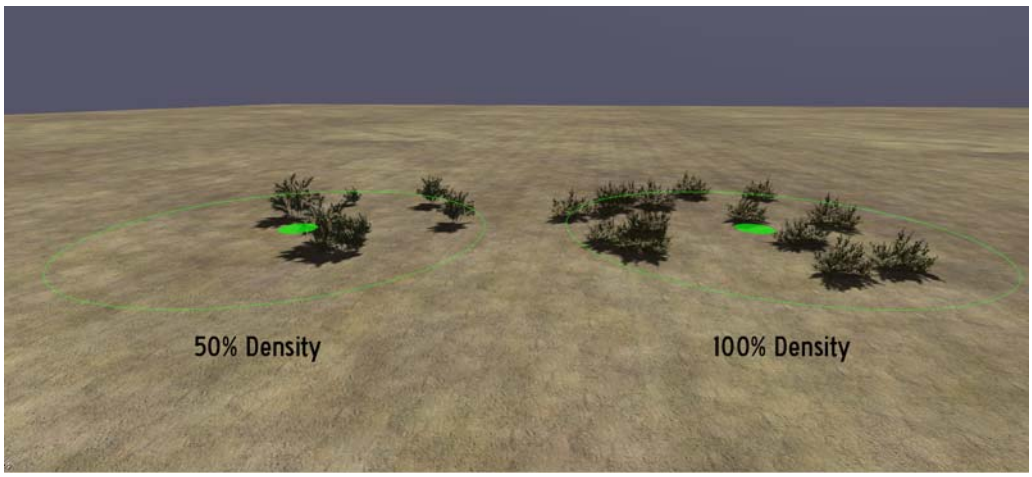
In the Prop Browser, use the Prop Sub-Widgets to change parameters that globally affect a prop. Some of these parameters, such as parameters related to LOD, influence the efficiency of terrain rendering. Other parameters affect randomness and other important properties.

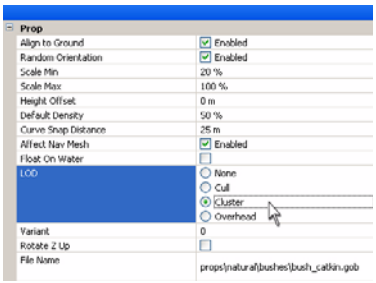
- **Align to Ground:** aligns the prop to the normal vector of the ground beneath it. Disable to align the prop to absolute vertical, regardless of the ground. (Some props, such as a trash can, should be aligned with the ground they sit on. Other props, such as trees, usually stand vertically regardless of the slope of the ground beneath them.)
- **Random Orientation**: randomly rotates all instances of the selected prop. Enable for trees and other organic props that typically vary in their orientation.
- **Scale Min:** sets the minimum size of the prop, as a percentage relative to the default size. Values below 100% cause the prop to vary randomly in size, between Scale Min and Scale Max. Enable for trees and other organic props that typically vary in their size.
- **Scale Max:** sets the maximum size of the prop, as a percentage relative to the default size. Values above 100% cause the prop to vary randomly in size, with the largest size being determined by Scale Max. Enable for trees and other organic props that typically vary in their size.

- **Height Offset:** raises or lowers the prop, relative to the ground. (Note: To fix a few isolated instances of a prop not sitting at ground level as it should, use the Absolute / Relative height widget for each troublesome instance of the prop. Otherwise, your changes here will affect all instances of the prop in all terrains using it.)



- **Default Density:** sets the default number of props in a given area. With most tools, the Main Widgets allow further refinement of prop density. (Note: If you select a prop in the Prop Browse, and then place a new prop circle or polygon in the open terrain, then CityScape assigns the *Default Density* to the props in the circle or polygon. Thereafter, you can use the Density parameter in the Main Widgets change the density of props in each instance of a prop circle or polygon.)





- **Curve Snap Distance:** Prop Points will snap to the closest road within the Curve Snap Distance. If no road is within this distance, the Point Prop will not snap.
- **Float on Water:** causes the prop to float on water (props sink by default)
- **Road Avoidance Radius:** If this parameter is set to a positive number (in meters), props will be placed at least this far from roads.

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- **LOD:** When the terrain is exported, levels of detail (LOD) of the prop will export according to your LOD selection here.
  - **None:** exports LODs of this prop according to their native (embedded) LOD definitions.
  - **Cull:** when the terrain enters its lowest LOD mode, then all instances of this prop will be culled (i.e. disappear).
  - **Cluster:** converts this prop into meshes (i.e. bakes the prop into the terrain) for the lowest LOD mode only. Enabling this option can greatly improve the rendering of large terrains from a distance.
  - **Overhead:** replaces this prop with a rendering that becomes part of the textures on the terrain in the lowest LOD mode only. Enabling this option can greatly accelerate the rendering of large terrains from a distance – a common need in flight simulators and some other applications.

- **Rotate Z Up:** Props should be modeled with the Y-axis as vertical (up), prior to import into CityScape. If your props have been modeled with the Z-axis up, then enabling this parameter corrects the problem by rotating them within CityScape.

- **File Name:** lists the file name and location of the source prop file on your hard drive.

10.3 IMPORTING NEW PROPS

Using the Prop Browser, you can import new props made in external 3D modeling software. When you import a 3D prop into CityScape, that prop is referenced by the project file associated with the terrain you have open at the time of import. This reference adds the prop to the Prop Library of the current project file, so the prop becomes available for placement in any open terrain associated with the same project file.

Before you can import props into CityScape, you must first create and Configure Props in External 3D Software such as Maya, and then export these props from your external software to the format that best preserves all the prop's data (including LOD data). Then can you Import Props into CityScape.

10.3.1 Configure Props in External 3D Software

To ensure 3D props can be imported into CityScape, you need to configure them in specific ways using external 3D modeling software. For details on the configuration procedure for Maya, 3D Studio Max, Creator, and SketchUp, see the sections below.

CONFIGURING PROPS IN MAYA

Run a Maya script (created and provided by PixelActive) that allows you to assign custom LOD attributes to Maya nodes. Each node should be grouped with the mesh with the corresponding LOD. When you export the prop to Collada format, Maya can thereby preserve these extra attributes and their

associations. When CityScape imports the prop, CityScape can convert these attributes to LOD nodes for the prop. For further information on PixelActive's LOD script for Maya, including a free copy of the script, please contact a PixelActive representative.

1. In Maya, model each prop in meters at the origin.
   – In Preferences > Settings, set the working units to meters and degrees.
   – Ensure the prop is frozen at the center of the world. The visible bottom of the prop should be flush with the XZ plane. The front of the prop (such as the facade of a building) facing the -Z axis. The top of the prop should be along the +Y axis.
   – Ensure that high, medium, and low LOD meshes overlap at the origin. Name each LOD using its attributes.
   – Map textures on the prop, as desired. The textures can be in any of the following formats: DirectDraw Surface (DDS), BMP, GIF, JPEG (and JPG, JPE, and JP2), PNG, PhotoShop (PSD), Silicon Graphics (SGI, BW, RGB, RGBA), TGA, TIF, and X Pixel Map (XPM)

2. Load the PixelActive MEL script to create an LOD node...

3. Export the prop to the Collada (DAE) format. Ensure the prop file, and its associated texture files, are exported together into the same folder under the master folder for your project.

Configuring Props in 3D Studio Max

Details coming soon...

Configuring Props in Creator

Set up LOD nodes using Creator's internal tools for that purpose, and export to OpenFlight format. Details coming soon...

Configuring Props in SketchUp

Details coming soon...

10.3.2 Import Props into CityScape

Once a prop has been created in external 3D modeling software, configured for maximum compatibility with CityScape, and exported to the appropriate exchange format (depending on which 3D authoring tool you're using), then you are ready to import the prop into CityScape.

Option 1: Import a Single Prop

1. In CityScape, open any terrain in your project, and activate any of the **Prop Tools**. The Prop Browser will appear.

2. In the Prop Tree View of the Prop Browser, LEFT-CLICK the target **folder** that will contain the new prop or props. (Note: The folders in the Prop Browser are totally separate from the folders on your hard drive.)

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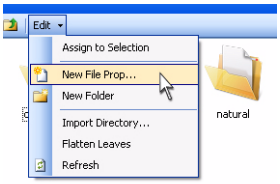
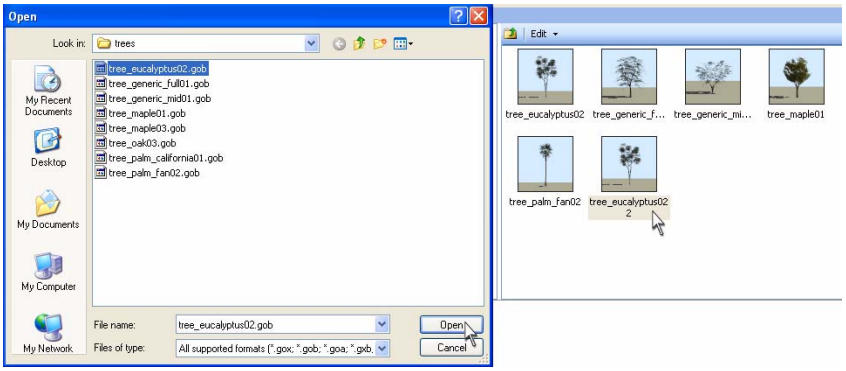
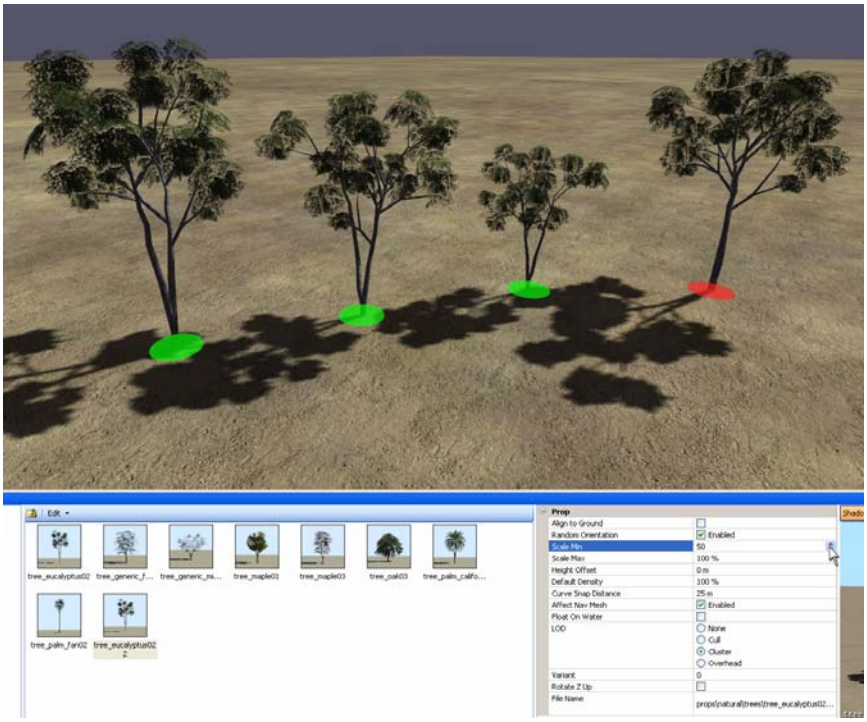


3. To import a single prop, LEFT-CLICK **Edit > New File Prop** in the Edit menu of the Icon View. (Shortcut: RIGHT-CLICK in the Prop Icon View, and select **New File Prop** from the drop-down menu.) A file browser will appear.

4. In the file browser, navigate to the **prop model file** (Collada, OpenFlight, or Filmbox format) and open it. The new prop and its textures will be imported into the prop library of your project. Notice that a new icon, using the same name as the 3D mesh file for the prop, appears in the Icon View.

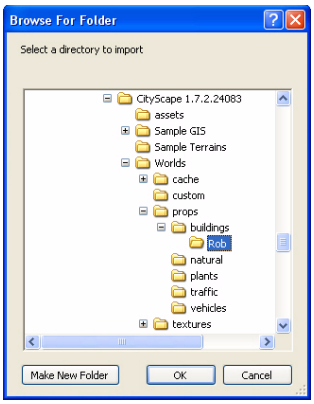


5. Use the Prop Sub-Widgets to customize the new prop.



Option 2: Batch Import Props

The process for importing all the props in a directory is recursive; it will import all props (in supported formats) in all sub-directories as well:



1. To import a directory full of props, LEFT-CLICK **Edit > Import Directory...** in the Edit menu of the Icon View in the Prop Browser. (Shortcut: RIGHT-CLICK in the Prop Icon View, and select **Import Directory** from the drop-down menu.) The *Browse for Folder* dialog appear (see figure).

2. Browse to the directory you wish to import in its entirety, select it, and choose **OK**. CityScape imports all the 3D models and associated textures in this directory (and sub-directories, if any), and adds the new props to the current project. The new props appear in the Icon View of the Prop Browser. The imported props are automatically added to the Prop Browser using the same names as the source directories and mesh files.

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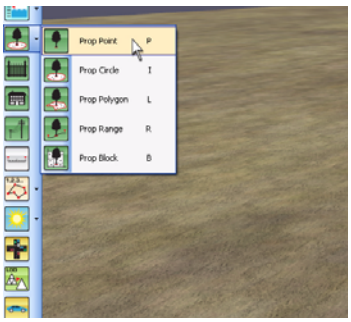
10.4 PLACING PROPS IN TERRAINS

You can place props into terrains using any of the following Prop Tools:

- Prop Point Tool
- Prop Circle Tool
- Prop Polygon Tool
- Prop Range Tool
- Prop Block Tool

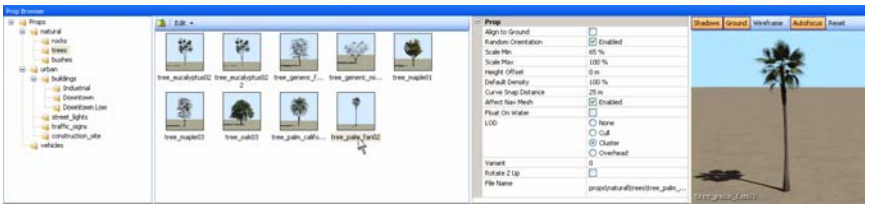
10.4.1 *Prop Point* Tool



You can place props individually using the *Prop Point Tool.* Here's how:

1. With a terrain open, activate the **Prop Point Tool**, which is part of the Prop Tool Group.

2. In the Prop Browser, use the Prop Tree and Prop Icons to locate and select an item in the prop library. The selected prop is previewed in the Prop Render View.



3. To place a selected prop in the world, RIGHT-CLICK the target location in the terrain.

4. To move a prop in the terrain, first LEFT-CLICK-HOLD its control point (typically at the base of the prop), and then DRAG the MOUSE (see the figure on the next page).

The Control Point for a placed prop turns red when it's actively selected.

The selected prop is highlighted in the Prop Icons.

PROP POINT WIDGETS

Use the Main Widgets for the Prop Point Tool to modify the properties of prop points that are actively selected in the terrain:

- **Name:** you can optionally give a name to the selected instances of the prop; this Name is totally different than the name of the prop in the Prop Library.
- **Position X, Y:** The (x, y) coordinates of a prop in the terrain, relative to the center of the terrain.
- **Rotation:** rotates the selected prop around its vertical axis.
- **Relative Height:** raises / lowers the prop relative to the ground.
- **Random Seed:** All props can be randomly rotated and scaled – within limits that you can set using the Random Orientation and Scale Min / Max parameters in the Prop Sub-Widgets.
  - Think of the Random Seed widget as the ID number of a unique random arrangement. By using the same seed for different props, you can duplicate all their random parameters.

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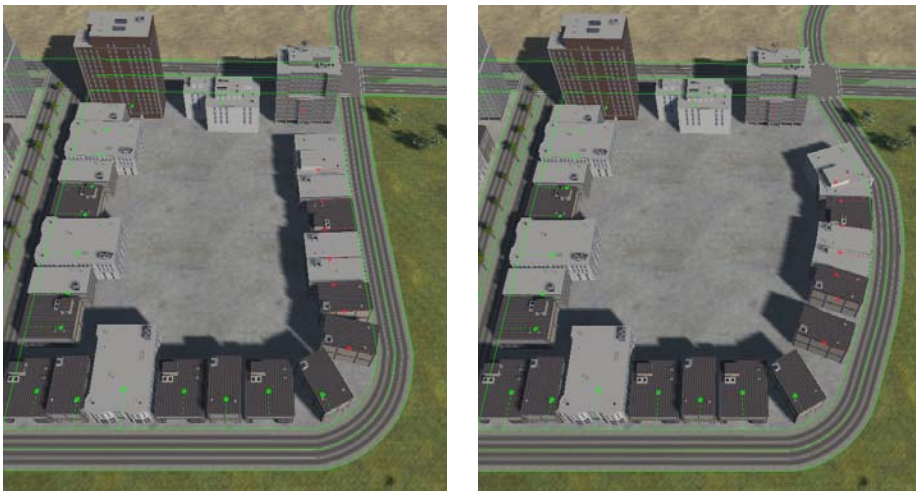
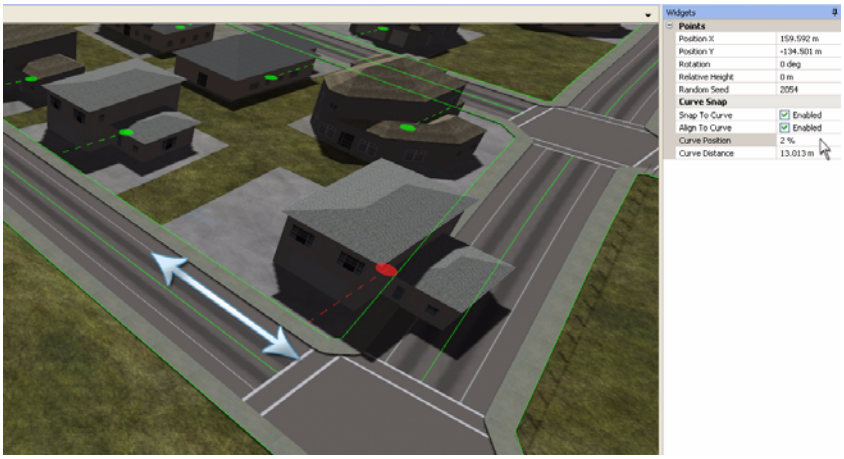
- **Snap to Curve:** when enabled, the selected prop snaps to a nearby road. Moving the road will also move associated props.
  - Note: Many streets in real cities are lined with rows of streetlights, trees, and/or buildings. Curve Snap lets you create rows of props automatically by "snapping" the props to a nearby road.

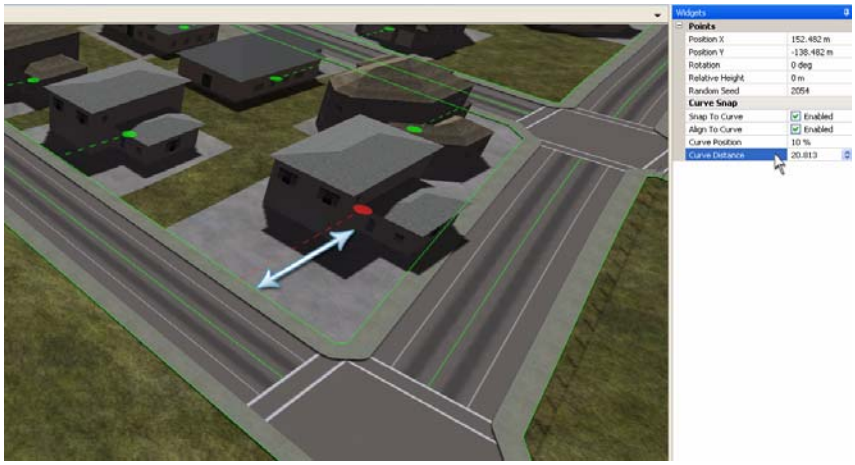


In the left figure, note the initial positions of the buildings with control points highlighted in red. In the right figure, moving the road causes buildings to automatically move and re-align with the road they are snapping to.

- **Align to Curve:** rotates the prop around its vertical axis so as to align the "front" of the prop with the associated road, as illustrated by the buildings in the figure above (notice all their facades are facing a road).
- **Curve Position**: slides the prop along the side of the associated road.

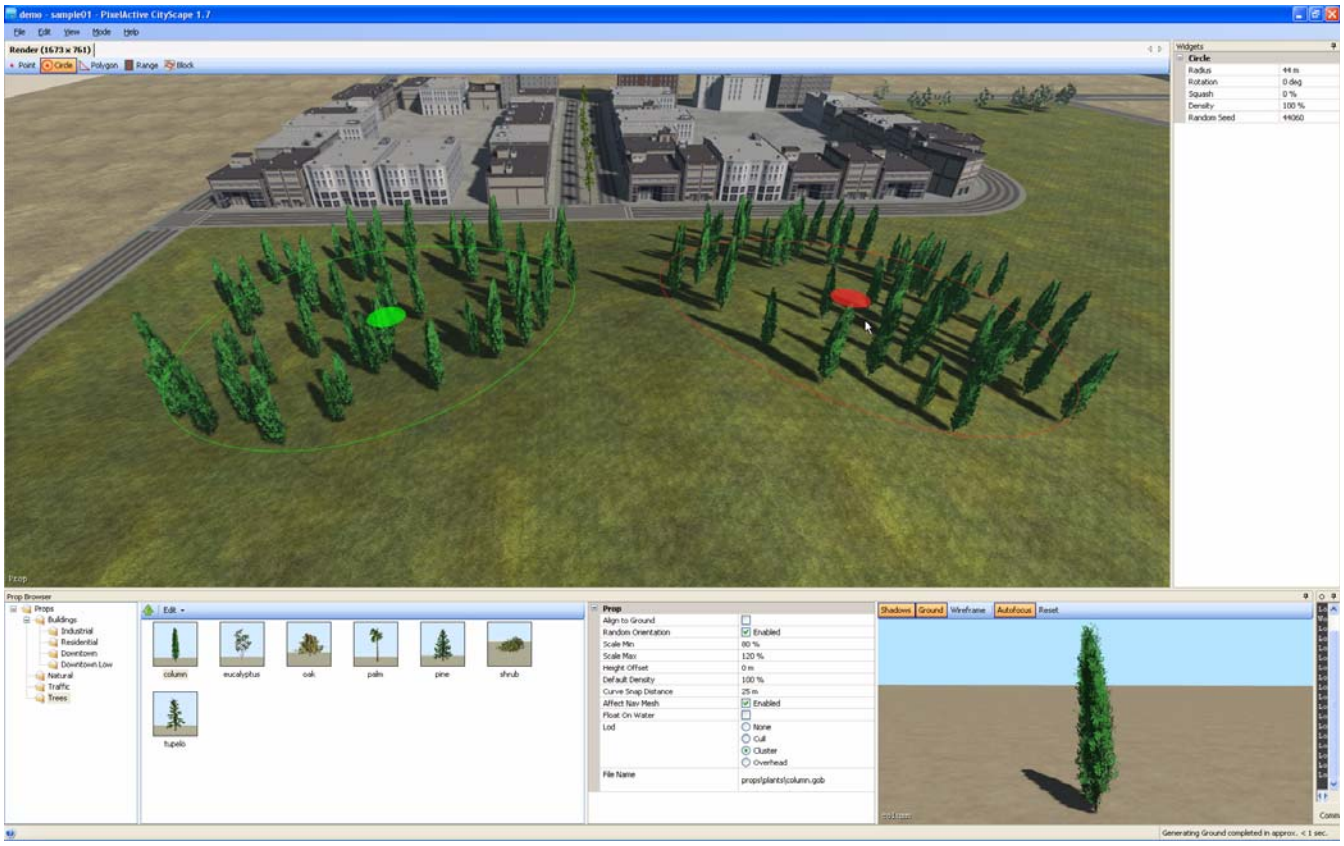


- **Curve Distance**: defines the distance between the prop and its associated road.



10.4.2 *Prop Circle* Tool

Use the *Prop Circle Tool* to distribute props within Circle Features, which can be handy for covering larger areas with trees or other props.



1. With a terrain open, activate the **Prop Circle Tool**, which is part of the Prop Tool Group.

2. In the Prop Browser, locate and select a tree from the prop library.

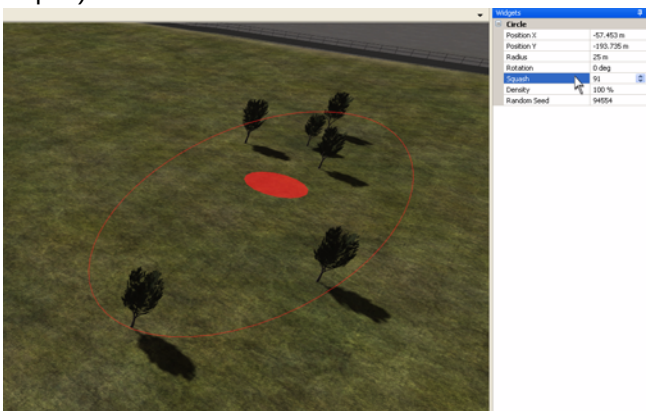
3. Use the Prop Circle Tool to place Prop Circles just as you would use The Geography Circle Tool. Multiple instances of the selected prop appear within each Prop Circle Feature.

4. To move a Prop Circle Feature, first LEFT-CLICK-HOLD on the center point of the circle, and then DRAG the MOUSE.

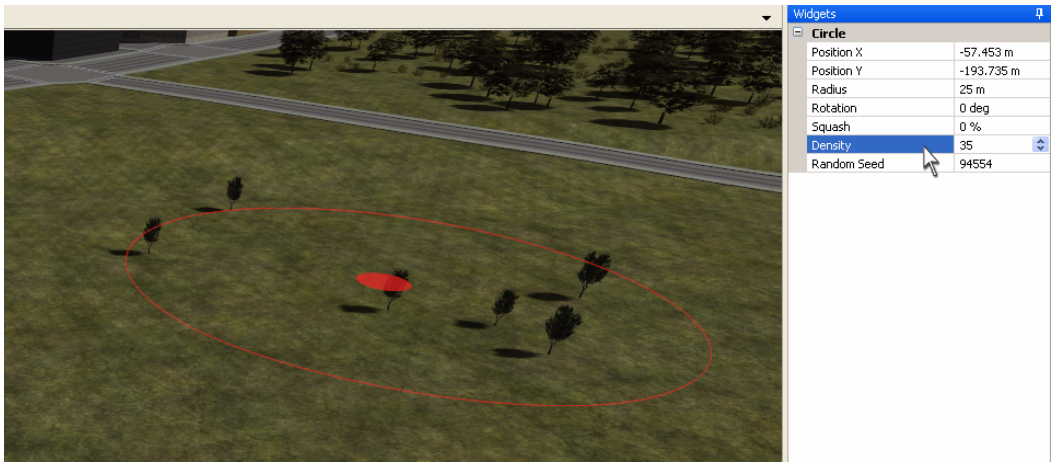
Prop Circle Widgets

With a Prop Circle Tool selected in the terrain, use the Main Widgets to modify its properties:

- **Position X, Y:** The (x, y) coordinates of the center of the Prop Circle, relative to the center of the terrain.
- **Radius:** sets the size of the circle containing the props
- **Rotation:** rotates the circle and the props it contains
- **Squash:** sets the eccentricity of the circle (non-zero values create an ellipse).



- **Density:** determines the number of props within the circle.

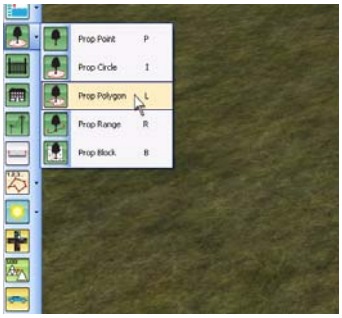


- **Random Seed:** All props can be randomly rotated and scaled – within limits that you can set using the Random Orientation and Scale Min / Max parameters in the Prop Sub-Widgets. Props in a circle feature can also be distributed randomly inside the circle.

  Think of the Random Seed widget as the ID number of a specific distribution of trees. By using the same seed, you can duplicate the same distribution of trees in two different circle features, for example.

10.4.3 *Prop Polygon* Tool

Use the *Prop Polygon Tool* to distribute props within Polygon Features.

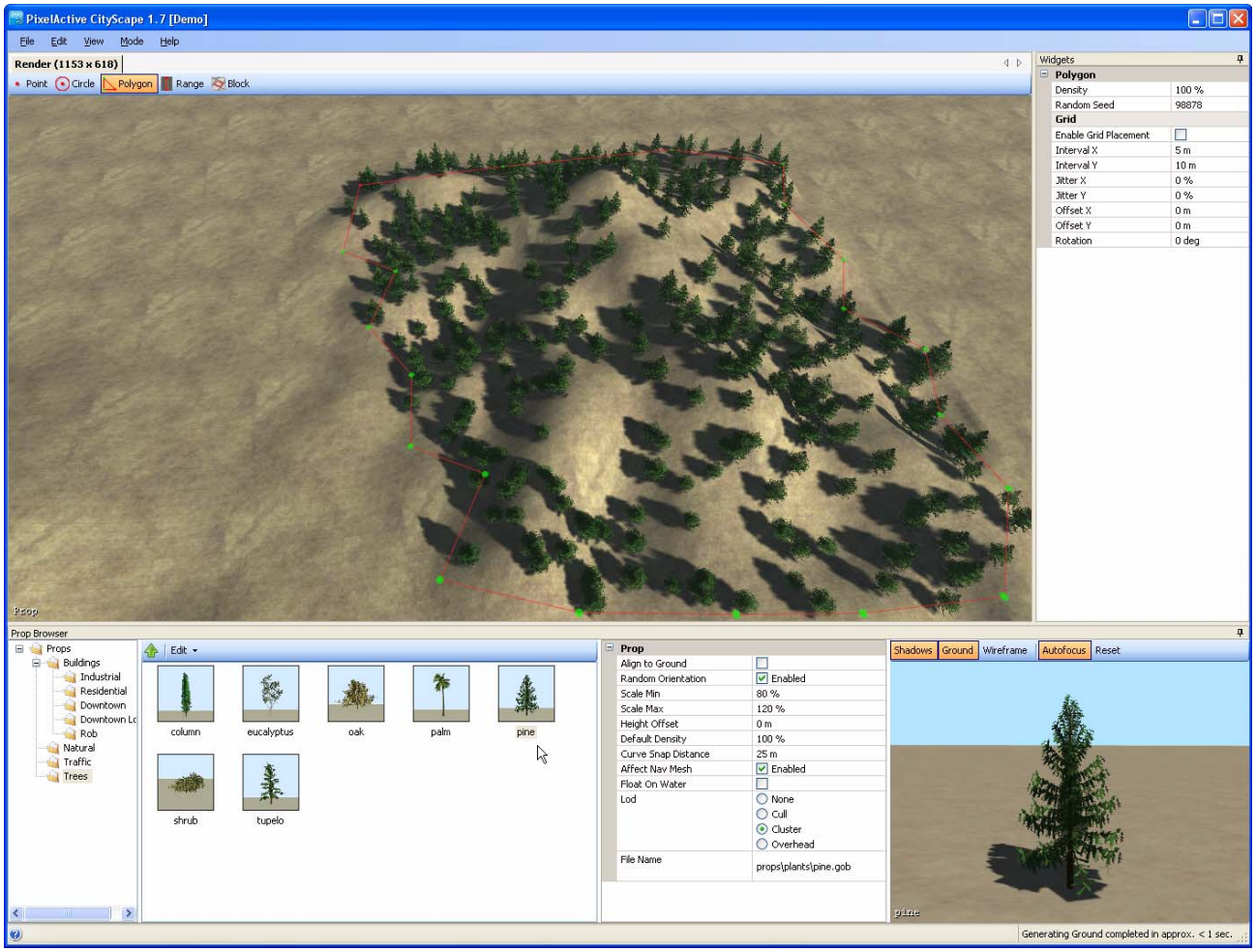


1. With a terrain open, activate the **Prop Polygon Tool**, which is part of the Prop Tool Group.

2. Using the Prop Browser, locate and select an item from the prop library.

3. Use the Prop Polygon Tool to place Prop Polygons in the terrain much like you would with The Geography Polygon Tool. Multiple instances of the selected prop appear within each polygon.

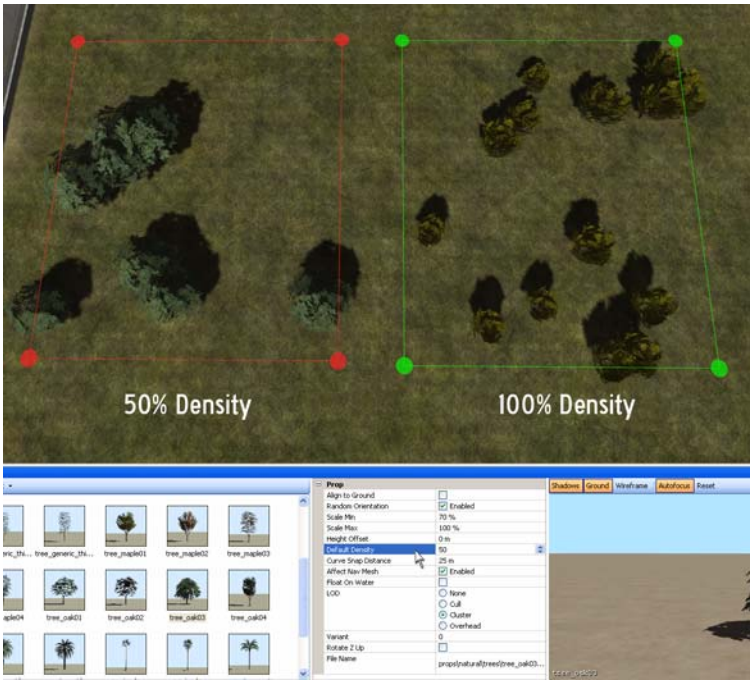
4. To move the entire polygon, DOUBLE-LEFT-CLICK-HOLD any point in the polygon, and then DRAG the MOUSE.



PROP POLYGON WIDGETS

Use the Main Widgets for the Prop Polygon Tool to modify the properties of prop polygons that are actively selected in the terrain:

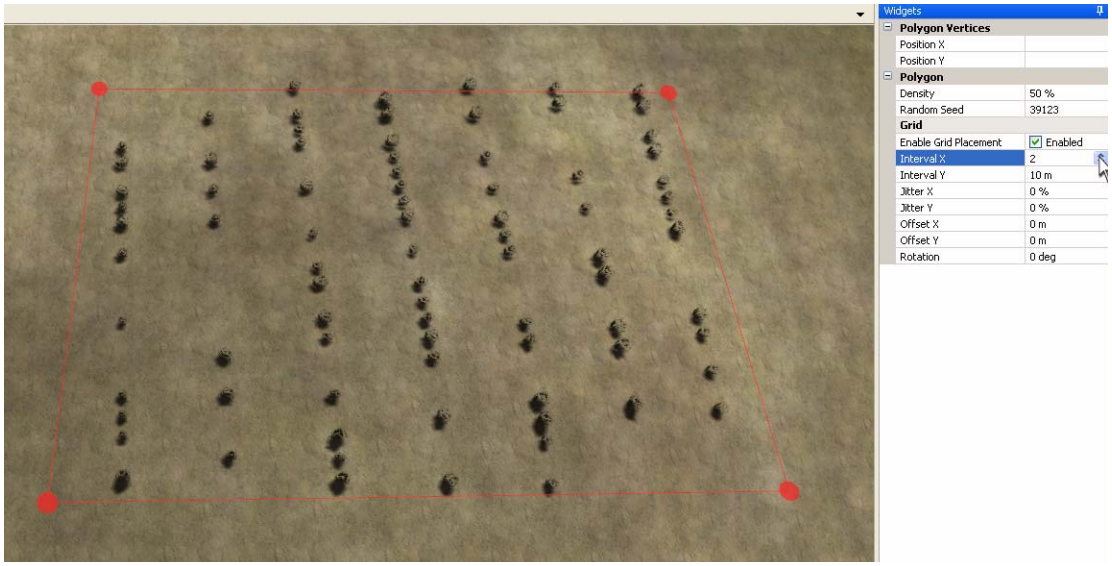
- **Position X, Y:** The (x, y) coordinates of the center of the Prop Polygon, relative to the center of the terrain.
- **Density:** sets the number of props within the polygon.



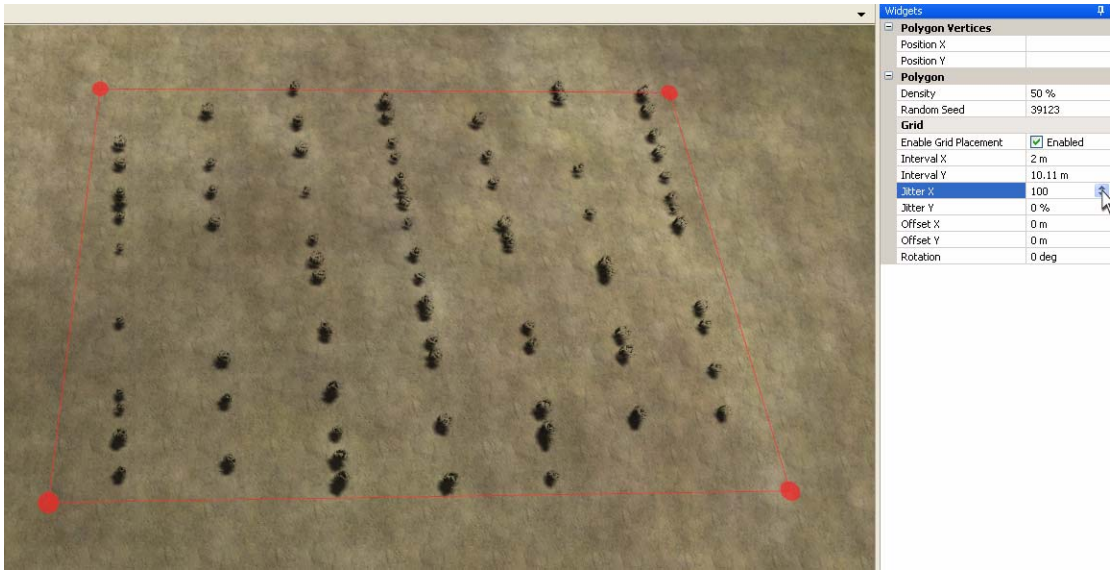
Both of these prop polygons use the same Random Seed value.

- **Random Seed:** All props can be randomly rotated and scaled – within limits that you can set using the Random Orientation and Scale Min / Max parameters in the Prop Sub-Widgets. Props in polygon features can also be distributed randomly inside the polygon.
  - Think of the Random Seed widget as the ID number of a specific distribution of trees. By using the same seed, you can duplicate the same distribution of trees in two different polygon features, for example (see figure to the left).
- **Enable Grid Placement:** snaps props to an implicit 2D grid inside a polygon. Enable this widget to create rows of crops in a field, for example, or a grid of tombstones.

- **Interval X, Y:** sets the distance between props along the X-axis and Y-axis, respectively.



- **Jitter X, Y:** sets the degree of randomness in the placement of props along the X-axis and Y-axis, respectively.



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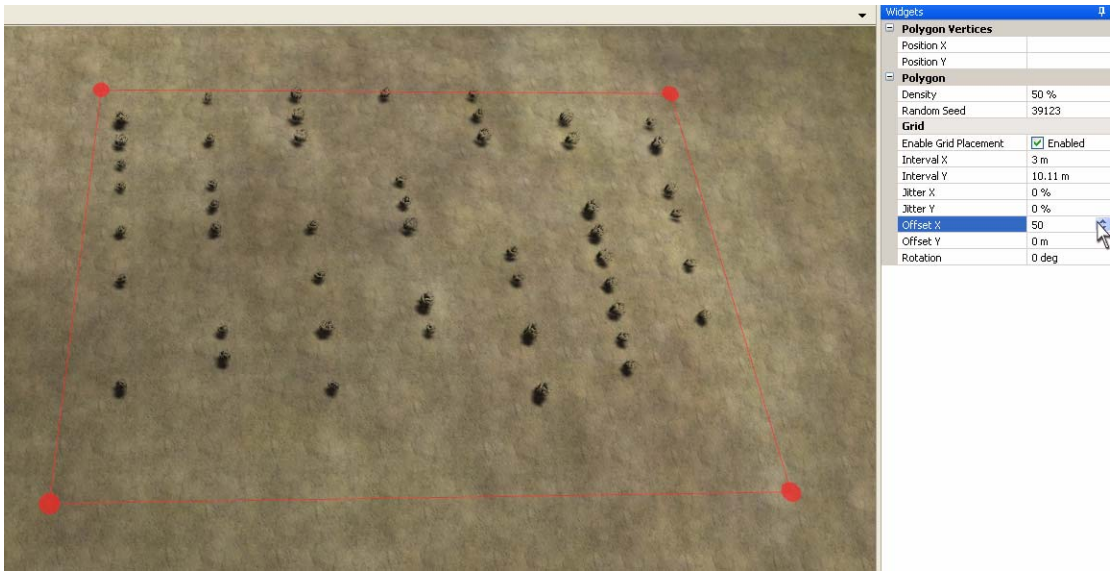
- **Offset X, Y:** offsets the position of all props along the X-axis and Y-axis, respectively.

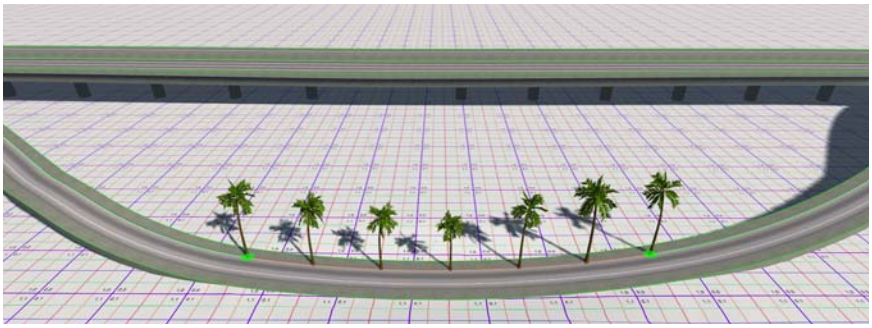


- **Rotation:** rotates the grid and all the props on it.

10.4.4 *Prop Range* Tool

You can use the *Prop Range Tool* to quickly place multiple instances of the same prop along roads. This tool is great for the common real-world situation of duplicating streetlights, trees, or similar objects at regular intervals along a road.

1. Ensure your open terrain contains at least one road.
2. Activate the **Prop Range Tool**, which is part of the Prop Tool Group.
3. In the Prop Browser, select a tree or streetlight from the prop library.
4. LEFT-CLICK a road **segment** to select it, and then RIGHT-CLICK that same segment to place the selected prop repeatedly along the segment. (In the figure below, the Random Orientation and Scale Min / Max parameters have been enabled, so the rotation and size of the tree varies.)

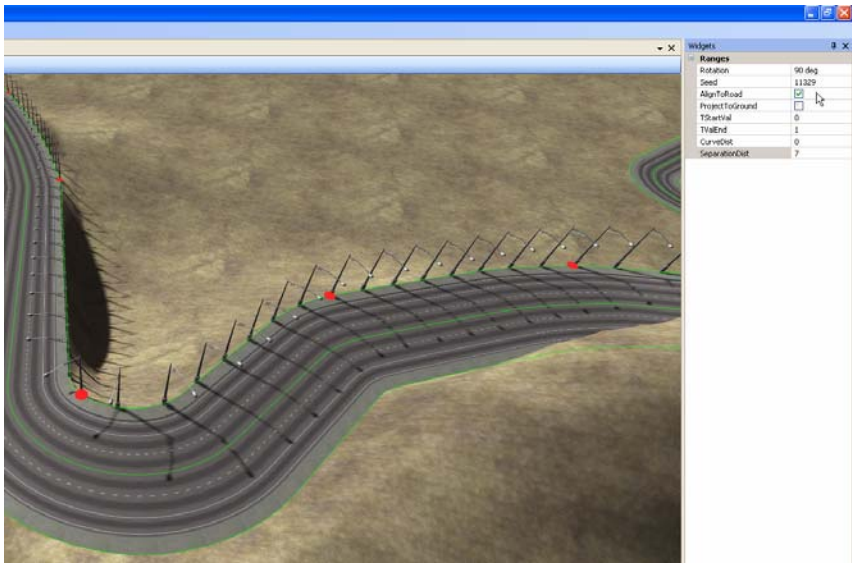


5. To contract a range of props, LEFT-CLICK-HOLD the road node at one end of the road segment for the prop range, and DRAG the MOUSE.

PROP RANGE WIDGETS

Use the Main Widgets for the Prop Range Tool to modify the properties of the prop range that's actively selected in the terrain. (To select an existing prop range, LEFT-CLICK either the start or end NODE of the associated road segment.)

- **Rotation:** rotates each prop in the range around its vertical axis
- **Seed:** All props can be randomly rotated and scaled – within limits that you can set using the Random Orientation and Scale Min / Max parameters in the Prop Sub-Widgets. Organic props, such as trees, look more realistic when groups of them are placed randomly. Think of the Random Seed widget as the ID number of a specific distribution of trees. By using the same seed, you can duplicate the same distribution of trees in two different ranges, for example.
- **Align to Road:** aligns the "front" of the prop to face the road. A common application is to align all streetlights to hang over the road, as illustrated below.



- **Project to Ground:** moves the prop vertically so it's base is always flush with the ground. The effects of this widget are not usually visible because most props align with the ground by default. If the road and ground are

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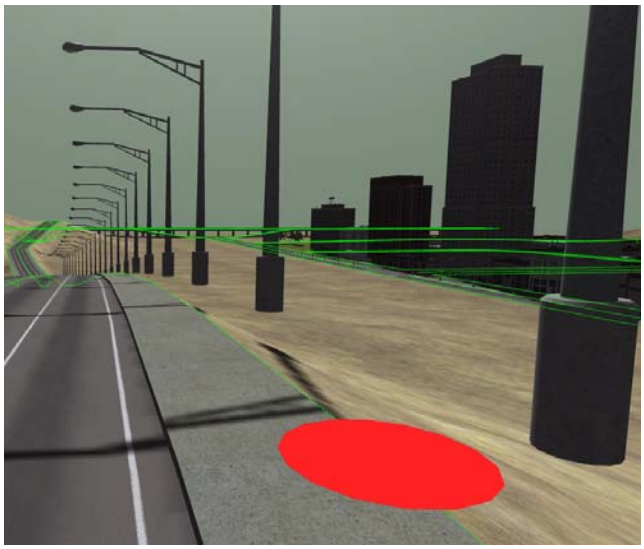
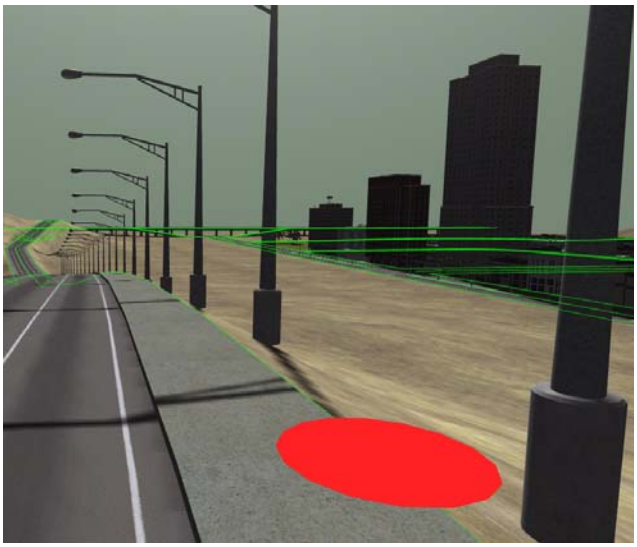
different heights, such as the ground on different sides of a road on a steep hill, then the prop would float in space (see the left figure, below) unless you enabled the Project to Ground widgets (see right figure).

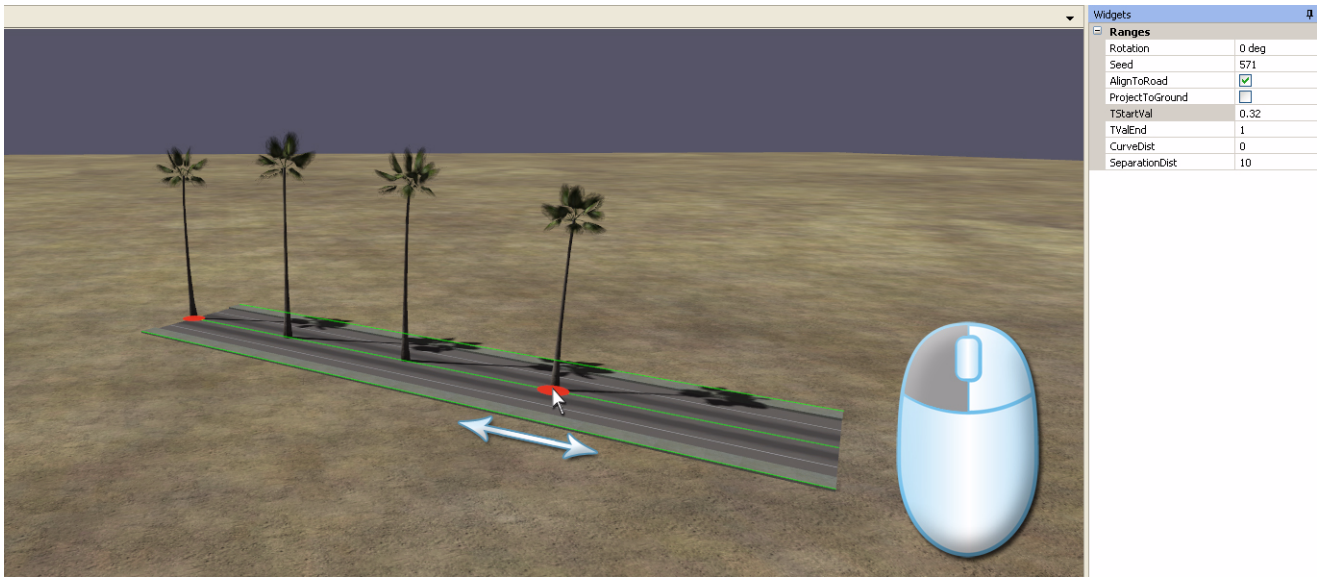


Project to Ground disabled.



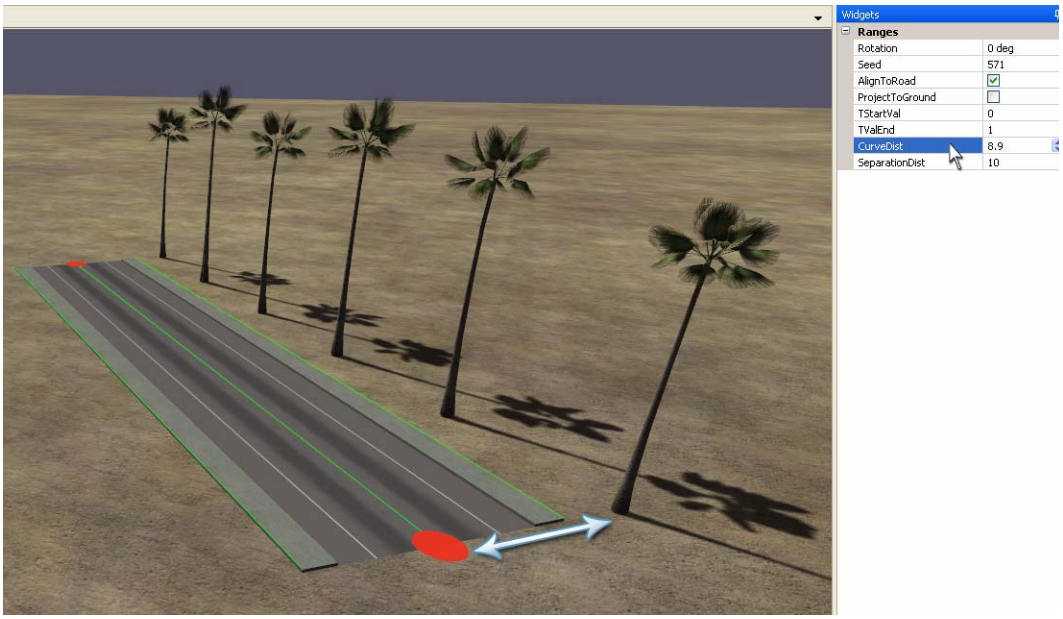
Project to Ground enabled.

- **TStartVal:** changes the starting point of the prop range, relative to the road node defining the start. You can also LEFT-CLICK-HOLD and DRAG the start node.

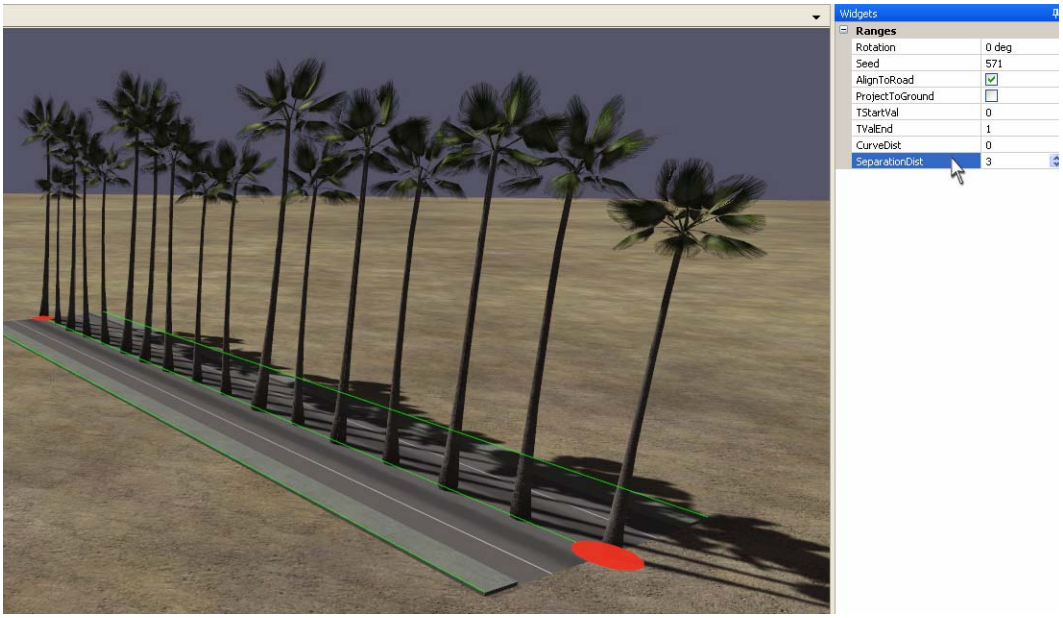


- **TValEnd:** changes the ending point of the prop range, relative to the road node defining the end. You can also LEFT-CLICK-HOLD and DRAG the end node.

- **CurveDist:** sets the distance between the prop range and the road segment it's associated with. This value allows you to offset the props back from the road, or into the road, by a specific distance relative to the side of the road.



- **SeparationDist:** defines the desired distance between props in the selected range. Small values result in a more dense distribution of props.



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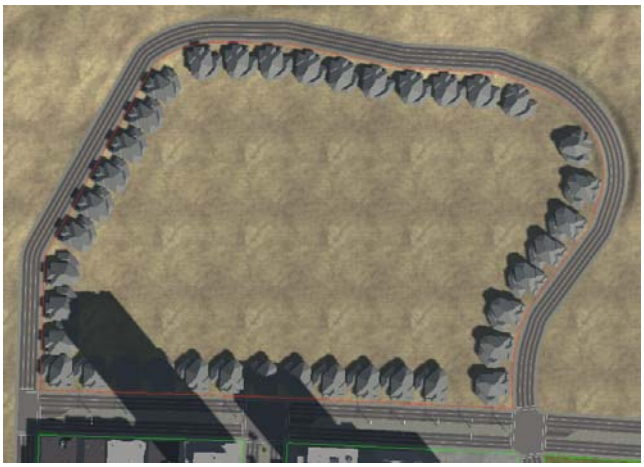
10.4.5 *Prop Block* Tool

With the *Prop Block Tool,* you can quickly place large numbers of props along the edges of any lot in the current terrain. (A "lot" is an area enclosed by roads, or by a polygonal area that you created in Lot Mode.) The Block Tool is particularly useful for lining city blocks with buildings.

1. Ensure your open terrain includes at least one **lot**.
2. Activate the **Prop Block Tool**, which is part of the Prop Tool Group.
3. LEFT-CLICK in the **target lot** to select it. When selected, the borders of the lot turn red.

4. Select the icon(s) of one or more props in the Prop Browser:
   - To select one icon, LEFT-CLICK the **icon**.
   - To select multiple icons individually, hold down the SHIFT-KEY and LEFT-CLICK **each icon** in succession.
   - To select a range of icons, LEFT-CLICK-HOLD and drag a selection box around the **icons**.
   - To select all the icons in a folder, LEFT-CLICK anywhere inside the **Icon View**, and then press **CTRL-A**.

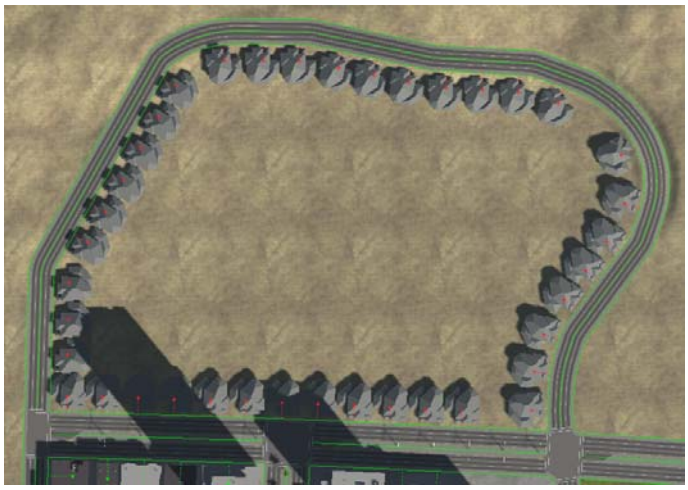
5. LEFT-CLICK **Fill Block** in the Main Widgets of the Block Tool. Instances of the selected prop(s) will be placed along the edges of the lot. (Note: Point props placed within *Curve Snap Distance* of a road will automatically snap to the road. Per instance of props placed in terrains, you can optionally turn off snapping in the Main Widgets).



6. After you use the Block Tool to place props along the edges of a lot, you can individually edit each prop with the Prop Point Tool.

PROP BLOCK WIDGETS

After the props have been placed in the lot, they become Prop Points that you can edit individually with the Prop Point Widgets.



The Prop Block Widgets control how props will be placed in a lot (i.e. future action). These widgets do not control props that have previously been placed using the Prop Block Tool. Once placed, these props can be edited only with the Prop Point Tool and its widgets.

- **Gap:** sets the minimum distance between props, down to a limit based on the sizes of the props (i.e. two props are not allowed to share the same space).

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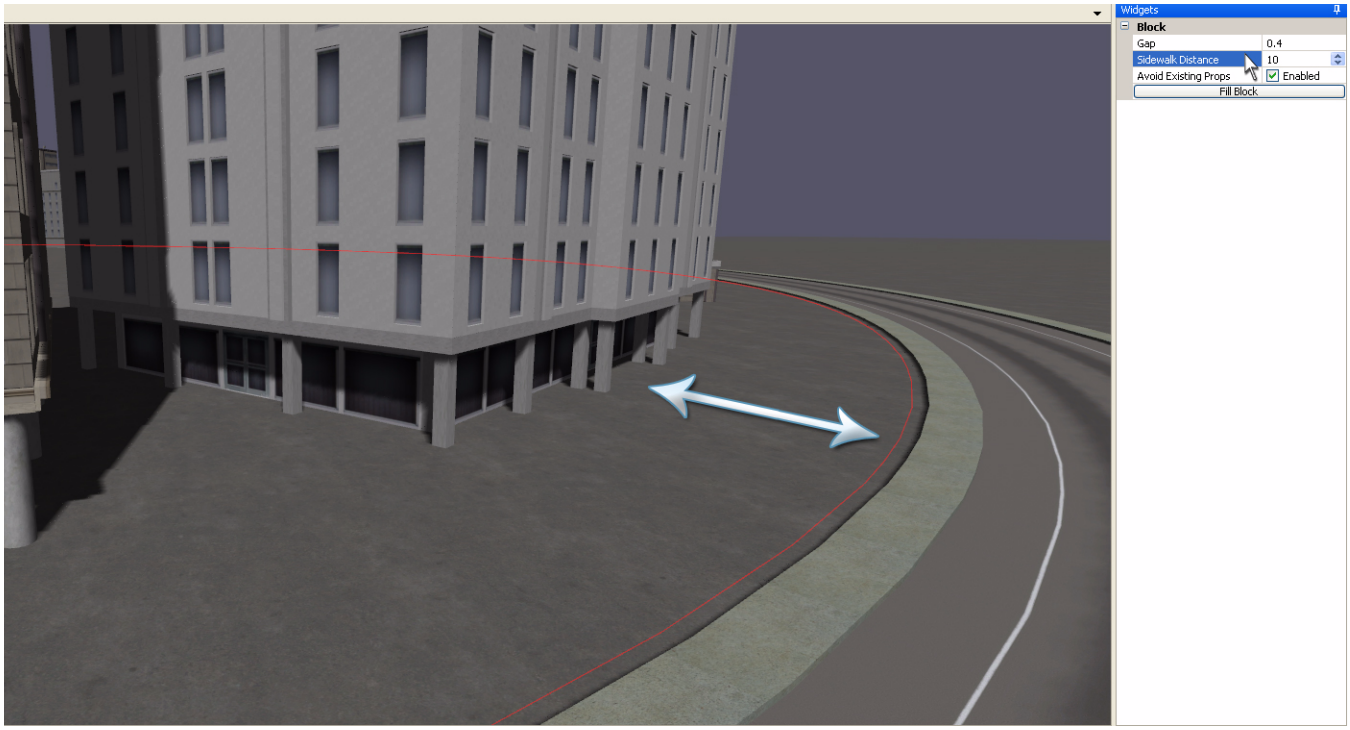
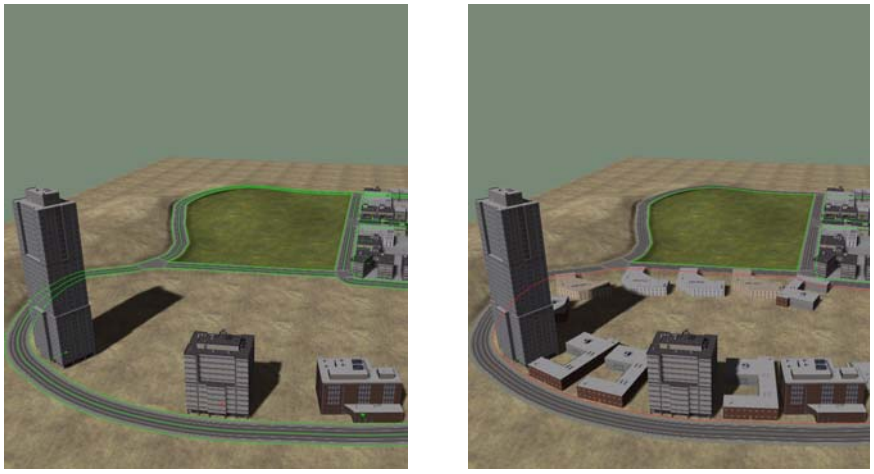
- **Sidewalk Distance:** sets the distance of the props from the nearest road or sidewalk. This widget is handy when placing buildings along streets.



- **Avoid Existing Props:** avoids props that already exist in the same lot. Notes:
  - Avoid Existing Props does not consider buildings created with Building Tools.)
  - This feature is powerful. For example, you can hand-place a few special props (e.g. left figure below), and then fill the rest of your block with more common props (e.g. right figure).



- **Fill Block:** places the selected props into the selected block.

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Chapter 11: Using Fence Tools

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Chapter 12: Using Building Tools

This chapter is in progress and will be released soon.

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Chapter 13: Using Power Line Tools

This chapter is in progress and will be released soon.

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Chapter 14: Using Measure Tools

This chapter is in progress and will be released soon.

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Chapter 15: Using Traffic Tools

This chapter is in progress and will be released soon.

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Chapter 16: Using Sky Tools

This chapter is in progress and will be released soon.

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Chapter 17: Using Physics Tools

This chapter is in progress and will be released soon.

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Chapter 18: Using Meta Data & Abstract Data Tools

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18.1 ABOUT META DATA AND ABSTRACT DATA TOOLS

CityScape lets you to associate data (such as a float or string) with some types of objects in terrains. To this end, CityScape provides a Meta Data Editor for Creating Meta Data Templates. You can use meta data templates to organize one or more meta data *elements* under a single name.

CityScape supports the following *types* of meta data elements: Boolean, Color, String, Integer, Float, 2D Vector, 3D Vector, 4D Vector, and Enumeration.

This chapter provides instructions for using the Meta Data Editor to select the Types of Meta Data Elements in a custom meta data template; and to Assign Meta Data Templates to Objects in the open terrain. You can then add unique values to each instance of meta data in the terrain.

This chapter also provides instructions for Using Abstract Data Tools to create invisible meta data objects in a terrain. You can assign meta data templates to these invisible objects in the same way as visible objects.

When you export a terrain, CityScape preserves your custom meta data so you use it however you wish. The possibilities are endless, depending on your unique needs.

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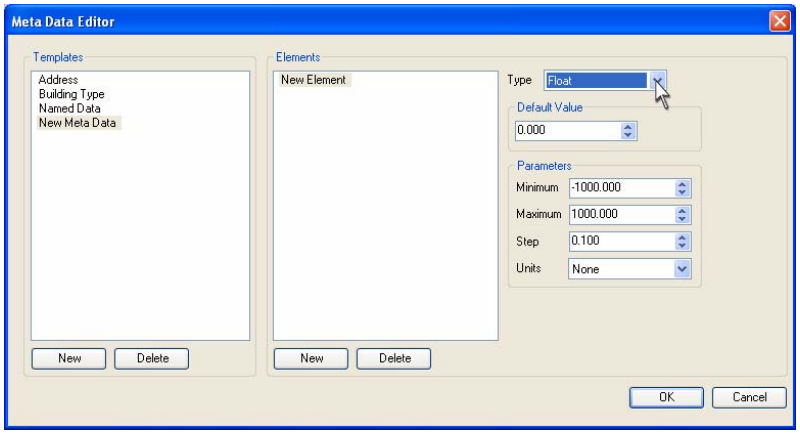
18.2 CREATING META DATA TEMPLATES

A meta data template is a collection of one or more data elements that you select and configure. CityScape provides you with the following types of meta data elements:

Boolean, Color, String, Integer, Float, 2D Vector, 3D Vector, 4D Vector, Enumeration

Here's how to construct a new meta data template:

1. With a terrain open in CityScape, select **Meta Data > Edit Meta Data Templates** in the Main Menu. (Shortcut: **Ctrl-T**) The Meta Data Editor appears.



2. On the Templates side of the Meta Data Editor, LEFT-CLICK the **New** button. A new template called "New Meta Data" will appear under Templates.
   – Optional: LEFT-CLICK the name of the new template, and type a different name for it.

3. On the Elements side of the Editor, LEFT-CLICK the **New** button. A new element called "New Element" will appear under Elements.
   – Optional: LEFT-CLICK the name of the new element, and type a different name for it.

For the next few steps, ensure that the name of the target element is highlighted in the Editor:

4. In the **Type** drop-down menu on the right, select the desired type for the new element. Each type of element has a unique set of parameters.

5. To configure your new element, enter default values for its parameters. (For details on elements and their parameters, see Types of Meta Data Elements below.)

6. To create and define another element in the same template, repeat step 3 and thereafter. To create a new template, repeat step 2.

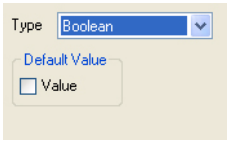
18.2.1 Types of Meta Data Elements

CityScape supports the following types of meta data elements:

- Boolean
- Color
- String
- Integer
- Float
- 2D Vector
- 3D Vector
- 4D Vector
- Enumeration

When you assign a meta data template to an object in a terrain, the object is assigned the default values of the elements in the template. Using the Main Widgets, you can customize the values of these elements separately per object. If you later wish to revert to a default value, RIGHT-CLICK the target element in the Main Widgets, and select **Reset Value**.

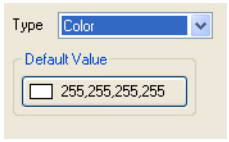
BOOLEAN



The Boolean data type appears in the Main Widgets as a check box that you can enable or disable.

- **Value:** the default value is either enabled or disabled.

COLOR



The Color data type appears in the Main Widgets as a series of numbers, between 0-255, alongside a swatch of the corresponding RGBA color. To change this color in the Main Widgets, LEFT-CLICK the drop-down menu in the Color Widget, and select the new color from the dialog that appears.

- **Color:** the default value can be any RGBA color (including alpha channel).

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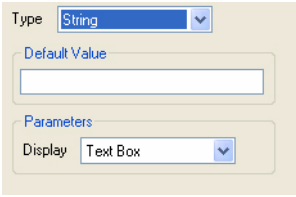
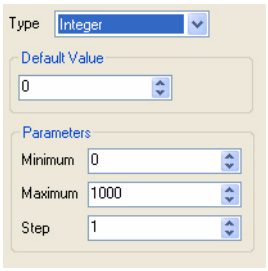


STRING

The String data type appears in the Main Widgets as either a text field in which you can type ASCII characters, or a button you can LEFT-CLICK to invoke a file loader for opening external files.

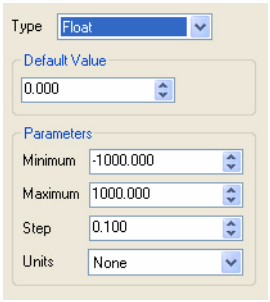
- **Default Value:** can be any combination of ASCII characters.
- **Display:** select between two ways this element can appear in the Main Widgets:
  - *Text Box* accepts the entry of any ASCII characters in a single line text field.
  - *File Loader* causes a file browser to appear when you click it in the Main Widgets, so you can locate and select a stored file. CityScape stores an absolute path to the file you select via a file loader.



INTEGER

The Integer data type appears in the Main Widgets as a signed whole number which you can customize within its Minimum and Maximum limits:

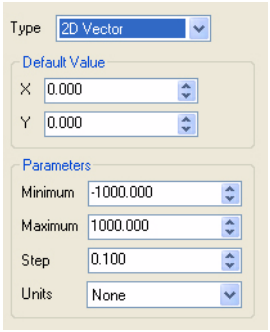
- **Default Value:** can be any integer.
- **Minimum:** the minimum allowable value
- **Maximum:** the maximum allowable value
- **Step:** the step size between integers when you scroll through possible values



FLOAT

The Float data type appears in the Main Widgets as a standard single precision floating point number which you can customize within its Minimum and Maximum limits:

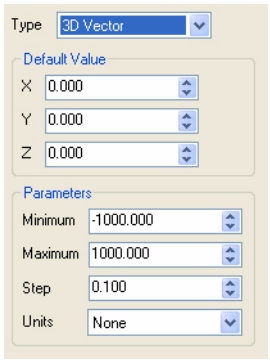
- **Default Value:** can be any floating point number
- **Minimum:** the minimum allowable value
- **Maximum:** the maximum allowable value
- **Step:** the step size between numbers when you scroll through possible values
- **Units:** choose from the following options:
  - *None:* leaves the meaning of the number unspecified
  - *Meters:* selects distance (in meters) as the meaning of the number
  - *Degrees:* selects rotation (in degrees) as the meaning of the number
  - *Percent:* selects a relative meaning of the number (i.e. relative to 100)



2D VECTOR

The 2D Vector data type appears in the Main Widgets as two numbers: (X,Y), which you can customize within its Minimum and Maximum limits:

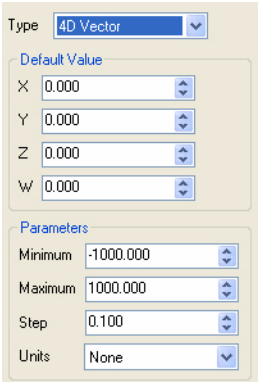
- **X:** the default value can be any floating point number
- **Y:** the default value can be any floating point number
- **Minimum:** the minimum allowable value of X or Y
- **Maximum:** the maximum allowable value of X or Y
- **Step:** the step size between numbers when you scroll through possible values
- **Units:** choose from the following options:
  - *None:* leaves the meaning of the number unspecified
  - *Meters:* selects distance (in meters) as the meaning of the number
  - *Degrees:* selects rotation (in degrees) as the meaning of the number
  - *Percent:* selects a relative meaning of the number (i.e. relative to 100)



3D VECTOR

The 3D Vector data type appears in the Main Widgets as three numbers: (X,Y,Z), which you can customize within its Minimum and Maximum limits:

- **X:** the default value can be any floating point number
- **Y:** the default value can be any floating point number
- **Z:** the default value can be any floating point number
- **Minimum:** the minimum allowable value of X, Y, or Z
- **Maximum:** the maximum allowable value of X, Y, or Z
- **Step:** the step size between numbers when you scroll through possible values
- **Units:** choose from the following options:
  - *None:* leaves the meaning of the number unspecified
  - *Meters:* selects distance (in meters) as the meaning of the number
  - *Degrees:* selects rotation (in degrees) as the meaning of the number
  - *Percent:* selects a relative meaning of the number (i.e. relative to 100)



4D VECTOR

The 3D Vector data type appears in the Main Widgets as four numbers: (X,Y,Z,W), which you can customize within its Minimum and Maximum limits:

- **X:** the default value can be any floating point number
- **Y:** the default value can be any floating point number
- **Z:** the default value can be any floating point number
- **W:** the default value can be any floating point number

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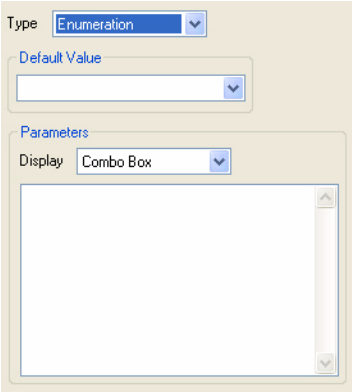
- **Minimum:** the minimum allowable value of X, Y, Z, or W
- **Maximum:** the maximum allowable value of X, Y, Z, or W
- **Step:** the step size between numbers when you scroll through possible values
- **Units:** choose from the following options:
  - *None:* leaves the meaning of the number unspecified
  - *Meters:* selects distance (in meters) as the meaning of the number
  - *Degrees:* selects rotation (in degrees) as the meaning of the number
  - *Percent:* selects a relative meaning of the number (i.e. relative to 100)

ENUMERATION



The Enumeration data type allows you to create an arbitrary list of values. In the Values field of the Meta Data Editor, each line is a separate value. Each value can be a string of any ASCII characters. In the Main Widgets, you can select any one of these values.

- **Default Value:** any one of the values listed in the *Value Field*.
- **Display:** select between two ways to display the list of values in the Main Widgets:
  - *Combo Box:* creates a drop-down list from which you can choose one value. (The default value will be initially selected.)
  - *Radio Buttons:* creates radio button for each data member of the enumeration. (The default value will be initially selected.)
- **Value Field:** the large field below the *Display* drop-down menu. In the Value Field, enter the actual values (one per line) of the enumeration.

18.3 WORKING WITH META DATA OBJECTS

Future versions of CityScape will allow you to assign meta data templates to almost every object that can exist in CityScape terrains. With CityScape 1.8, you can assign meta data templates to the following types of objects in terrains:

- **Buildings** created using the Building Tool in CityScape
- **Water** created using Water Polygon Tool in CityScape
- **Power lines** created using Power Line Tool in CityScape
- **Abstract data (AD) objects** created Using Abstract Data Tools in CityScape

You can also customize the values of meta data separately per assigned object, and remove meta data templates from assigned objects.

18.3.1 Assign Meta Data Templates to Objects

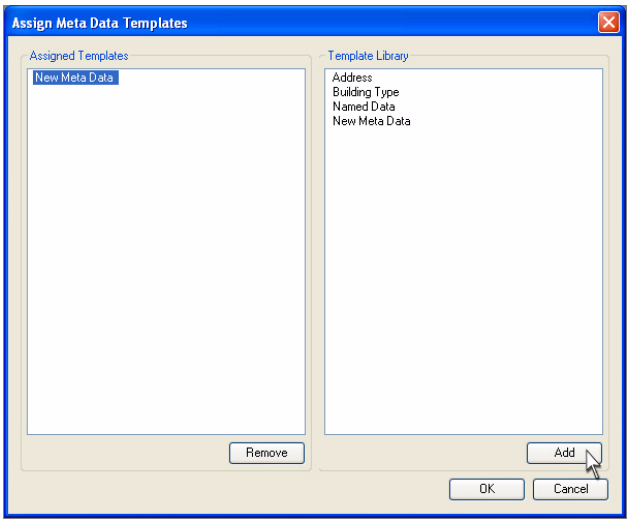
You use the same overall process to assign meta data templates to buildings, water, power lines, and invisible AD objects:

1. Ensure a terrain is open in CityScape, and at least one meta data template already exists in your project. (To create a new meta data template, see "Creating Meta Data Templates" on page 173.)

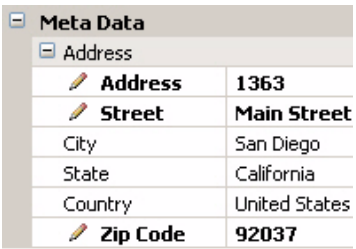
2. Select one existing building polygon, water polygon, power line, or AD object in your terrain. (Or create a new object.) This step requires that you first activate the tool corresponding to the type of target object (i.e. for buildings, activate the Building Tool, etc).

3. Choose **Meta Data > Assign Meta Data Templates** from the Main Menu (shortcut: **Ctrl-M**). The *Assign Meta Data Templates* dialog appears.

4. In the dialog, LEFT-CLICK the name of the **relevant template** in the Template Library, and then LEFT-CLICK **Add**. The template will be added to the Assigned Templates list. Notes:
   – Shortcut: DOUBLE-CLICK the template in the Template Library to add it directly to the Assigned Templates list.
   – You can assign more than one template to the same object, provided the templates are different.



5. In the dialog, LEFT-CLICK **OK** to complete the assignment.



Whenever you select an object in the terrain that has been assigned a meta data template, the elements in this instance of the template appear in the Main Widgets. If the default value of an element has been replaced with a custom value in this instance, then the modified element / value will appear in bold – as do the **Address**, **Street**, and **Zip Code** elements in the figure, which illustrates meta data that can be assigned to a building.

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Assigning Templates to Multiple Objects

The general instructions in Assign Meta Data Templates to Objects tell you to assign a meta template to one object at a time. You can optionally select many objects and assign a single template to all of them at once, but with some caveats:

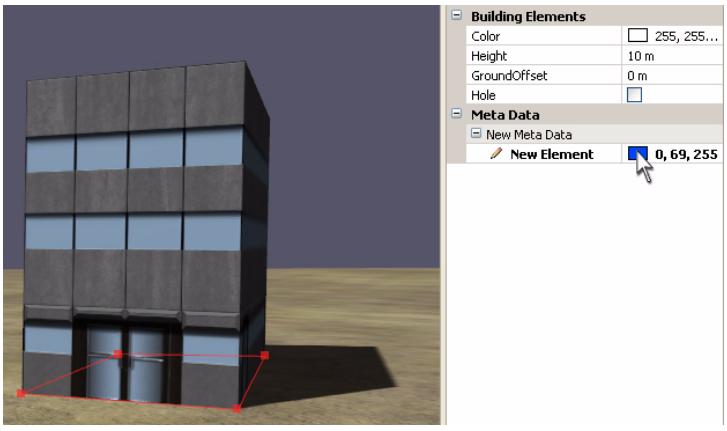
- A building or body of water can optionally be constructed from multiple polygons that are grouped together. You can assign a single meta data template to an entire group of polygons, but you cannot assign templates separately to each polygon in the group.
- If you select multiple objects with different meta data templates already assigned to them, and then display the *Assign Meta Data Templates* dialog, CityScape will warn you about the impending conflict. If you then assign a new template to multiple selected objects, the new template will replace all previously assigned templates.

18.3.2 Customize Meta Data for Objects

When you select an object in the terrain, the meta data elements assigned to that object appear in the Main Widgets, so you can customize these values.

The first step is to activate the tool that will enable you to select the target object. For example, you cannot select a water polygon with the Road Tools, or any other tools except the Water Polygon Tool. In general, your choice of tool must match the type of object you wish to select.

1. Select the target object in the terrain, and then change the values of meta data elements appearing in the Main Widgets. Notes:
   – A default value appears in plain text with, without a pencil next to it.
   – A customized value appears in bold text, with a pencil next to it.
   – To revert back to a default value, RIGHT-CLICK the target element in the Main Widgets, and select **Reset Value**.



18.3.3 Fixing Dead References

Dead references can arise for a variety of reasons. This section focuses only on dead references that relate to meta data.

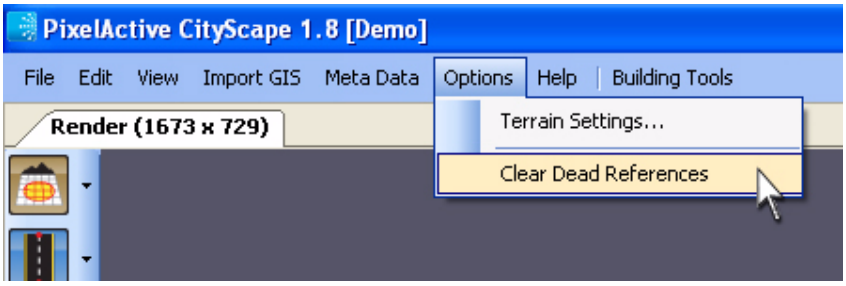
If you assign a meta data template to at least one object in a terrain, doing any of the following things at any time thereafter could cause a dead references message to appear:

- In the Meta Data Editor, you deleted an element with a custom value in an instance assigned to an object in the terrain.
- In the Meta Data Editor, you changed an element so it's no longer compatible with a custom value in an assigned object.
  - CityScape will attempt to convert the value, if possible. For example, if you change a Float to an Integer, CityScape will truncate the decimal, so a number like 27.46 becomes 27.)
  - Some types cannot be converted (e.g. Float to Vector), which will result in a dead references warning.
- In the Meta Data Editor, you deleted a meta data template with one or more custom values in an instance assigned to an object in the terrain.
- You associated the terrain with a different project file that does not contain the meta data template(s) have been assigned to object(s) in your terrain.

Clearing Dead References

You can fix dead meta data references in the following ways, neither of which can be undone:

- When CityScape prompts you to clear dead references (e.g. when you open a terrain with dead references), select **Yes**. Bear in mind that you will permanently lose the custom meta data that created the issue.
- Select **Edit > Clear Dead References** to clear dead references manually. This option will also permanently delete invalid custom meta data.



Preserving Dead References

Preserving dead references allows you to correct the source of the problem without losing any custom meta data.

18.3.4 Remove Meta Data Templates from Objects

You can remove a meta data template from an object in the terrain:

1. In CityScape, open the terrain to be edited, and select the target object.
2. Press **Ctrl-M** to display the *Assign Meta Data Templates* dialog.
3. In the dialog, LEFT-CLICK the name of the **target template** in the Assigned Templates list. Then LEFT-CLICK the **Remove** button followed by the **OK** button. The target meta data template will be removed from the selected objects.

18.4 USING ABSTRACT DATA TOOLS

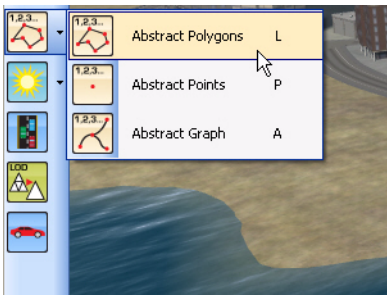
You can use the Abstract Data (AD) Tools to create special polygons, points, and graphs. The resulting AD objects are always invisible in CityScape terrains – except when you activate the Abstract Data Tools used to create the AD objects. Because AD objects do not interact with visible objects in terrains, AD objects are useful solely for assigning meta data templates to.

Assigning meta data templates to invisible objects has many uses. For example, invisible meta data could indicate the location and point value of a power-up in a video game; or the location and power of a land mine in a training simulation. The potential uses are limitless.

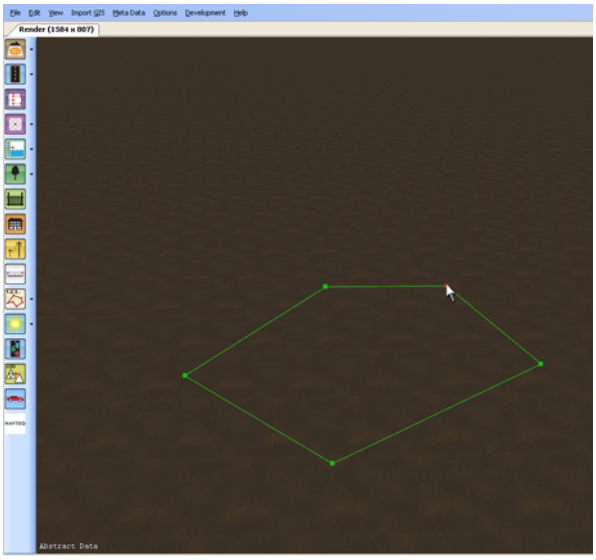
CityScape provides three Abstract Data (AD) Tools for creating invisible objects that you can assign meta data templates to, just like visible objects. For details, see the sections below.

18.4.1 Abstract Polygon Tool

The Abstract Polygon Tool lets you create invisible polygons in terrains. AD polygons are very similar to building polygons and Water Polygon, except for two things: AD polygons are visible only in Abstract Data Mode, and do not support the polygon groups.

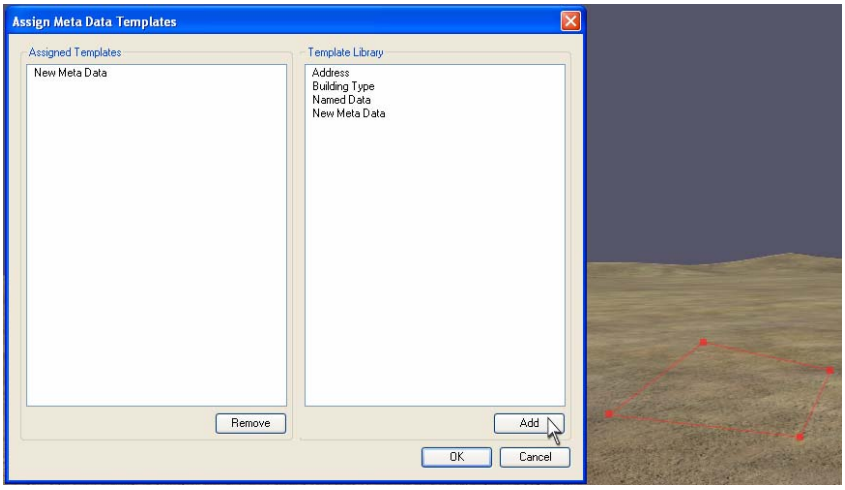


1. Ensure a terrain is open in CityScape; at least one meta data template already exists in your project; and the **Abstract Polygon Tool** is actively selected from the Tools Palette.



2. RIGHT-CLICK in the terrain to place nodes in a new polygon using the same process as The Geography Polygon Tool.

3. Ensure the new polygon is actively selected with the Abstract Polygon Tool. Then assign an existing meta data template to this polygon in the same way as a visible building or lake. For details, see “Assign Meta Data Templates to Objects” on page 178.



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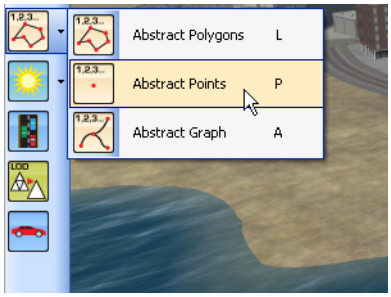
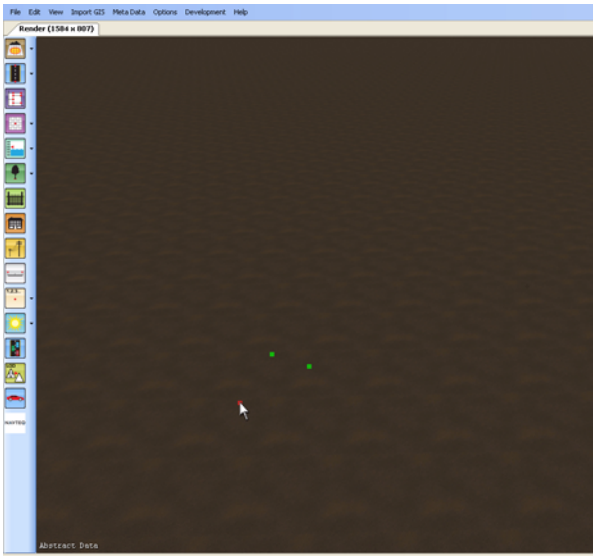
18.4.2 Abstract Point Tool

The Polygons AD Tool lets you place invisible points in terrains. These points are visible only in Abstract Data Mode.

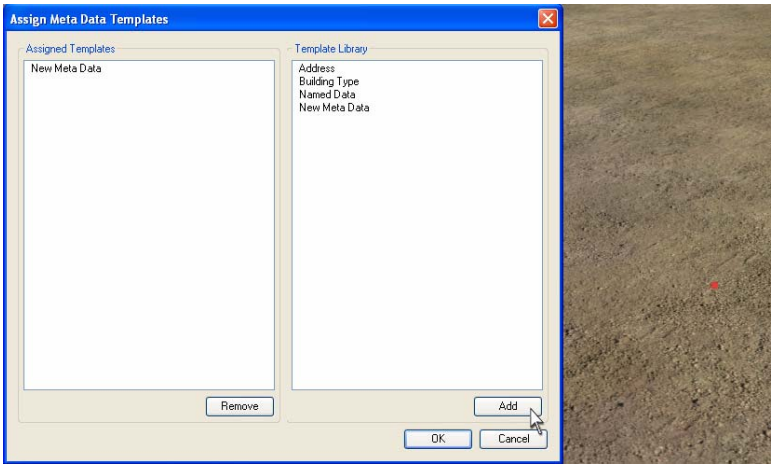


1. Ensure a terrain is open in CityScape; at least one meta data template already exists in your project; and the **Abstract Point Tool** is actively selected.



2. RIGHT-CLICK in the terrain to place a point using the same process as The Prop Point Tool. The obvious difference is that the meta data point can have invisible meta data assigned to it. The prop point represents a visible prop, but cannot have meta data assigned to it (not yet anyway).

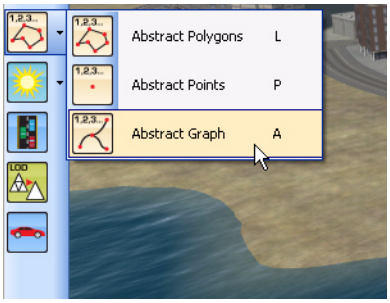
3. Ensure the new point is actively selected with the Abstract Point Tool, and then assign an existing meta data template to it in the same way as assigning a template to a visible object. For details, see “Assign Meta Data Templates to Objects” on page 178.



18.4.3 Abstract Graph Tool

The Abstract Graph Tool lets you place invisible graphs in terrains. A "graph" is a network of nodes and line segments that are more flexible than a polygon or road. Abstract Graphs are visible only when the Abstract Graph Tool is active.

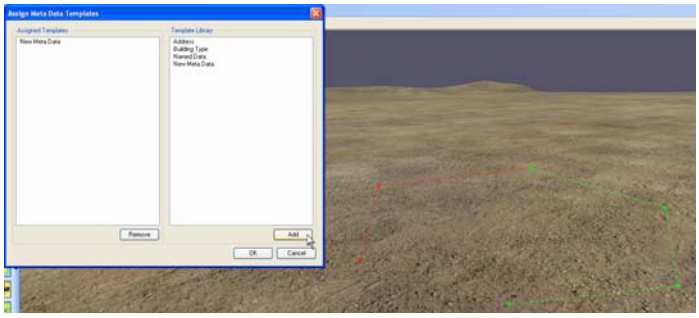
CityScape regards each node in a graph as a separate object; and each segment as a separate object. Hence, you can optionally assign a separate meta data template to each and every node and segment in a graph – unlike polygons, which can have at most one template assigned to the them.



1. Ensure a terrain is open in CityScape; at least one meta data template already exists in your project; and the **Abstract Graph Tool** is actively selected from the Tools Palette.

2. RIGHT-CLICK a few times in the terrain to place some nodes with connecting segments. The behavior of the Abstract Graph Tool is very similar to The Geography Polygon Tool. Experiment with creating a network of nodes and segments.

3. Use the Abstract Data Tool to actively select at least one new node and/or segment in the graph, and then assign an existing meta data template to it in the same way as with visible objects.



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Chapter 19: Scripting

This chapter is in progress and will be released soon.

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PART 3: IMPORTING AND EXPORTING

Chapter 20: Global Settings Affecting Import / Export

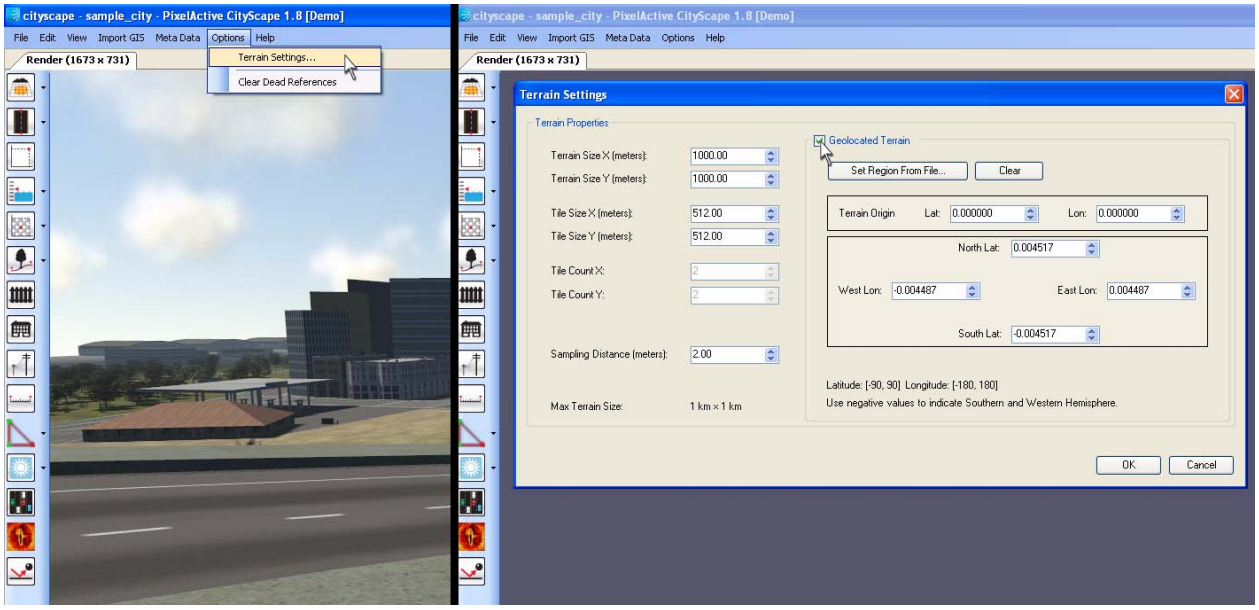
The following settings control fundamental attributes of the open terrain – such as size and geo-location, as well as level of detail (LOD) setting.

- Terrain Settings
- Setting LOD Transitions for Entire Terrains

20.1 TERRAIN SETTINGS

You can use the Terrain Settings dialog to redefine the following properties of the open terrain: terrain size, tile size, sampling distance, and geo-location prior to export.

1. Select **Options > Terrain Settings** in the Main Menu. The Terrain Settings dialog appears.



2. If you do not wish to geo-locate your terrain, or your terrain is already geo-located, skip to the next step. If you wish to geo-locate an existing terrain, fill in the **Geolocated Terrain** side of the Terrain Settings dialog according to the instructions in "Geo-Locating Your Terrain" on page 201 (be sure to skip the first step calling for a new terrain, since you are now working with an existing terrain).

3. The Terrain Properties side of the Terrain Settings dialog automatically fills in when your terrain is geo-located. If you wish to override these geo-located values, or your terrain is not geo-located, then you can customize these values as follows:
   - **Terrain Size:** Lets you manually change the size of the terrain, so as to crop geo-specific data to a smaller region, for example. If you increase *Terrain Size*, CityScape creates blank areas in the terrain beyond the old borders, up to the indicated maximum terrain size.
   - **Tile Size:** The size of each export tile underlying your terrain.
   - **Tile Count:** CityScape calculates *Tile Count* to ensure that there are enough export tiles to allow the entire *Terrain Size* to be exported at a given *Tile Size*. In cases where *Tile Size* does not divide evenly into *Terrain Size*, the tiles along the edges will extend beyond the terrain.
   - **Sampling Distance:** The distance between points that CityScape samples to determine the exact shape of the geography, as well as the exact way roads integrate with the geography. Shorter sample distances mean more points, which will more accurately simulate terrains with large variations in elevation over short distances.

4. In the New Terrain dialog, LEFT-CLICK **OK** to implement your changes in the open terrain.

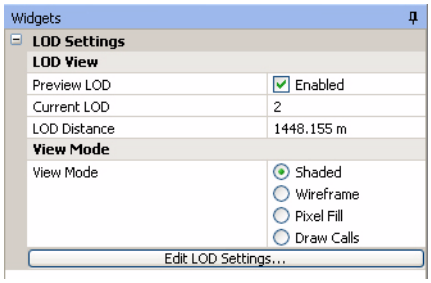
20.2 SETTING LOD TRANSITIONS FOR ENTIRE TERRAINS

This section documents LOD Settings that control LOD transitions based on camera distance, and related functions, for nearly everything in a terrain except props. LODs for imported 3D props are handled differently, as described in “Importing New Props” on page 137.

To modify the LOD settings for the open terrain, follow these steps:

1. Activate the **LOD Tool** by LEFT-CLICKING its icon in the Tools Palette.

2. In the Main LOD Widgets, edit the following values as desired:



   – **Preview LOD:** When enabled, CityScape displays the specific LOD you select via the *Current LOD* parameter or *LOD Distance* parameter, regardless of the position of the camera, so you can preview a specific LOD at various distances.
   – **Current LOD:** Lets you select a specific LOD by number. The highest (and therefore most detailed) LOD is always LOD 0. LOD numbers increment from there (i.e. as the LOD # increases, the geometry becomes less and less detailed).
   – **LOD Distance:** Lets you select a specific LOD by distance, based on the *Transition Distance* parameter in the LOD Settings dialog (see the next step).
   – **View Mode:** Let you choose from the following render modes:
      » **Shaded:** Renders the terrain as a tessellated and shaded surface (i.e. standard rendering).
      » **Wireframe:** Shows 3D meshes as triangles, with minimal representation of textures and shading.
      » **Pixel Fill:** Provides a quick visual way to identify excessive overlapping of near and far objects, relative to the realtime location of the camera. The more opaque the white areas, the greater the pixel fill. By moving around the camera, you can eyeball the pixel fill from various perspectives. Excessive pixel fill could make heavy demands on your target hardware.
      » **Draw Calls:** Provides a quick visual way to identify geometry sharing the same draw call. Items with the same color might share the same draw call (there aren’t nearly enough distinct colors to uniquely tag each draw call). Relates to the *Global Settings* > *Enable Clustering* option in the LOD Settings dialog.

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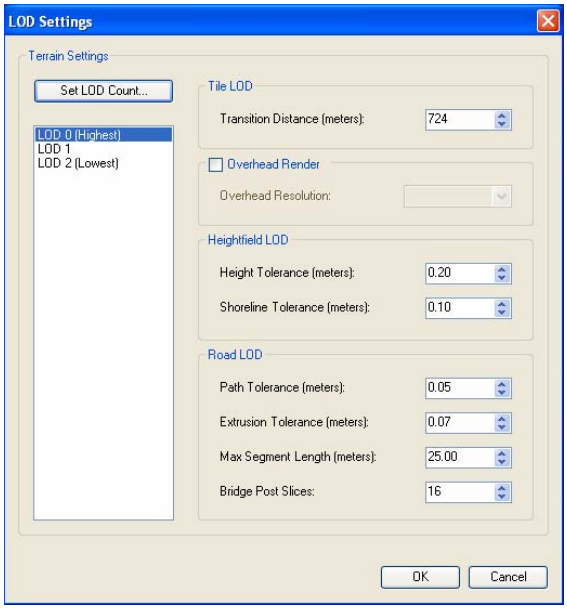
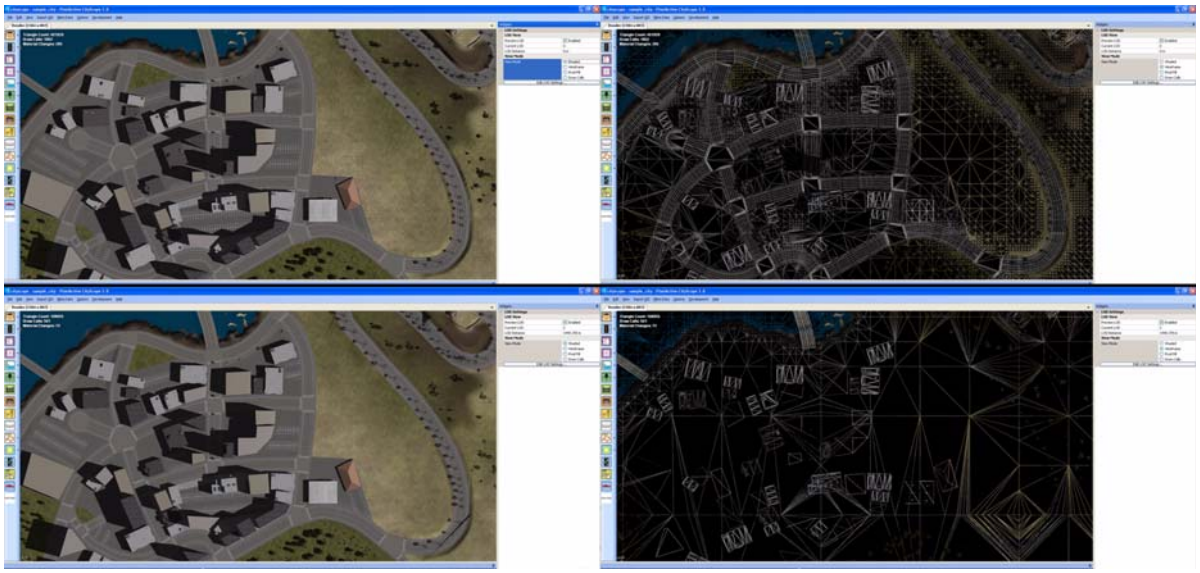
3. In the Main Widgets, LEFT-CLICK the **Edit LOD Settings** button to display the *LOD Settings* dialog. Edit the values in the dialog, as explained below.



- **Set LOD Count:** CityScape assigns sequential integers to LODs based on the value you enter here. For example, if *Set LOD Count* = *3*, then CityScape assigns the numbers 0, 1, and 2 to the highest, medium, and lowest detail LOD, respectively. You must have at least one LOD. The LODs you select will appear in the field below the Set LOD Count button.

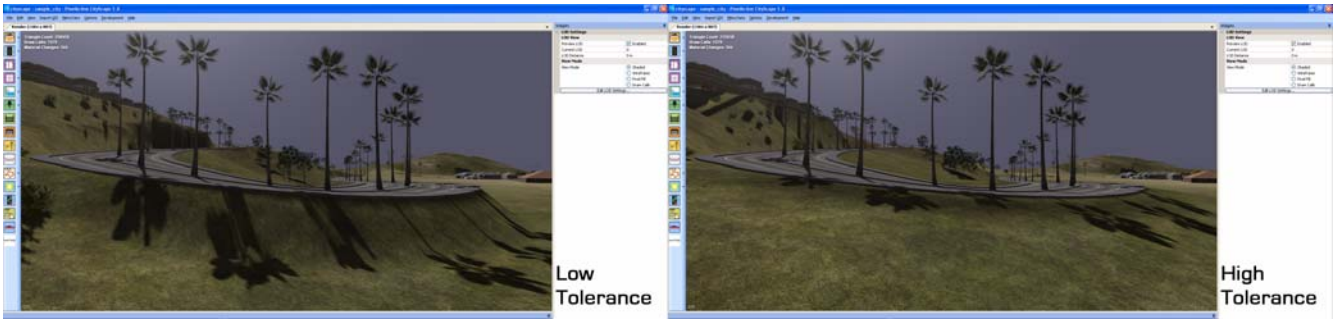
- **Tile LOD**
  - **Transition Distance:** Per terrain tile and LOD, this is the distance between the camera and the center of the tile. based on this Transition Distance, CityScape switches between LODs. Hence, as the camera moves around a terrain, different tiles can potentially use different LOD distances.
    To actively edit the *Transition Distance* for a specific LOD, LEFT-CLICK the target LOD in the field below the *Set LOD Count* button, and then enter a value for *Transition Distance*. Per terrain tile:
    - » When the distance between the camera and the tile containing the camera is less than the *Transition Distance* for LOD 0 (the lowest LOD), then CityScape renders LOD 0 in that tile.
    - » When the distance between the camera and the tile containing the camera is less than the *Transition Distance* for LOD 1, but greater than LOD 0, then CityScape renders LOD 1 in that tile.
    - » (per tile) When the distance between the camera and the tile containing the camera is less than the *Transition Distance* for LOD 2, but greater than LOD 1, then CityScape renders LOD 2 on that tile. Etc.

- **Overhead Render:** If enabled, CityScape renders the selected LOD to a single texture. Specifically, CityScape tessellates almost everything in the open terrain – except for bridges, lakes, and props, which remain as 3D geometry – and blends all textures into a single texture, as seen from an overhead (i.e. orthogonal) perspective. Then CityScape applies the overhead texture to the entire underlying terrain. In general, use *Overhead Render* only for the lowest detail LOD.



  - **Overhead Resolution:** Select the resolution of the Overhead Render texture from the drop-down list.
  - **Trim Roads:** Removes the roads from the creation of the overhead texture.
  - **Apply Global Detail Texture:** Applies the global detail material (i.e. the detail material set for the entire terrain) to the generation of the overhead texture. The final overhead texture preserves the colors blended from all textures, but uses the bump maps and such from the global detail texture only.

- **Heightfield LOD**



  - **Height Tolerance:** Sets the error (in meters) allowed when tessellating the geometry into triangles, relative to mathematically exact source geometry. Higher tolerance allows the triangles to deviate further from the source, so that fewer triangles can be used.

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  - **Shoreline Tolerance:** Same as *Height Tolerance*, but for shorelines only. Allows you to retain smooth shorelines (i.e. low *Shoreline Tolerance*) even when you use a high *Height Tolerance*.

- **Road LOD**
  - **Path Tolerance:** Determines the number of samples CityScape uses to approximate a curved road segment, based on the deviation of the the tessellated (polygonal) segment relative to the mathematical source curve.
  - **Extrusion Tolerance:** Similar to *Path Tolerance*, but takes into account secondary attributes such as lane counts and left turn lanes.
  - **Max Segment Length:** Sets the maximum distance between samples along a curved road segment. Use this parameter to force more samples that CityScape would otherwise use based on the values of the other parameters in the *LOD Settings* dialog.
  - **Bridge Post Slices:** Sets the number of samples CityScape uses to approximate the curved surfaces of bridge posts.

- **Global LOD Settings**
  - **Prop Packing:** Packs all instances of the same prop into a single big prop, to save draw calls. We recommend enabling this option for the lowest LOD only.

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Chapter 21: Importing GIS Data

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21.1 OVERVIEW OF GIS MODE

Using the GIS Tools in CityScape, you can import geo-specific data into CityScape. The original data remains unaffected.

Importing geo-specific data converts the data to standard CityScape objects. In CityScape, you can edit geo-sourced objects in the same ways as objects created in other ways. Specifically, you can edit geo-sourced elevation data with CityScape's Geography Tools; geo-sourced props with Prop Tools; geo-sourced roads with Road Tools; geo-sourced buildings with Building Tools; and geo-sourced water with Water Tools. You can then use CityScape to export the terrain to standard 3D formats such as Collada and OpenFlight, among others.

Converting geo-specific data into 3D visualizations has many practical applications, including but not limited to:

- GPS navigation
- 3D simulations of driving, flying, and tactical training
- Urban planning (3D visualizations of buildings, roads, train tracks, parks, etc.)
- Cartography (3D views of digital maps)
- Video games

21.1.1 About Coordinate Systems

All geo-specific data uses a coordinate system (in specific units, such as meters and degrees) to project geographic coordinates (e.g. latitude/longitude) onto an ellipsoid approximating the earth's surface.

CityScape can import geo-specific data that defines its units internally, and uses one of the following coordinate systems:

- UTM (Universal Transverse Mercator)
- TM (Transverse Mercator)
- Geographic (latitude, longitude)
- LCC (Lambert Conformal Conic)
- US State Plane (LCC and TM)

CityScape assumes that Elevation Files define their own location and coordinate system internally, and that Shapefiles include associated PRJ files defining their coordinate system. See the linked sections for details.

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21.2 SUPPORTED FILE FORMATS

CityScape can import the following categories of geo-specific files. For details, see the linked sections:

- Elevation Files for height fields
- Shapefiles for roads, buildings, props, and water
- OpenStreetMap Files for roads, buildings, props, and water
- Orthographic Imagery, such as satellite photos, provides an optional visual reference for the correct locations of roads, lakes, and other key objects.

In a typical workflow, you would import elevation files first to define the elevations of the geography. Then you can import either Shapefiles or OpenStreetMap files to bring in roads, buildings, etc.

21.2.1 Elevation Files

Elevation files typically contain a height fields that mathematically denote elevations at regularly spaced points in a geographic region. Elevation files typically define their own coordinate system internally.

CityScape can import the following types of elevation files, and convert this data into 3D hills and valleys in a PixelActive terrain: DEM, DTED, GeoTiff, and GridFloat.

DEM (Digital Elevation Model)

To import one region, you must start with one DEM file that uses one of the following coordinate systems: geographic, UTM, or State Plane.

DTED (Digital Terrain Elevation Data)

To import a single region, you must start with one DT0 or DT1 or DT2 file.

GeoTiff

To import a single region, you must start with one geo-located and fully tagged TIF file that uses one of the following coordinate systems: geographic, UTM, or State Plane.

GridFloat

To import one region, you can start with one binary (FLT) file and a corresponding data file (HDR) using the same file name (except for extension) in the same folder.

Alternately, you can import GridFloat in the form of a single ascii file (ASCII).

21.2.2 OpenStreetMap Files

Just one OSM file can contain the locations and associated properties of roads, buildings, and water in a region. You would need 9 Shapefiles (a set of 3 for roads, another set of 3 for buildings, etc) to equal the data that can be in one OSM file.

Continuously updated OSM data is freely available from the open street map web site: www.openstreetmap.org. (OpenStreetMap uses the XML format with fairly standardized elements, but less so with attributes, due to the open source nature of this format. CityScape does its best to accommodate the variations.)

OpenStreetMap files always use geographic coordinates (latitude and longitude) to specify the origin, and bounding box, or a region.

21.2.3 Shapefiles

Shapefiles define the locations of points, line segments, and polygons. For import into CityScape, a Shapefile is actually 3 associated files that are identically named (except for their file extensions), and stored together in the same folder. Specifically, CityScape requires a Geometry Shapefile (SHP) be associated with a dBASE file (DBF) and a projection file (PRJ). For details, see the following sections:

- Geometry Shapefiles (SHP) define the locations of points representing props, line segments representing roads, or polygons representing the footprints of buildings or borders of water. (A single geometry Shapefile cannot mix types of geometry, but can define only points or only lines or only polygons.)
- dBASE Files (DBF) contain attributes associated with the geometry in their companion SHP files. Import options let you to map custom attributes in DBF files to features in CityScape.
- Projection Files (PRJ) tell CityScape how to interpret the spatial data in an associated SHP file. The format is plain text.

GEOMETRY SHAPEFILES (SHP)

A single SHP file mathematically defines the positions of either points, line segments, or polygons in a standard 3D coordinate system. You cannot mix different types of geometry in the same SHP file.

SHP For Roads

CityScape uses a SHP file containing *line segments only* to deterine the center lines of roads in terrains. The SHP file does not contain instructions for connecting the line segments, but rather depends on the associated DBF for connectivity data. (For details on roads, see Chapter 6: Using Road Tools.)

- Cityscape assists with importing incomplete road data:
  - Crossing roads are automatically identified as intersections and will be properly connected.
  - Spacing algorithms smooth vertical kinks and ensure that bridges have sufficient space above the roads below them.
  - Road data is smoothed to avoid sharp corners and jagged roads.

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Once you have imported a road shapefile (i.e. CityScape has converted the data to roads in a geo-located terrain), you can edit the resulting roads with the Road Tools in CityScape.

SHP for Buildings

CityScape uses a SHP file containing *polygons only* to determine the footprints of buildings in terrains. Buildings generated from a SHP file are not treated as "props" by CityScape. Rather, CityScape used the polygons to define the footprints of buildings, and then extrudes these footprints into textured 3D models that are compatible with Using Building Tools (not Prop Tools).

SHP for Props

CityScape uses a SHP file containing *points only* to determine the locations of prop points (i.e. one prop per point). CityScape uses a SHP file containing *polygons only* to determine the locations of prop polygons (i.e. polygons containing multiple instances of a prop).

Once you have imported a prop shapefile (i.e. CityScape has converted the data to props in a geo-located terrain), you can edit the resulting props with the Prop Tools in CityScape. Specifically, you can edit point props with the Prop Polygon Tool, and polygon props with the Prop Polygon Tool.

SHP for Water

CityScape uses a SHP file containing *polygons only* to determine the shores of a lake or other body of water. Once you have imported a water shapefile (i.e. CityScape has converted the data to polygonal water regions in a geo-located terrain), you can edit the water with Using Water Tools in CityScape.

dBASE FILES (DBF)

A DBF file contains attributes associated with specific geometry in the companion SHP file. In general, DBF files can contain almost any type and organization of data. PixelActive supports custom DBF attributes by allowing you to map some attributes in DBF files with specific features in CityScape. For example, line segments with the "freeway" attribute could be mapped to a Freeway Road Style in CityScape.

DBF for Roads

A DBF file for roads usually contains connectivity data for the line segments in the associated SHP file. The DBF file can optionally contain custom attributes of these line segments, as explained below.

- CityScape lets you map the following custom attributes in DBF road files to key road features in CityScape:
  - **Road Style:** Lets you map any attribute used in the road DBF file to a particular style in the Road Style Library associated with the open terrain.
  - **Lane Counts:** Lets you map attributes used in the road DBF file to *Right Lanes* and *Left Lanes* parameters in CityScape, so as to control the number of lanes in the affected road.
  - **Custom connectivity:** Gives you some control of how CityScape converts the lines in the Shapefiles to nodes and line segments that CityScape connects into roads, intersections, bridges, and tunnels.

For details on mapping custom attributes for roads, see “Import Roads” on page 205.

DBF for Buildings

A DBF file for buildings contains the attributes of polygons in the corresponding SHP file. You can import shapefiles containing custom attributes of buildings by mapping these attributes to the corresponding functionality in CityScape:

- **Building Style:** Lets you map any attribute used in the building DBF file to an existing building style in the Building Style Library associated with the open terrain. Building styles define the roof type, wall texture, and roof texture of a building, among other things.
- **Height:** Lets you map any attribute used in the building DBF file to the *Height* parameter controlling the height of the building.

For details on mapping custom attributes for buildings, see “Import Buildings” on page 211.

DBF for Props

A DBF file for props contains the attributes of the points or polygons in a corresponding SHP file. You can map the following prop attributes to indicated functionality in CityScape:

- **Prop Style:** Lets you map any attribute in the prop DBF file to a particular prop in the open Prop Library.

For details on mapping custom attributes for props, see “Import Prop Points” on page 213.

DBF for Water

A DBF file for bodies of water contains the attributes of polygons in the corresponding SHP file. No custom attributes are currently supported for water, but support will be added in a future release.

PROJECTION FILES (PRJ)

A projection file (PRJ) tells CityScape how to interpret the spatial data in an associated SHP file. Specifically, a PRJ file defines the coordinate system, projection datum, and units used by the associated SHP file. (For details on coordinate systems, see “About Coordinate Systems” on page 196.)

21.2.4 Orthographic Imagery

Orthographic images (e.g. satellite photos, etc) can be readily downloaded from various web sites, such as seamless.usgs.gov

In CityScape, you can import any ortho image in GeoTIFF format. Once imported, an ortho image overlays your terrain visually, but does not interact with anything. It’s just a visual reference for you. Hence, you can import an ortho image into your terrain at any point in the import process.

Importing ortho images is optional, but satellite photos are particularly handy for visual reference. As you import roads, buildings, water, and prop landmarks, you can verify that CityScape positions these elements in the same placed as they appear on the ortho image. You can also double-check that all the geo-data is complete and has been imported.

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21.3 GIS IMPORT PROCESS

The GIS Tools in CityScape allow you to import the position, scale, and elevations of geographic regions; props like trees; road networks; buildings and bodies of water. When importing GIS data, you must first geo-locate your PixelActive terrain. Then you can import geo-specific data at this location with the following process:

1. Geo-Locating Your Terrain. You must specify a geographic location for a terrain before you can import elevations or any other geo-specific data into the terrain.

2. Import an Elevation File to bring in height fields that CityScape uses to generate terrain with corresponding hills and valleys.

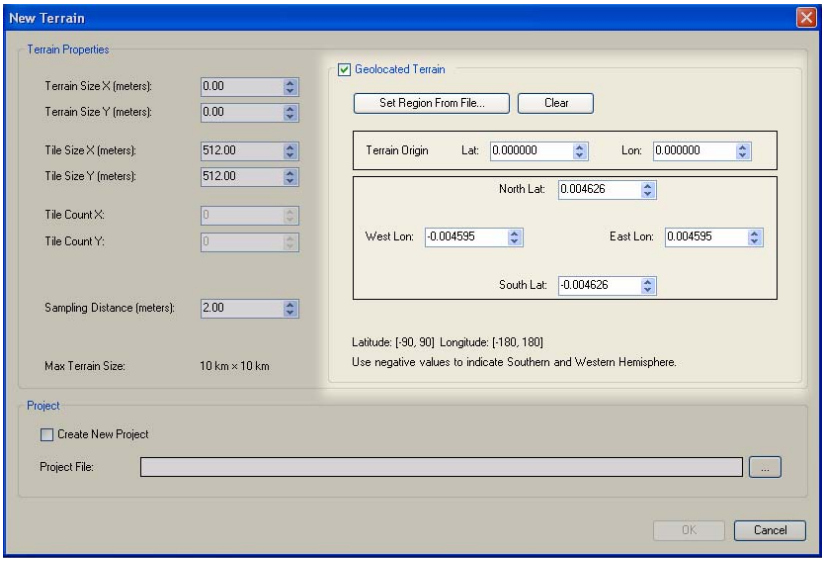
3. Either Import an OpenStreetMap File, or Import a Set of Shapefiles, to bring roads, buildings, props, and water into the terrain.

4. Optionally Import Orthographic Imagery to serve as a visual reference for roads, buildings, water, and props.

21.3.1 Geo-Locating Your Terrain

If you have an existing terrain with custom contents, and this existing terrain has been geo-located, CityScape will let you import geo-specific data into the terrain. CityScape will attempt to merge the imported data with the existing data in the terrain. The results are difficult to control, and may be undesirable. Hence, we strongly recommend that you import geo-specific data into a new (i.e. empty) geo-located terrain. In either case, the process of geo-locating the terrain is nearly identical.

1. To geo-locate a new terrain, press **Ctrl-N** to display the New Terrain dialog. Or to geo-locate the open terrain, select **Options > Terrain Settings** in the Main Menu to display the Terrain Settings dialog. The two dialogs are nearly identical.



2. In the dialog, LEFT-CLICK the **Geo-located Terrain check box** to enable the right side of the dialog, which pertains to the geo-location of the terrain.

3. Now you can set the geo-location of the terrain in either of two ways:
   – (Recommended) LEFT-CLICK the **Set Region from File** button: In the dialog that appears, do the following:
     a. Locate and load any Shapefile, OpenStreetMap file, or elevation file in your target region. CityScape centers the terrain at the geo-location of the loaded Shapefile. CityScape also bases the size of the terrain on the borders of the loaded Shapefile – up to the maximum size supported by your version of CityScape.
   – (Manual Alternative) Set a precise **Terrain Origin** to locate the terrain on the globe. The size of the terrain is determined by its borders, up to the maximum allowable size. Here's how:
     a. Type in the latitude and longitude of your *Terrain Origin.*
     b. Type in the latitude and longitude of your terrain's borders in the *North Lat, South Lat, West Lon,* and *East Lon* fields.

4. In the dialog, CityScape updates the Terrain Size (X, Y) fields to match its geo-location and borders from the previous step. You can optionally customize any or all the **Terrain Properties**. Notes:
   – If the geo-specific region you wish to import is larger than the maximum size that CityScape allows, then CityScape will crop the data to its max size. The cropped terrain will be centered at the specified *Terrain Origin*.

5. In the Project section of the dialog, select either **Create New Project** to create a new project file (CSP); or select an existing Project File by LEFT-CLICKING the browse button **[...]** and locating the target CSP file on your hard drive.

6. In the dialog, LEFT-CLICK **OK** to create a blank terrain that's geo-located. You can now import geo-specific data.

The next step is to Import an Elevation File.

21.3.2 Import an Elevation File

You should import an elevation file for your target region prior to importing anything else.

1. In CityScape, open a blank geo-located terrain, or Geo-Locating Your Terrain.

2. Select **Import GIS > Import Elevation > format** from the Main Menu, where "format" is the file format of your elevation file. (For a list of supported formats, see "Elevation Files" on page 197.)

3. Your next action depends on the format of your elevation file:
   – **DEM:** In the *Import Elevation from DEM* dialog, LEFT-CLICK the browse button **[...]** to display a file loader, locate the target DEM file, and LEFT-CLICK OK to complete the import.

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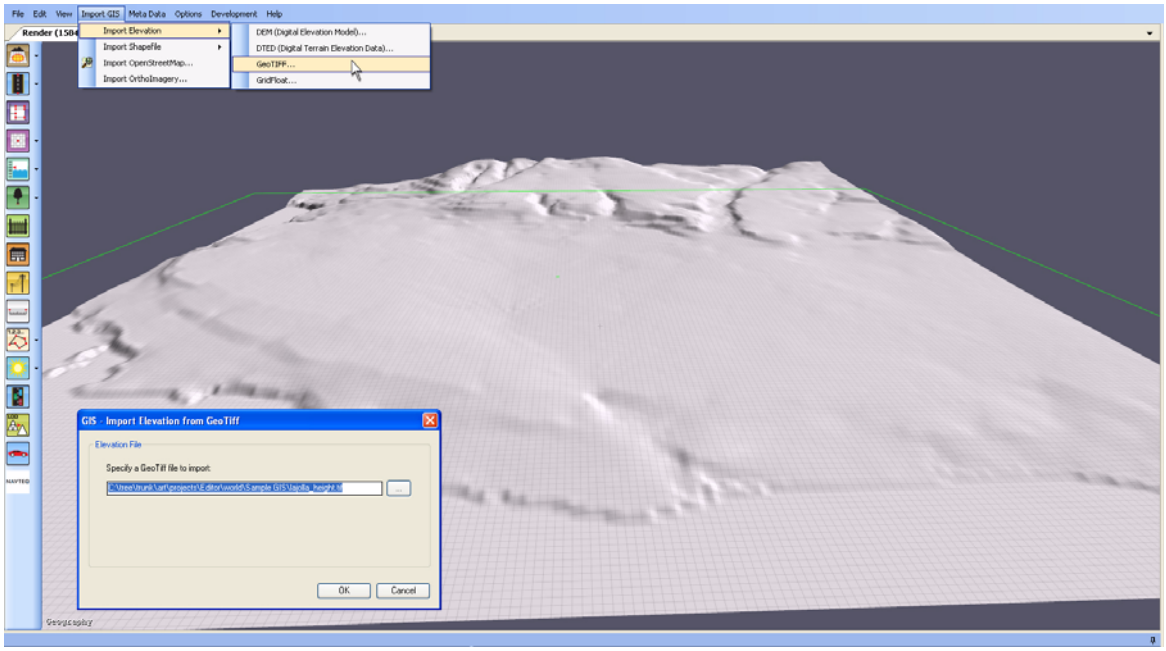
- **DTED:** In the *Import Elevation from DTED* dialog, LEFT-CLICK the browse button to display a file loader, locate the target DTED file (DT0, DT1, or DT2 extension), and LEFT-CLICK OK to complete the import.
- **GeoTiff:** In the *Import Elevation from GeoTiff* dialog, LEFT-CLICK the browse button to display a file loader, locate the target GeoTiff file (TIF extension), and LEFT-CLICK OK to complete the import.
- **GridFloat:** In the *Import Elevation from GridFloat* dialog, LEFT-CLICK the browse button to display a file loader, locate the target GridFloat file (FLT or ASC extension), and LEFT-CLICK OK to complete the import.



The next step is to either Import an OpenStreetMap File, or to Import a Set of Shapefiles. The data contained in OSM files and Shapefiles are mutually exclusive, so you need to import only one of these formats.

21.3.3 Import an OpenStreetMap File

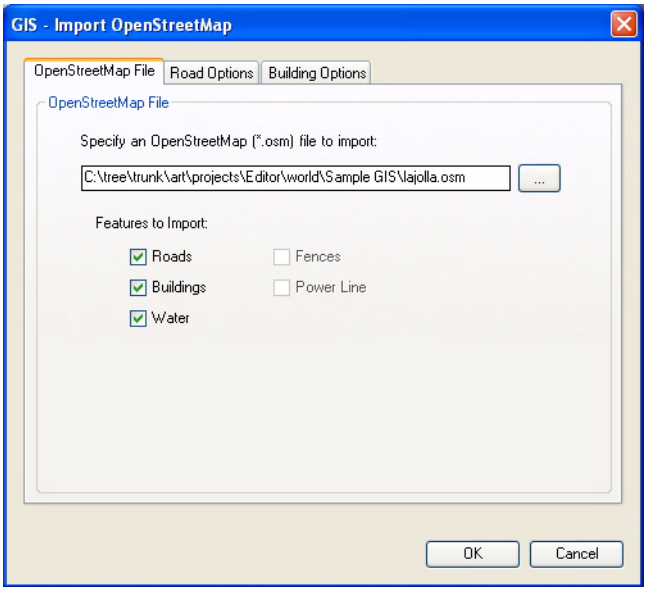
If your geo-specific data is in OSM format, then you will not need to import any Shapefiles. You should import an OpenStreetMap (OSM) file for your target region after importing an elevation file for the same region.

1. In CityScape, open the geo-located terrain, plus elevation data, that you prepared in Import an Elevation File.

2. Select **Import GIS > Import OpenStreetMap** from the Main Menu. The *Import OpenStreetMap* dialog appears.

3. In the dialog, LEFT-CLICK the browse button **[...]** to display a file loader, locate the **target OSM file**, and LEFT-CLICK **OK**.

4. In the dialog, select the *Features to Import*. (In most cases, just select all supported features: Roads, Buildings, Water. Additional features will be supported in future versions of CityScape.) LEFT-CLICK **OK** to finish the import.



21.3.4 Import a Set of Shapefiles

If your geo-specific data is in Shapefiles, then you do not need to import any OpenStreetMap files. You should import Shapefiles for your target region after importing an elevation file for the same region.

You have the option to import Shapefiles for all of the following features, or just one feature, or any combination of features in any order:

- Import Roads
- Import Buildings
- Import Prop Points
- Import Prop Polygons
- Import Water

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IMPORT ROADS

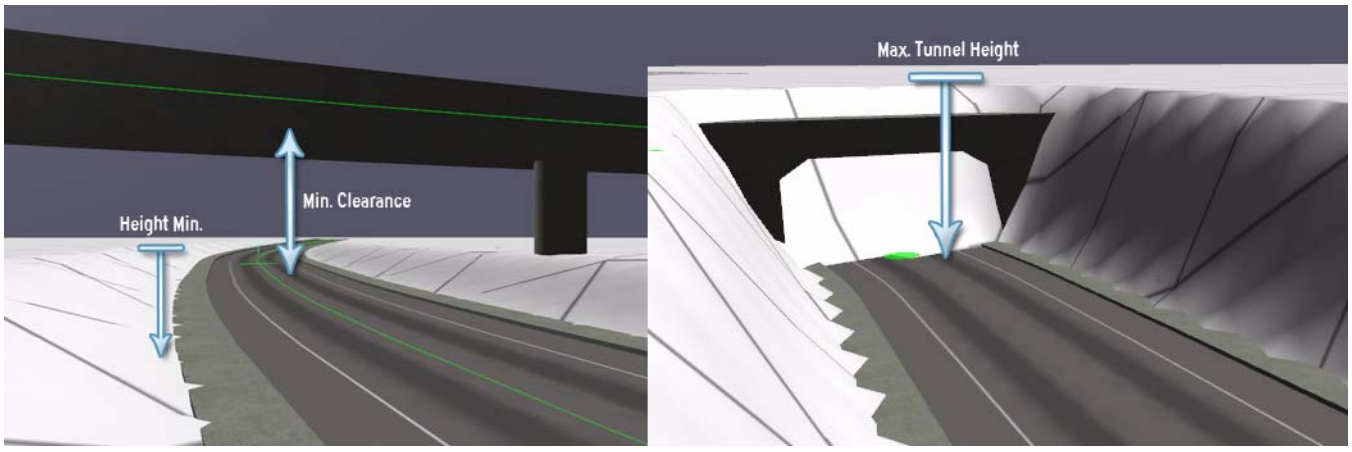
Here's how you import a 3-file set of Shapefiles containing road data:

1. In CityScape, open an existing geo-located terrain that already contains elevation data. (To add elevation data to a terrain, see "Import an Elevation File" on page 202.)

2. Select **Import GIS > Import Shapefile > Roads** from the Main Menu. The Import Roads dialog appears.

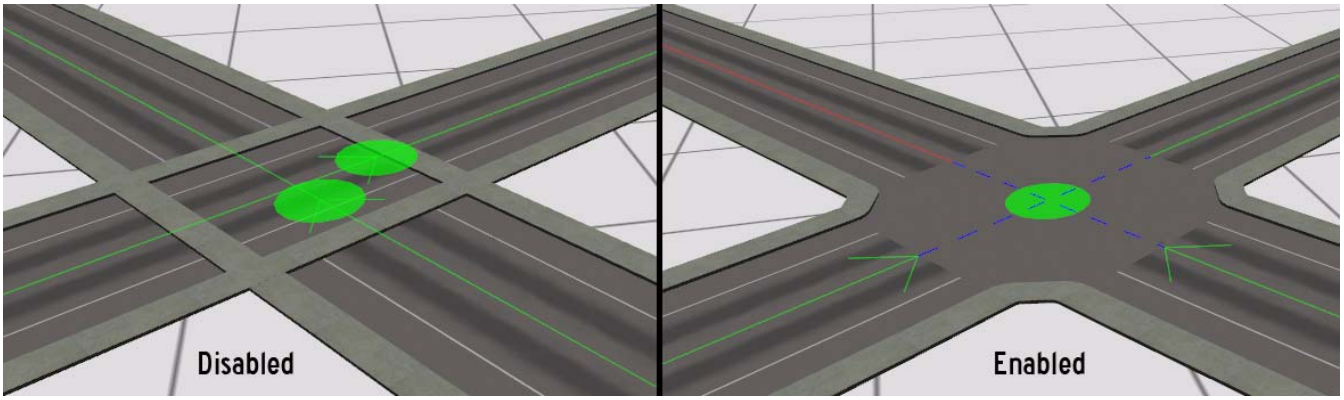
3. In the *Shapefile* tab of the Import Roads dialog, do the following:
   i. To select the Shapefile (SHP) containing the road geometry you wish to import, LEFT-CLICK the **[...]** button to display a file loader, so you can locate and load the **target SHP file**.
   ii. Under Road Attributes, either select *Use Default Attributes*, or select *Use Custom Attributes* and click the **Customize** button to display the *GIS – Import Roads – Custom Attributes* dialog. If you select the Customize option, Map Custom Road Attributes to CityScape features now.

4. (Optional) Edit the following parameters in the *Road Options* tab to control how CityScape interprets crossing roads:
   – **Auto-Resolve Crossings:** When enabled, CityScape can accurately resolve crossings into intersections, bridges, and tunnels only when each road has a shapefile point at the crossing, and the roads have Z-level attributes to indicate their relative heights. Otherwise (or if Auto-Resolve Crossings is disabled), CityScape will interpret all crossings as intersections.
   – If you enable Auto-Resolve Crossings, the following parameters influence how CityScape handles bridges and tunnels:
      » **Height Min:** the distance CityScape will push down a road to create at least *Min Clearance* space between it and the bridge above it.
      » **Min Clearance:** the minimum distance between a road and a bridge over it.
      » **Max Tunnel Height:** the minimum depth of the tunnel under ground.



   – **Resolve Quality:** TBD

– **Merge Overlapping Segments:** If this option is enabled, CityScape will merge too-close nodes into one node and a control point, so an intersection can be formed. If disabled, CityScape will preserve both nodes and let you fix the intersections manually later.



5. In the main *Import Roads* dialog, LEFT-CLICK **OK** to import the roads in the source Shapefiles. CityScape will generate roads in the open terrain using the geometry and attributes defined in the source road Shapefiles.

Map Custom Road Attributes

In CityScape, an "attribute map" is a set of associations between custom attributes in a DBF Shapefile and specific functionality in CityScape. For example, a "freeway" functional class in a DBF file could be mapped to a Freeway road style in CityScape.

When you import road Shapefiles into CityScape, you can optionally choose to create a custom Attribute Mapping Template, which is a plain text file containing two types of things:

1. an attribute map;

2. options for converting the connectivity data in the road DBF file to actual roads in the PixelActive 3D terrain.

In CityScape, you create a custom Attribute Mapping Template in the *GIS – Import Roads - Custom Attributes* dialog, which you can display by LEFT-CLICKING the **Customize** button in the Shapefile tab of the main Import Roads dialog. See the subsections below for details on the parameters you can edit in the Custom Attributes dialog.

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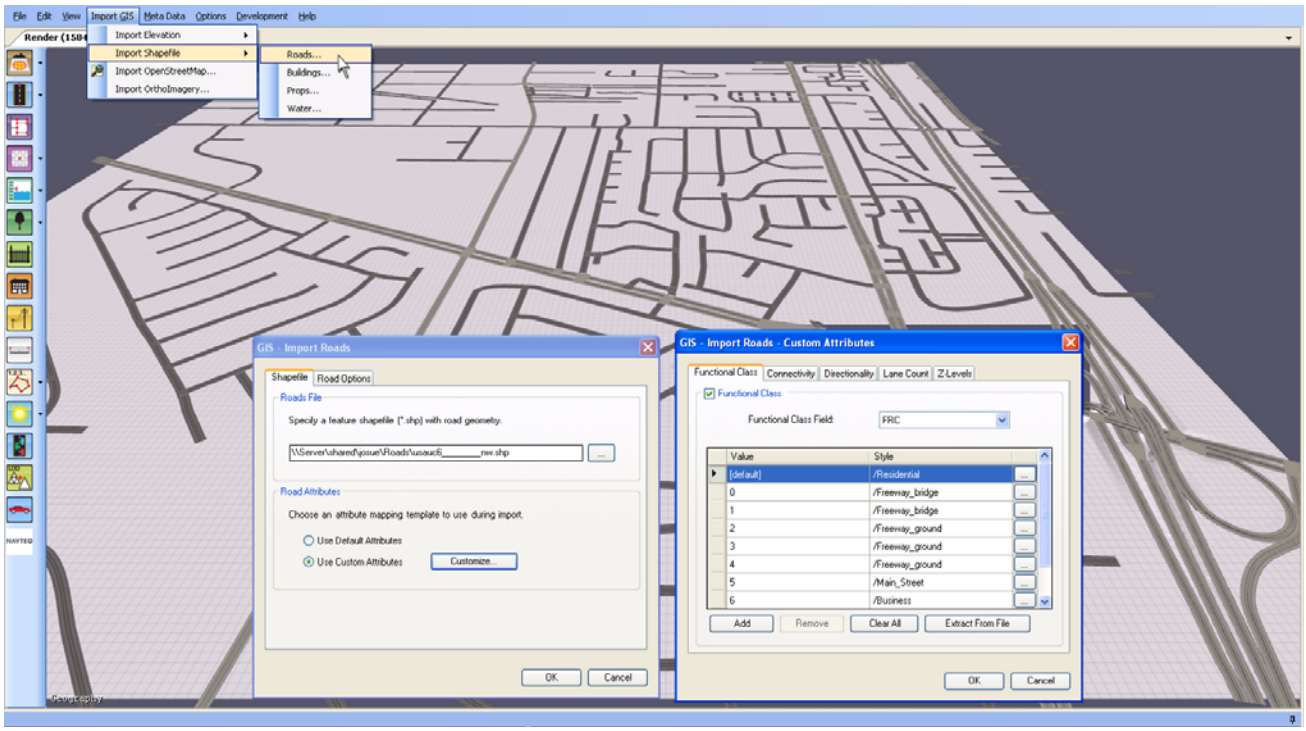
FUNCTIONAL CLASS TAB

If you enable the Functional Class checkbox, you can map the Functional Class attribute (in the associated road DBF file) to a road style in CityScape. The "functional class" is the type of road, such as a freeway, main street with median, or small residential street. Here's how to use this dialog:



1. Enable the **Functional Class** checkbox in the *Custom Template* dialog.

2. **Functional Class Field:** In the drop-down menu, choose the specific DBF attribute containing values that identify the functional classes of the roads being imported.

3. LEFT-CLICK the **Extract from File** button: CityScape displays the values of the selected DBF attribute, as they appear in the DBF file.

4. (Optional) Add additional values that are not in this DBF file, but might be in other DBF files that you will eventually import using this template:
   i. LEFT-CLICK the **Add** button to add a new value to the table.
   ii. LEFT-CLICK the **default name** of the new value that appears in the Values column, and type a custom name for the value.
   iii. Repeat for additional values.

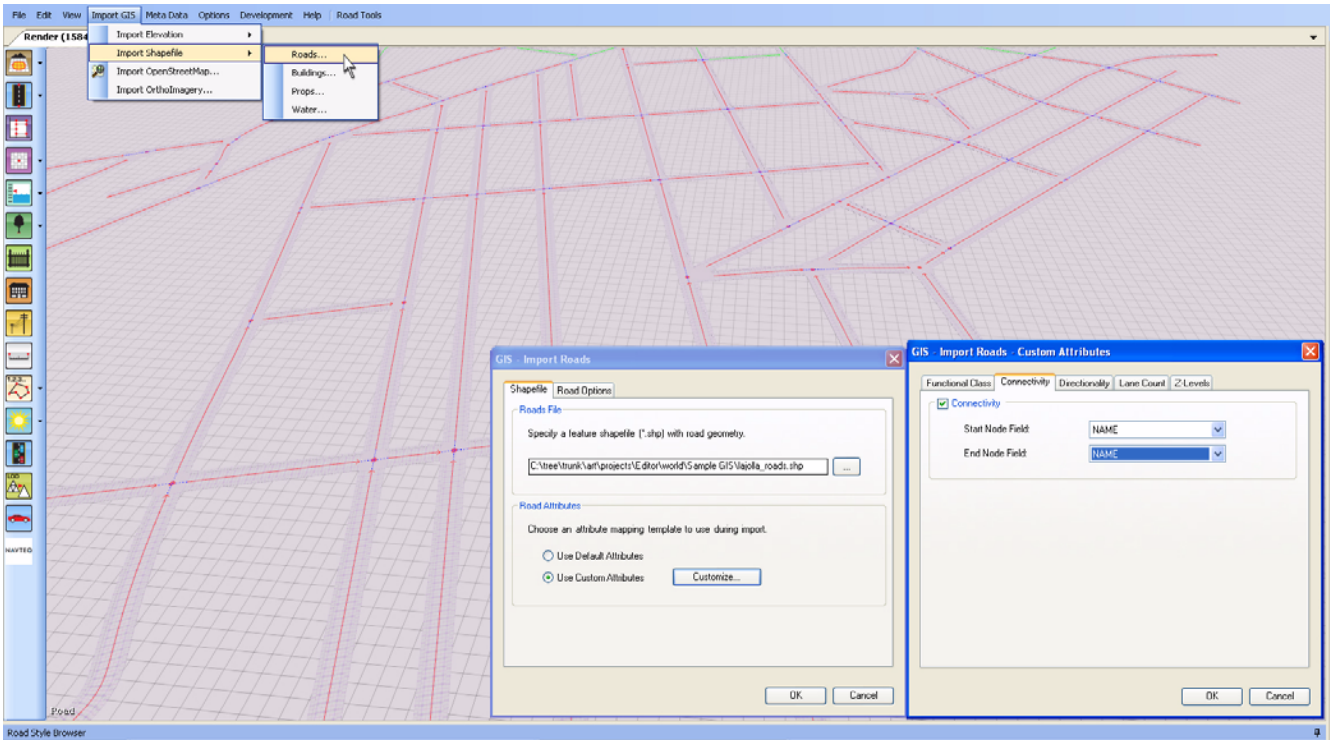
5. (Optional) Delete a value:
   i. Select the **row** of the doomed value in the table.
   ii. LEFT-CLICK the **Remove** button. The target value is nuked.

6. Starting with the *default* value:
   i. LEFT-CLICK the browse button **[...]** on the right end of each row to display the Choose a Road Style dialog.
   ii. Select the **road style** to map to this value.

iii. Repeat until all the values are mapped to road styles. If a value is left unmapped, CityScape will assign the **default** road style to that road. That's why it's important that you map at least the *default* value to an existing road style.

CONNECTIVITY TAB

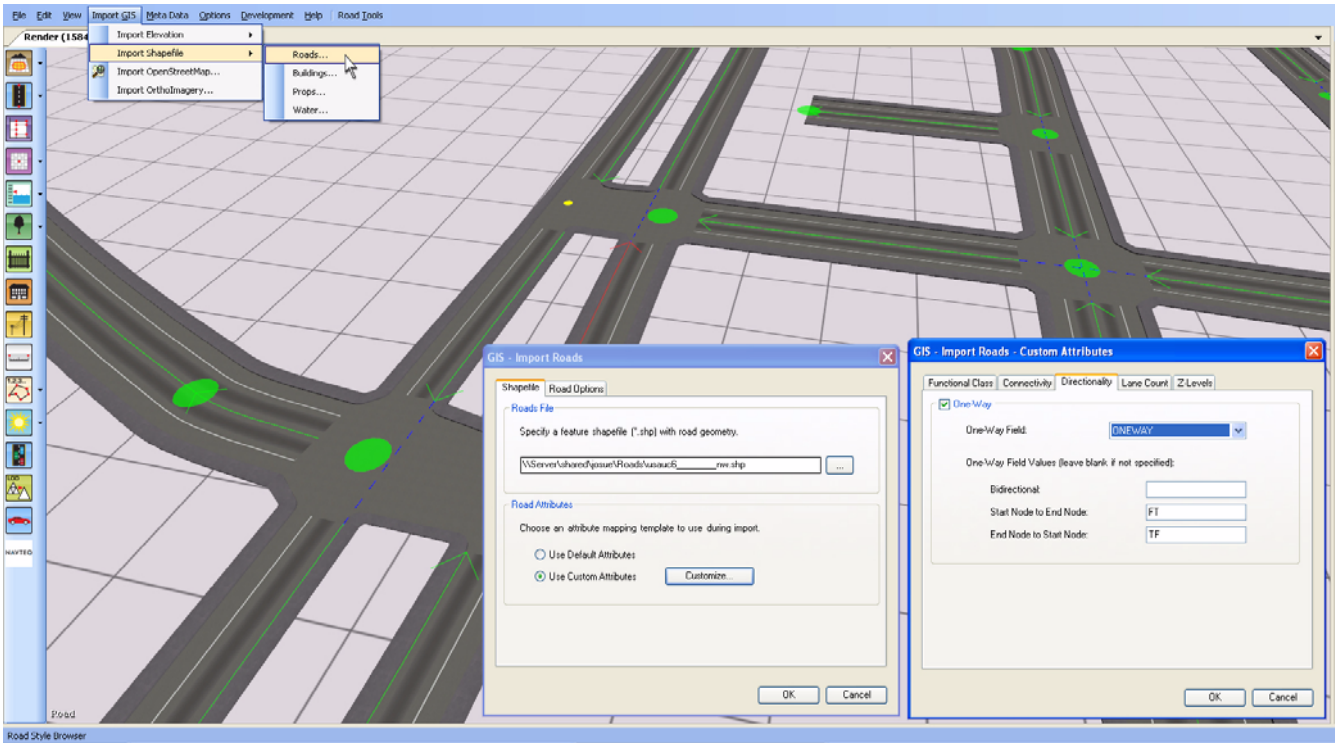
Edit the following parameters in the *Connectivity* tab of the *Custom Template* dialog, so as to control aspects of how CityScape connects the geometry in the SHP file to form a road in a PixelActive terrain.



- **Connectivity:** Enabling this checkbox exposes the parameters below, so you can identify the start and end points of each set of road points in the source Shapefile. This information helps CityScape convert the source data into PixelActive nodes and line segments for intersections, bridges, and tunnels. In particular, *Connectivity* helps distinguish between intersections, bridges, and tunnels in roads that cross from an overhead of view. If disabled, CityScape inserts intersections at all road crossings.
  - **Start Node Field:** Choose the unique ID for the Start Node of the road.
  - **End Node Field:** Choose the unique ID for the End Node of the road.

DIRECTIONALITY TAB

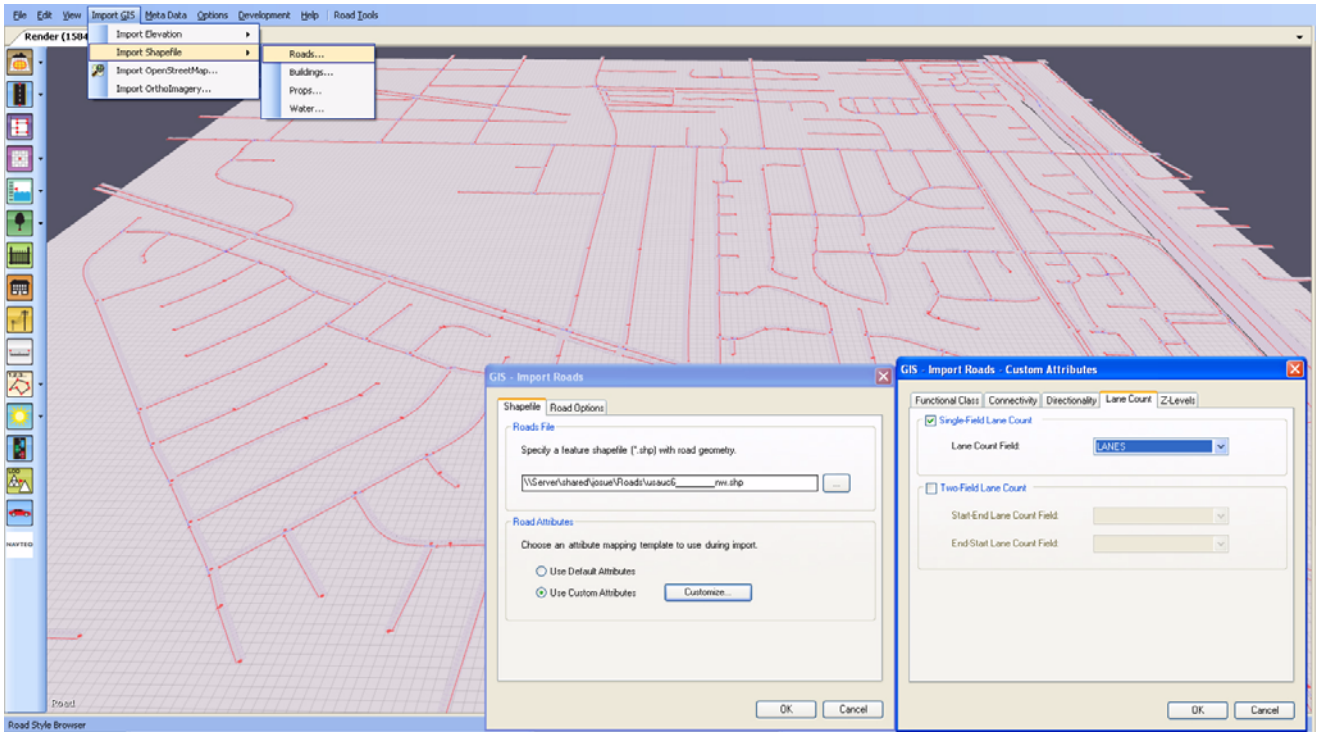
Edit the following parameters in the *Directionality* tab of the *Custom Template* dialog, so as to distinguish between one-way and bi-directional roads.



- **One-Way:** Enabling this checkbox exposes the parameters below, so you can identify the Start Node and End Node of one-way and bi-directional roads:
  - **One-Way Field:** select the directionality field in the source DBF file. This field contains a value that identifies one-way and bi-directional sections of road.
  - **Bidirectional:** type in the value identifying the road as bi-directional.
  - **Start Node to End Node:** type in the value identifying the road as one-way in the direction from the Start node to the End node.
  - **End Node to Start Node:** type in the value identifying the road as one-way in the direction from the Start node to the End node.

LANE COUNT TAB

Edit the following parameters in the *Lane Count* tab of the *Custom Template* dialog, so as to determine the number of lanes on the right and left sides of the associated road.



- **Single Lane Field Count:** Enabling this checkbox exposes the parameter below, so you can edit the DBF field containing a single value for the number of lanes on each side of the road:
  - **Lane Count Field:** select the field designating the number of lanes on each side of the road (e.g. a value of "3" means the road had 3 lanes are on each side).

- **Two-Lane Field Count:** Enabling this checkbox exposes the parameters below, so you can choose the DBF fields containing values that identify the associated section of road as having a different number of lanes on each side:
  - **Start-End Lane Count Field:** defines the number of lanes on the right side only, in the direction from the Start node to the End node of the affected node.
  - **End-Start Lane Count Field:** defines the number of lanes on the left side only, in the direction from the End node to the Start node of the affected node.

Z-LEVELS TAB

- **Z-levels:** relative elevations (i.e. which road goes above or below) are normally defined at the start or end point of a section of road – as well as the point where two roads cross.
  - **Start Node Z-Level Field:** select the DBF field containing Z-level data.

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– **End Node Z-Level Field:** select the DBF field containing Z-level data.

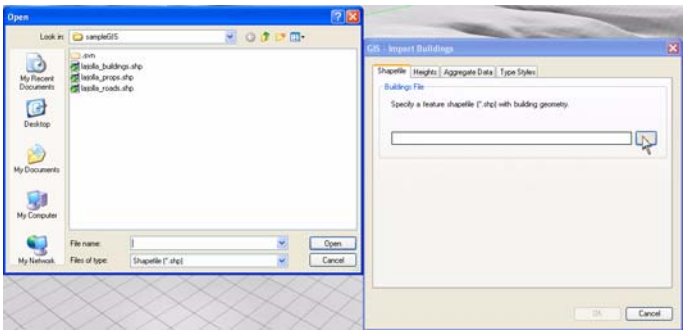
IMPORT BUILDINGS

The overall process for importing Shapefiles with building data is similar to roads, but the editable parameters are different:

1. In CityScape, open an existing geo-located terrain that already contains elevation data and optionally road data. (To add elevation data to a terrain, see "Import an Elevation File" on page 202.)

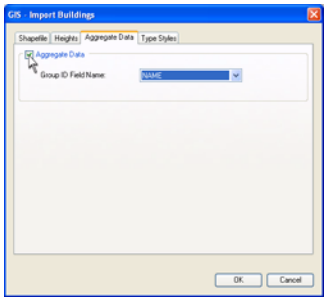
2. Select **Import GIS > Import Shapefile > Buildings** from the Main Menu. The Import Buildings dialog appears.

3. In the *Shapefile* tab of the Import Buildings dialog, LEFT-CLICK the **[...]** button to display a file loader, so you can locate and load the **target SHP file** containing building data.



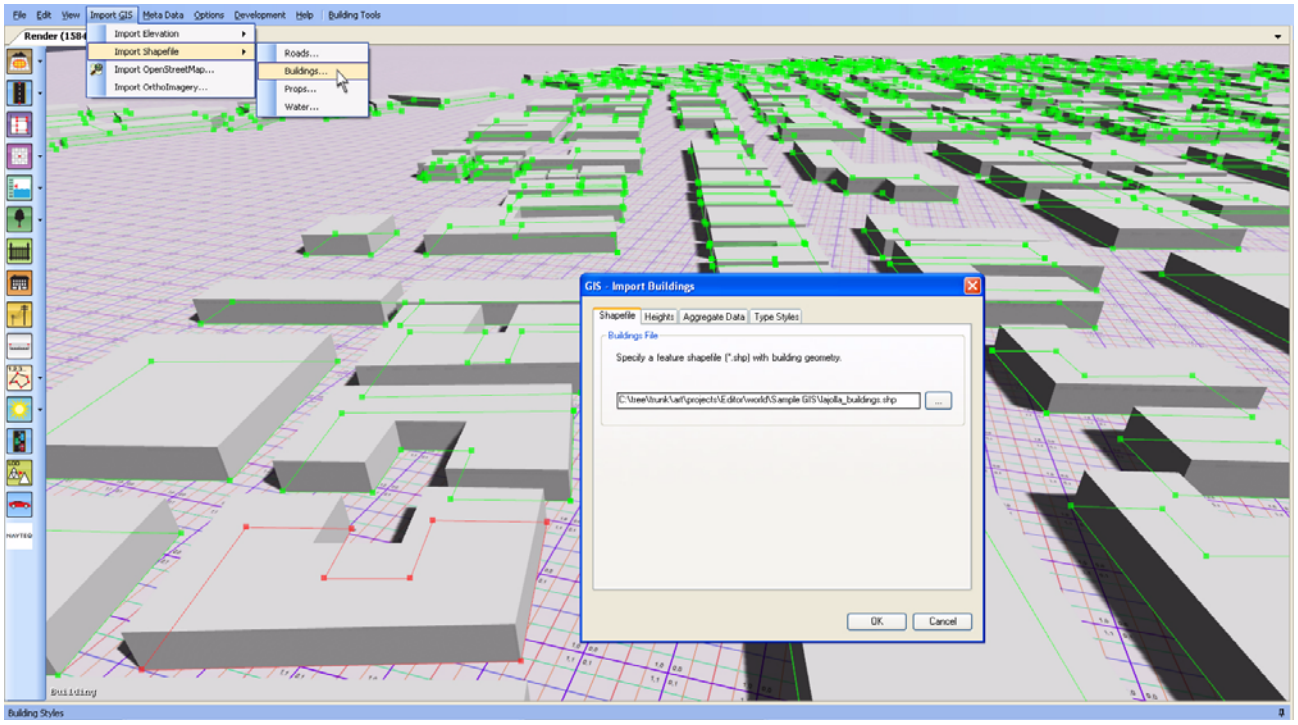
4. In the *Heights* tab of the Import Buildings dialog, either Import Heights from the DBF file, or CityScape will randomize the heights of generated buildings within the min/max values you specify:
   – **Random Heights Range**
      » Min Height: the lower bounds (in meters) of randomized heights for generated buildings
      » Max Height: upper bounds (in meters) of randomized heights for generated buildings
   – **Import Heights:** enable this option to load the heights of buildings from the building DBF file, and expose the parameters below:
      » **Height Field:** select the DBF field containing the height values of the buildings.
      » **Units:** Select the units of the Height Field. Options include *meters*, *feet*, *inches*, or *custom*. If you select Custom, you can enter a custom value for converting to meters.

5. In the *Aggregate Data* tab of the Import Buildings dialog, enable **Group ID Field Name** to tell CityScape that buildings with the same ID should be mapped to the same building styles in CityScape.



6. In the *Type Styles* tab of the Import Buildings dialog, enable the Building Types checkbox to map Building Type attributes in the DBF file to CityScape building styles. Using this dialog is similar to using the Functional Class Tab dialog for importing roads, except you are now mapping building styles rather than road styles, of course.

7. In the main *Import Buildings* dialog, LEFT-CLICK **OK** to import the locations of buildings in the terrain. CityScape will generate buildings in the terrain using the polygons and attributes defined in the Shapefiles for buildings. (Note: If you do not enable the Type Styles tab, or you do not specify a default style to use, then CityScape will use the white texture style shown in the figure below.



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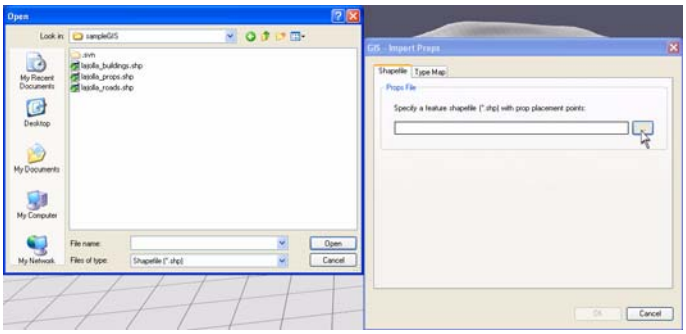
IMPORT PROP POINTS

When you import Shapefiles with point data, you are identifying the locations to place point props. The overall process is the same as for roads, but the editable parameters are different:

1. In CityScape, open an existing geo-located terrain that already contains elevation data, and optionally road and buildings too. (To add elevation data to a terrain, see “Import an Elevation File” on page 202.)

2. Select **Import GIS > Import Shapefile > Prop Points** from the Main Menu. The Import Prop Points dialog appears.

3. In the *Shapefile* tab of the Import Props dialog, LEFT-CLICK the **[...]** button to display a file loader, and locate the **target SHP file** containing prop point data.



4. In the *Type Map* tab of the Import Props dialog, enable the Prop Type Mapping checkbox to map Prop attributes (in the source DBF file) to prop styles in CityScape. Using this dialog is similar to using the Functional Class Tab dialog while importing roads, except you are now mapping prop styles rather than road styles, of course.

5. In the main *Import Props* dialog, LEFT-CLICK **OK** to import the locations of specific props in the terrain. CityScape will place props in the terrain using the points and attributes defined in the Shapefiles for props.

IMPORT PROP POLYGONS

When you import Shapefiles for props with polygonal data, you are identifying polygons within which to place props. The overall process is the same as for prop points, but the editable parameters are different:

1. In CityScape, open an existing geo-located terrain that already contains elevation data, and optionally other geo-data.

2. Select **Import GIS > Import Shapefile > Prop Polygons** from the Main Menu. The Import Prop Polygons dialog appears.

3. In the *Shapefile* tab of the Import Props dialog, LEFT-CLICK the **[...]** button to display a file loader, and locate the **target SHP file** containing prop polygon data.

4. In the main *Import Props* dialog, LEFT-CLICK **OK** to import the locations of specific props in the terrain. CityScape will place props in the terrain using the points and attributes defined in the Shapefiles for props.

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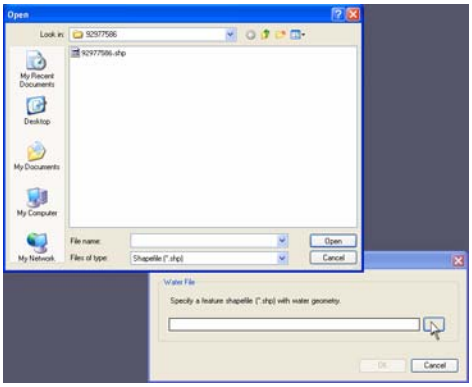
IMPORT WATER

The overall process for importing Shapefiles with water data is the same as for roads, but the editable parameters are different:

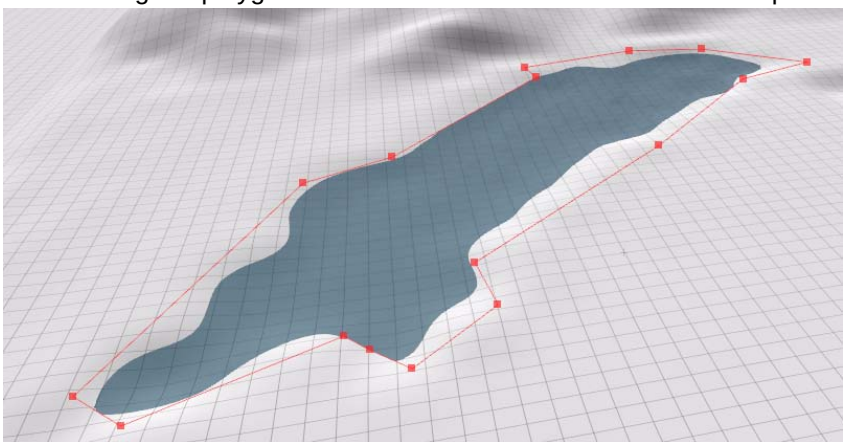
1. In CityScape, open an existing geo-located terrain that already contains elevation data. (To add elevation data to a terrain, see "Import an Elevation File" on page 202.)

2. Select **Import GIS > Import Shapefile > Water** from the Main Menu. The Import Water dialog appears.

3. In the *Shapefile* tab of the Import Water dialog, LEFT-CLICK the **[...]** button to display a file loader, so you can locate and select the **target SHP file** containing water data.



4. In the main *Import Water* dialog, LEFT-CLICK **OK** to import the locations of water in the terrain. CityScape will generate bodies of water in the terrain using the polygons and attributes defined in the water Shapefile.



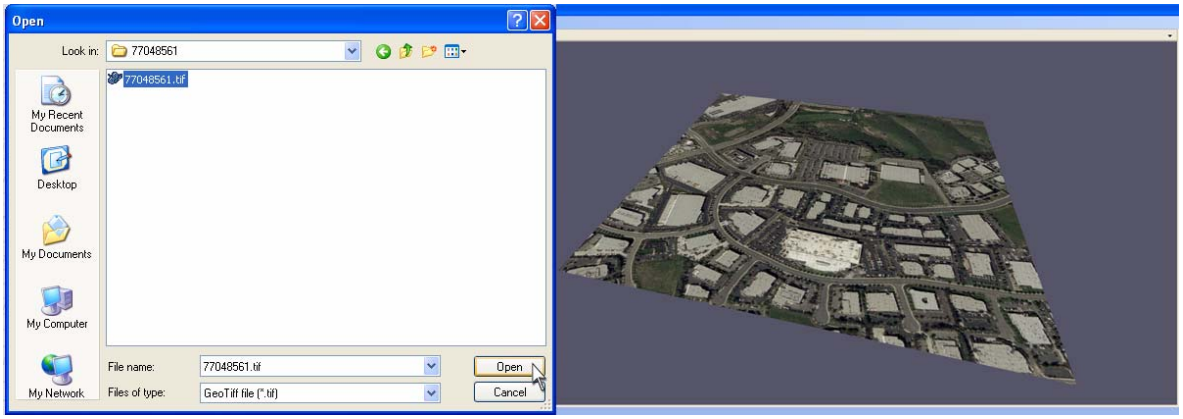
21.3.5 Import Orthographic Imagery

You can optionally import orthographic imagery for this geo-location, so as to compare a photo with the 3D data you are about to import. You can import ortho images into a terrain any time after geo-locating the terrain.

1. Download aerial and satellite photos for your target region, and save the images on your hard drive. (Note: For orthographic images, CityScape currently supports only the GeoTiff format with standardized geo-data meta tags. A standard Tiff file will not work.) Try this web site: seamless.usgs.gov

1. Select **Import GIS > Import OrthoImagery > From File** from the Main Menu. The *Import Ortho Image* dialog appears.

2. In the dialog, LEFT-CLICK the browse button **[...]** to display a file loader, locate the **target Ortho Image file** on your hard drive, and LEFT-CLICK **OK**. The ortho image will overlay your terrain in non-tessellated modes.



3. You can show or hide the ortho image with the **View > Show Overlay** toggle in the Main Menu.

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Chapter 22: Exporting Terrains

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22.1 OVERVIEW

CityScape exports terrains to various standard formats, such as Collada among others. CityScape can also export to the PixelActive Interchange Format, which anyone can view in the free CityScape Viewer. CityScape can even be customized to export to proprietary formats, as needed by individual customers.

Exporting terrains is straightforward, as explained in "The Export Process" section below. The "Exported Data Formats" section of this chapter describes the exported data formats in detail.

22.2 THE EXPORT PROCESS

CityScape provides three different export paths. For details, see the following sections:

- Exporting Terrains to PixelActive Interchange Format
- Exporting Terrains to Standard Formats
- Exporting Meta Data to CSV Format

22.2.1 Exporting Terrains to PixelActive Interchange Format

Exporting to the PixelActive Interchange Format lets anyone view your CityScape terrains using the *PixelActive Viewer*, which can be downloaded for free from PixelActive's web site.

Exporting to the PA Interchange format is easy:

1. Select **File > Export > PA Format** from the Main Menu. The *PixelActive Export Options* dialog appears.
2. In the dialog, you should usually accept the default options, and then LEFT-CLICK **OK** to complete the export. To customize the export options, refer to the next section.

CUSTOMIZING PIXELACTIVE INTERCHANGE OPTIONS

The following options appear in the *PixelActive Export Options* dialog:

Platforms

Currently, we support export to the PC platform only. You will soon be able to export directly to the Xbox 360, PS3, and Wii.

Export Range

- **All:** Exports all tiles in the open terrain.
- **Tiles:** Exports only the tiles you specify.

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Types

Select all of the Types options, unless you want to suppress the export of a particular type of object.

- **Export Level:** Exports the entire terrain, minus certain traffic-related data, to the proprietary LVV PixelActive Interchange format that can be read by the PixelActive Viewer.
- **Export Traffic:** Exports certain traffic data to PTS format. This traffic data supplements the road splines in the LVV file, so in the Viewer vehicles can move around the terrain in realtime.

Game Objects

- **Export All:** If enabled, CityScape exports everything in the open terrain to the PixelActive Interchange format. If disabled, a list will unlock, so you can select the specific types of objects (such as props) for export.

Export Path

Enter the path to the folder where you want CityScape to save the exported files. To display a file browser, LEFT-CLICK the browse button **[...]** to the right of the Export Path text field.

22.2.2 Exporting Terrains to Standard Formats

CityScape can export terrains to the following standard formats:

- Collada 1.4.1 (DAE)
- 3D Studio (3DS)
- Autodesk AutoCAD (DXF)
- Filmbox (FBX)
- Alias (OBJ)
- Gamebryo 2.6 (NIF)
- OpenFlight 16.2 (FLT)
- PixelActive (XML)

To export a terrain to one of the above-listed formats, follow this procedure:

1. In CityScape, open the terrain you wish to export.
2. Select **File > Export > Export...** from the Main Menu. The Export Options dialog appears.
3. At the top of the Export Options dialog, select the target export format from the **Exporter** drop-down menu.
4. In the “Scene Options” tab, set the options that you prefer. For details, see “Scene Options” on page 220.
5. In the “Advanced Options” tab, set the options that you prefer. Some of the Advanced options depend on the Scene options, so set the Scene options first. For details, see “Advanced Options” on page 223.

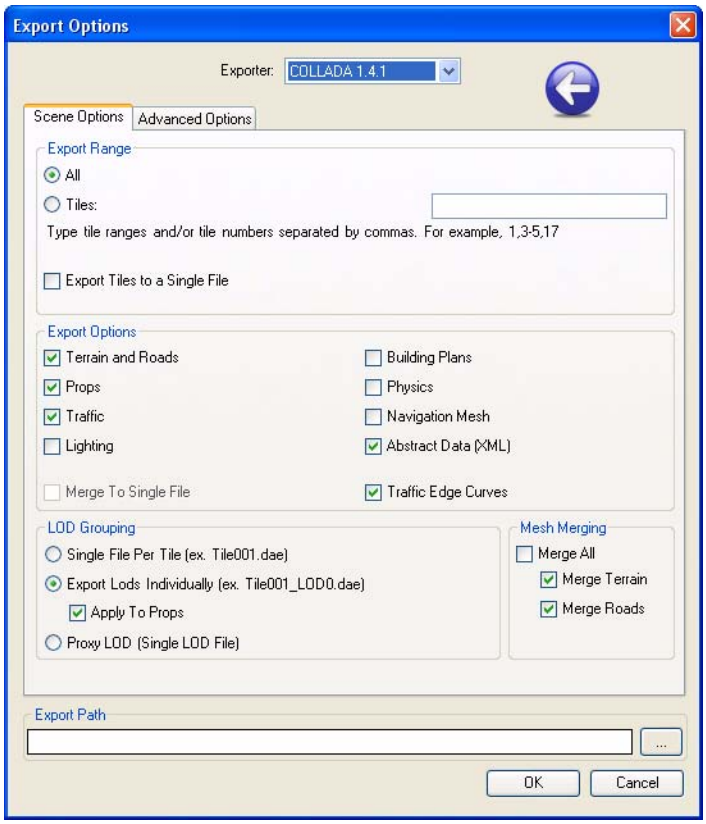
6. Some export formats (including OpenFlight and Filmbox) have additional format-specific export options. If you choose one of these export formats, an additional tab will appear in the Export Options dialog. For details, see "Format-Specific Options" on page 226.

7. After selecting all export options, choose your export path and a master filename for the exported files:
   i. The export path is an absolute path where you want exported files to be saved. To select this path, LEFT-CLICK the **Export Path button** ... at the bottom of the Export Options dialog to display the "Choose File Path to Save Exported Files" file loader, so you can select (or create) the target output folder.
   ii. In the file loader, navigate to the target folder for the exported files. If this folder does not exist, create it using the dialog.
   iii. In the "File name" field of the file loader, type a master filename for the exported files. All exported files will start with this filename; the files will be automatically appended with a suffix and file extension that depend on the type of file. For details on filenames, and the contents of files, see "Exported Data Formats" on page 227.

8. LEFT-CLICK **Save** in the file loader, and then LEFT-CLICK **OK** in the "Export Options" dialog.

Scene Options

Review the descriptions below to determine the appropriate settings in the Scene Options tab, given your specific needs. Some formats do not support all options; unsupported options will be greyed out.



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Export Range

- **All:** If selected, exports all tiles in the open terrain. Generates one set of exported files per tile. Which files are in a "set" depend on some of your other export options.

- **Tiles:** If selected, exports only the tiles that you specify.

- **Export Tiles to a Single File:** If selected, all the tiles in a terrain will be exported to a single file per Export Option. (I.e. when you select *Export Tiles to a Single File,* export options that normally operate "per tile" will operate "per terrain".)

Export Options

Selecting any of the following options will cause CityScape to export files as explained below. The format of exported files corresponds to your selection of target format in the Exporter drop-down menu, except as indicated below.

- **Terrain and Roads:** Per tile, CityScape exports at least one file containing meshes for the 3D geography and roads in the tile. CityScape exports more than one file per tile only if you select the "Export LODs Individually" option.

- **Props:** Per tile, CityScape exports one file defining the location, orientation, and scale of each instance of props used in the tile. For each type of prop (with one or more instances in the tile), the exported file also contains either a reference to the prop's source file, or the actual definition of the prop's mesh, depending on the target export format and your choices of Prop Options in the Advanced Options tab.

- **Traffic:** For the entire terrain (not per tile), CityScape exports one file containing traffic data. This exported traffic data defines the 3D locations of roads (via splines in the center of lanes), and road parameters affecting the movement of traffic (e.g. width of lanes, speed limits, density of traffic, turning options, etc.).

- **Lighting:** Per tile, CityScape exports lighting data for the lights that light that tile.

- **Building Plans:** Per tile, CityScape exports one file containing meshes for buildings plans. (Note: Building plans should not be confused with props that take the form of buildings: You create building plans with the Building Tools of CityScape, while you create props in external software and import them into a prop library for placement in CityScape terrains.)

- **Physics:** Per tile, CityScape exports one file containing a single collision mesh for all the geography and roads in the tile. CityScape handles this collision mesh as an invisible LOD with low detail. If you use standard 3D software to open a collision mesh that you have exported from CityScape, you would see a simple mesh without materials.

- **Abstract Data:** Per tile, CityScape exports the type and value of every instance of abstract data that you assigned to objects in the exported terrain. The exported data also includes the position and connectivity of the objects associated with the abstract data.

- **Traffic Edge Curves:** If selected, CityScape exports splines for the edges of roads, along with the other data that's normally exported when you select the primary *Traffic* export option (above).

- **Merge to Single File:** If selected, all of the Export Options are merged into a single file, if supported by your choice of export format. (Currently, only Filmbox format supports everything, including textures, in a single file.) Data that is not directly supported by a given format will be exported to PixelActive XML format. (For example, traffic data is not supported in Collada format, so it will be exported in XML format.) If you combine this option with the "Export files to a Single File" option, then everything in the terrain gets dumped into a single giant file (except when your target format doesn't support the embedded data).

Mesh Merging

The following Mesh Merging options influence the grouping of meshes in exported Terrains and Roads files. Mesh Merging options are relevant only if you have selected the "Terrain and Roads" export option, of course.

- **Merge Terrain:** If selected, CityScape merges (into one mesh per tile) all 3D geography created with the Geography Tools of CityScape. If you have set up more than one LOD, each LOD becomes a separate mesh.

- **Merge Roads:** If selected, CityScape merges (into one mesh per tile) all road segments created with the Road Tools of CityScape. If you have set up more than one LOD, each LOD becomes a separate mesh.

- **Merge All:** If selected, CityScape merges (into one mesh per tile) all 3D geography and roads into one mesh. If you have set up more than one LOD, each LOD becomes a separate mesh. This option overrides the other Mesh Merging options.

- If you do not select any Mesh Merging options:
  - Terrain (per tile, per LOD): Each "lot" you defined with the Lot Tool becomes a separate mesh. Each section of terrain enclosed by roads becomes a separate mesh. All contiguous terrain that is neither a lot nor enclosed by roads becomes a separate mesh.
  - Roads (per tile, per LOD): Each road segment becomes a separate mesh.

LOD Grouping

CityScape exports Terrains and Roads files that contain definitions of meshes for all LODs of geography and roads in the terrain. CityScape does not export more than one LOD for physics meshes or building plan meshes.

- **Single File Per Tile:** If selected, CityScape outputs meshes for each LOD of geography and roads into one Terrain and Roads file per tile. Each file contains multiple definitions of each mesh, with each definition corresponding to a specific LOD.

- **Export LODs Individually:** If selected, CityScape exports LODs of geography and roads into separate Terrain and Roads files per tile. For example, if you have set up 3 LODs in a terrain, then CityScape exports 3 Terrain and Roads files per tile. Each file contains the definitions of all

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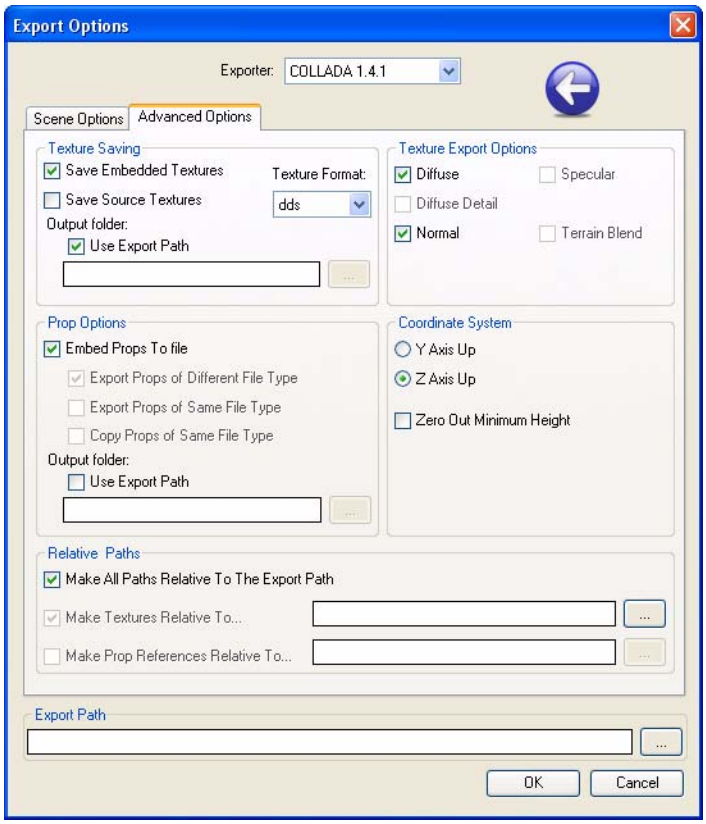
meshes at the same LOD. (Note that, if you choose the "Embed Props to File" option in the Advanced Options tab, then all prop meshes will be merged into the highest LOD file only.)

  – **Apply to Props:** If selected, CityScape also exports LODs of props into a separate Props file per tile.

- **Proxy LOD (Single LOD File):** If selected, CityScape also exports a single file that contains the lowest LOD of the terrain with everything (including props) baked into it.

ADVANCED OPTIONS

Review the descriptions below to determine the appropriate settings in the Advanced Options tab, given your specific needs. Some formats do not support all options; unsupported options will be greyed out.



Texture Saving

CityScape converts textures used in the terrain into the file format that you select in the *Texture Format* drop-down menu (below), and saves those textures to the Texture Output Path (below).

- **Save Embedded Textures:** If selected, CityScape exports embedded textures to files you can access. (Note: CityScape uses certain textures internally; you have no access to the source files for these "embedded" textures, unless you select this option.)

- **Save Source Textures:** If selected, CityScape saves all the texture files used in the terrain to the *Output Folder* (see option below). The format of saved textures is determined by the *Texture Format* option (below). CityScape updates references to the new location and format of exported texture files.
  - If de-selected, exported terrains reference texture files in their original locations and formats.

- **Texture Format:** Use the drop-down list to select the format of textures that CityScape saves when you also select the *Save Source Textures* option (above).

- **Output Folder:** When you select the *Save Source Textures* option (above), Output Folder allows you to specify the folder in which the exported textures are saved:
  - **Use Export Path:** If selected, CityScape saves textures to the main *Export Path* used by most other exported files.
  - **text field:** If *Use Export Path* is de-selected, then type your preferred path in the provided text field, or LEFT-CLICK the browse button **[...]** and navigate to the target folder for exported textures.

Texture Export Options

The *Texture Export Options* let you to choose the types of textures to export. Only the texture used in the open terrain will be exported with that terrain. If you also select the *Save Embedded Textures* option, CityScape exports embedded textures that it dynamically generates, as well as textures you have imported, directly or along with props

- **Diffuse:** If selected, CityScape exports diffuse textures in the materials used in the exported terrain.

- **Diffuse Detail:** If selected, CityScape exports diffuse detail textures (i.e. close-up diffuse textures) in the materials used in the exported terrain.

- **Normal:** If selected, CityScape exports normal maps in the materials used in the exported terrain.

- **Specular:** If selected, CityScape exports specular textures in the materials used in the exported terrain.

- **Terrain Blend:** Texture blending is supported in OpenFlight format only. If this option is selected, CityScape blends the base terrain texture with up to three other textures that you specify according to your selections in the OpenFlight Options tab of the Exporter Options dialog.

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Prop Options

The following options are available only if you have selected the **Props** export option in the Scene Options tab.

- **Embed Props to File:** If selected, CityScape converts props to meshes in the "Terrain and Roads" file. If de-selected, CityScape exports each prop mesh to the folder determined by the *Output Folder* option below. The following sub-options let you further refine how props are exported:
  - **Export Props of Different File Types:** This option applies only to props in different formats than the export format for the terrain. If this option is selected, CityScape converts props from their source formats to the same export format as the terrain, and saves the converted prop files to the *Output Folder*.
  - **Export Props of Same File Types:** This option applies only to props in the same format as the export format for the terrain. (Note: Props exported in this way have been processed by CityScape, which can make subtle alterations to the prop.) If this option is selected, CityScape exports props to the Output Folder.
  - **Copy Props of Same File Types:** If selected, CityScape saves copies of prop files to the Output Folder (below). In this case, CityScape does not update paths to textures used by these props.

- **Output Folder:** Determines the output folder for prop files (above):
  - **Use Export Path:** If selected, CityScape saves props to the same location as everything else, using the main *Export Path*.
  - **text field:** If *Use Export Path* is de-selected, then type your preferred path in the provided text field, or LEFT-CLICK the browse button **[...]** and locate the target folder for exported props.

Coordinate System

If you choose a formats that supports only one coordinate system, the up-axis options below will be greyed out:

- **Y Axis Up:** Standard coordinate system for Maya:
  - X-axis is positive to the right
  - Z-axis is positive coming out of the screen (towards you)
  - Y-axis is positive going up

- **Z Axis Up:** Standard coordinate system for 3D Studio Max:
  - X-axis is positive to right
  - Y-axis is positive coming out of the screen (toward you)
  - Z-axis is positive going up

- **Zero Out Minimum Height:** Sets the lowest point in your terrain to zero elevation, so you don’t have any negative elevations.

Relative Paths

If all of these options are de-selected, all paths are absolute.

- **Make All Paths Relative to Export Path:** If selected, CityScape ensures that all paths to the *Output Folder* for exported **props** and **textures** are relative to the main *Export Path* – overriding the two other options below. If de-selected, your selections for the options below determine the export paths for props and textures separately.

- **Make Textures Relative To:** If selected, CityScape ensures that all paths to the *Output Folder* for exported **textures** are relative to the main *Export Path*. If de-selected, CityScape uses absolute paths to the *Output Folder*.

- **Make Prop References Relative To:** If selected, CityScape ensures that all paths to the *Output Folder* for exported **props** are relative to the main *Export Path*. If de-selected, CityScape uses absolute paths to the *Output Folder*.

Format-Specific Options

Filmbox and OpenFight are currently the only formats that call for format-specific options on a separate tab of the Export Options dialog. Some format-specific options depend on Scene Options and Advanced Options, so you should set them first.

Filmbox Options

- **Embed Textures:** If enabled, CityScape embeds textures into the other file(s) being exported. (Filmbox is the only export format that supports embedded textures in exported terrain files.) If disabled, CityScape exports textures according to your other selections in the Export Options dialog.

OpenFlight Options

- **Use Meshes:** If enabled, CityScape uses meshes according to the specs of OpenFlight 16.2. If disabled, CityScape adheres to the older non-mesh format for backward compatibility.

- **Flip Texture Coordinates in Y:** If enabled, CityScape flips the texture coordinates in the Y-axis, so that textures render correctly with DirectX. If disabled, CityScape retains its default texture coordinates, so that textures render correctly with OpenGL.

- **Palette File List:** If enabled, CityScape exports a list of texture palettes (assuming you are using multiple palettes) for rendering engines that support dynamic palette swapping.

- **Diffuse Texture Maps:** If the *Terrain Blend* option is selected (in the Advanced Options tab), CityScape blends the diffuse texture in the base terrain material (i.e. the material applied to the terrain via the Material Seed tool) with up to 3 other textures. The three Diffuse Effect channels (below) allow you to specify the index numbers (in the OpenFlight Texture Record) that uniquely identify these other textures.
  - **Diffuse 2 Effect #**
  - **Diffuse 3 Effect #**
  - **Diffuse 4 Effect #**

- **Detail Texture Maps:** Same as Diffuse Texture Maps (above), but for the detail diffuse texture used in the base material.
  - **Detail 2 Effect #**
  - **Detail 3 Effect #**
  - **Detail 4 Effect #**

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22.2.3 Exporting Meta Data to CSV Format

When you select **File > Export > PA Format**, CityScape exports the type and value of every instance of abstract data that you assigned to objects in the terrain.

The exported data uses the comma separated values (CSV) format with columns for the **types** and **values** of meta data assigned to objects in the open terrain. The CSV does not include position and connectivity data for these objects.

22.3 EXPORTED DATA FORMATS

This section provides additional information on the format of data that CityScape exports to Collada and OpenFlight standards. This information is intended to help programmers and technical artists import this data into another 3D tool or 3D realtime game engine.

CityScape exports to the following 3D and 2D formats. For details, see the associated subsections:

- Collada Data Formats use DAE for meshes and DDS, BMP, or SGI for textures
- OpenFlight Data Formats use FLT for meshes and DDS or SGI for textures
- Gamebryo Data Formats use NIF for meshes and DDS or BMP for textures
- PixelActive XML Data Formats: If you choose the export in the PixelActive XML format, or when any of the following data types are not supported by another format, CityScape will export to the PixelActive XML format:
  - traffic data
  - references to prop meshes (not the meshes themselves)
  - game data
- Additional data formats will be described in the next draft of this document.

22.3.1 Collada Data Formats

If you use CityScape's Collada exporter, and select all export options, the following types of exported files will adhere to the Collada format, which is a Collada-specific flavor or XML:

- Collada Terrain and Roads Format
- Collada Physics Format
- Collada Props Format
- Collada Building Plans Format
- Collada Lighting Format

The exceptions are traffic data and game data, which will both be exported in PixelActive XML Data Formats. See the subsections below for details.

All files output by the Collada exporter will be output to the folder you specify in the "Export Path" field of the Export dialog box.

Collada Terrain and Roads Format

Per tile, CityScape exports one or more Terrain and Roads files containing definitions of meshes for all 3D geography and roads in the tile. The internal format of data in Terrain and Roads files adheres to Collada standards for defining the tessellated positions of each vertex, the RGB color of each vertex, references to material textures, and other mesh-related data. Normal maps are exported using the FCOLLADA profile standard for normal map declaration. Duplicate objects will be instanced to the fullest capabilities of the Collada format.

You can open Terrain and Roads files in any 3D software (such as Maya, 3D Studio Max, and Creator) that can import the Collada format. OpenCollada is the best Collada importer to bring PixelActive exports into Maya 2010 or Max 2010. http://www.opencollada.org/

Beyond the issue of format, the exact content of Terrain and Roads files depends on several export options:

- The "Mesh Merging" export options influence the grouping of meshes.
- The "LOD Grouping" export options influence the number of output files, as well as the grouping of meshes defined in those files.
- Exporting textures is optional, and you have several choices of texture formats to export. For details, see the Texture Saving options of the Advanced Options tab in the Export dialog.
- File names:
  - If you select the "Single File Per Tile" export option: Per tile, the file name takes the form of MyFile###.dae, where you specify "MyFile" and "###" is the tile number. Example: *MyFile007.dae*.

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– If you select the "Export LODs Individually" export option: Per tile, file names take the form of MyFile###_LOD#.dae, where you specify "MyFile" and "###" is the tile number, and "LOD#" increments for each LOD in the terrain. Example for 3 LODs: *MyFile004_LOD0.dae, MyFile004_LOD1.dae, MyFile004_LOD2.dae*.

COLLADA PHYSICS FORMAT

Per tile, CityScape exports exactly one XML file in Collada format containing the definition of a single collision mesh for all the geography and roads in the tile. The file name of this Physics file takes the form of

MyFile###_physics.dae

where you specify "MyFile" and "###" is the tile number. Example: *MyFile004_physics.dae*.

In LOD Mode of CityScape, you can define the resolution of collision meshes much as you would for any other LOD.

Since a collision mesh has exactly one dedicated LOD by definition, the LOD Grouping export options have no effect on the exported physics file. Neither do the Mesh Merging export options. Further, collision meshes have no associated textures, no normals, and no texture coordinates per vertex. Hence, the data format for exported Physics files is an even simpler form of Collada than exported Terrain and Roads files.

If you want your 3D realtime engine to use physics files to detect collisions, you must create a means for your engine to distinguish the exported collision meshes from visible meshes.

COLLADA PROPS FORMAT

The Collada format references files using Collada's node instancing (i.e. a mechanism in Collada for using the scene graph to reference external files in this case). The props file stores a matrix for the position of each prop instanced in the terrain. Opening the props file in Maya, or other compatible program, will load the props into the terrain.

The file format takes the form of *MyFile###_props.dae*.

COLLADA BUILDING PLANS FORMAT

Per tile, CityScape exports one XML file in Collada format containing the definition of meshes for the building plans in the tile. (You create building plans with the Building Tools of CityScape, which distinguished them from props that resemble buildings.) The file name of each Building Plans file takes the form of

MyFile###_bldgs.dae

where you specify "MyFile" and "###" is the tile number. Example: *MyFile000_bldgs.dae*.

Since Building Plans files have exactly one LOD, they are not affected by the LOD Grouping export options or the Mesh Merging export options. Each building is always exported as a separate mesh. Further, a building mesh can have at most one texture mapped to it. Hence, the data format for Building Plans is a simple form of the Collada format for meshes.

COLLADA LIGHTING FORMAT

The Lighting data exports as a light node in a collada file that is named *MyFile###_lights.dae*. If you open this file in Maya, Creator, 3D Studio Max, or other programs that support Collada, the lights will load as light objects.

22.3.2 OpenFlight Data Formats

If you use CityScape's OpenFlight exporter, all but one of the exported files will adhere to the OpenFlight format, which is binary. The exceptions are traffic data and game data, which will both be exported in PixelActive XML Data Formats.

CityScape's OpenFlight exporter outputs all files to the folder you specify in the Export Path option, or to another folder depending on your choice of Prop Options. For each export, a single OpenFlight file named

MyFile_master.flt

is created. This file contains references to all the Terrain and Roads files being exported.

OPENFLIGHT TERRAIN AND ROADS FORMAT

Per tile, CityScape exports exactly one Terrain and Roads files containing definitions of meshes, and scene graphs, for the 3D geography and roads in the tile. Internally, this file uses a binary OpenFlight format which can be opened in any 3D software that can import OpenFlight files. The output file name takes the form of

MyFile###.flt

where you specify "MyFile" and "###" is the tile number. Example: *MyFile001.flt*.

The OpenFlight exporter in CityScape outputs the same types of information as the Collada exporter, plus scene graphs that include group nodes and LOD nodes. Hence, with OpenFlight there is no need to export LODs into separate files.

Depending on your selections in the Texture Saving options, CityScape will optionally export textures used on the geography and roads in the tile.

OPENFLIGHT PHYSICS FORMAT

Per tile, CityScape exports one binary file in OpenFlight format containing the definition of a single collision mesh for all the geography and roads in the tile. The file name of a Physics file takes the form of

MyFile###_physics.flt

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where you specify "MyFile" and "###" is the tile number. Example: *MyFile005_physics.flt*.

Except for the difference in internal format, an exported OpenFlight Physics file contains the same type of information as an exported Collada Physics file.

OpenFlight Props Format

Props for OpenFlight export according to the options you choose in the Prop Options dialog. The format uses file referencing.

OpenFlight Building Plans Format

Per tile, CityScape exports one binary file in OpenFlight format containing the definition of meshes for the building plans in the tile. The file name of each Building Plans file takes the form of

MyFile###_bldgs.flt

where you specify "MyFile" and "###" is the tile number. Example: *MyFile002_bldgs.flt*.

Except for the difference in internal format, an exported OpenFlight Building Plans file contains the same type of information as an exported Collada Building Plans file.

OpenFlight Lighting Format

OpenFlight support for lighting will be added soon.

22.3.3 Gamebryo Data Formats

If you use CityScape's Collada exporter, and select all export options, the following types of exported files will adhere to the Collada format:

- Terrain and Roads Format
- Building Plans File
- Lighting Format

The exceptions are physics, traffic, props, game data. Physics is not currently supported. Traffic, props, and game data are exported in PixelActive XML Data Formats.

All files output by the Gamebryo exporter will be output to the folder you specify in the "Export Path" field of the Export dialog box.

Gamebryo exports with the Y-axis up. Internally, mesh data is similar in its capabilities to the OpenFlight format, except in Gamebryo format. Gamebryo can instance mesh data, but not the unique scene graphs.

22.3.4 PixelActive XML Data Formats

When a data type is not supported by another format, or when you choose to export to XML, CityScape will export to PixelActive XML format:

– PixelActive XML Traffic Data Format

- PixelActive XML Props Format (references to external meshes)
- PixelActive Abstract Data Format

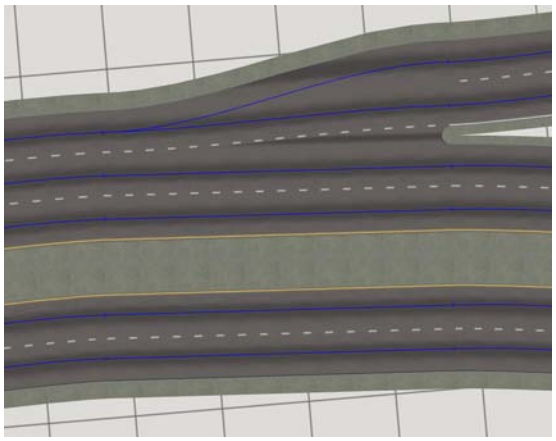
PixelActive XML Traffic Data Format

For one entire terrain (not per tile), CityScape exports one PixelActive XML file containing traffic data based on roads therein. This traffic file uses a flavor of XML that PixelActive developed. The filename takes the form of *MyFile_traffic.xml*.

Exported traffic data has two primary components: Base Curve Data and High-level Contextual Data. See the following subsections for details.

Base Curve Data

Base curves are 3D Hermite splines defined by the spatial position and tangent vector at the start and end of each curve. These curves are procedurally generated to match the lanes of roads in the terrain. (In the figure below, the splines are blue lines.)



CityScape exports these curves as a directed graph of curve segments. Each segment consists of a strip of connected splines as well as references to the previous and next segments. If desired, you could use this directed graph by itself, without high-level traffic data, to define the locations of lanes in roads for your own traffic library to use.

High-level Contextual Data

In addition to the curve data, the exported XML for traffic also contains contextual data that provides higher-level information of the traffic system, which very closely resembles the road network created in the editor. This high-level contextual data is comprised of the following components:

- **Road Segments:** A traffic segment defines a section of road that is terminated at each end by intersections. Segments are bidirectional containers of traffic lanes, and contain information such as the speed limit and desired density of traffic on those lanes.

- **Intersections:** A traffic intersection terminates one or more traffic segments, and also contains a list of traffic corridors and traffic phases. Intersections come in four flavors: Intersection, Join, MergeSplit, and Culdesac:

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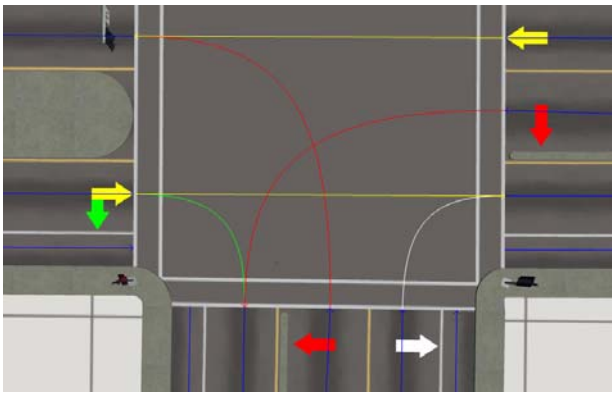
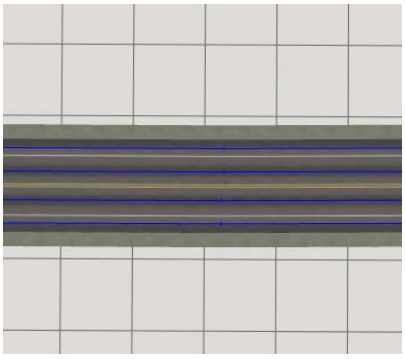
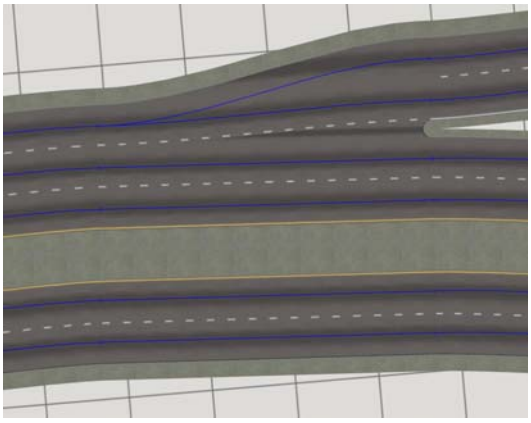
– *Basic intersections* are the most familiar, wherein vehicles obey various traffic laws to determine right of way.



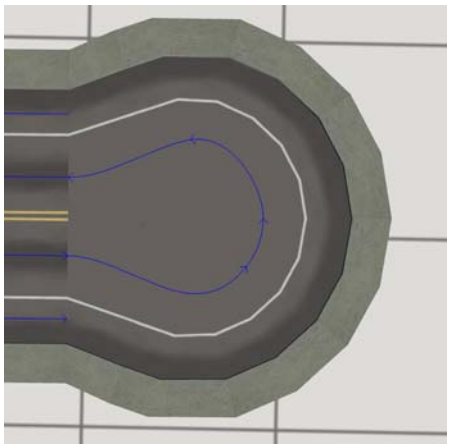
– *Join intersections* are special case nodes that directly connect two traffic segments with no gap. In other words, the lanes in one segment begin at the exact points where the lanes in the other segment end. Corridors in a join intersection contain curve segments with no Hermite curves within them.



– *MergeSplit intersections* are another special case where the corridors are roughly parallel and are grouped in two directions. The corridors in a MergeSplit physically resemble the lanes in a traffic segment.

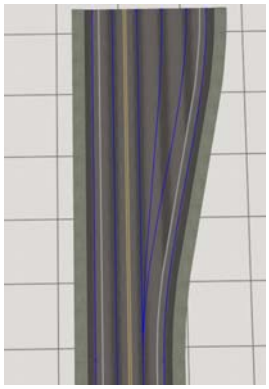


– *Culdesac* intersections are connected to only one segment and have rounded corridors that travel along the perimeter of the circular geometry.



– *Traffic Signals:* The position of traffic signals is not currently exported, but it will be in the near future.

- **Lanes:** Lanes are stored in traffic segments. Each lane contains a single strip of connected base curves defining the center of a lane. This strip also refers to any corridors connected at each end. Lanes are categorized as one of four types: straight, starting, ending, and left turn:
  – *Straight lanes* are, well, straight.
  – *Starting lanes* are those that begin at the start of the traffic segment.



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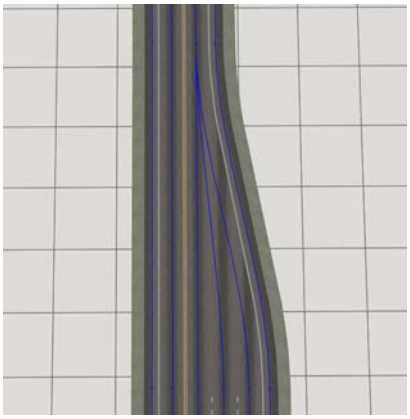
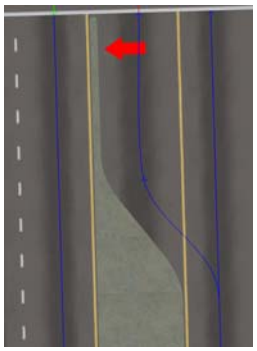
– *Ending lanes* are those that terminate at the end of the segment.



– *Left turn lanes* are designated for left turns only. The first spline for a left turn lane starts at a point where the left turn lane diverges from the straight lane.



- **Corridors:** Corridors are contained in traffic segments. Each corridor contains a single strip of connected base curves defining the center of the corridor, as well as references to the lanes at the beginning and end of the corridor. Each corridor contains a list of merging corridors, which begin or end in the same lane. Each corridor also stores a list of crossing corridors, which physically cross it. There are five types of Corridors:
  – Straight
  – LeftTurn
  – RightTurn
  – UTurn
  – Culdesac

- **Phases:** Each traffic intersection contains a list of phases for traffic signals. These phases define the rules of the traffic signal at the intersection. During a traffic simulation, the intersection would progress through phases in a sequential cycle. Each phase stores the amount of time during which the light is green while active, and yellow before the next phase starts. Each phases also includes a list of protected and permitted corridors:
  – *Protected corridors* are those which may be freely traversed during a phase (i.e. drivers would not need to watch for any crossing traffic).

  - *Permitted corridors* are those which can only be traversed during the phase if all crossing and merging corridors are clear (e.g. a right-turn on red, or unprotected left turn, etc.).
  - *Unlisted corridors* are considered to have red lights.

- **Banking:** The exported traffic data does not include any information on the banking (i.e. camber) of roads.

PixelActive XML Props Format

Per tile, CityScape exports one PixelActive XML file defining the location, orientation, and scale of each instance of every type of prop in the tile. For each type of prop, this XML file also contains a reference to the source file of the prop. If you wish to import this data into another 3D tool, or a 3D game engine, then you will need to write a script that can parse the custom XML in the PixelActive Props file.

The file name of this Props file takes the form of

MyFile###.xml

where you specify "MyFile" and "###" is the tile number. Example: *MyFile007.xml*.

- The PixelActive XML exporter does not export the actual definitions of prop meshes – unless you choose "Export Props to File" in Prop Options, in which case the exporter will embed the meshes of props into the highest LOD of the terrain and roads file. By default, CityScape only exports references to the source files for props. Since you must have created props in external software, you can import the source prop files directly into your real-time engine independently of CityScape.

- If you want props to have collision meshes, build theses meshes into the props when you create them, and configure your game engine to recognize these collision meshes as unique.

PixelActive Abstract Data Format

This XML file defines the positions of points and polygons for game data objects in the terrain file. These game objects can be associated with separate attribute files that define whatever properties you wish to assign to each game object.

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